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(54) **MODIFIED O-METHYLTRANSFERASES
USEFUL FOR PRODUCTION OF
O-METHYLATED PHENOLIC NATURAL
PRODUCTS**

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Publication Classification

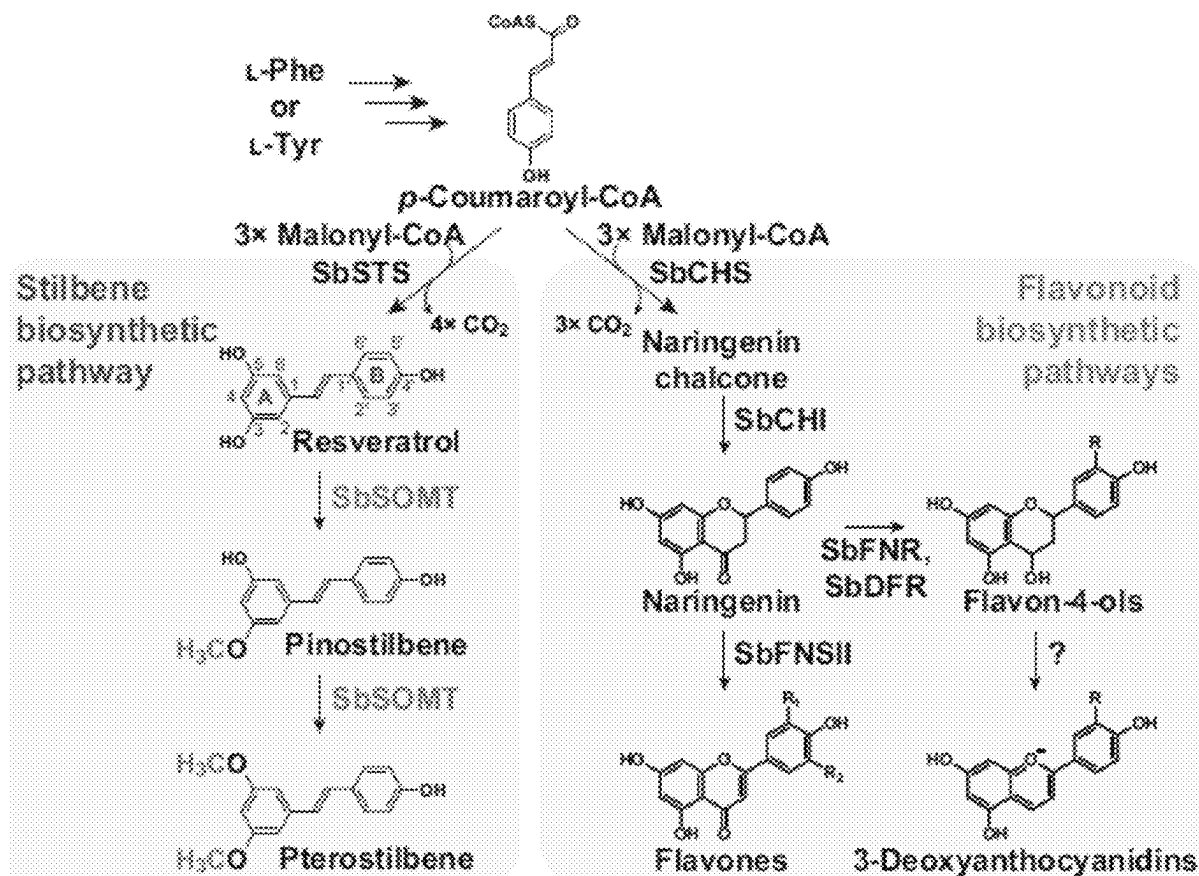
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C12P 7/22 (2006.01)
C12P 7/42 (2006.01)
(52) **U.S. Cl.**
CPC **C12N 9/1007** (2013.01); **C07C 37/06** (2013.01); **C07C 41/01** (2013.01); **C07C 51/377** (2013.01); **C12N 15/8243** (2013.01); **C12P 7/22** (2013.01); **C12P 7/42** (2013.01)

(57) **ABSTRACT**

Disclosed are modified Type 1 O-methyltransferases (OMTs) and methods of using such modified OMTs. In some forms of the modified OMTs, the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid. In some forms of the modified OMTs, the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

Specification includes a Sequence Listing.



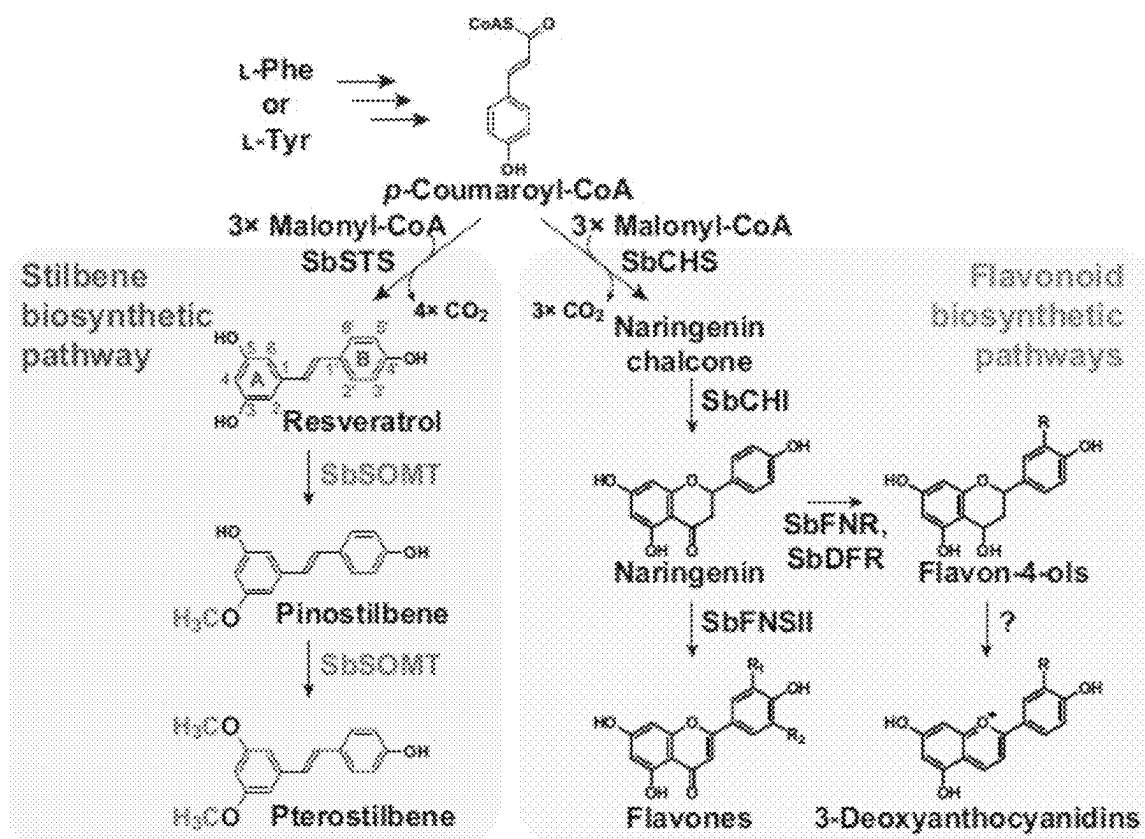


FIG. 1A

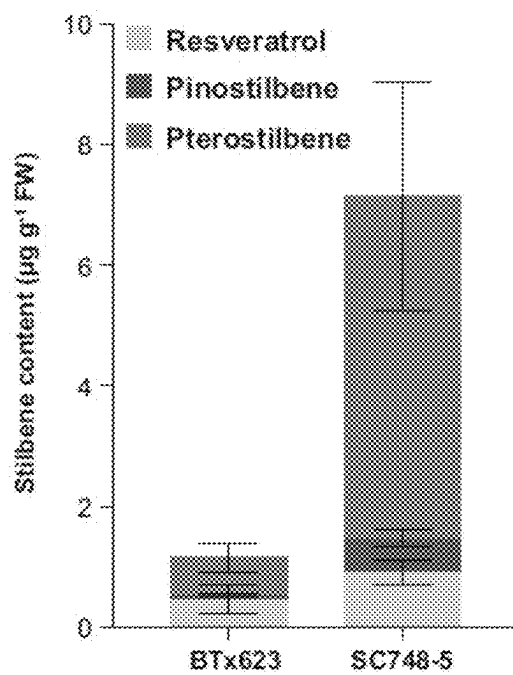


FIG. 1B

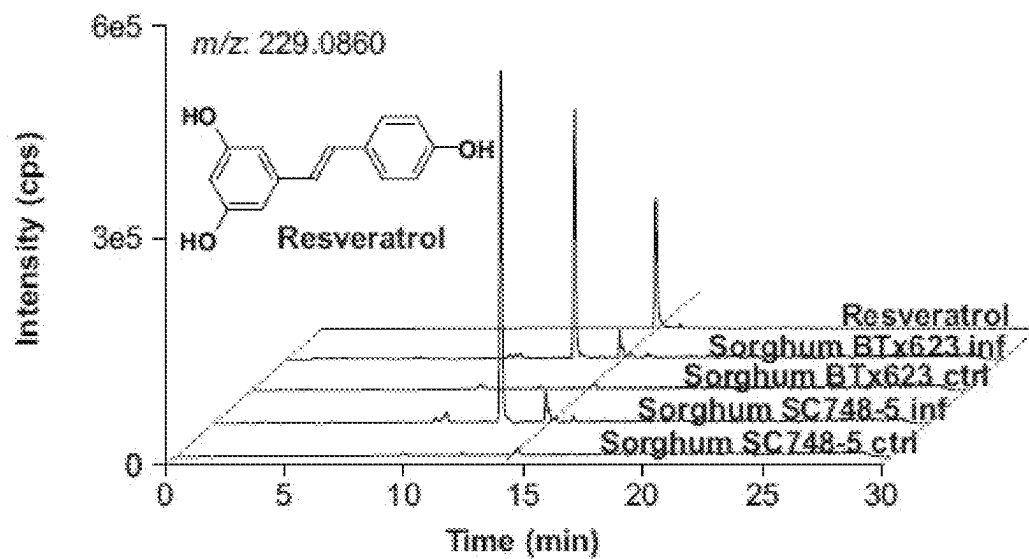


FIG. 1C

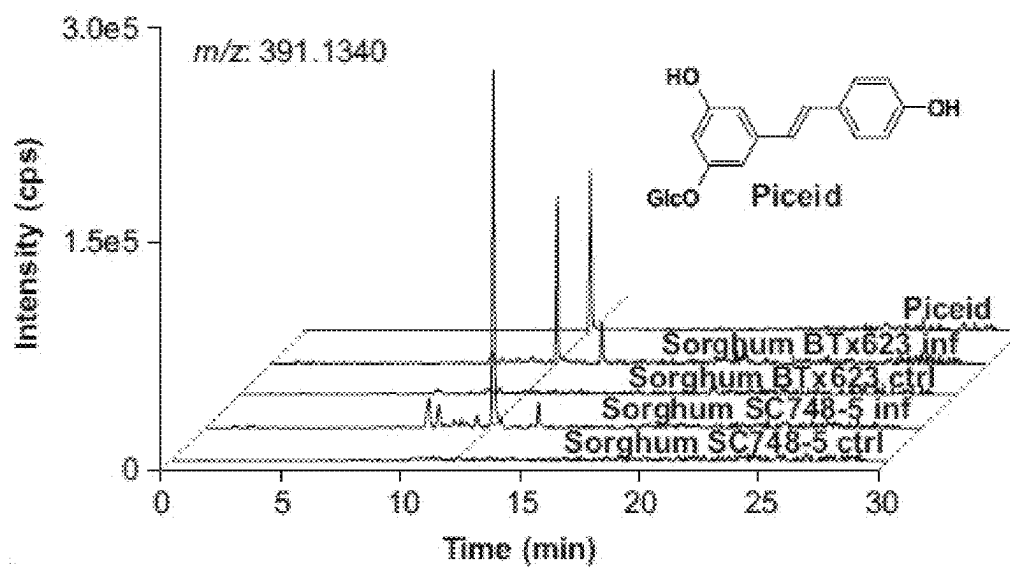


FIG. 1D

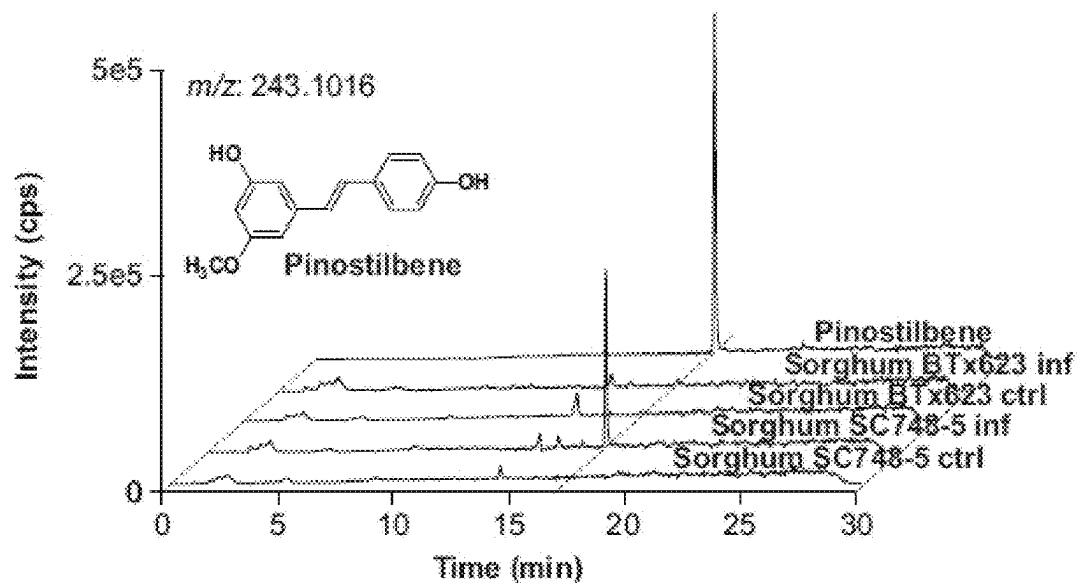


FIG. 1E

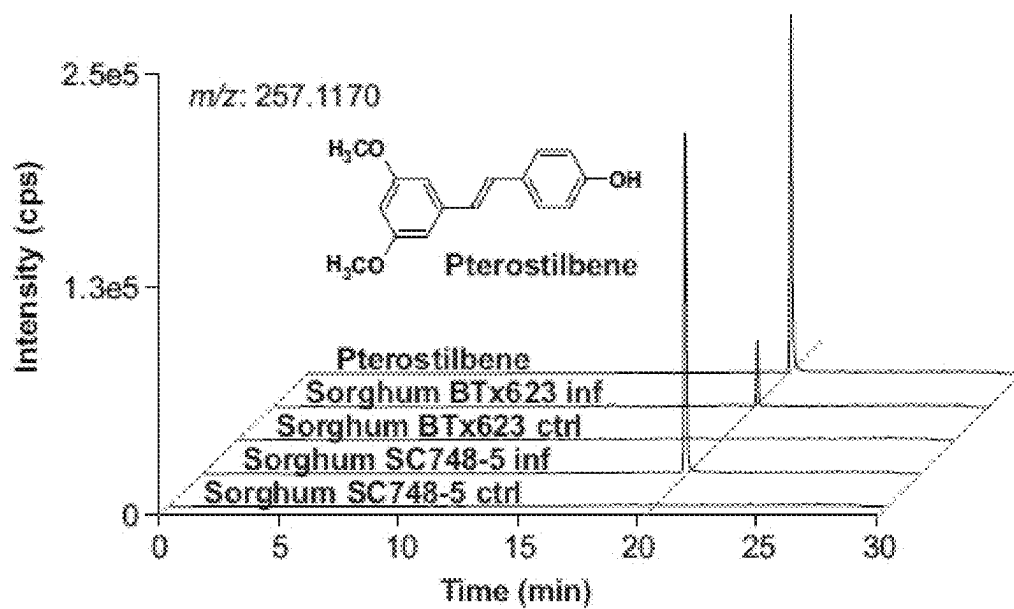


FIG. 1F

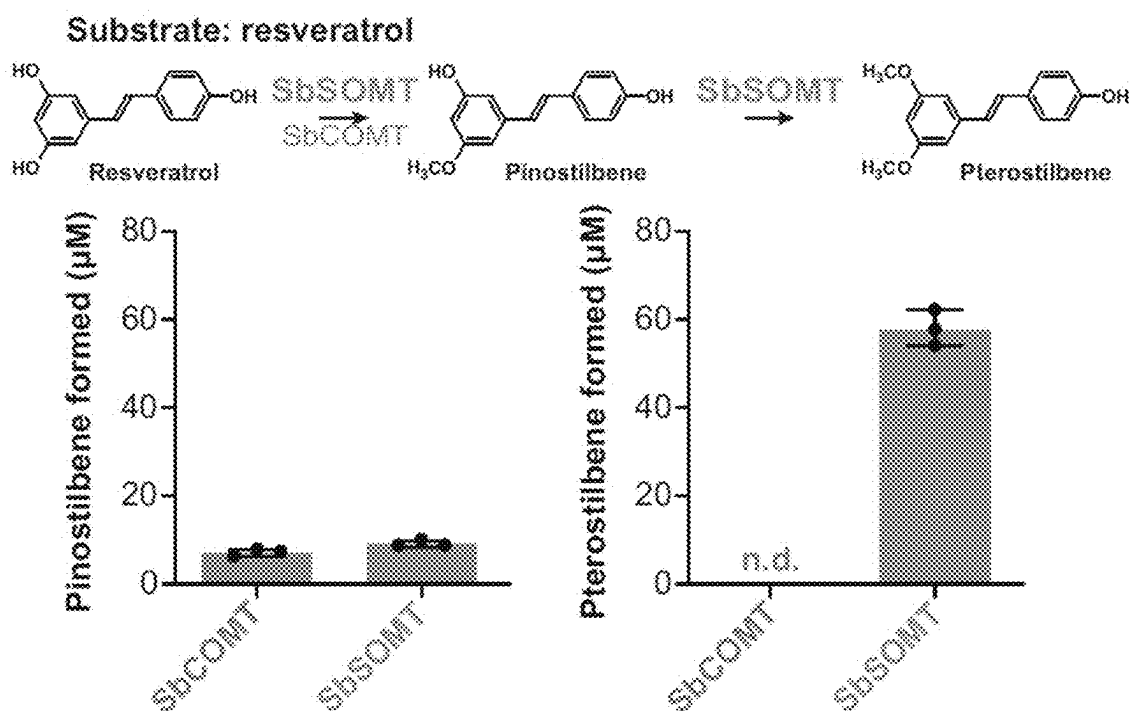


FIG. 2A

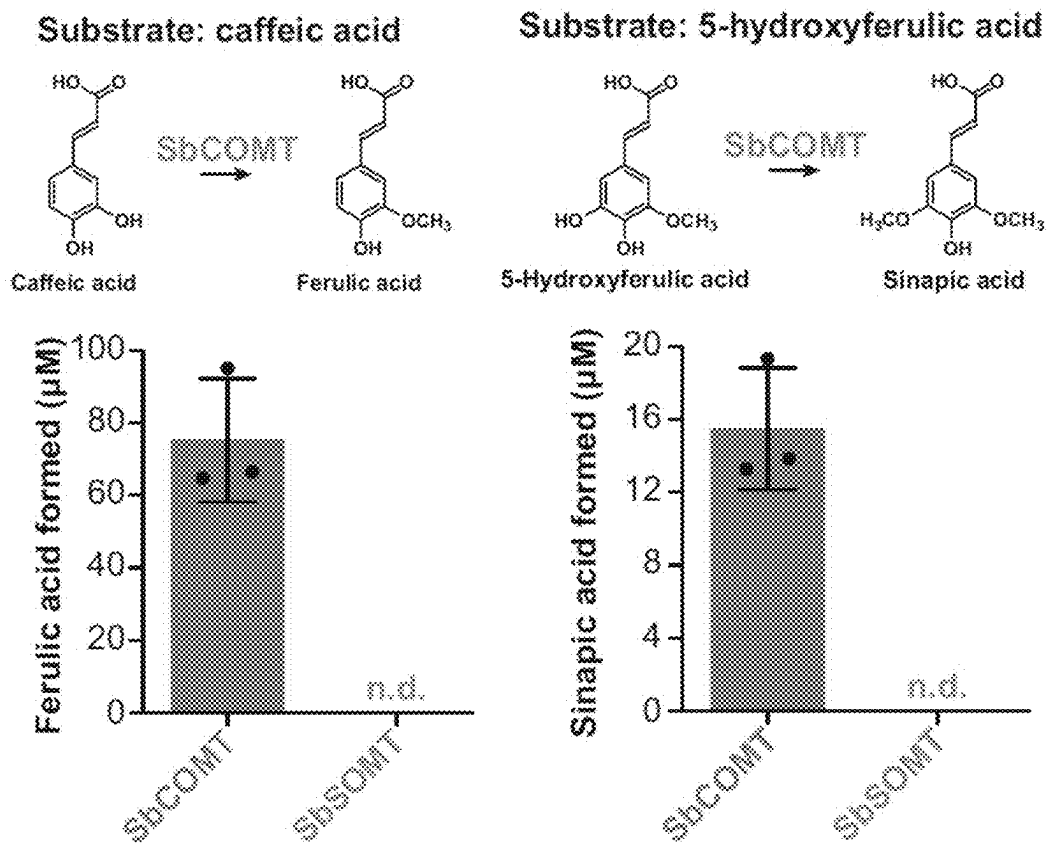


FIG. 2B

FIG. 2C

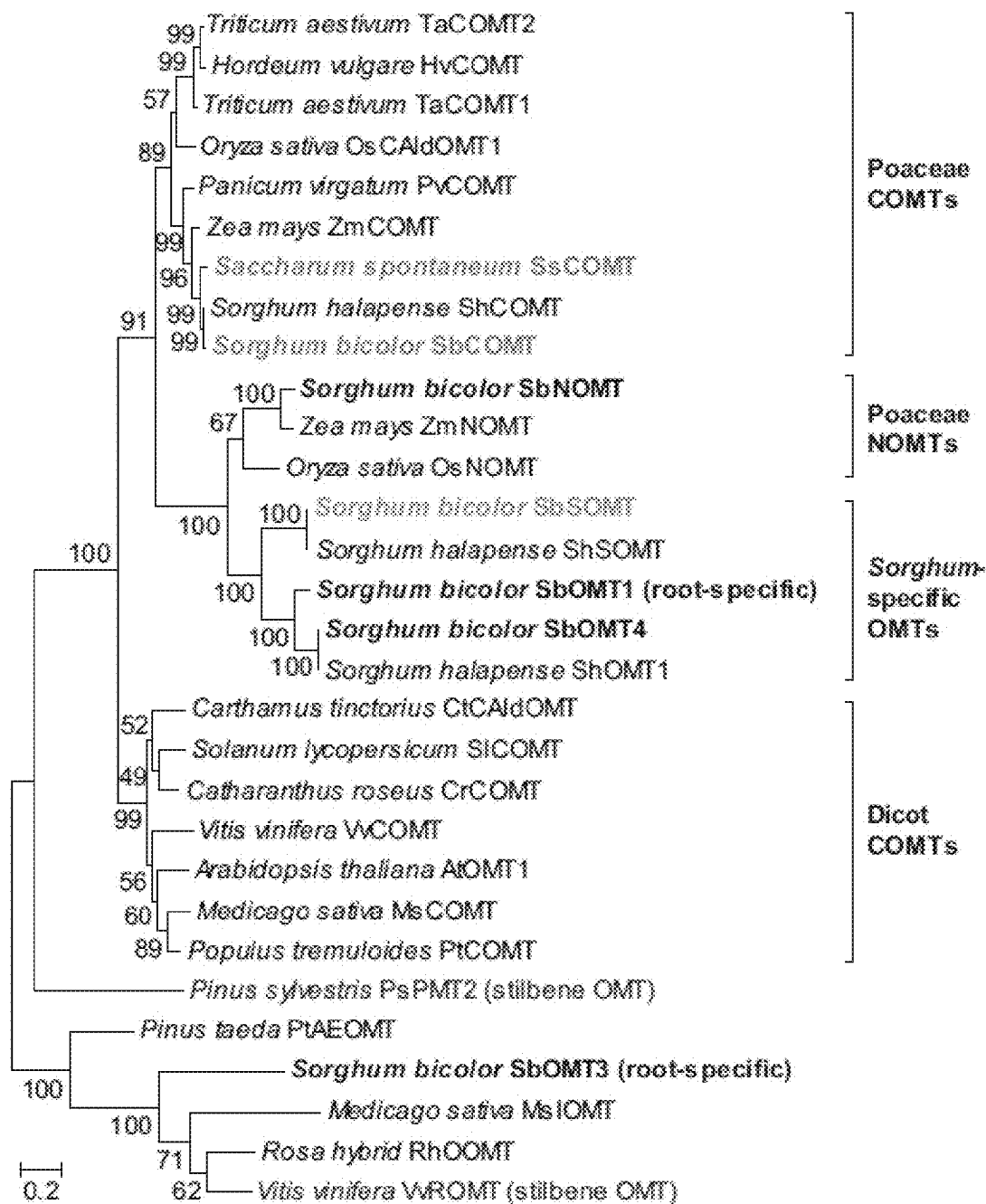


FIG. 2D

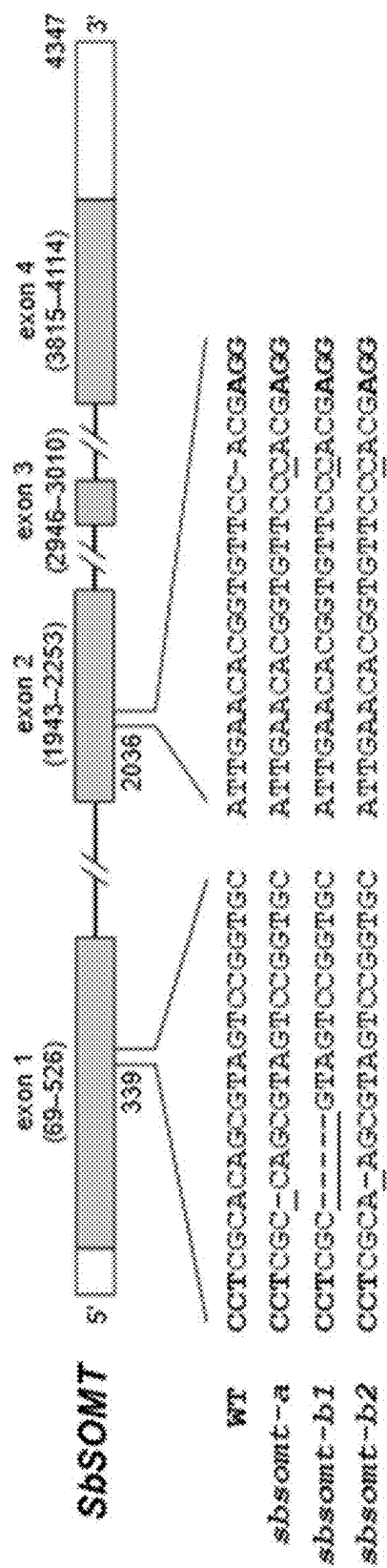


FIG. 3A

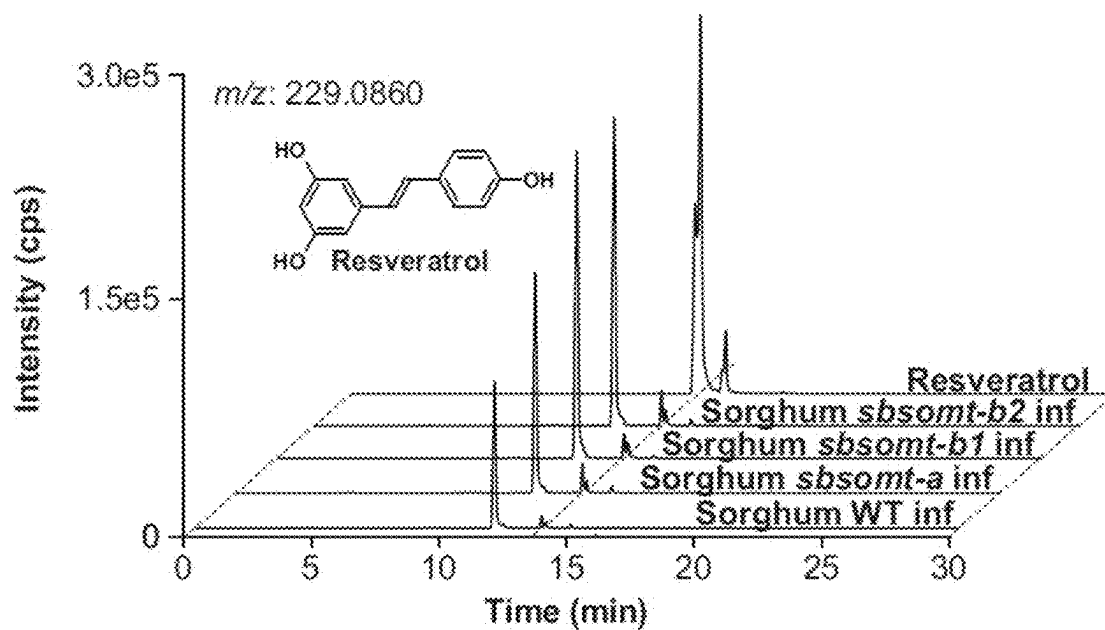


FIG. 3B

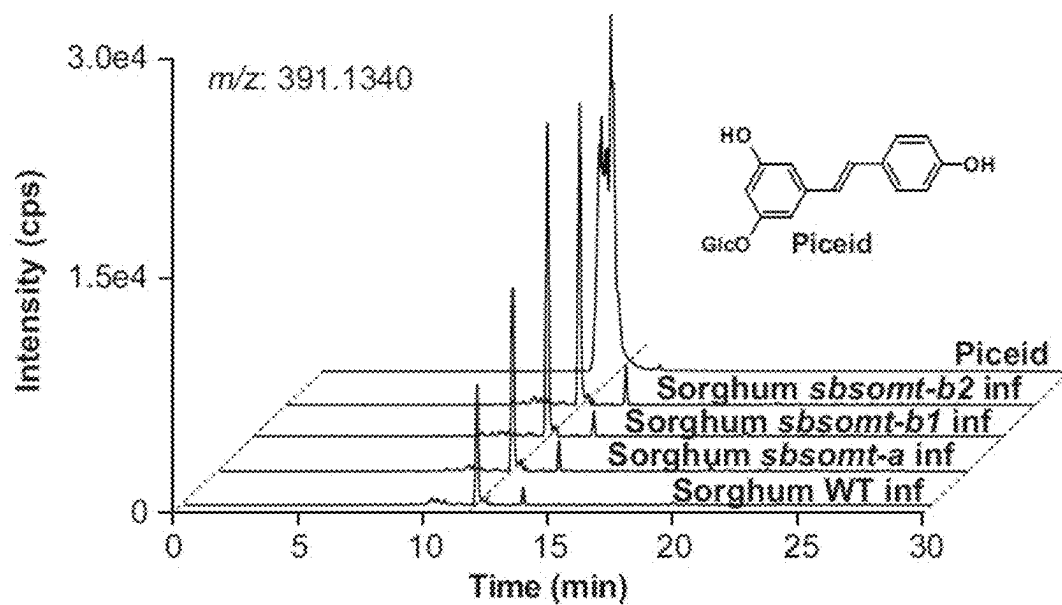


FIG. 3C

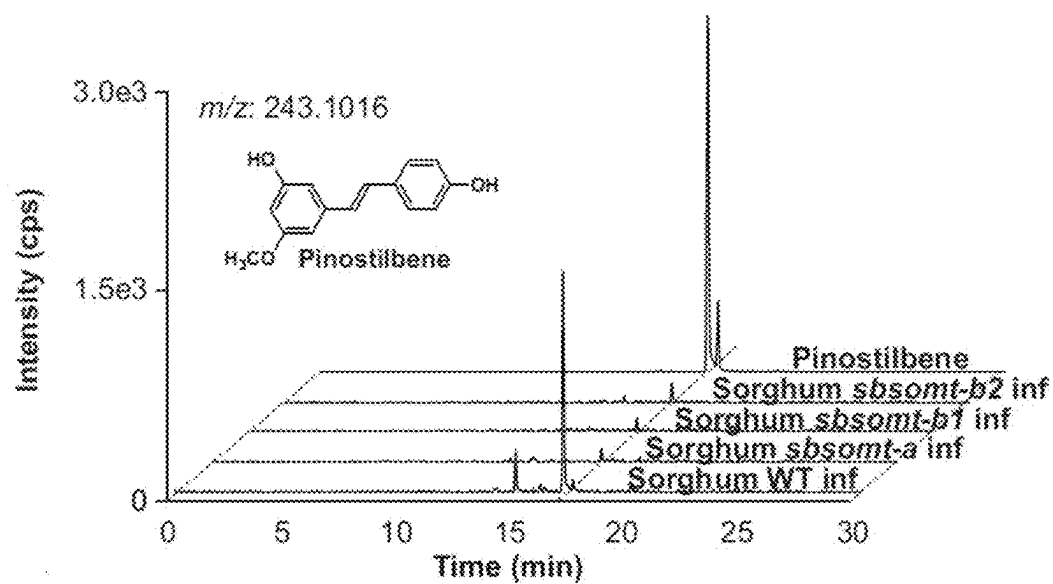


FIG. 3D

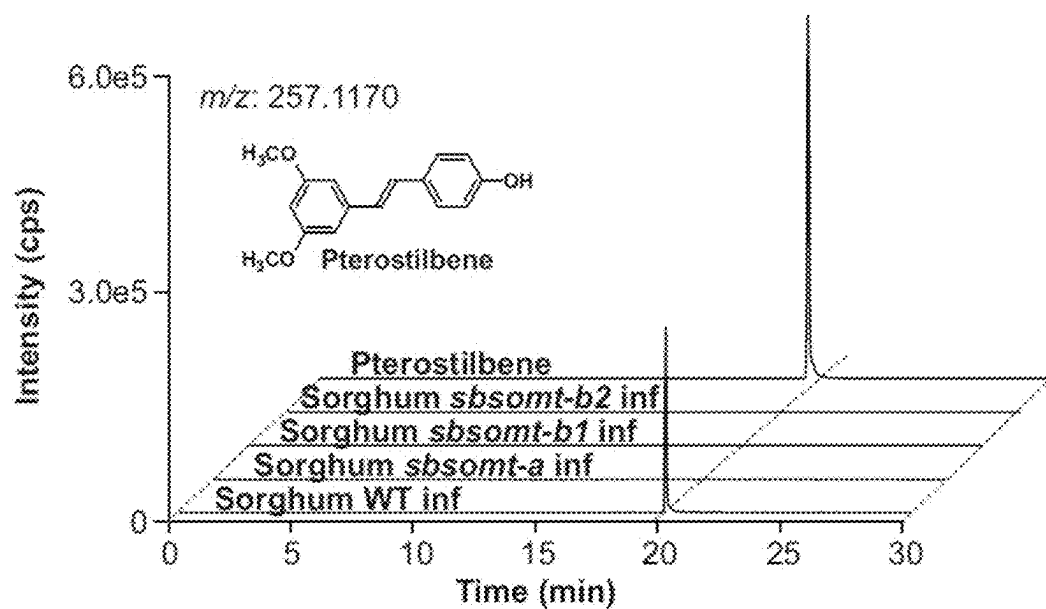


FIG. 3E

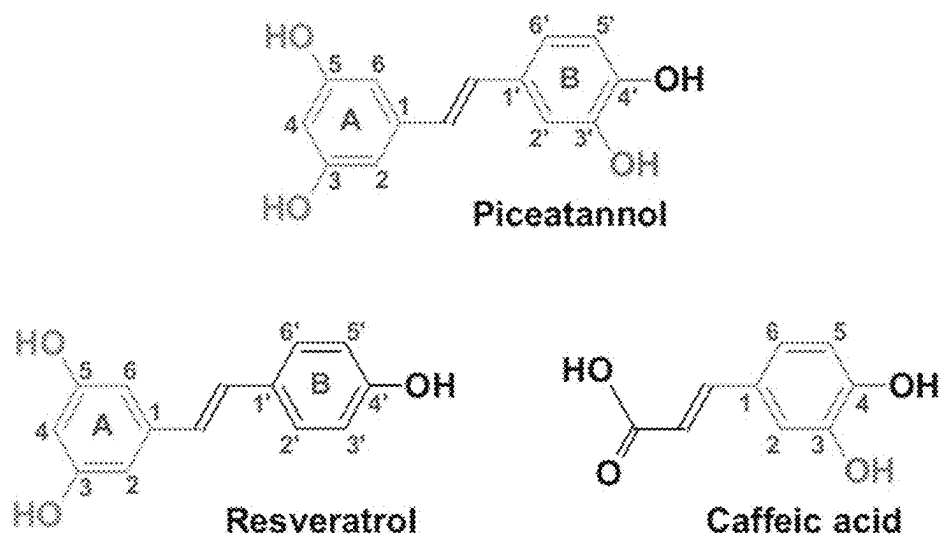


FIG. 4A

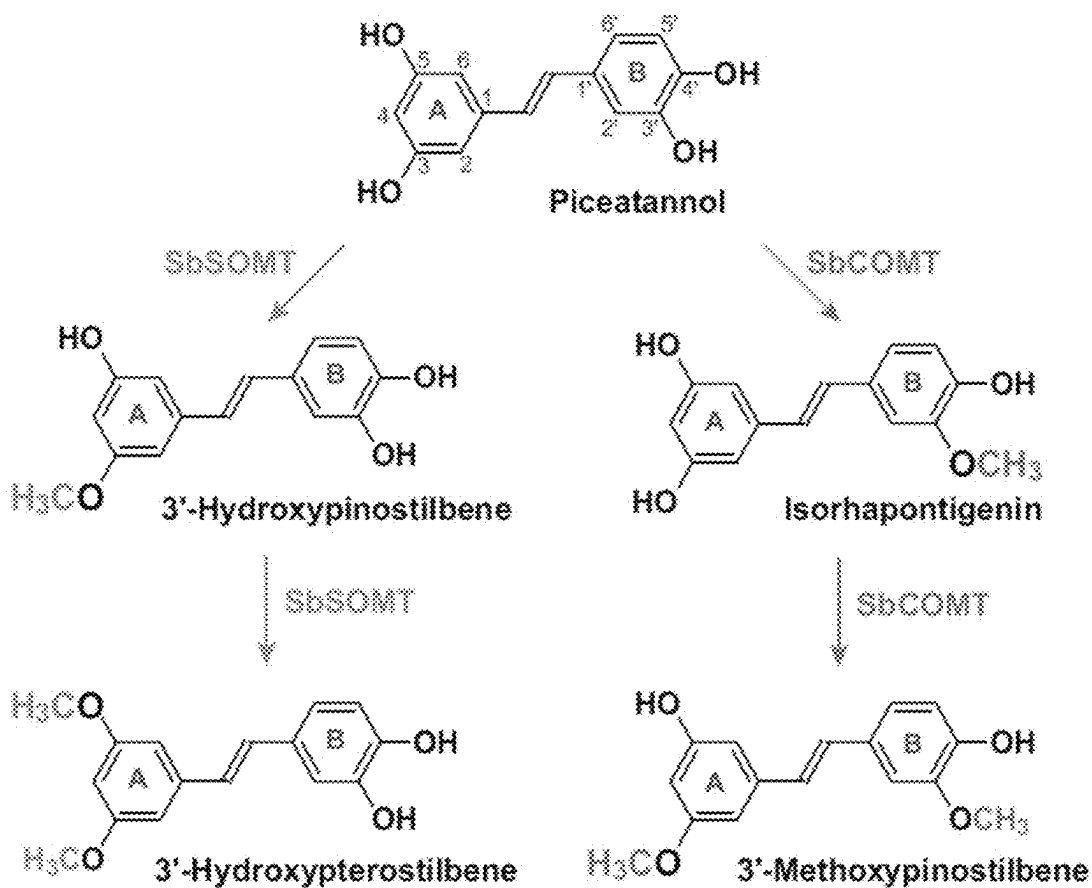


FIG. 4B

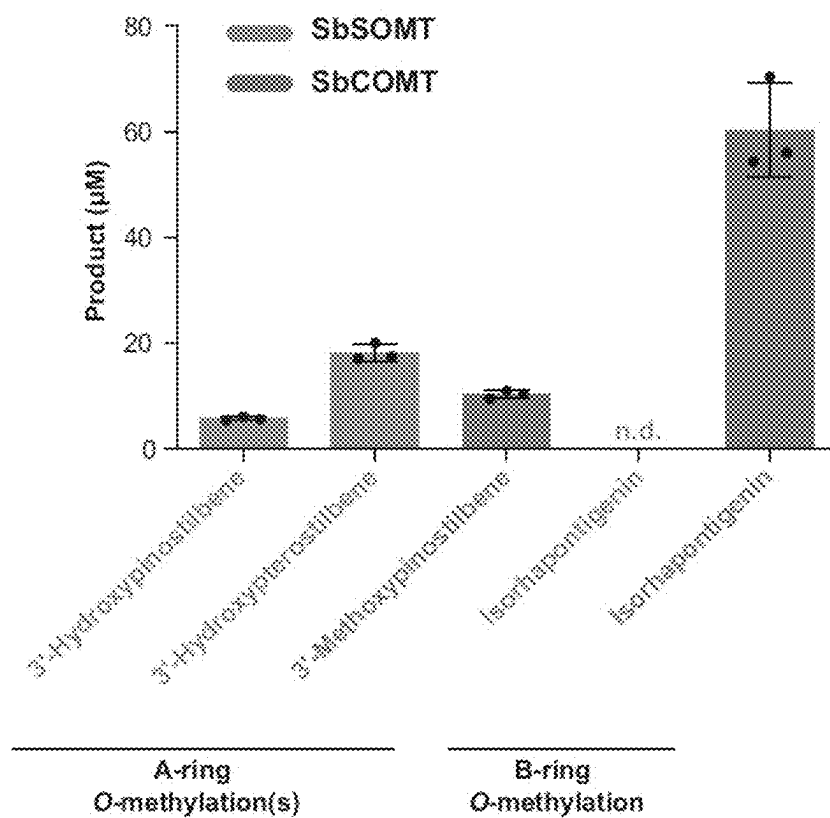


FIG. 4C

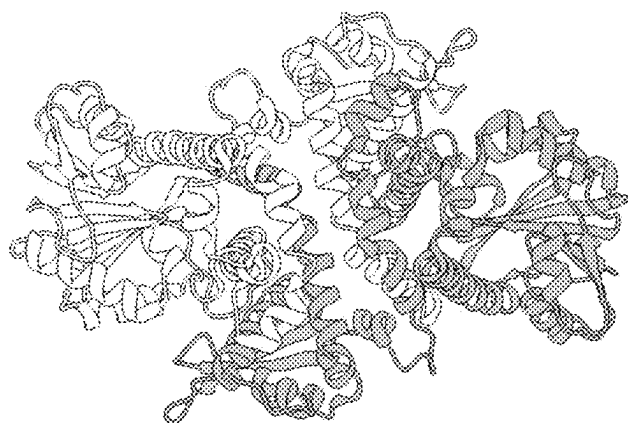


FIG. 4D

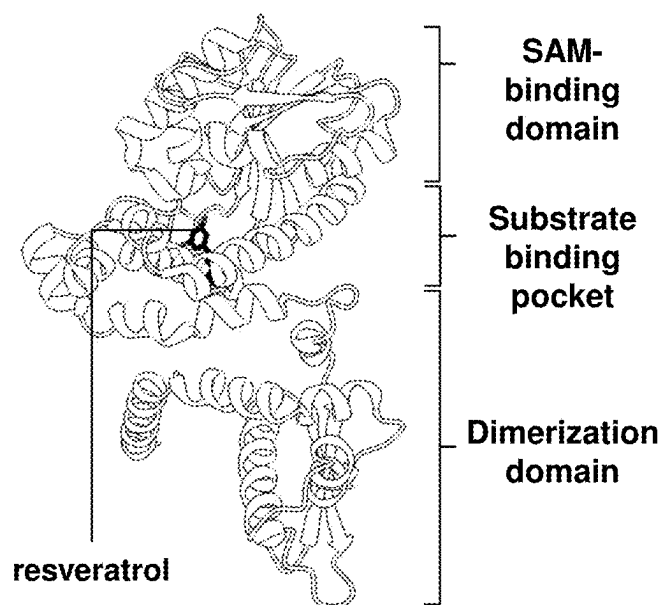


FIG 4E

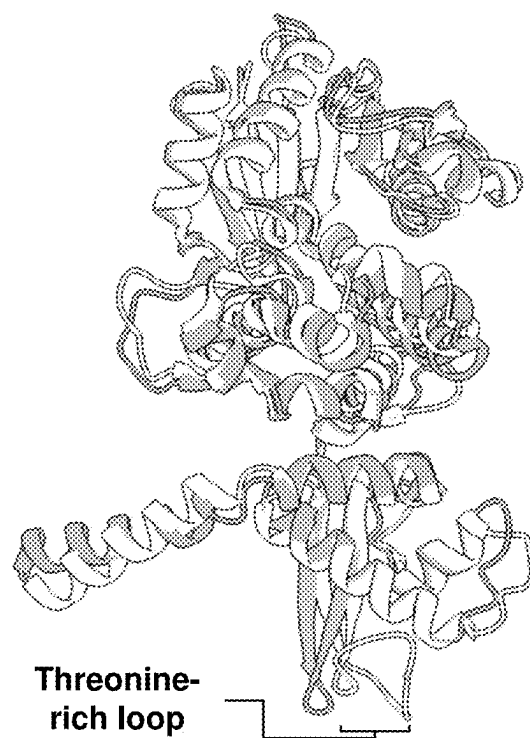


FIG. 4F

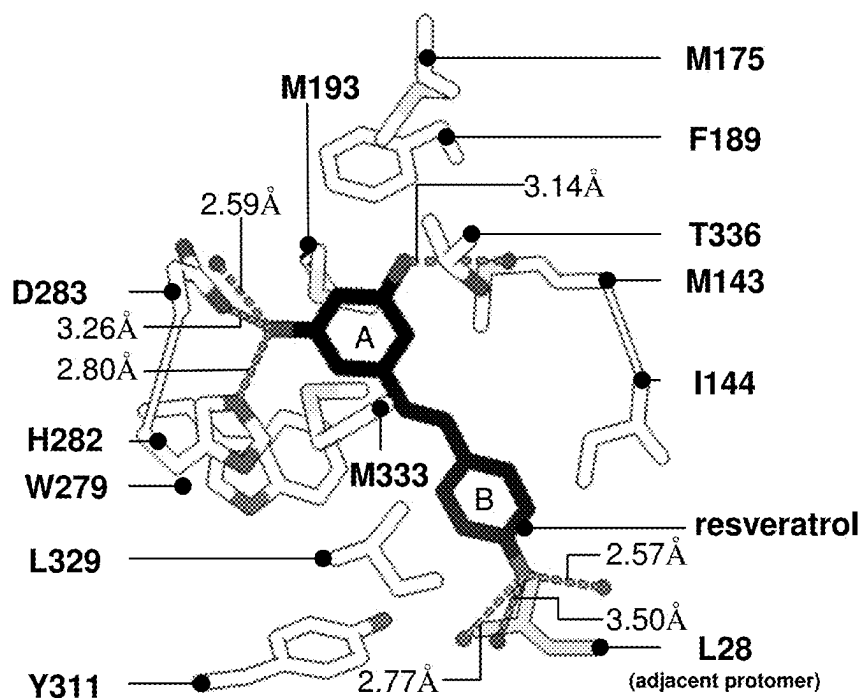


FIG. 5A

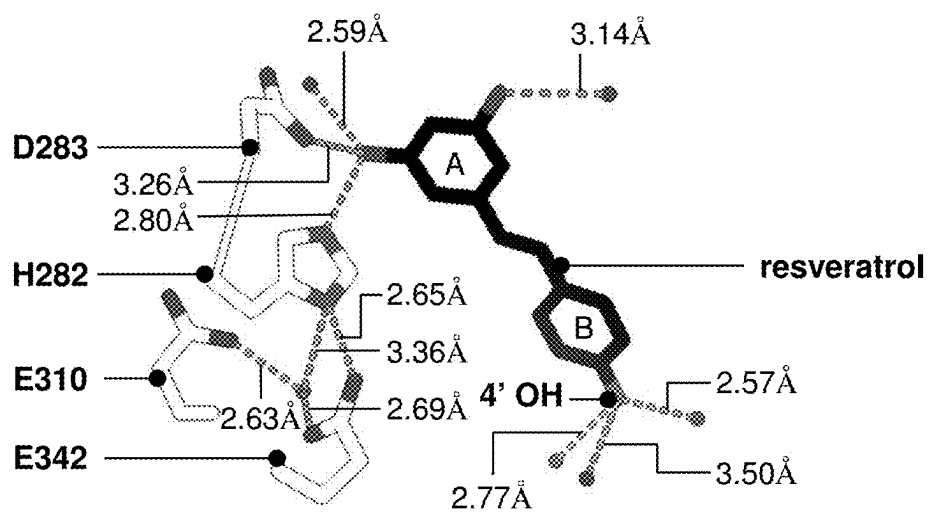


FIG. 5B

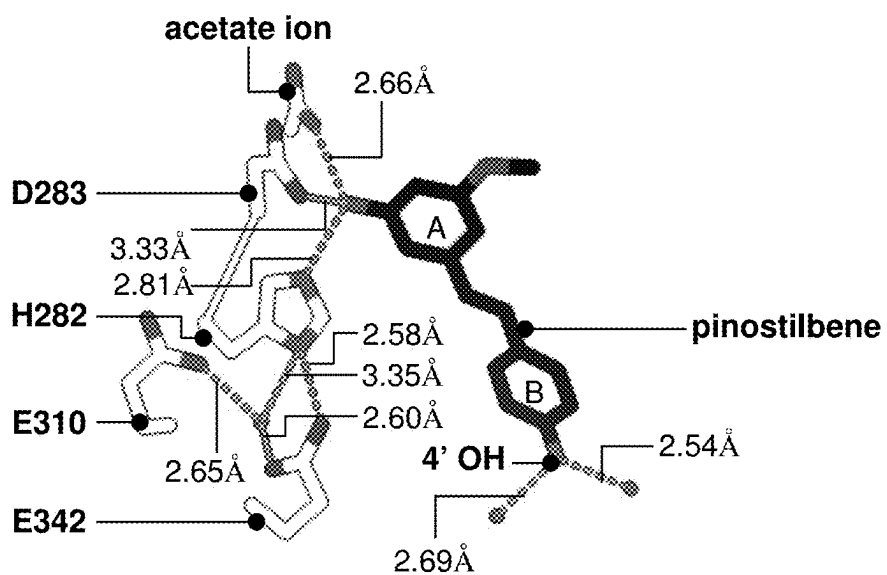


FIG. 5C

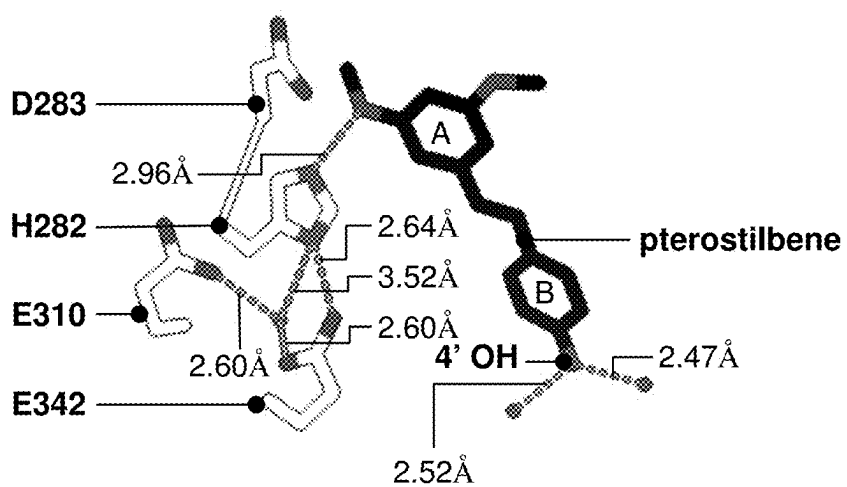


FIG. 5D

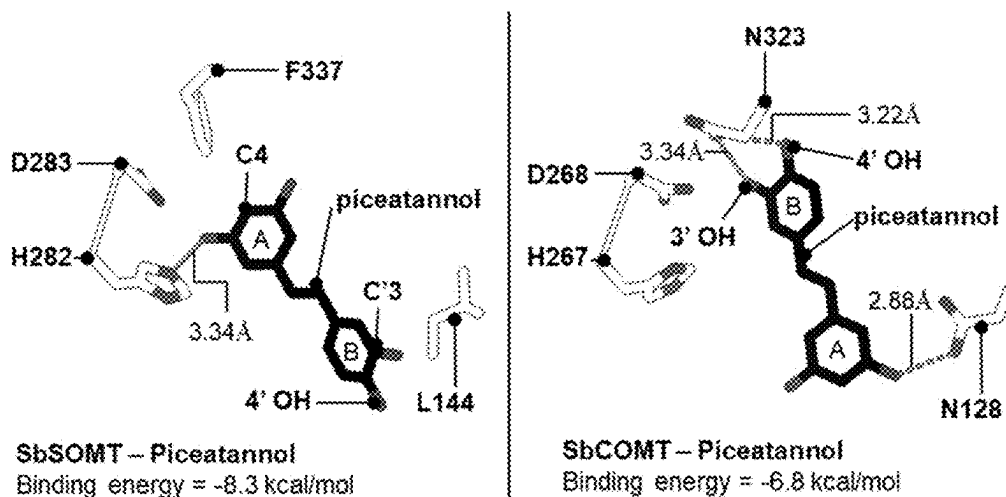


FIG. 5E

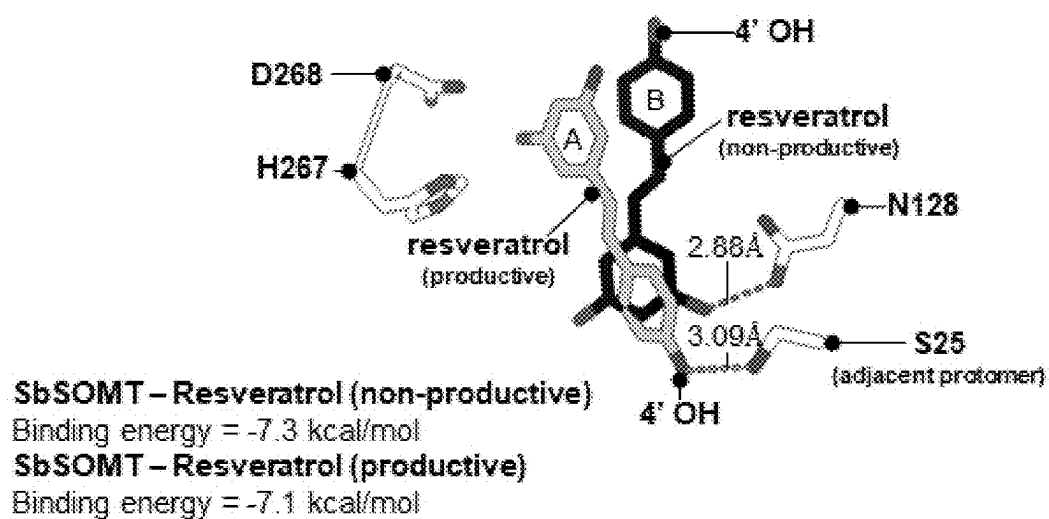


FIG. 5F

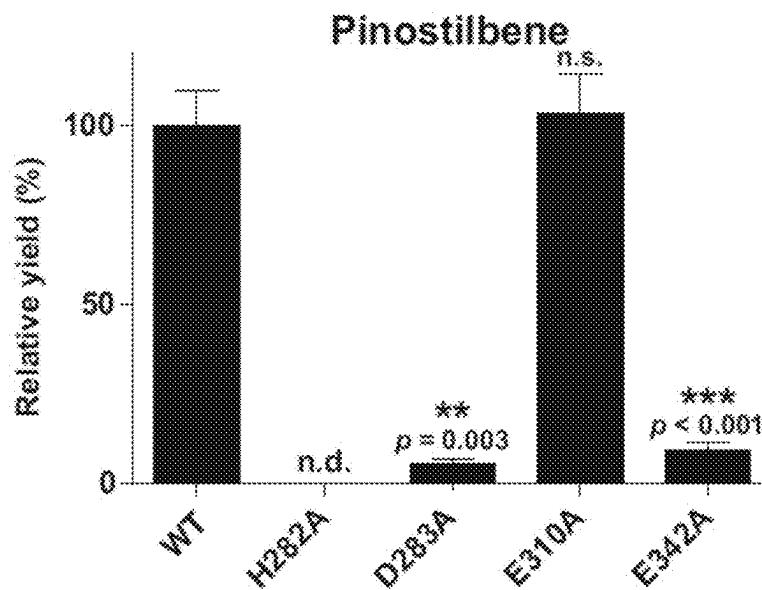


FIG. 6A

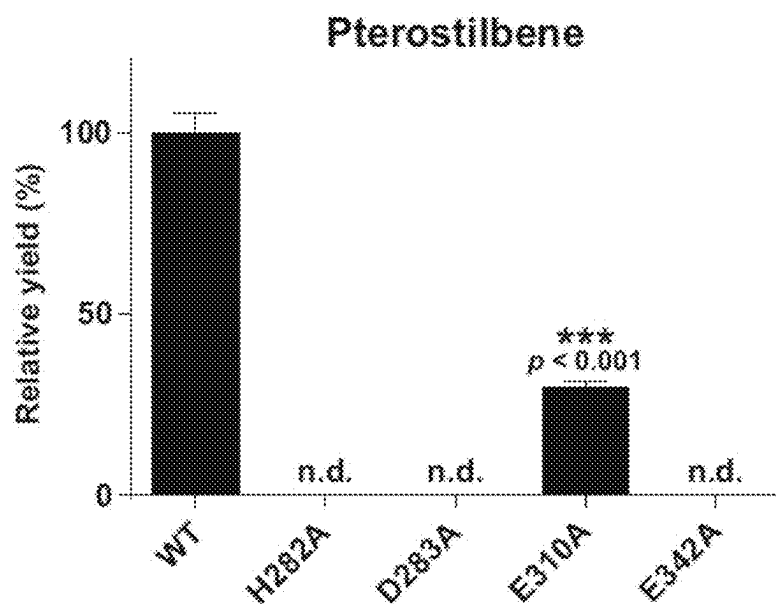


FIG. 6B

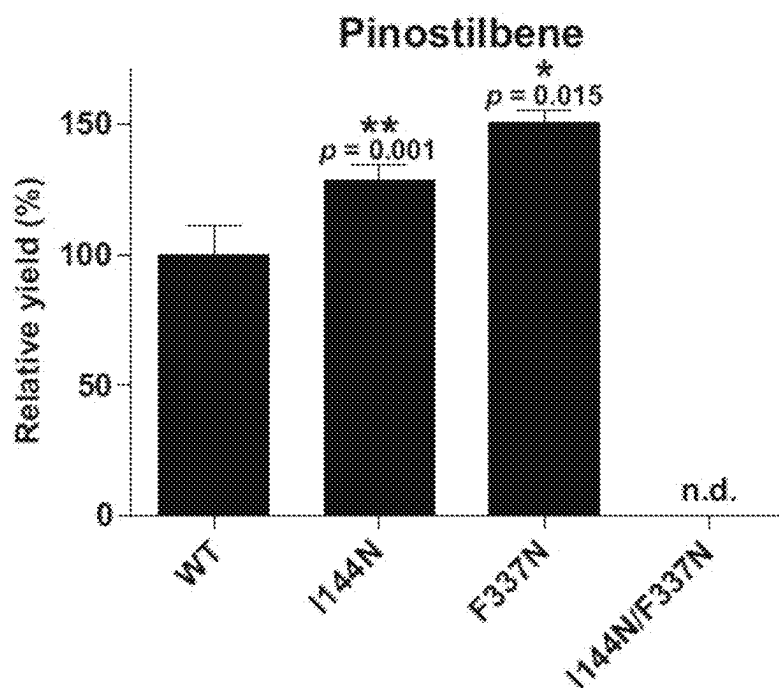


FIG. 6C

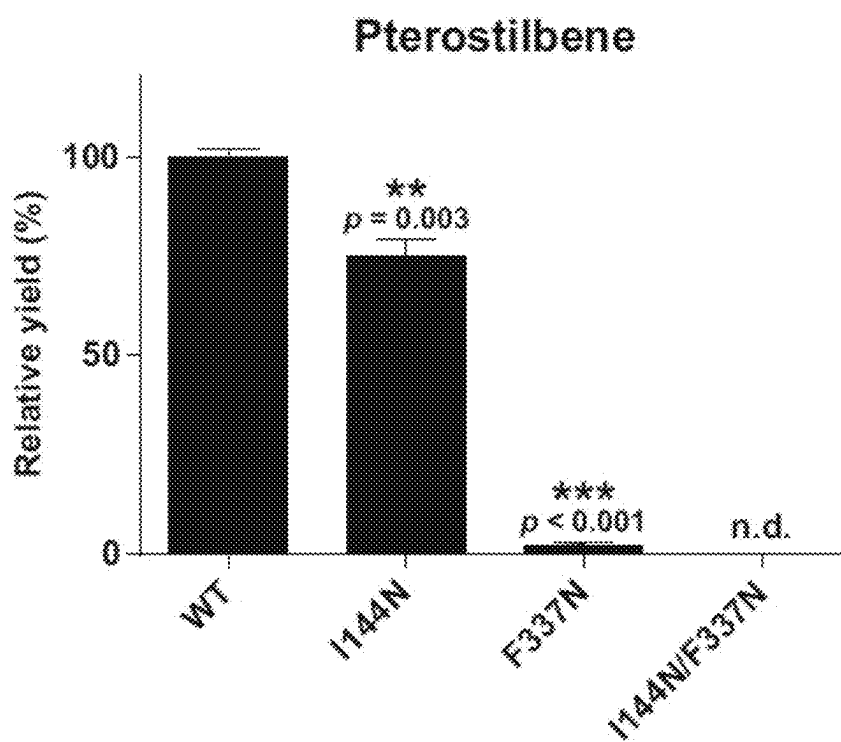


FIG. 6D

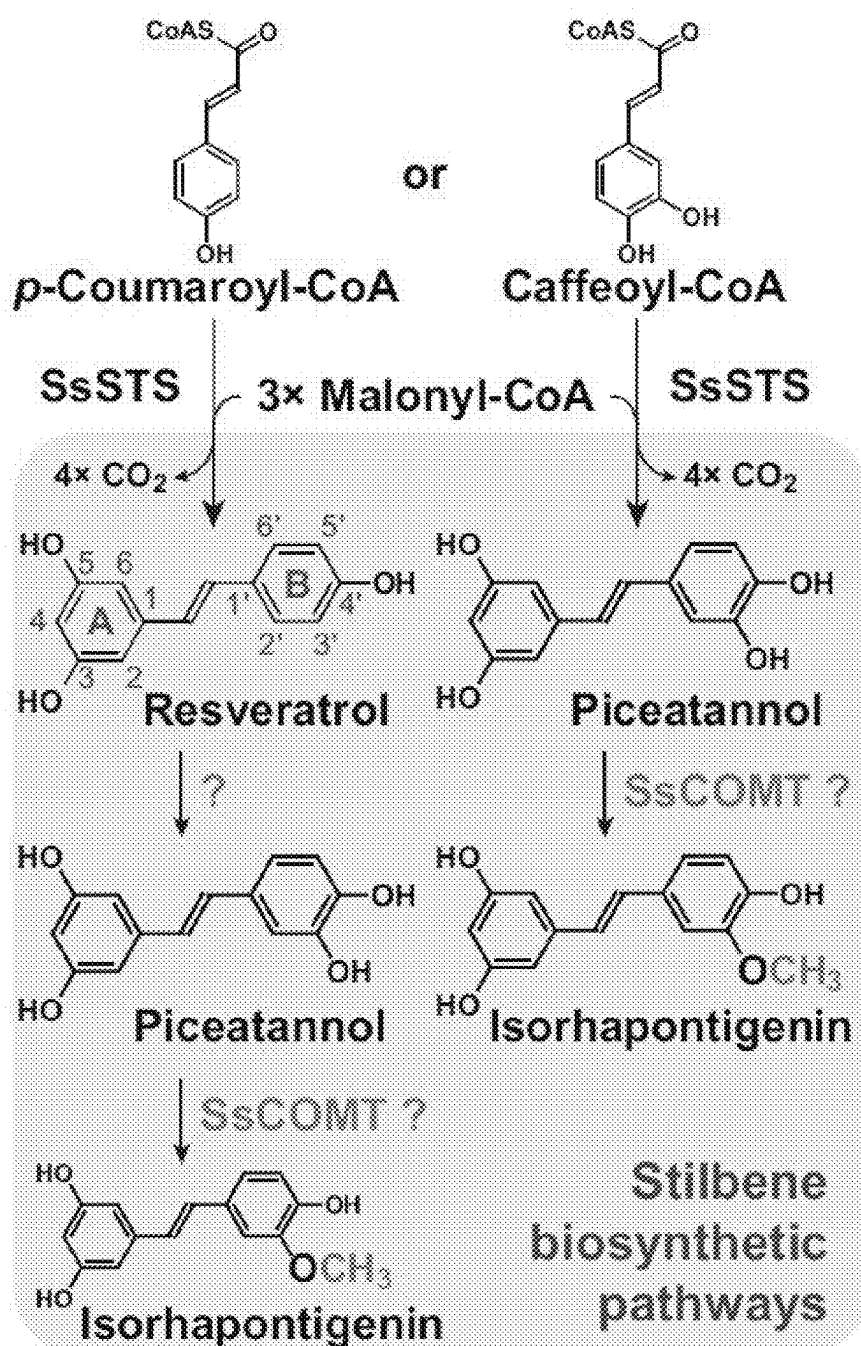


FIG. 7A

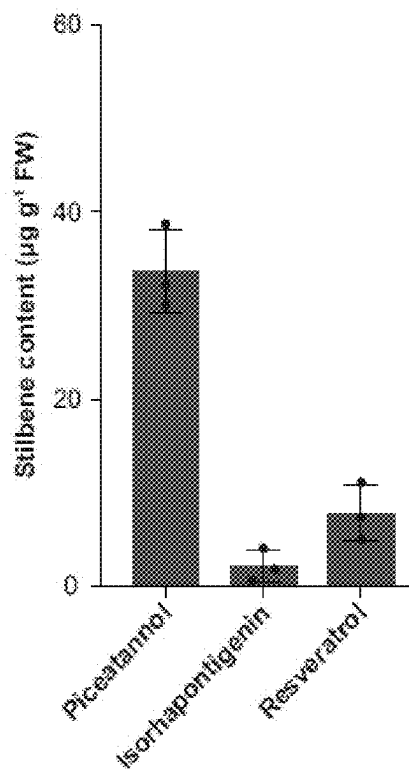


FIG. 7B

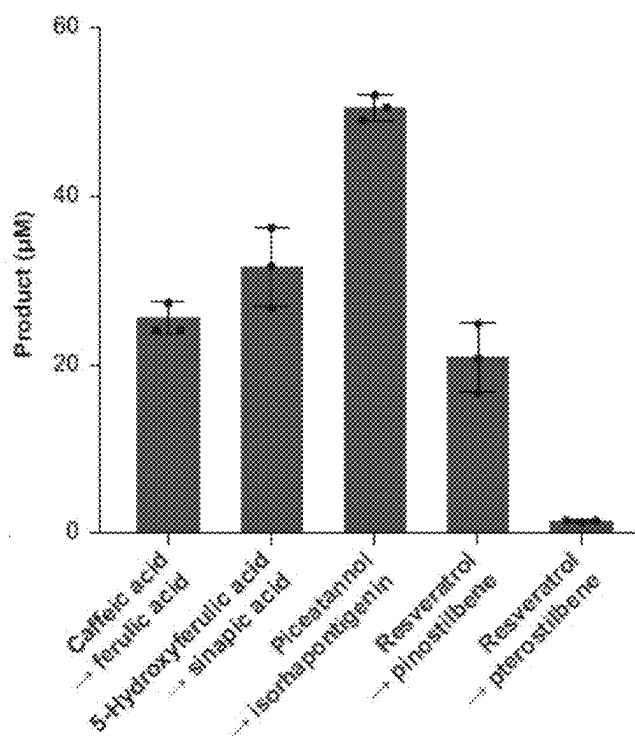


FIG. 7C

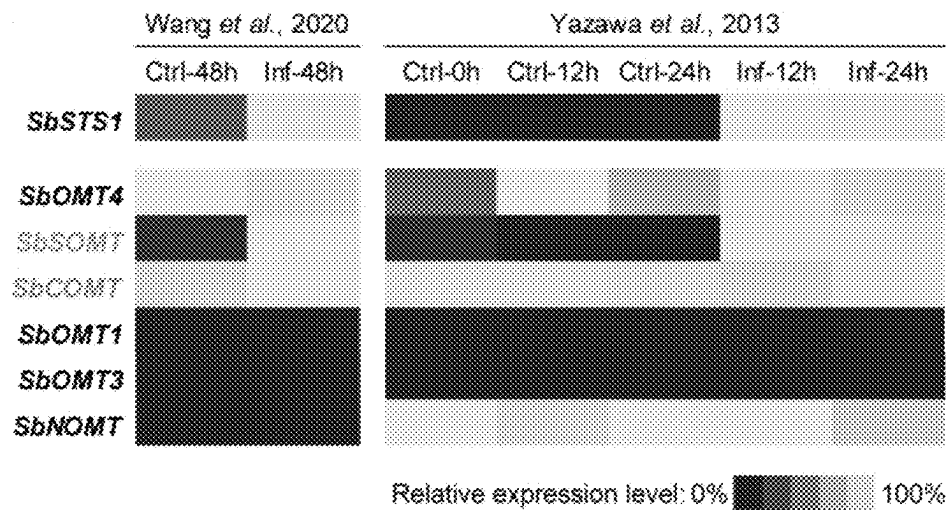


FIG. 8A

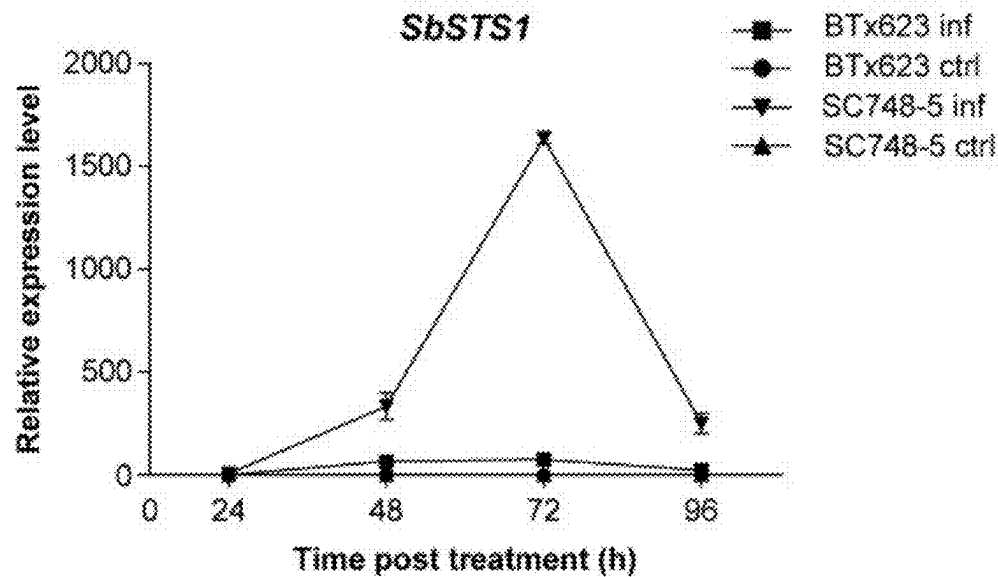


FIG. 8B

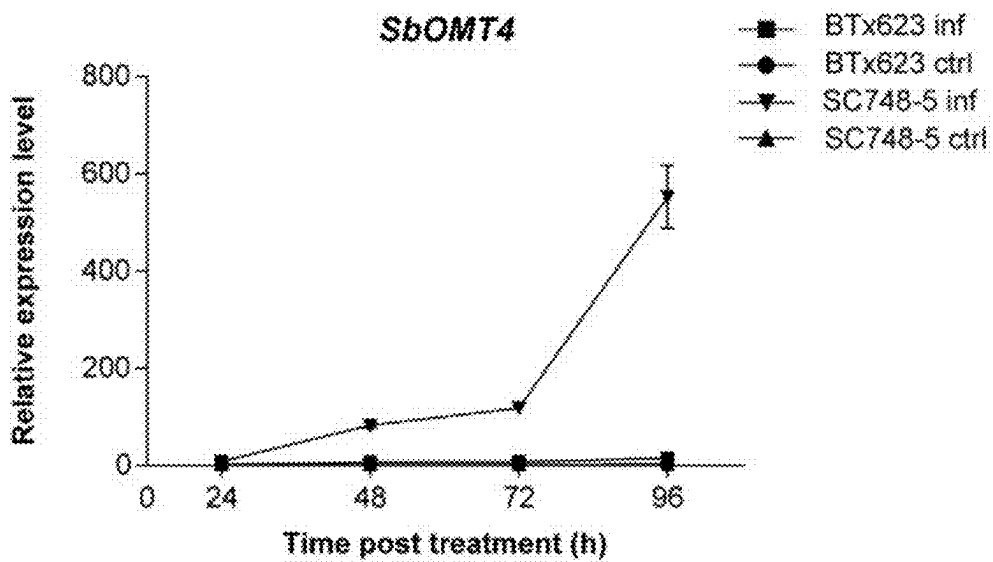


FIG. 8C

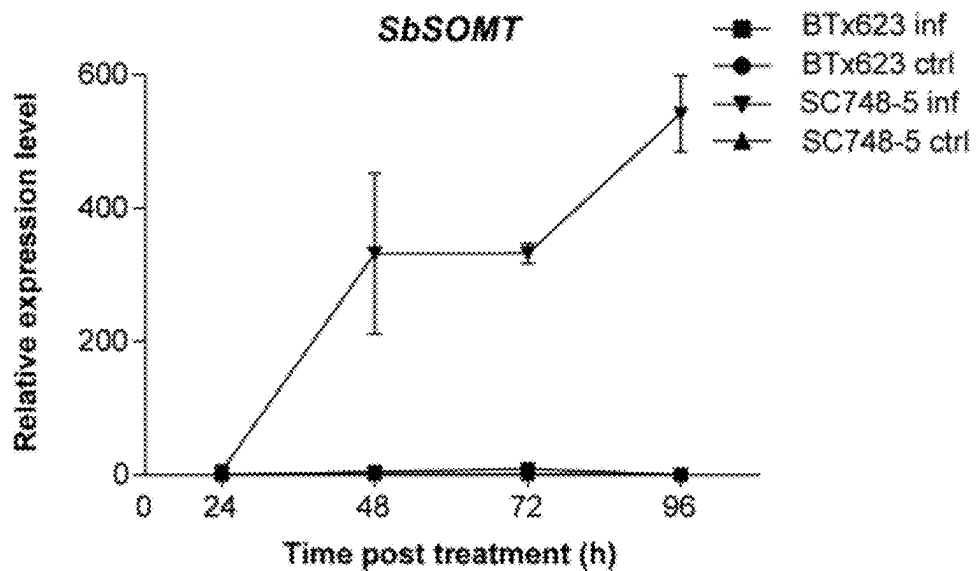


FIG. 8D

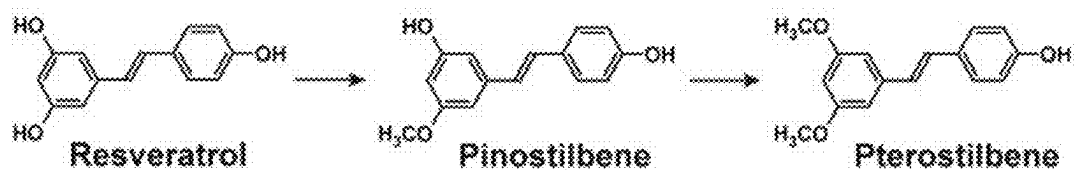


FIG. 9A

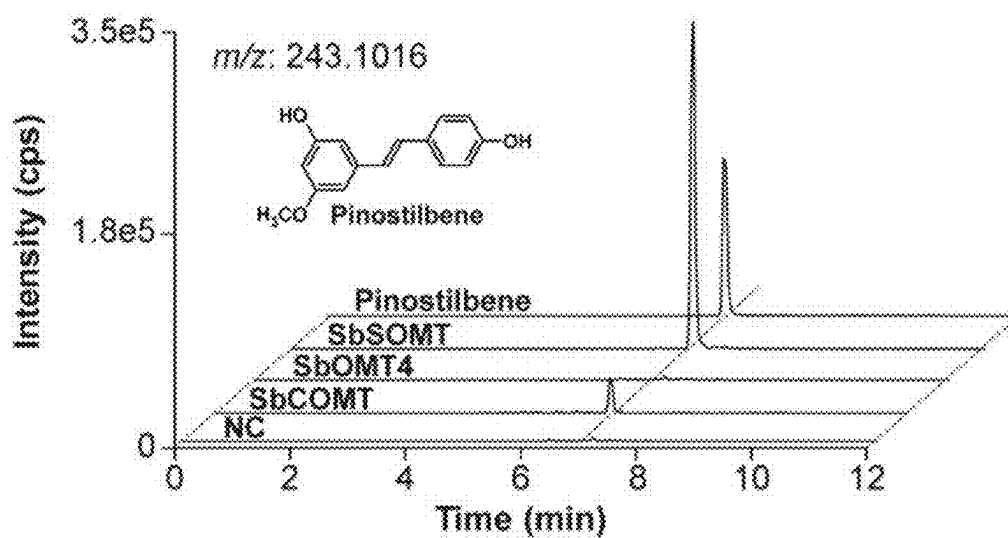


FIG. 9B

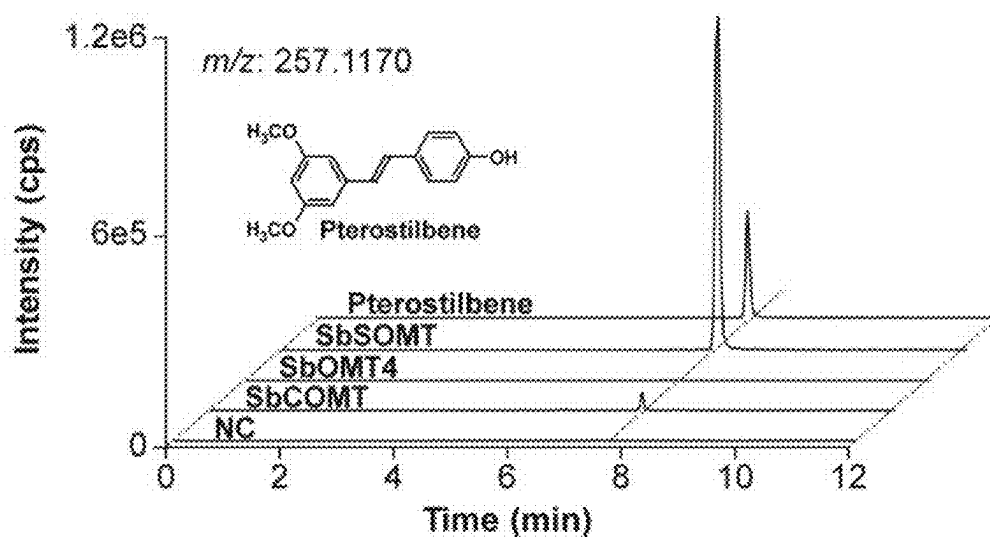


FIG. 9C

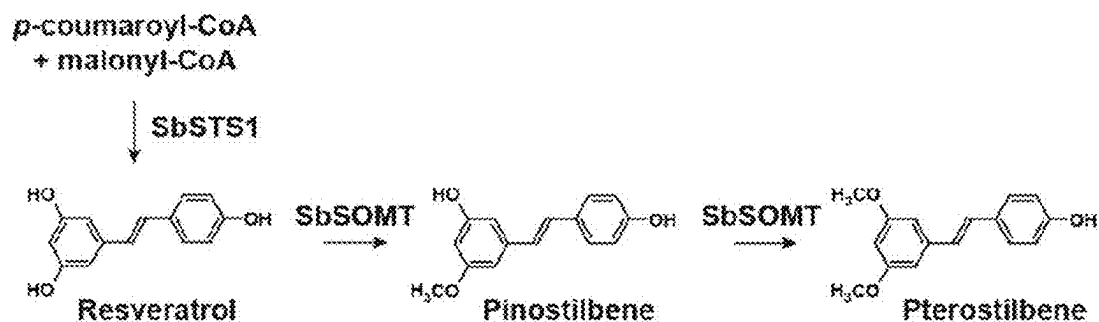


FIG. 10A

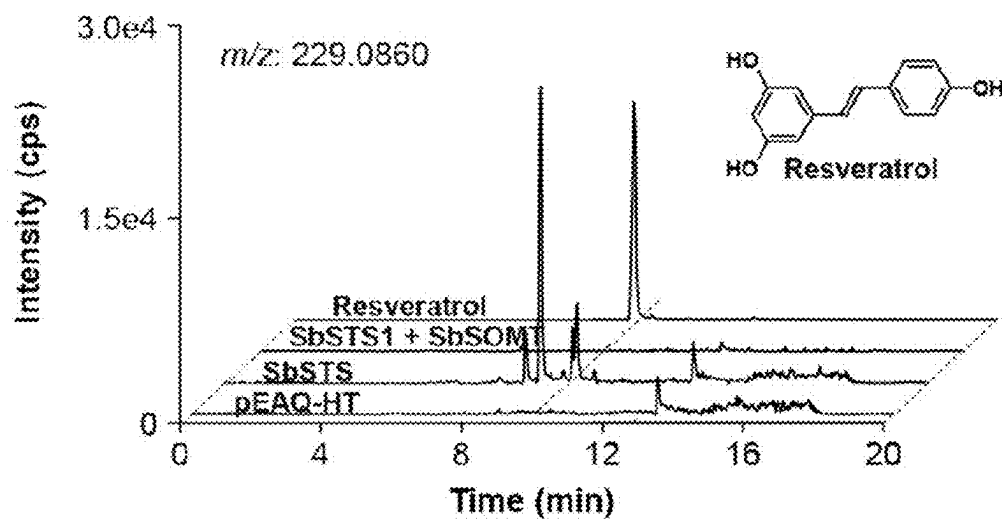


FIG. 10B

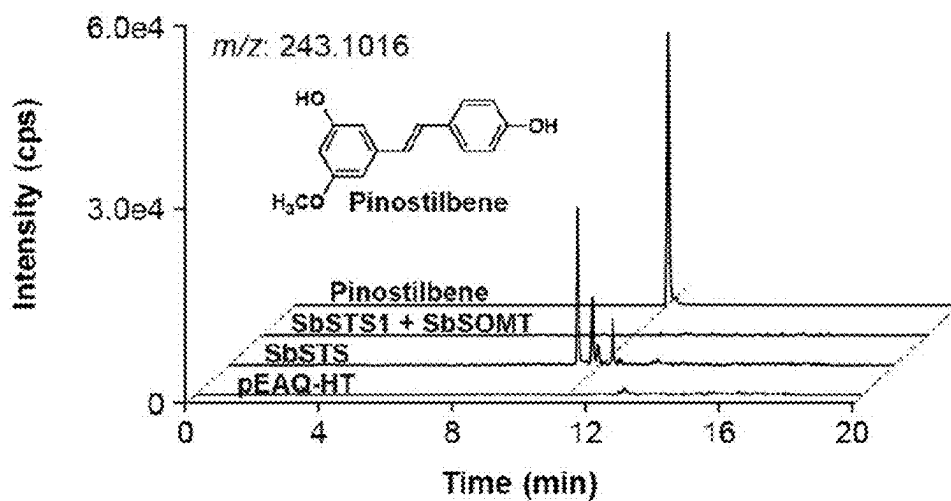


FIG. 10C

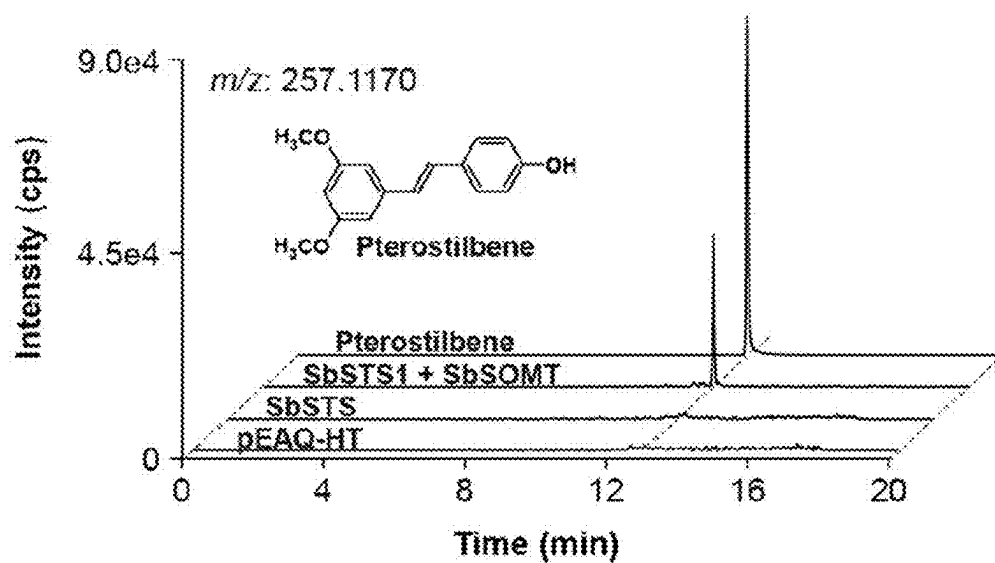


FIG. 10D

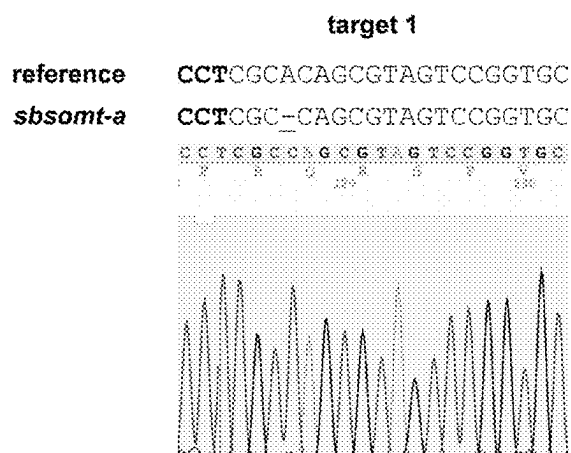


FIG. 11A

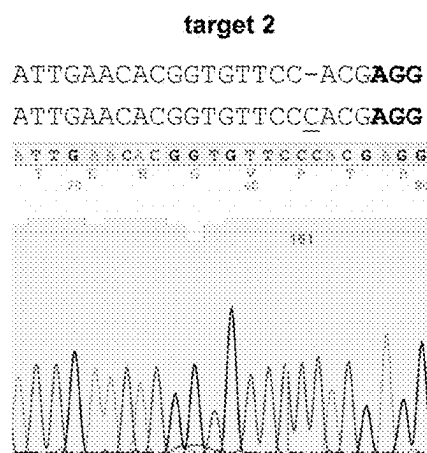


FIG. 11B

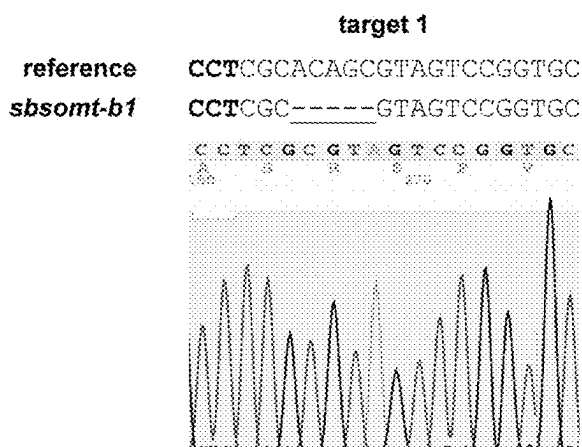


FIG. 11C

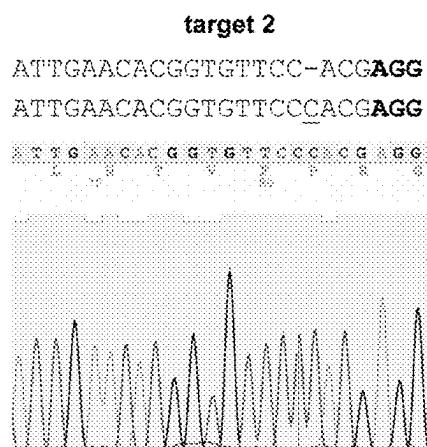


FIG. 11D

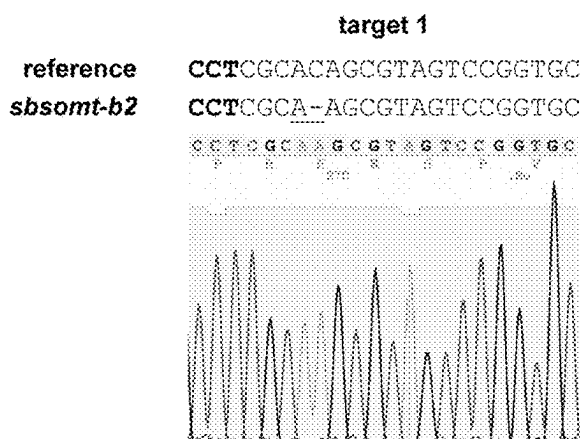


FIG. 11E

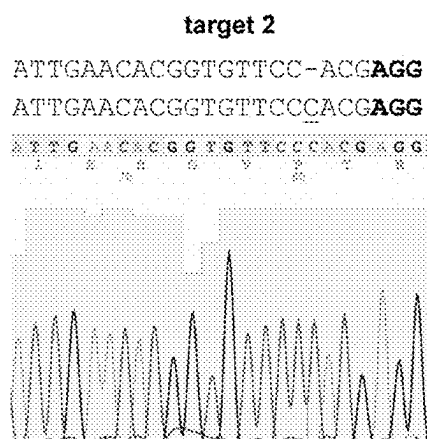


FIG. 11F

>SbSOMT			
		†	
WT	MGSYDSSSSSSNDSSARNEEDESCMFALKLLGGFAVPFT IKAVIELGVMDQLLTAERAMS		60
sbscmt-a	MGSYDSSSSSSNDSSARNEEDESCMFALKLLGGFAVPFT IKAVIELGVMDQLLTAERAMS		60
sbscmt-b1	MGSYDSSSSSSNDSSARNEEDESCMFALKLLGGFAVPFT IKAVIELGVMDQLLTAERAMS		60
sbscmt-b2	MGSYDSSSSSSNDSSARNEEDESCMFALKLLGGFAVPFT IKAVIELGVMDQLLTAERAMS		60
WT	AEELVAAVAQAQLPRPEVACTMVDRLLRFLASHSVVRCTIEVVVGTD-----		106
sbscmt-a	AEELVAAVAQAQLPRPEVACTMVDRLLRFLASPA*SGARPRWWARITP--PPPPAAGA		117
sbscmt-b1	AEELVAAVAQAQLPRPEVACTMVDRLLRFLASRSPVHDRGGGGGGRHHHLLPELREV		120
sbscmt-b2	AEELVAAVAQAQLPRPEVACTMVDRLLRFLASQA*SGARPRWWARITP--PPPPAAGA		117
		††	
WT	-----ATTTCCRRSYAASFVCMWFARNGVDSVLPLGMMILNKTFLDSWON		155
sbscmt-a	TPRHPSASGSPGTASRIRCFLG**S*TRHSWTAGT-----LRMQCWKE		159
sbscmt-b1	TRL-----QVVRQERRRGFG-----ASAWED		141
sbscmt-b2	TPRHPSASGSPGTASRIRCFLG**S*TRHSWTAGT-----LRMQCWKE		159
WT	ITDAVLEGAAPFEKTYGMPMFYLSNGLPLNT-----		187
sbscmt-a	QH-----HLRKPTGQCST*VQTDH*TR-----		181
sbscmt-b1	DFKQDIP--GQLAKHYGCSVGRSSTI*ENLRDANVEVPKYKRTIEHGVPTROWQIIR*L*		196
sbscmt-b2	QH-----HLRKPTGQCST*VQTDH*TR-----		181
		† †	
WT	---VTHEAMANHSMI-ITKLLKFFRGFEGLEVLVDVGGGNGTTLQMIRGQYKMRGINY		243
sbscmt-a	---CSHEAMANHSMI-ITKLLKFFRGFEGLEVLVDVGGGNGTTLQMIRGQYKMRGINY		237
sbscmt-b1	PRNCSSSSAASKALMCWST*AAATAFRCK*LEVNIRI*EA*TTTFILMSLRRLSLQKLVNM		252
sbscmt-b2	---CSHEAMANHSMI-ITKLLKFFRGFEGLEVLVDVGGGNGTTLQMIRGQYKMRGINY		237
WT	DLPH--VIAQ-----AAPVEGVEHVGGSMFDNIPRGN----		273
sbscmt-a	DLPH--VIAQ-----AAPVEGVEHVGGSMFDNIPRGN----		267
sbscmt-b1	*VAACSIIIFHAEMQFCSSGFFMIGTTTRRASRS*RIATQLSM*EAR*SFNSTLFRMNQNL		308
sbscmt-b2	DLPH--VIAQ-----AAPVEGVEHVGGSMFDNIPRGN----		267
		† † †	
WT	---AVLLKWILHDWDDKACIKILKNCYTALHVRGKVIVLEYVVPDEPEPTLAAQGAFELD		330
sbscmt-a	---AVLLKWILHDWDDKACIKILKNCYTALHVRGKVIVLEYVVPDEPEPTLAAQGAFELD		324
sbscmt-b1	LQLRVPSNWTSPCWSRLAVVVRGH-----RGSS-----PSSPW---RPASLESLR		350
sbscmt-b2	---AVLLKWILHDWDDKACIKILKNCYTALHVRGKVIVLEYVVPDEPEPTLAAQGAFELD		324
		† †	
WT	LTMLVTFGSGKERTQREFSELAMEAGFSREFKATYIFANVWALEFTK*	377	
sbscmt-a	LTMLVTFGSGKERTQREFSELAMEAGFSREFKATYIFANVWALEFTK*	371	
sbscmt-b1	LRISLP-TSGPLSSQS-----	365	
sbscmt-b2	LTMLVTFGSGKERTQREFSELAMEAGFSREFKATYIFANVWALEFTK*	371	

FIG. 12

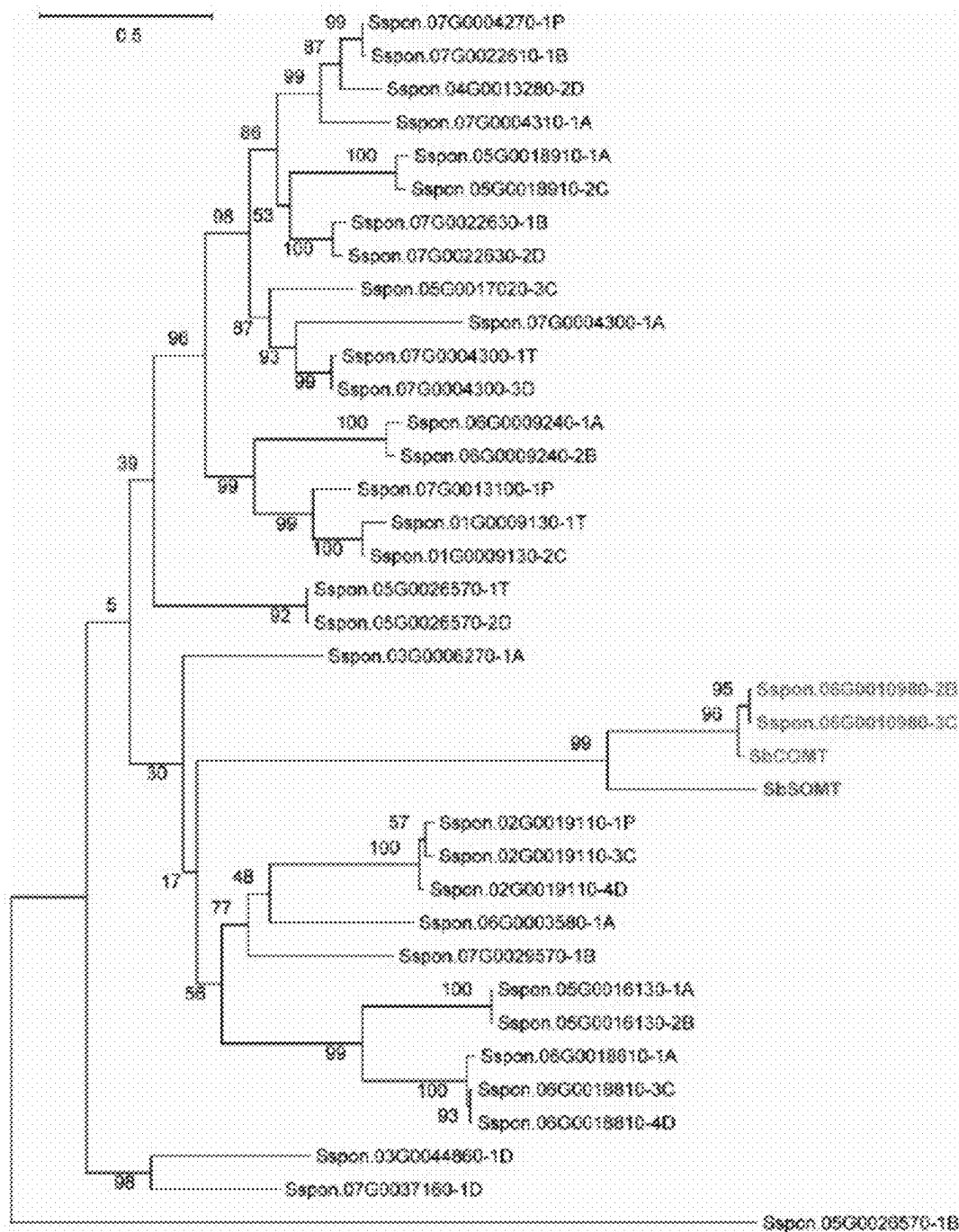


FIG. 14

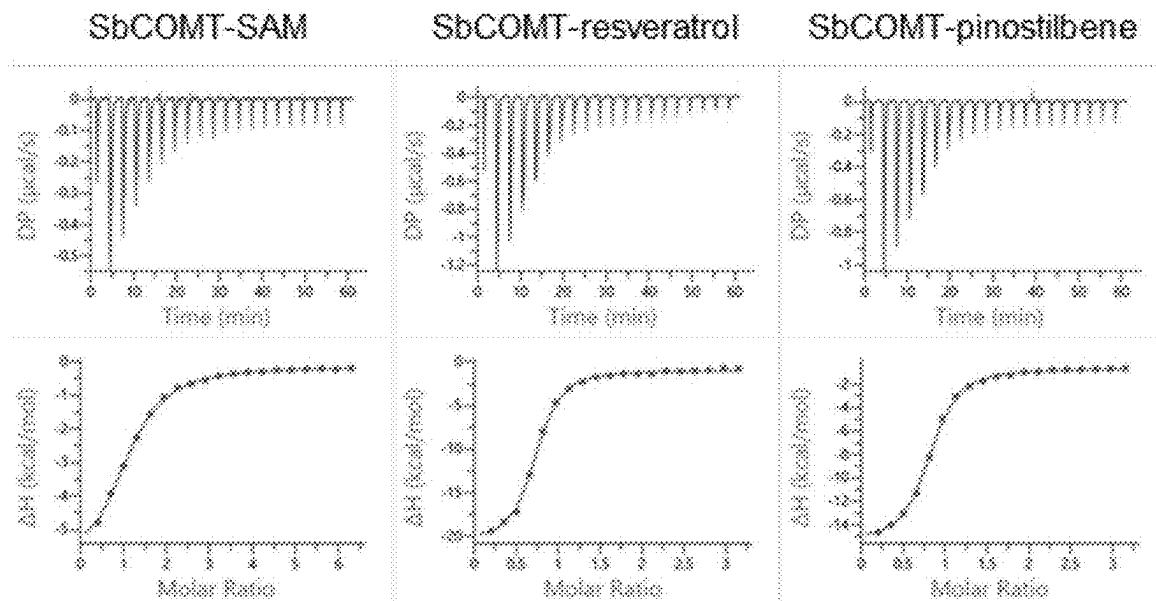


FIG. 15A

FIG. 15B

FIG. 15C

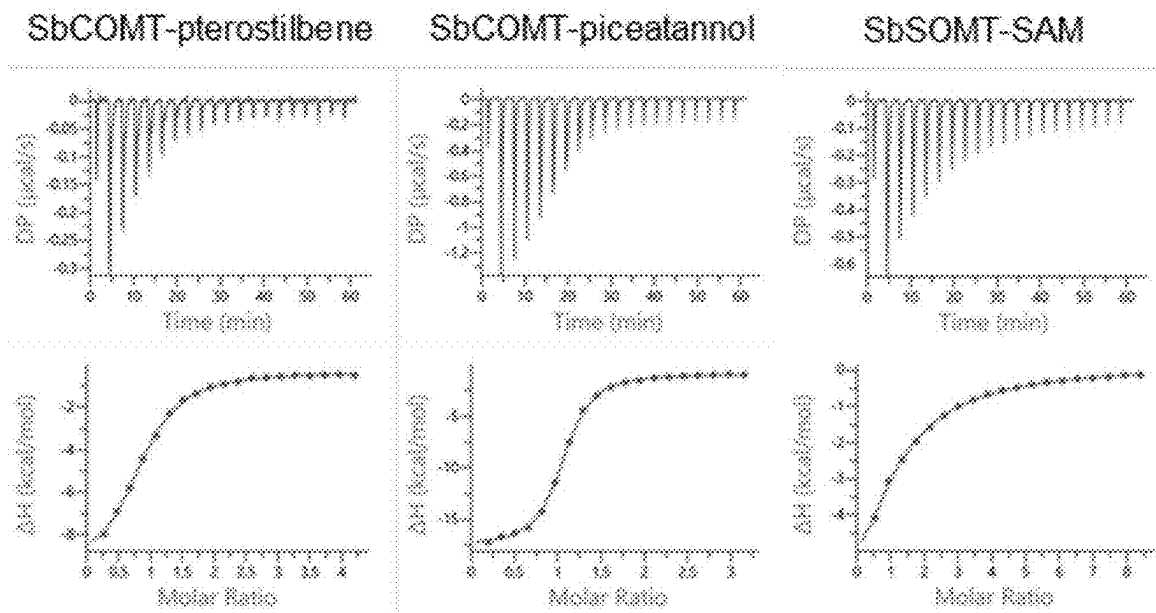


FIG. 15D

FIG. 15E

FIG. 15F

SbSOMT-resveratrol

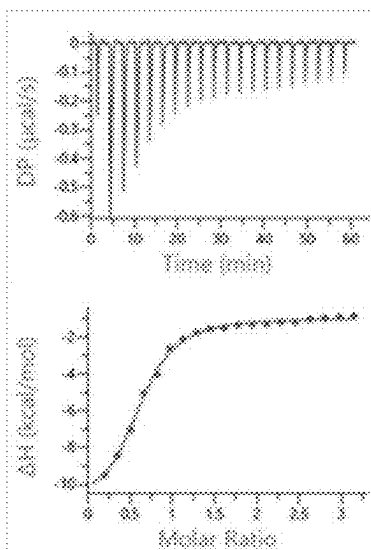


FIG. 15G

SbSOMT-pinostilbene

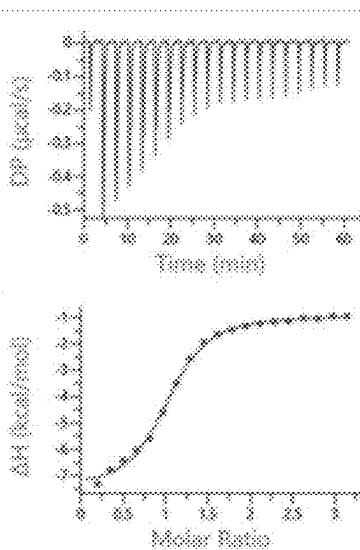


FIG. 15H

SbSOMT-pterostilbene

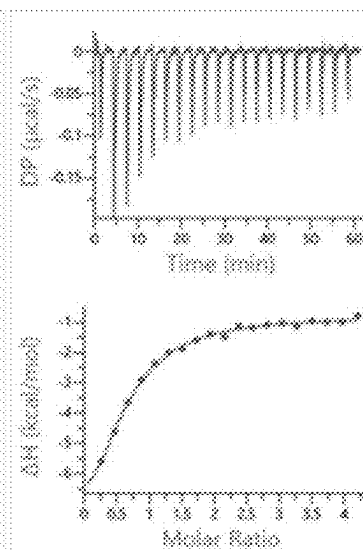


FIG. 15I

SbSOMT-piceatannol

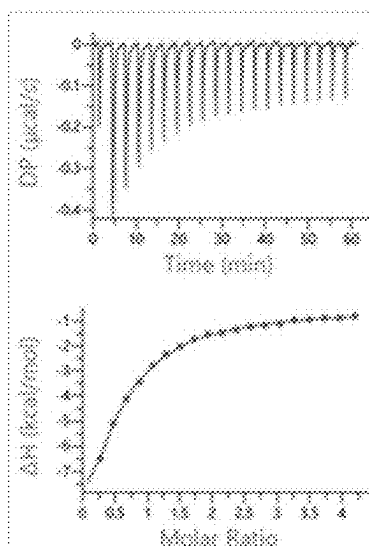


FIG. 15J

I144N-resveratrol

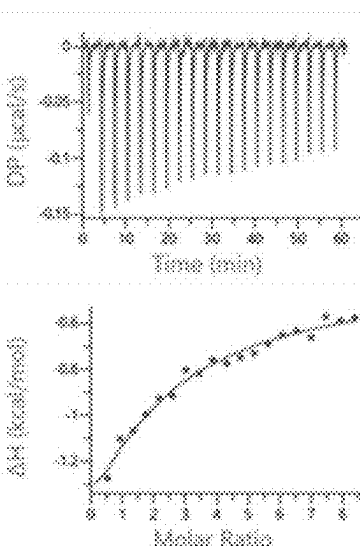


FIG. 15K

D283A-resveratrol

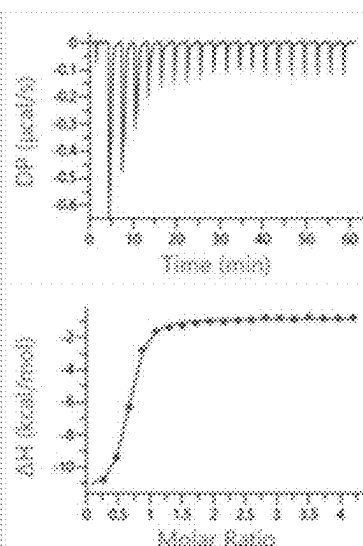


FIG. 15L

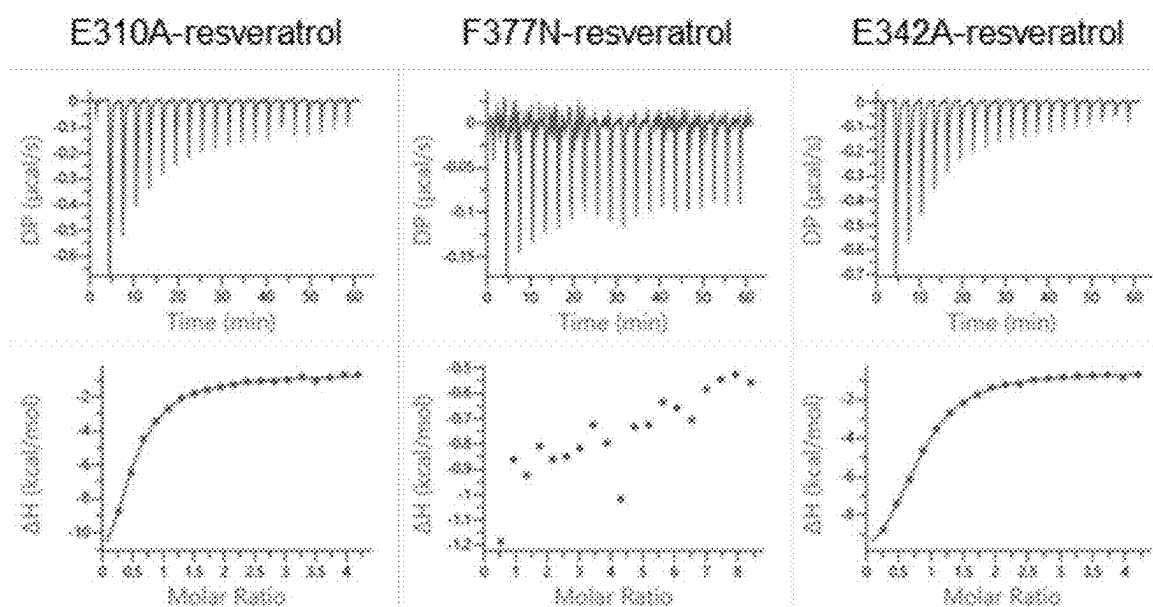


FIG. 15M

FIG. 15N

FIG. 15O

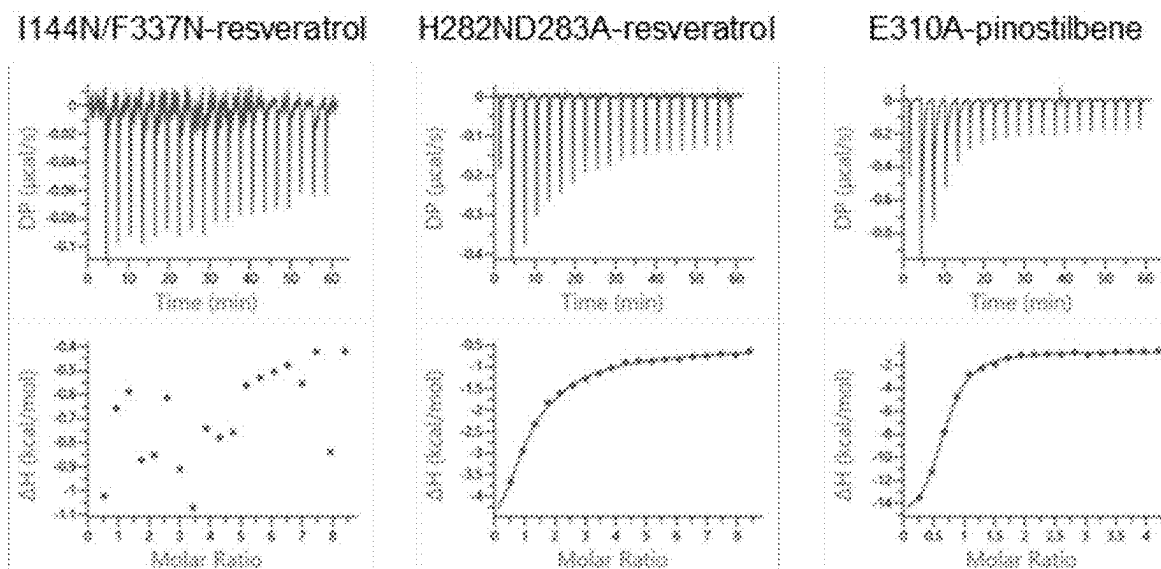


FIG. 15P

FIG. 15Q

FIG. 15R

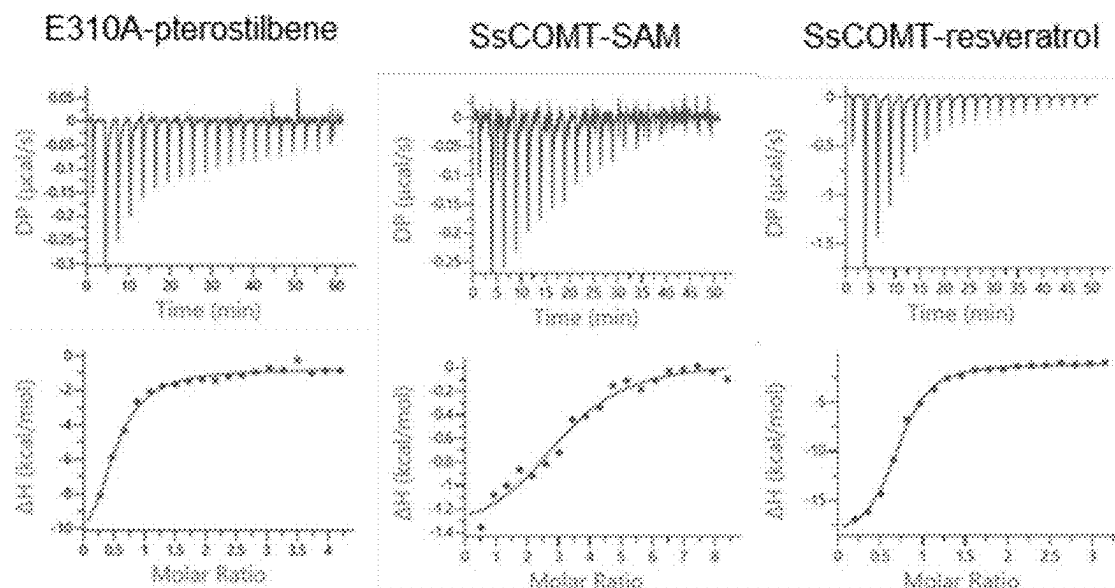


FIG. 15S

FIG. 15T

FIG. 15U

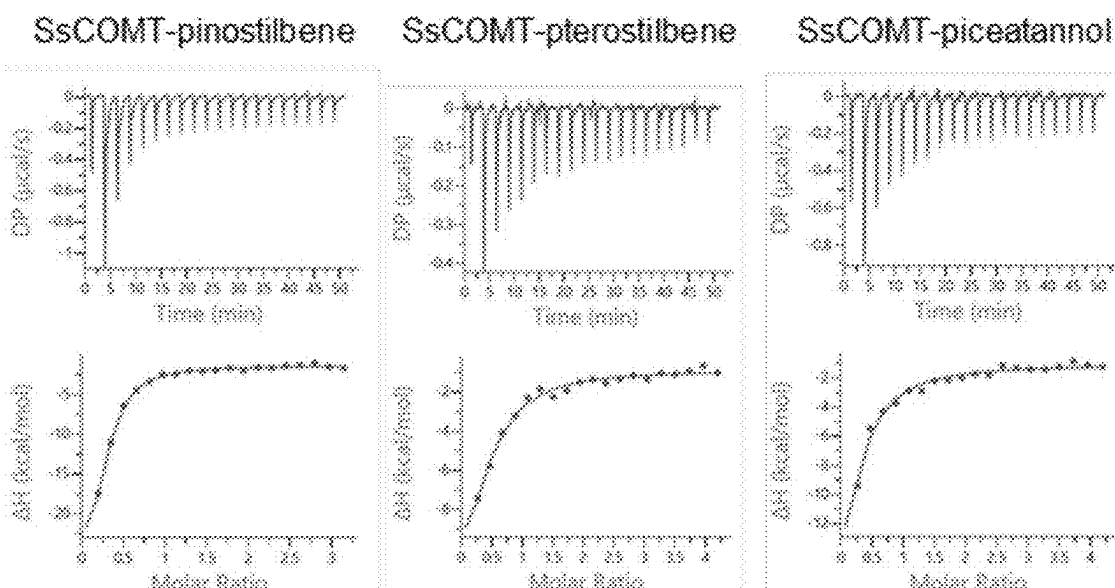


FIG. 15V

FIG. 15W

FIG. 15X

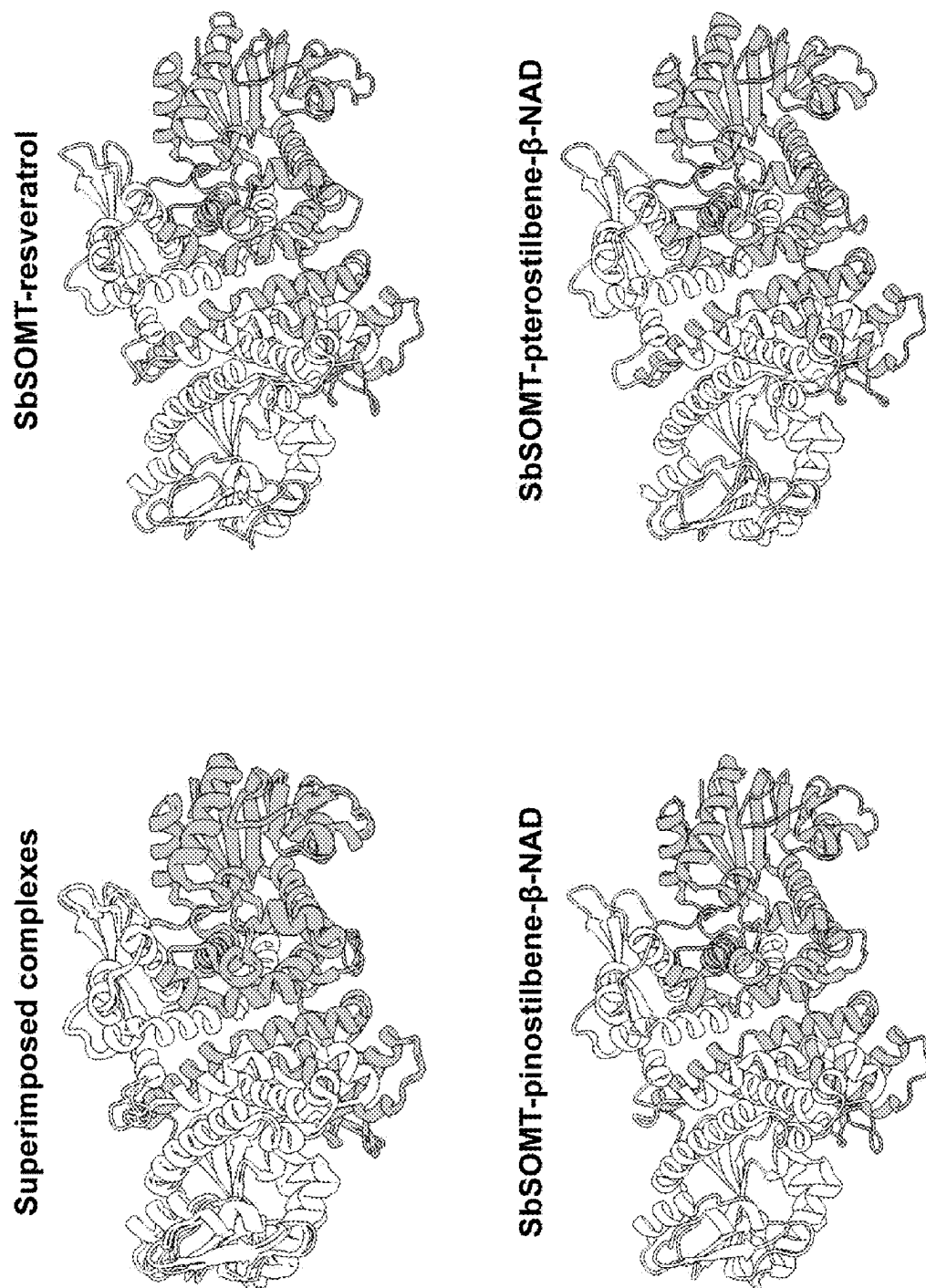


FIG. 16A

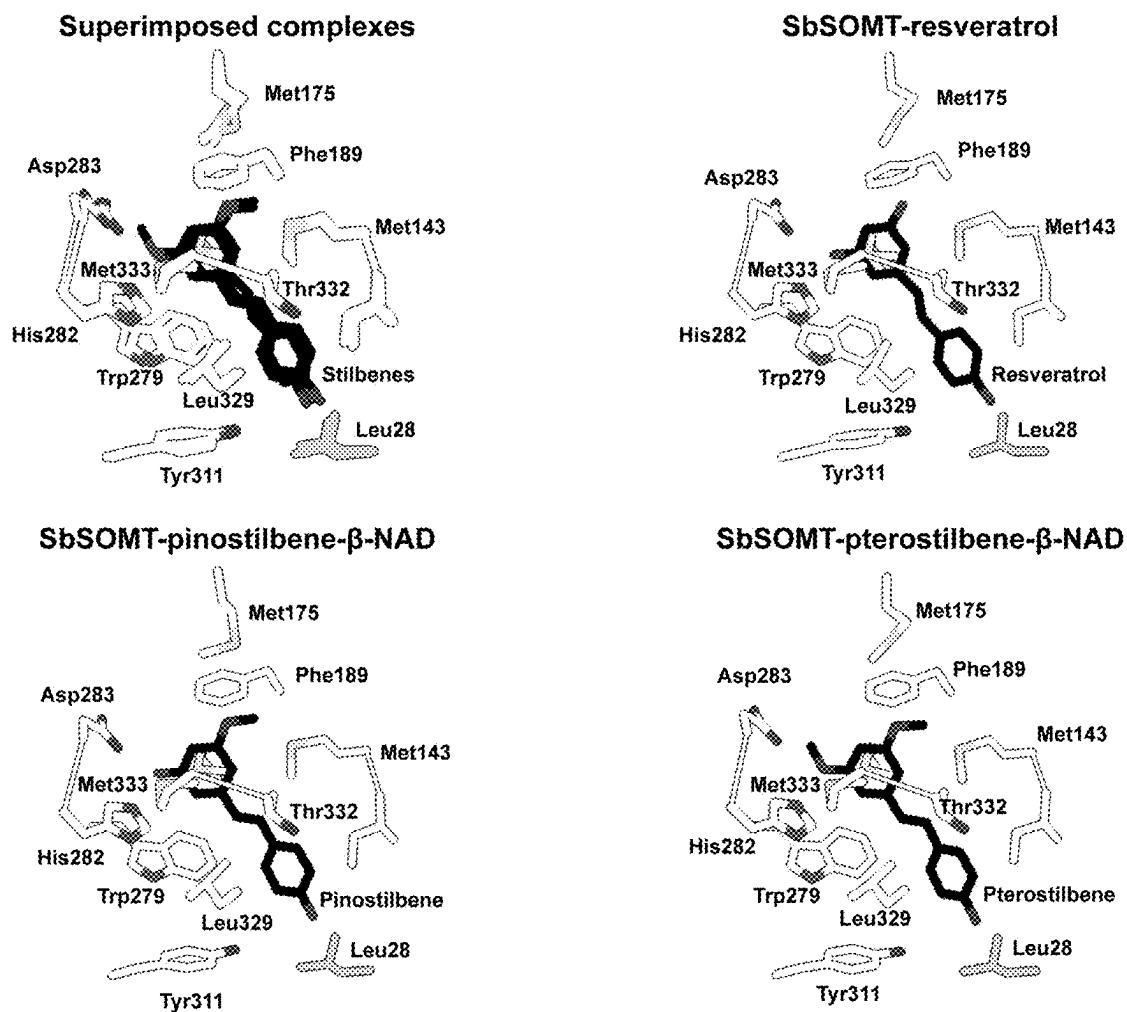


FIG. 16B

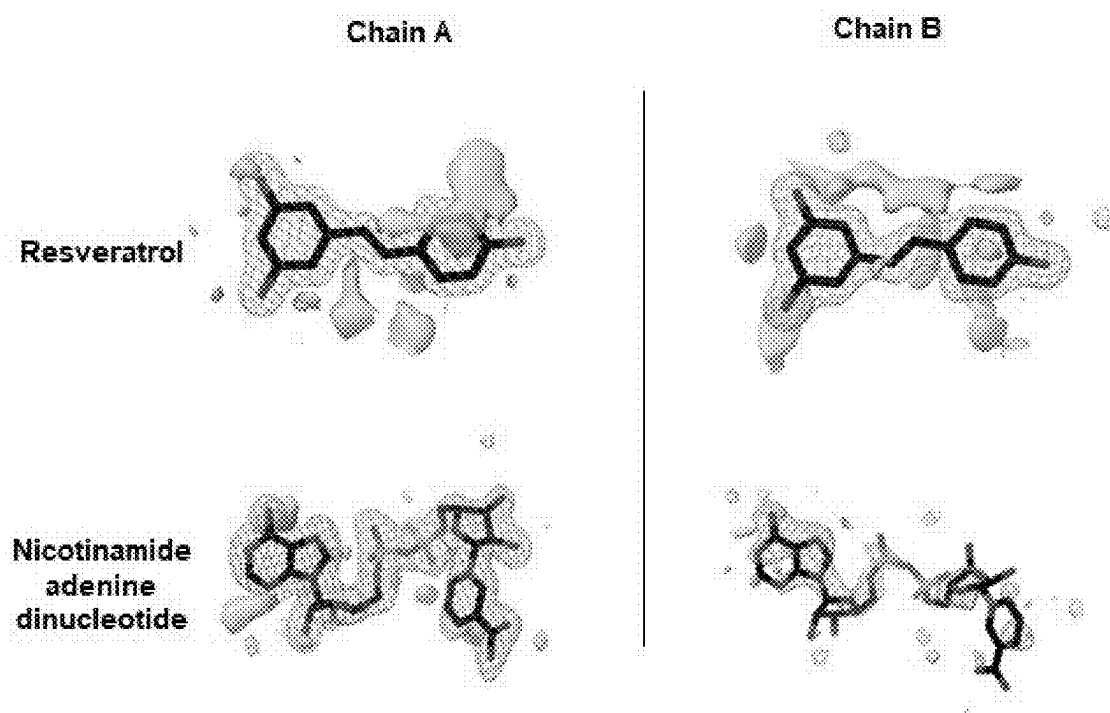


FIG. 16C

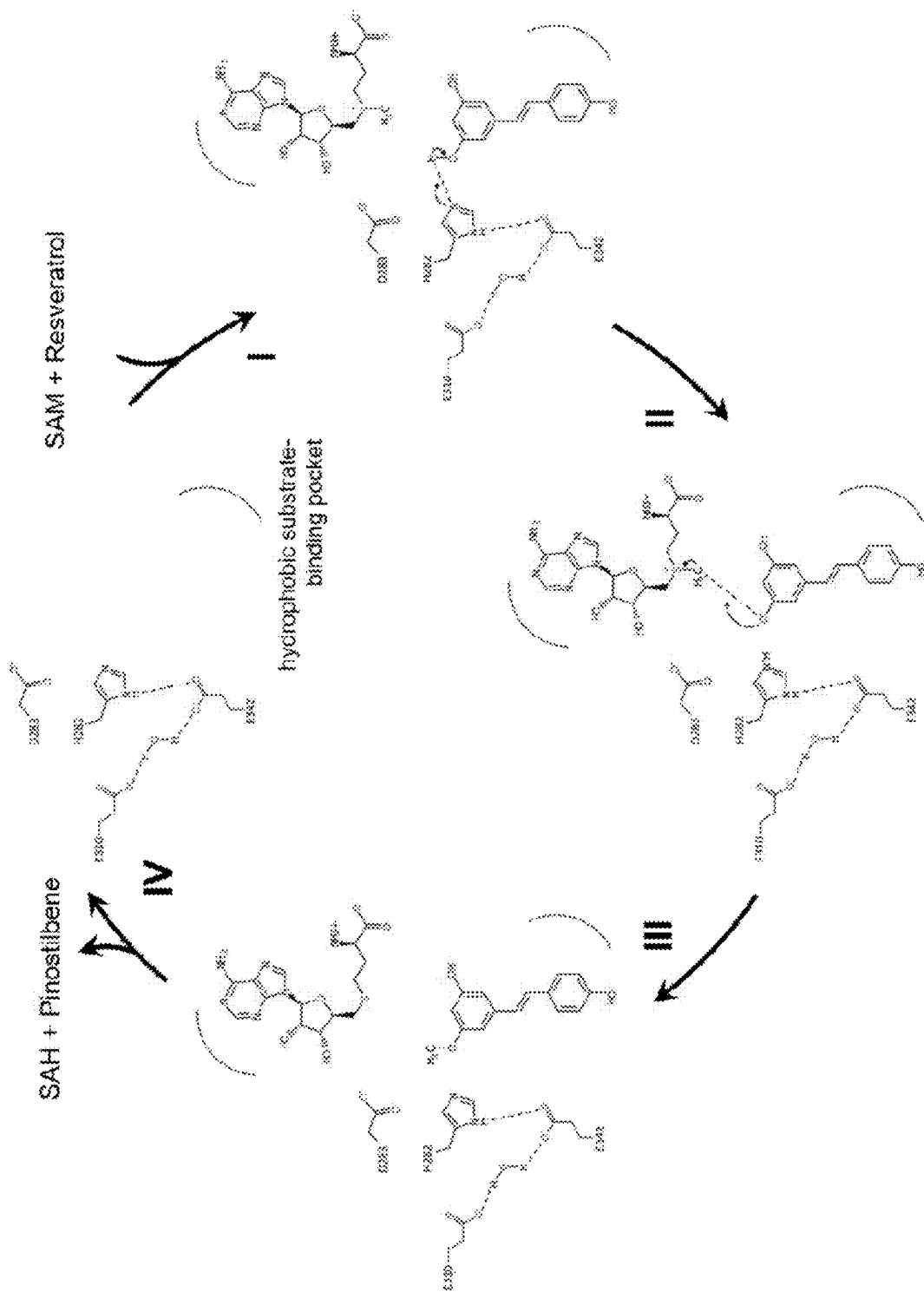


FIG. 17

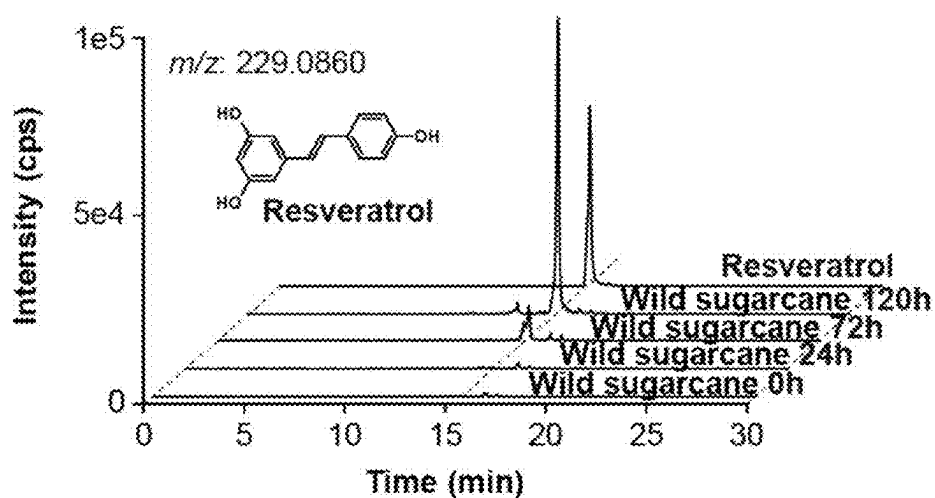


FIG. 18A

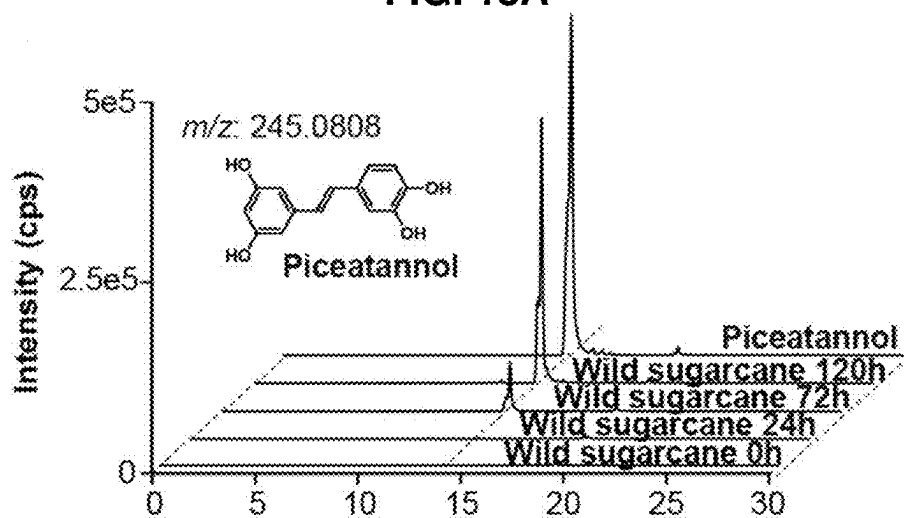


FIG. 18B

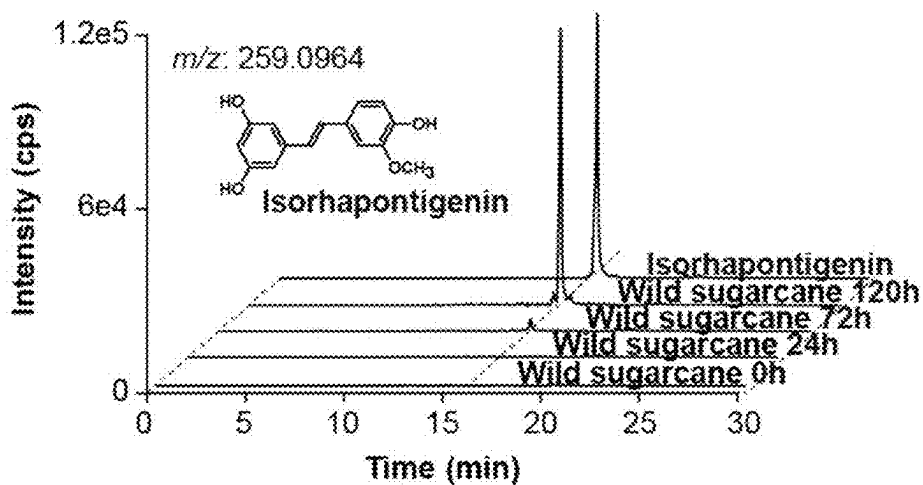


FIG. 18C

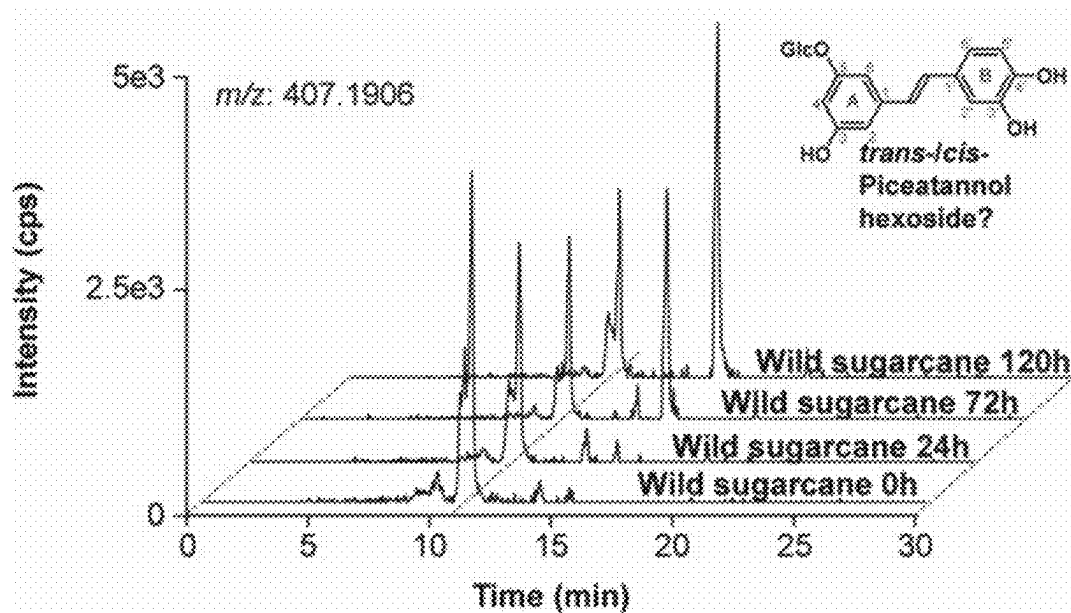


FIG. 19A

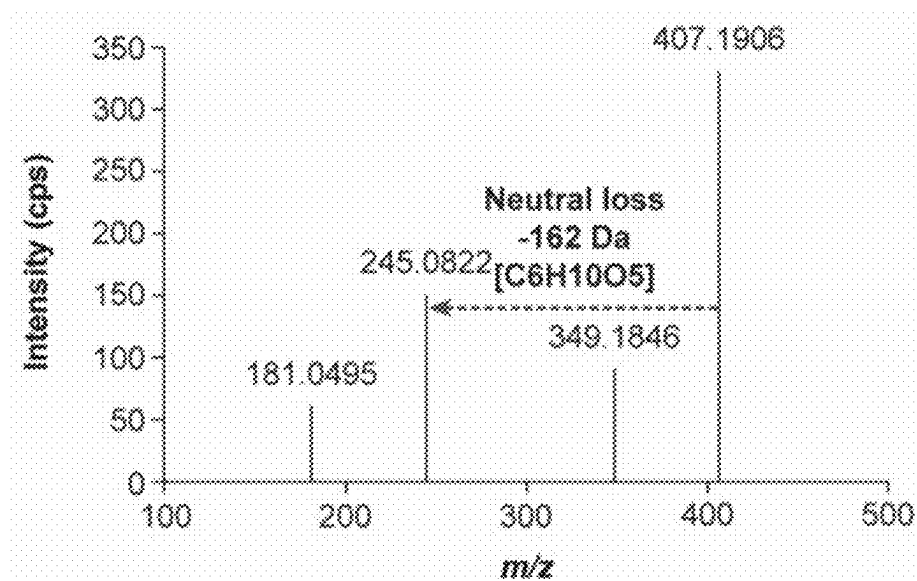


FIG. 19B

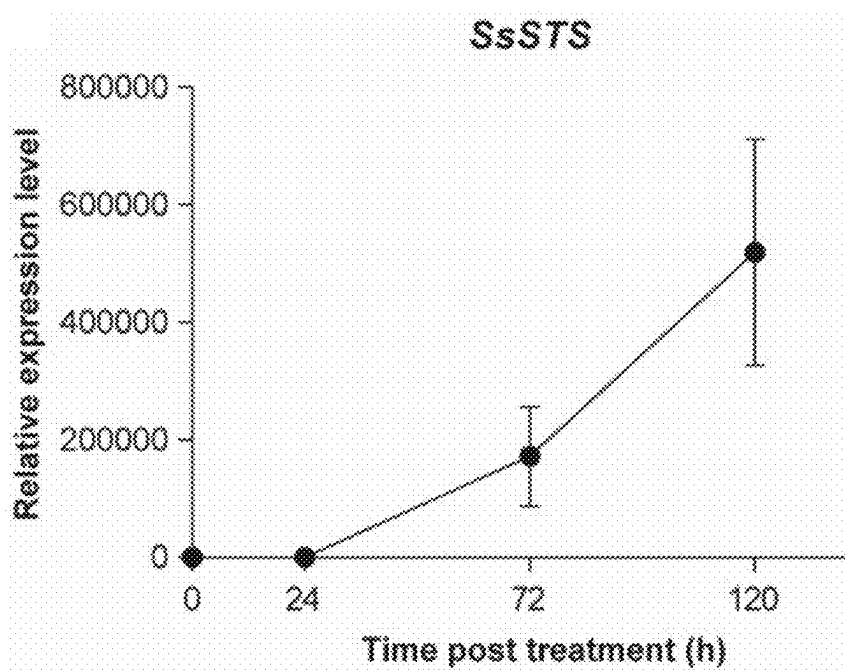


FIG. 20A

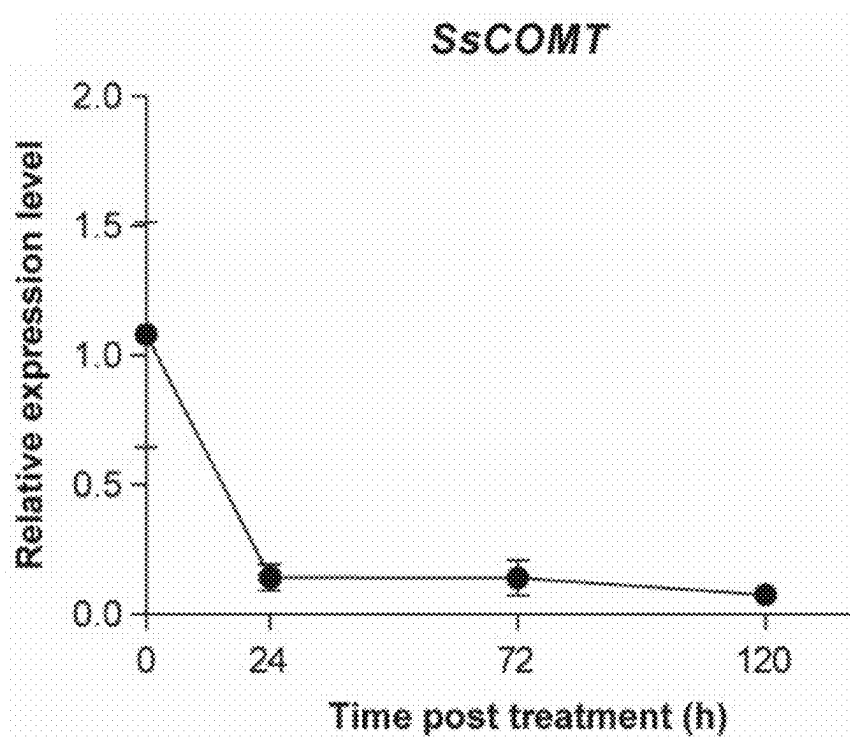


FIG. 20B

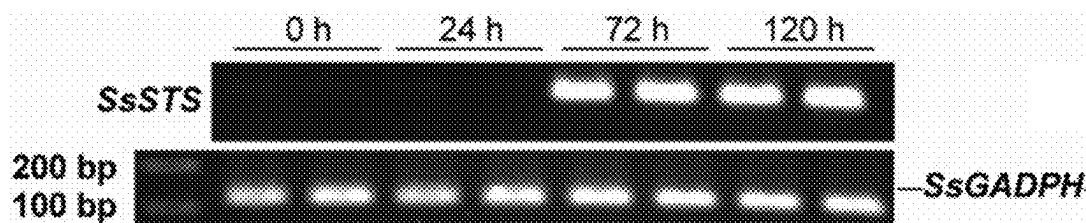


FIG. 20C

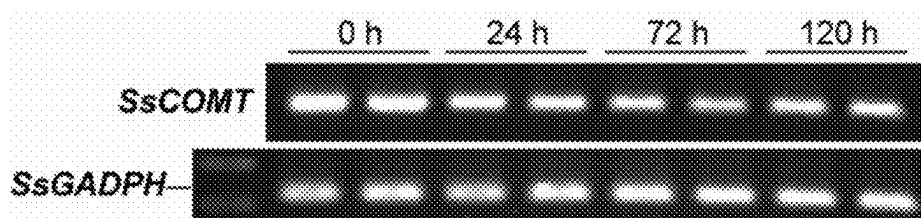


FIG. 20D

SbSTS1	MTTGKVTLEAVRKAQRAEGPATVLAIGTATPANCVYQADYPDYVFRVTKSEHLTDLKEKF	60
SsSTS	-MTGKVTLGAVRKAQRAEGSAAVLAIGTATPANCVYQADYPDYVFRVTKSEHLTDLKEKF	59

SbSTS1	KRICHKSMIRKRYMHLTEDILEENPNMSSYWAPSLDARQDILIQEIPKLGAEAAEKALKE	120
SsSTS	KRICHKSMITKRYMHLTEGFLQENPNMSSYSAPSLDARQDILIEEVPKLGAAAAEKALKE	119

SbSTS1	WGQPRSRIITHLVFCTTSGVDMPGADYQLIKLLGLCPVNRAMMYHQGCFAGGMVLRLLAKD	180
SsSTS	WGQPRSQIITHLVFCTTSGVDMPGADYQLIKLLGLSLVNRAMMYHQGCFAGGMVLRLLAKD	179

SbSTS1	LAENNRGARVLIVCSEITVVTFRGPSESHLDSL VGQALFGDGAAAVIVGADPSEPAERPL	240
SsSTS	LAENNRGARVLIVCSEITAVTFRGPSESHLDSL VGQALFGDGAAAVIVGADPSA-AEWPL	238

SbSTS1	FHLVSASQTILPDSEGAIEGHLREVGLTFHLQDRVPQLISMNIERLLEDAPPLGISDWN	300
SsSTS	FQLVSASQTILPDSEGAIEGHLREVGLTFHLQDRVPQLISTNIERLLEDAPPLGISDWN	298

SbSTS1	SIFWVAHPGGPAILNMVEAKVGLDKARMCATRHILAEYGNMSSVCVLFILDEMRNRSAKD	360
SsSTS	SIFWVAHPGGPAILNMVEAKAGLDKARLCATRHILAEYGNMSSACVLFILDEMRNKSAED	358

SbSTS1	GHTTTGEGMEWGVLF6FGPGLTVETIVLHSPITTVAA*	398
SsSTS	GHTTTGEGMEWGVLF6FGPGLTVETIVLQSVPTITV*--	394

FIG. 21

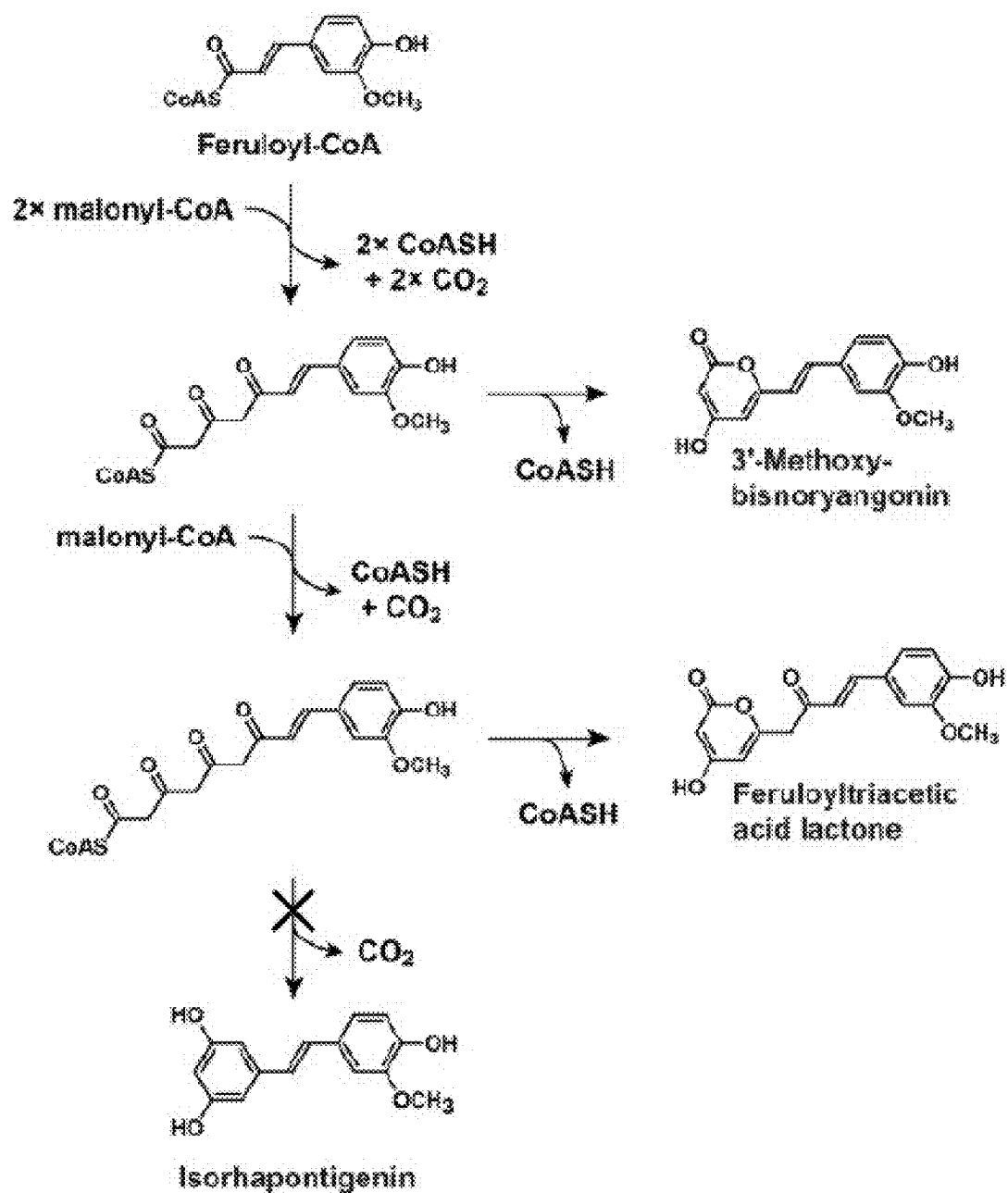


FIG. 22A

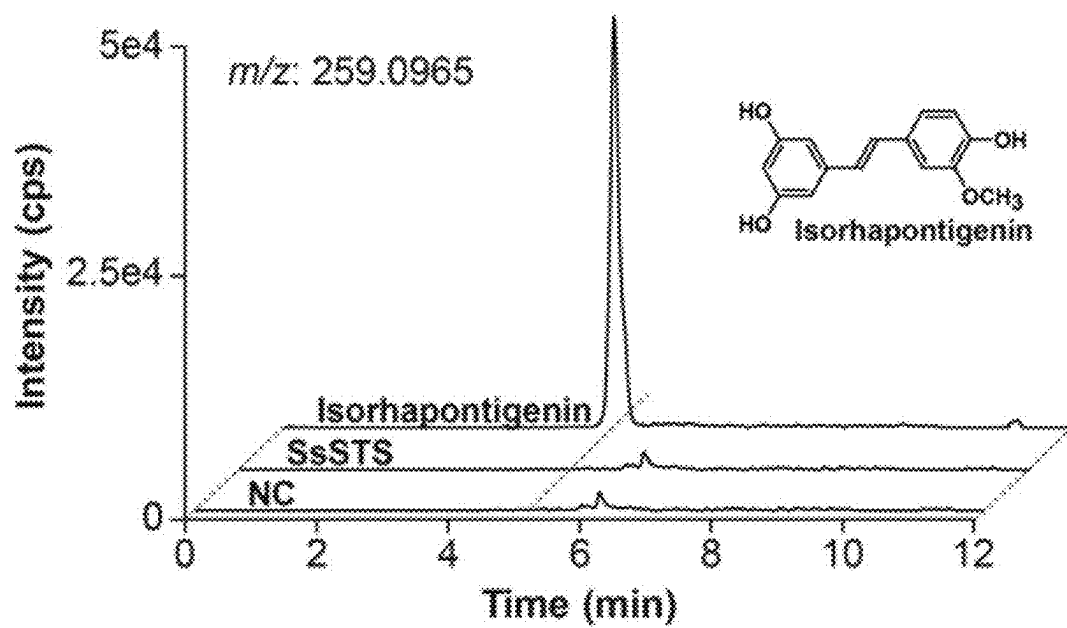


FIG. 22B

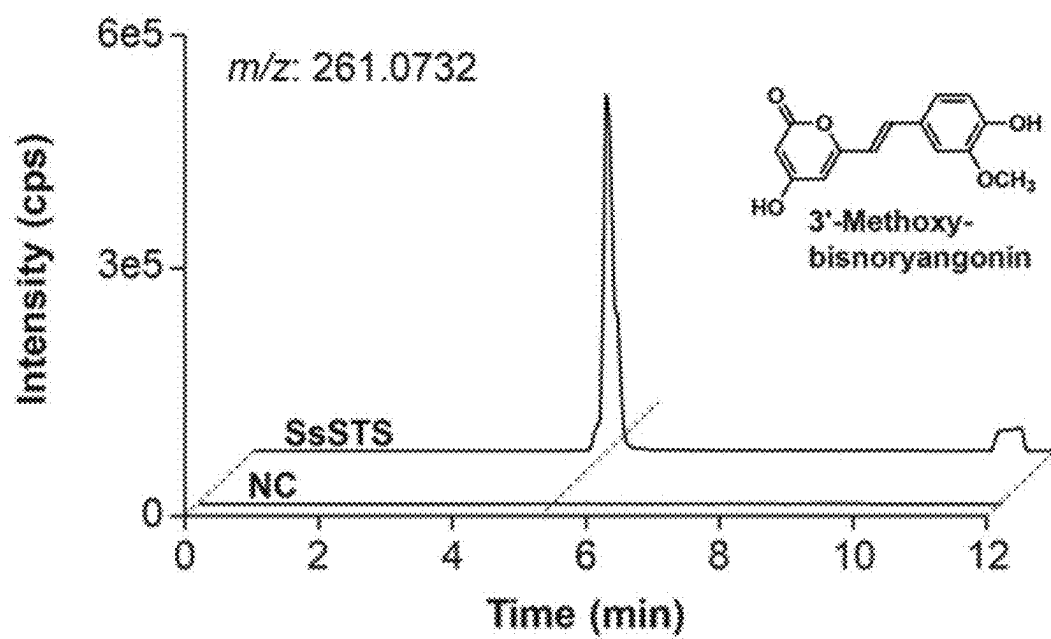


FIG. 22C

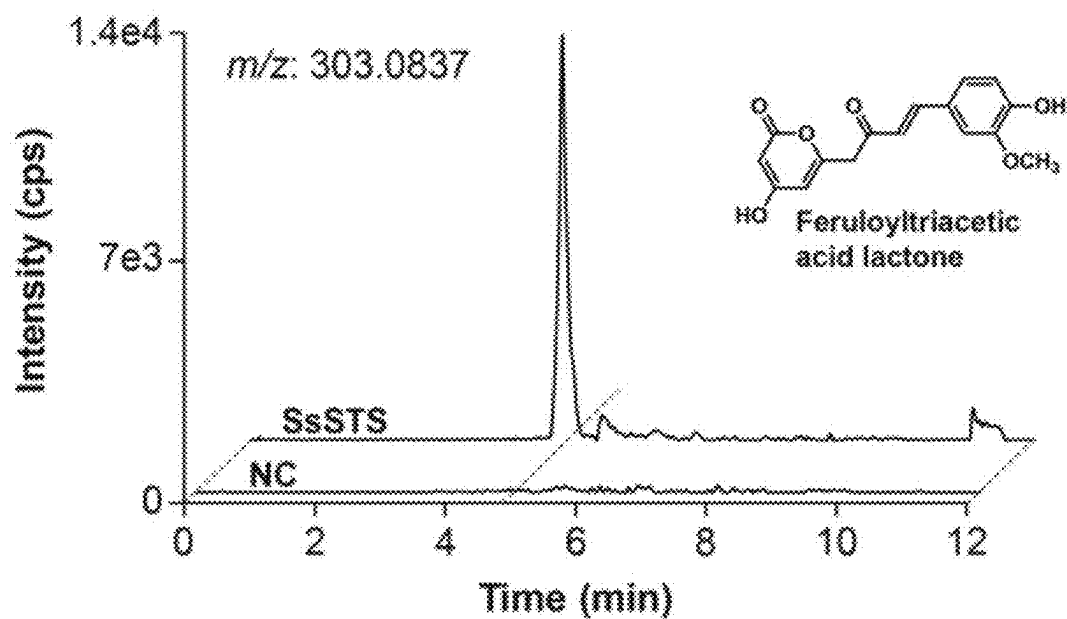


FIG. 22D

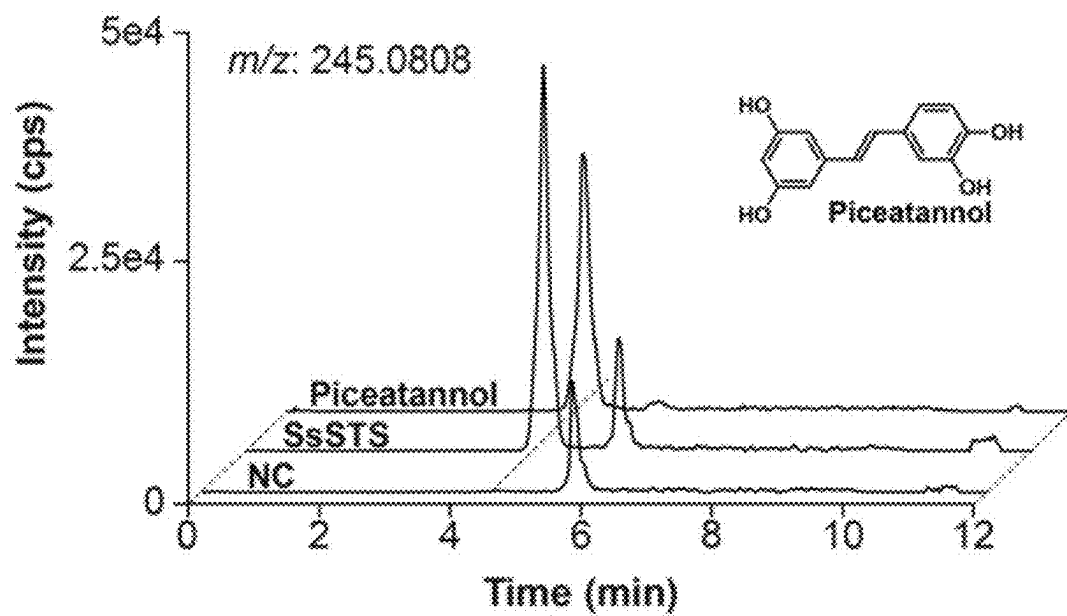


FIG. 22E

SbSOMT	MQSYDSSSSSSNDSSARNEEDESCMFALKLGGFAVPFTIRAVIELOVMDQLLTAE---R	57
SbCOMT	-----MOSTAEDVAAVADEEACMYAMQLASSSILPMTLRNALELOLLEVLQKD---AGE	51
SsCOMT	-----MOSTAEDVAAVADEEACMYAMQLASASILPMTLRNALELOLLEVLQAEAPAGK	53
	, : , * : (*):** : (* : , : (*):** ,)**** : : * :	
SbSOMT	AMSAEELVAAAVAAQLPRPEVACTMVDRLRLFLASHSVVRCCTTEVVGTDDAITTTTCRR	117
SbCOMT	ALAAEEVVARL--FVAPTNPAAADMVDRLRLRLASYDVVKCQMEDKDG-----KYER	101
SsCOMT	ALAPEEVVARL--FVAPNIPDAADMVDRLRLRLASYDVVKCQMEDKDG-----KYER	103
	* :) ** : ** * * , **** : (** : ** : , ** : * * * , *	
SbSOMT	SYAASFVCKWFARNGVEDSVLFLGMLLNKTFLDSWQNTDAVLEGAAPFEKTYGMPMFE	177
SbCOMT	RYSAAPVGRWLTPTNEDGVSMALALNNQDKVLMESWYYLRDAVLDGGIPFNKAYOMTAFE	161
SsCOMT	RYSAAPVGRWLTPTNEDGVSMALTLNNQDKVLMESWYYLRDAVLDGGIPFNKAYOMTAFE	163
	* : (** : ** : * * : * (* (* , (:)** (, **** : * , ** :)**** **	
SbSOMT	YLTNGPLNTVFHEAMAMHSEMIITKLLKFFRQFE-GLDVLVDVGGGNGTTLQMIHQQYE	236
SbCOMT	YHGTDFRFNRVFNESMKNHSEVIITKELLEFTYGFDESSTLVVDVGGGIGATLHAITSHE	221
SsCOMT	YHGTDFRFNRVFNESMKNHSEVIITKELLEFTYGF-GVSTLVVDVGGGIGATLHAITSHE	222
	* , * : (* ** : * , * **** : **** : * : ** : , : , , **** * : ** : * , : ,	
SbSOMT	NMRGINYDLPHVIAQAAPVEGVHVGGSMDNI PRGNVLLKWLHWDWDDKACIRILKNC	296
SbCOMT	HIROVWFDLPHVISEAPFFPGVQHVGGDMFKSVFAGDAILMKWILHWDSDAHCATLLKNC	281
SsCOMT	QIKGINFELPHVISEAPFFPGVQHVGGDMFKSVFAGDAILMKWILHWDSDAHCATLLKNC	282
	: : : ** : **** : * * , ** : **** , ** , : * * : ** : **** , * * , : ****	
SbSOMT	Y TALHVR-GKVIVLEYVVPDEPEPTLAAQGA FELDTMLVTFGSGKERTQREFSELAMEA	355
SbCOMT	YDALPEKGGKVIIVVECVLPVTTDAVFKAQGVFHVDMIMLAHNPGGREYEREFRLAKAA	341
SsCOMT	YDALPEN-GKVIIVECVLPVNTAEVFKAQGVFHVDMIMLAHNPGGREYEREFHDLAKGA	341
	* ** , **** : * * : : , ** , * , :) ** , , * : ** : *** (** *	
SbSOMT	GFSREFKATYIFANWALEFTK*	377
SbCOMT	GFS-GFKATYIYANAWAIEFIK*	362
SsCOMT	GFS-GFKATYIYANAWAIEFIK*	362
	*** **** : ** , ** : ** **	

FIG. 23

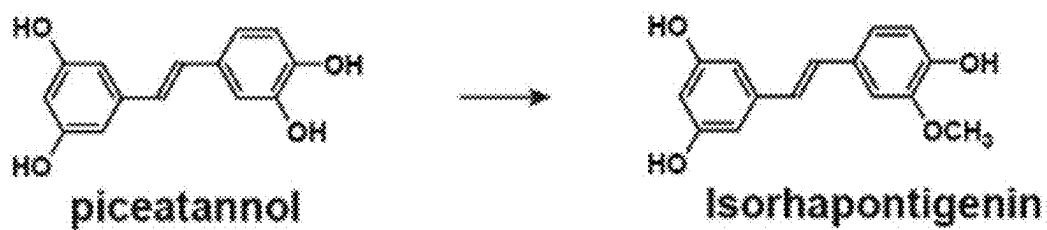


FIG. 24A

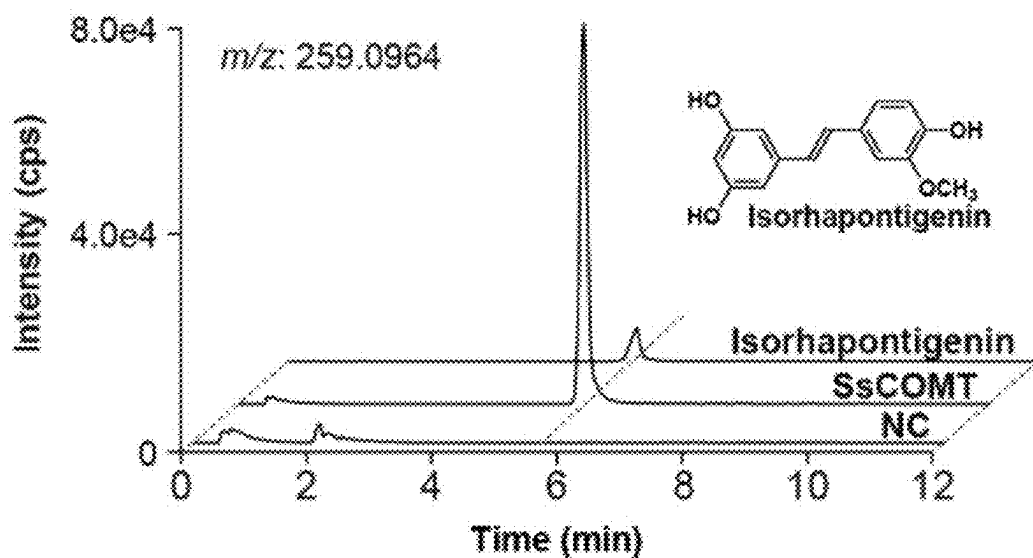


FIG. 24B

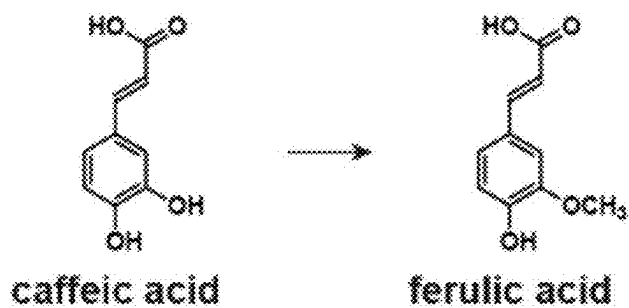


FIG. 24C

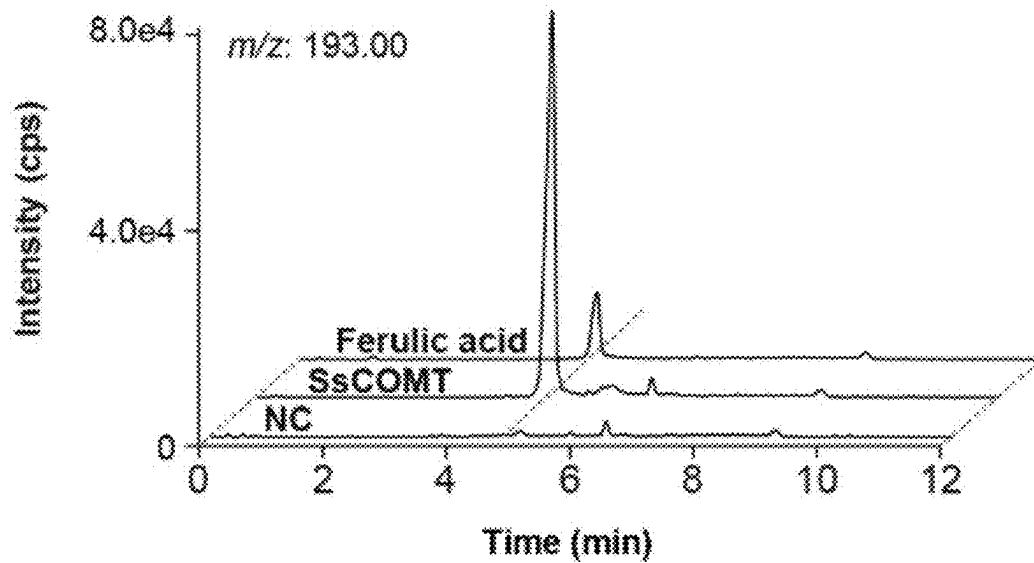


FIG. 24D

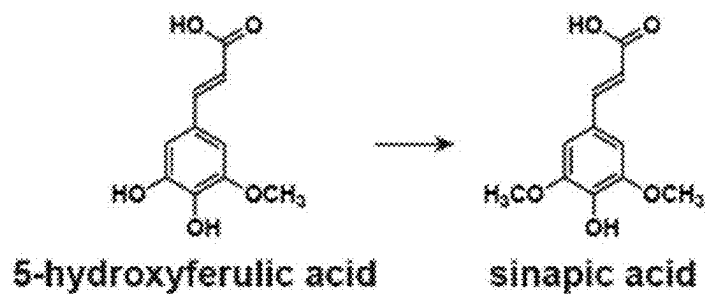


FIG. 24E

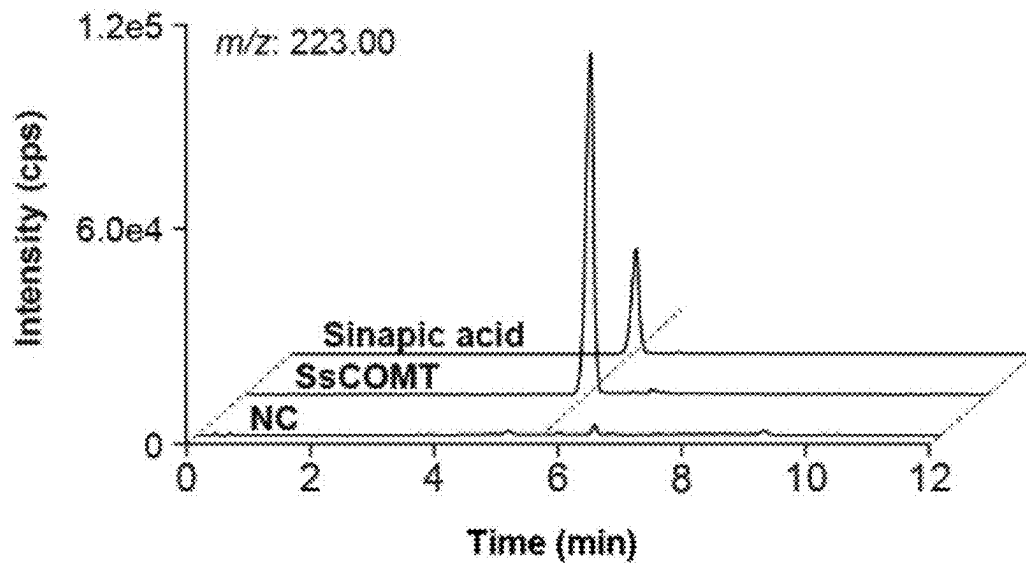


FIG. 24F

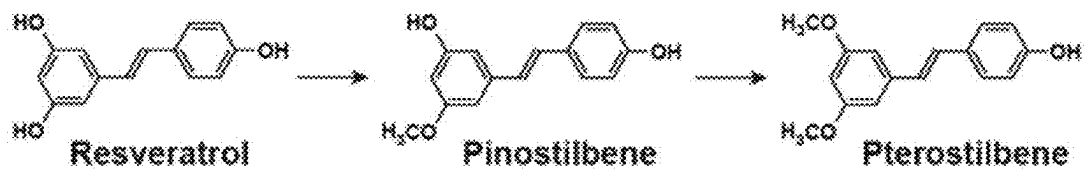


FIG. 24G

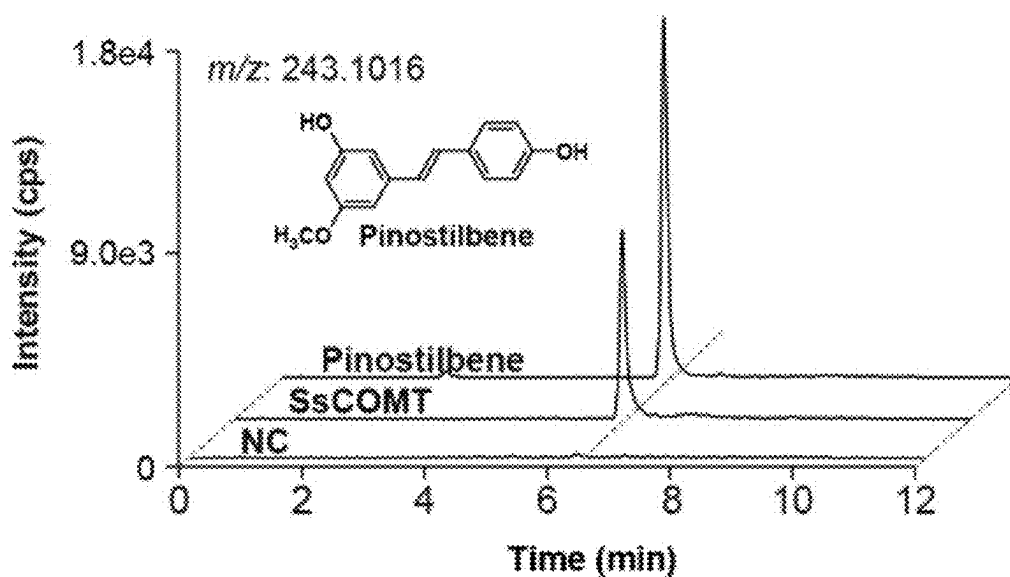


FIG. 24H

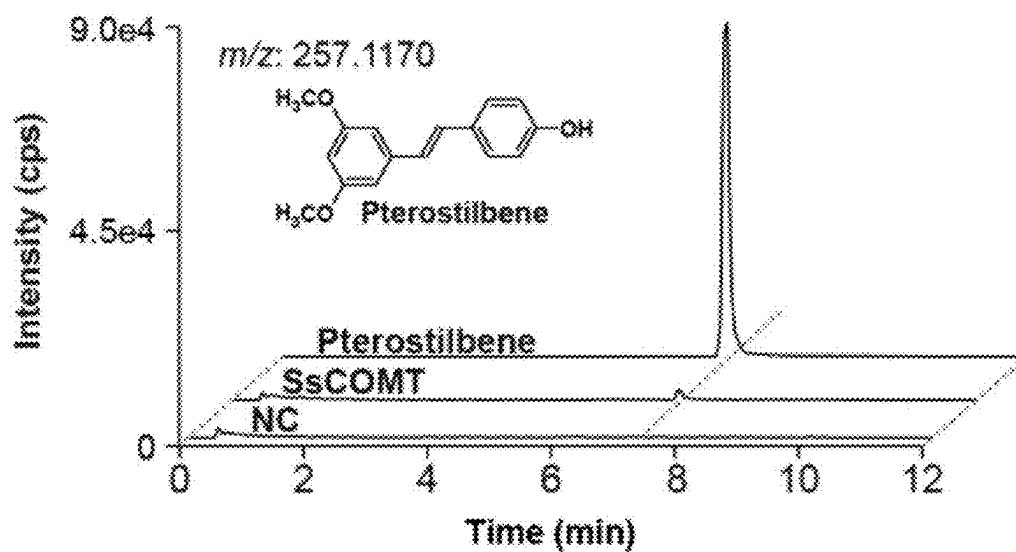


FIG. 24I

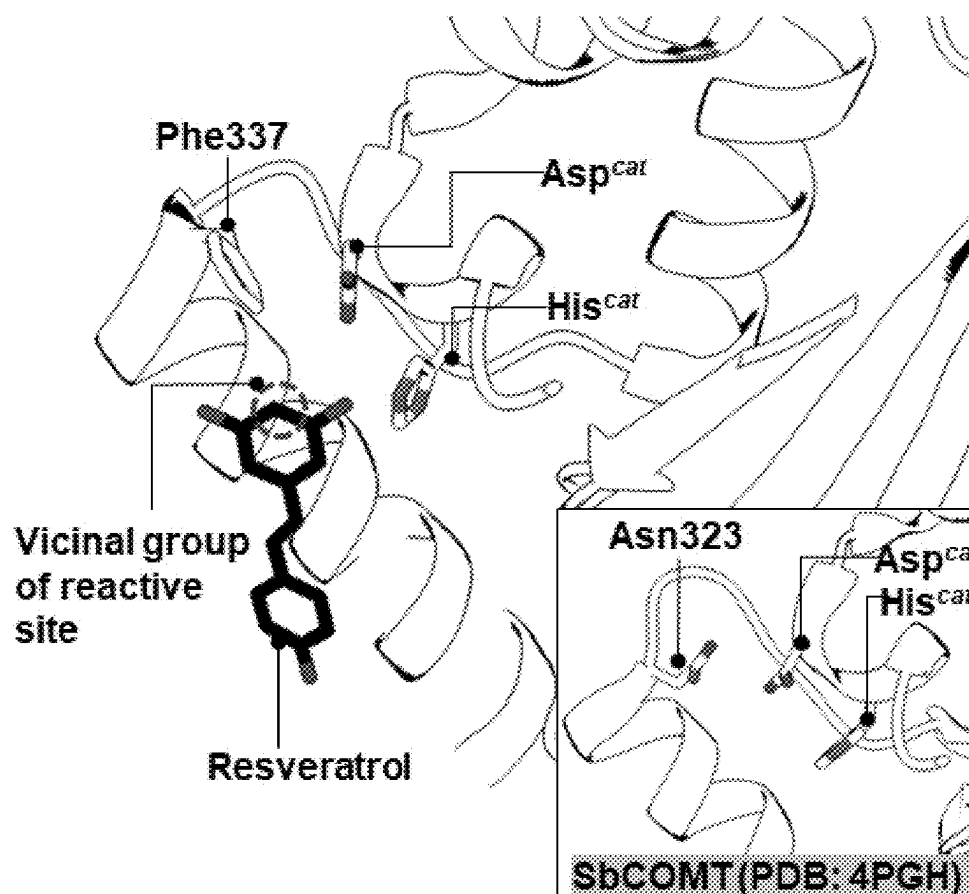


FIG. 25

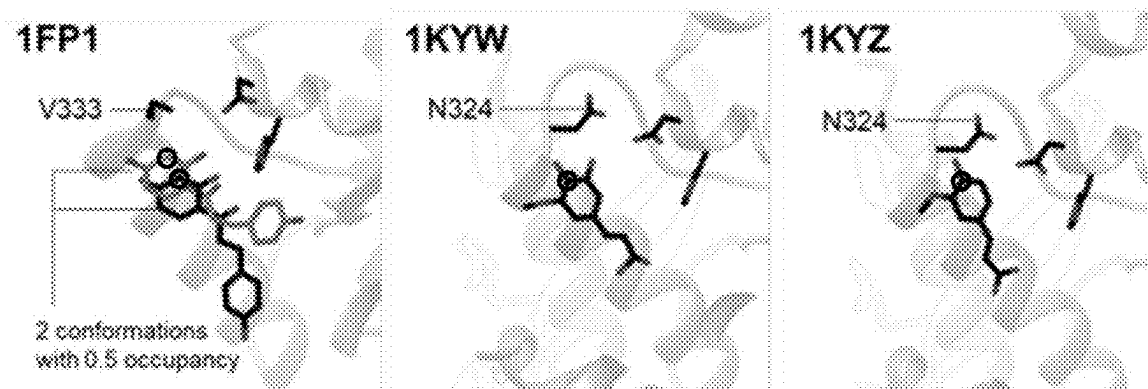


FIG. 26A

FIG. 26B

FIG. 26C

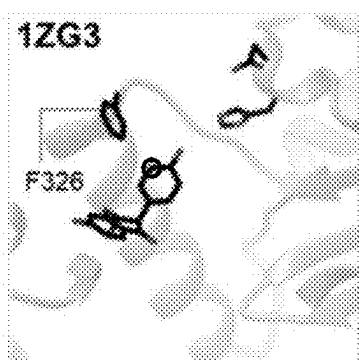


FIG. 26D

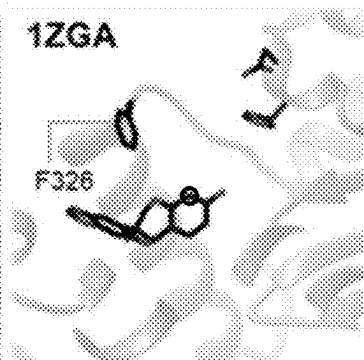


FIG. 26E

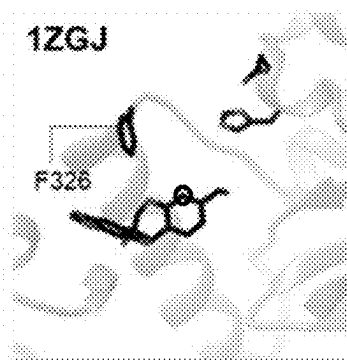


FIG. 26F

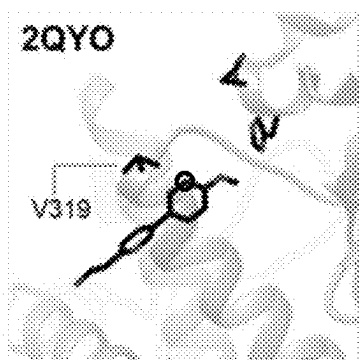


FIG. 26G

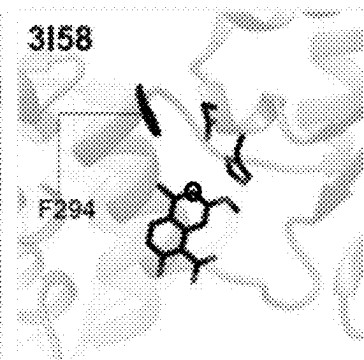


FIG. 26H

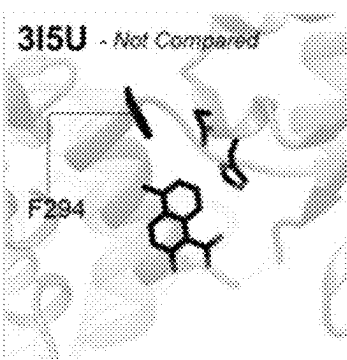


FIG. 26I

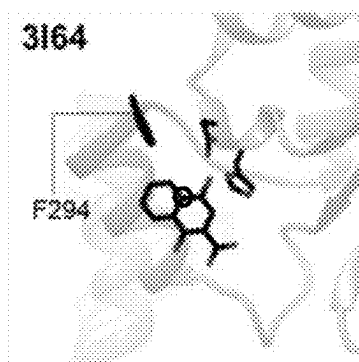


FIG. 26J

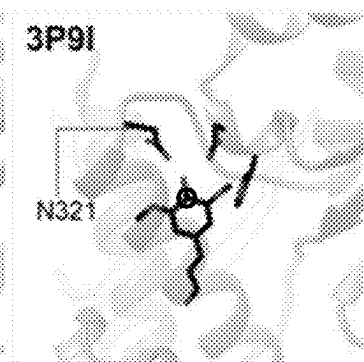


FIG. 26K

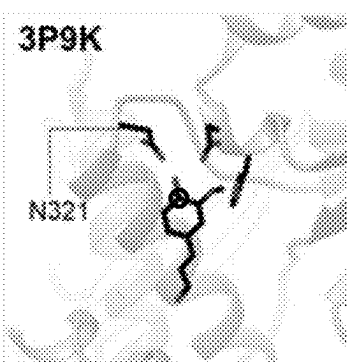


FIG. 26L

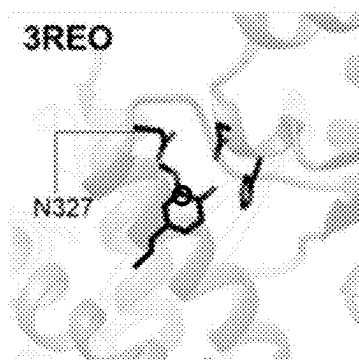


FIG. 26M

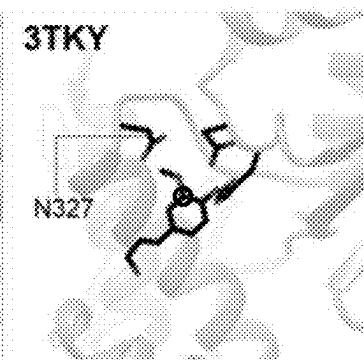


FIG. 26N

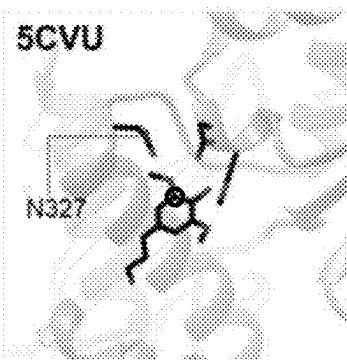


FIG. 26O

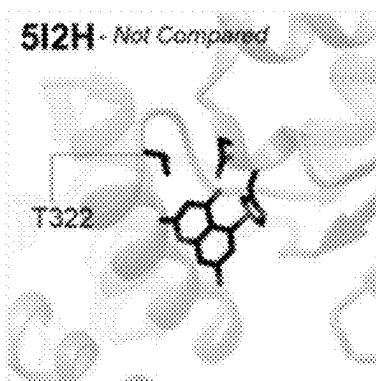


FIG. 26P

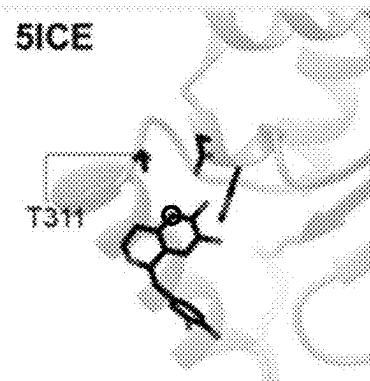


FIG. 26Q

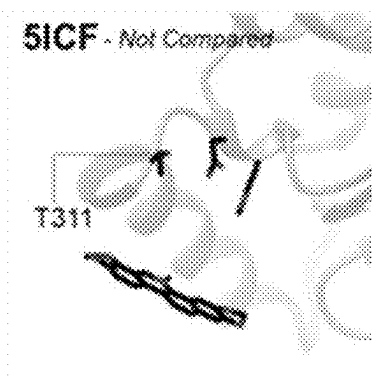


FIG. 26R

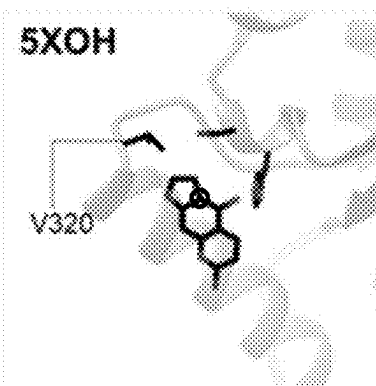


FIG. 26S

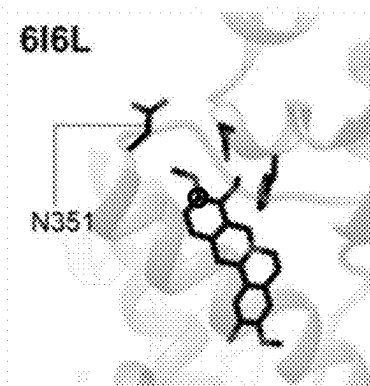


FIG. 26T

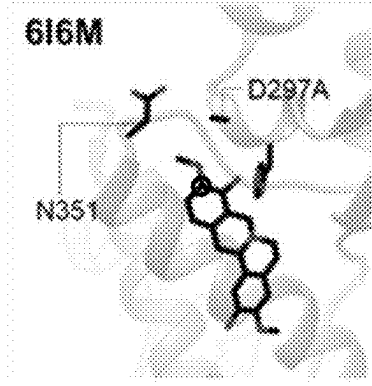


FIG. 26U

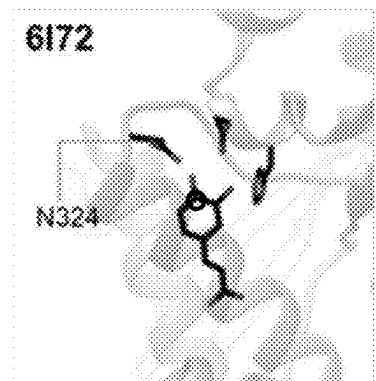


FIG. 26V

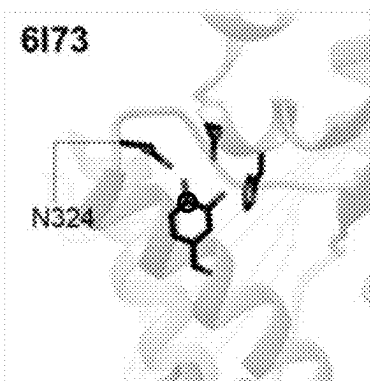


FIG. 26W

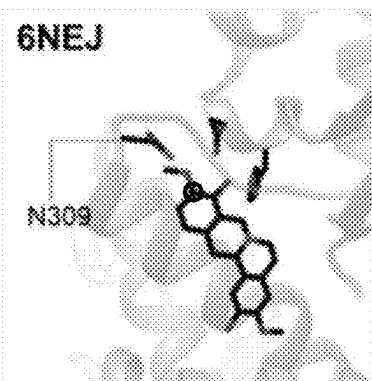


FIG. 26X

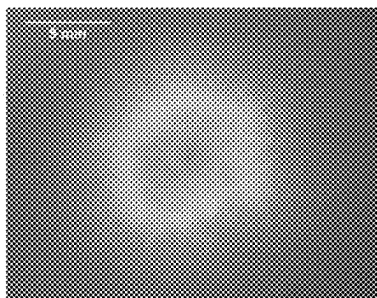


FIG. 27A

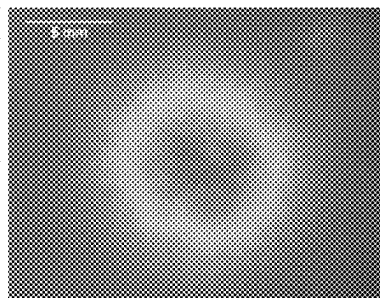


FIG. 27B

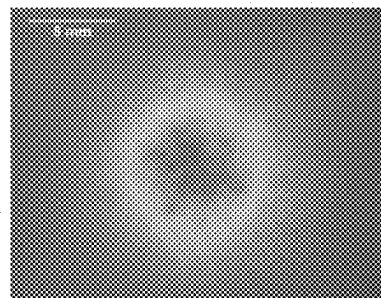


FIG. 27C

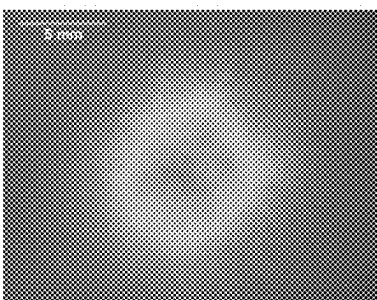


FIG. 27D

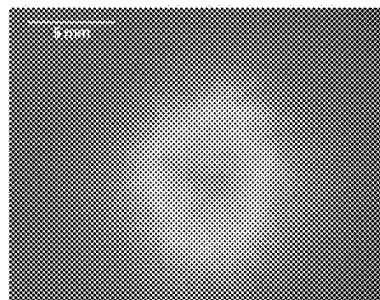


FIG. 27E

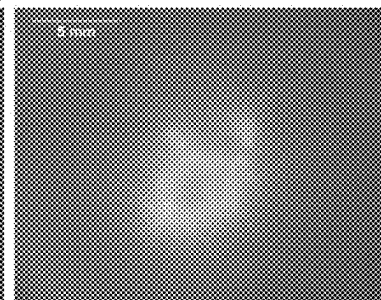


FIG. 27F

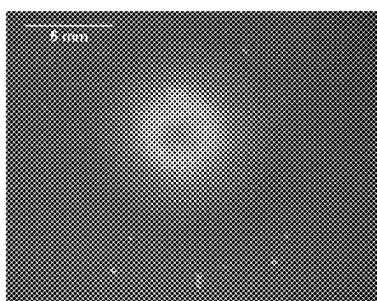


FIG. 27G

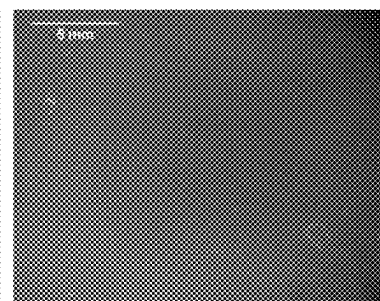


FIG. 27H

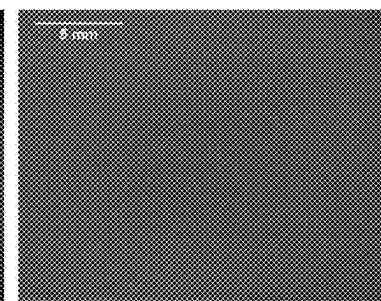


FIG. 27I

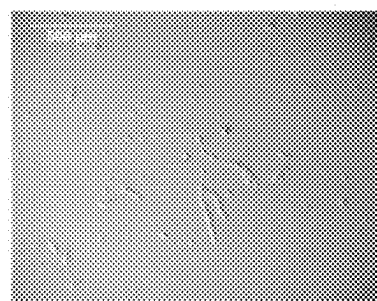


FIG. 27J

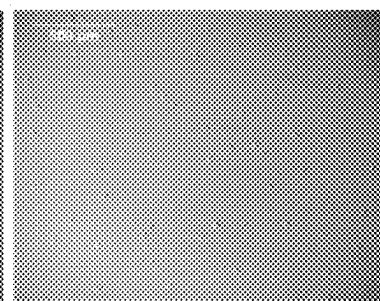


FIG. 27K

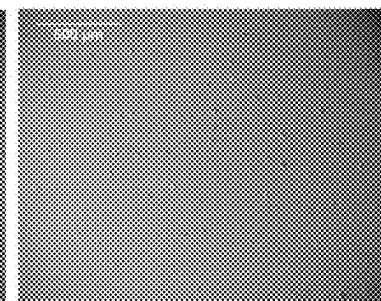


FIG. 27L

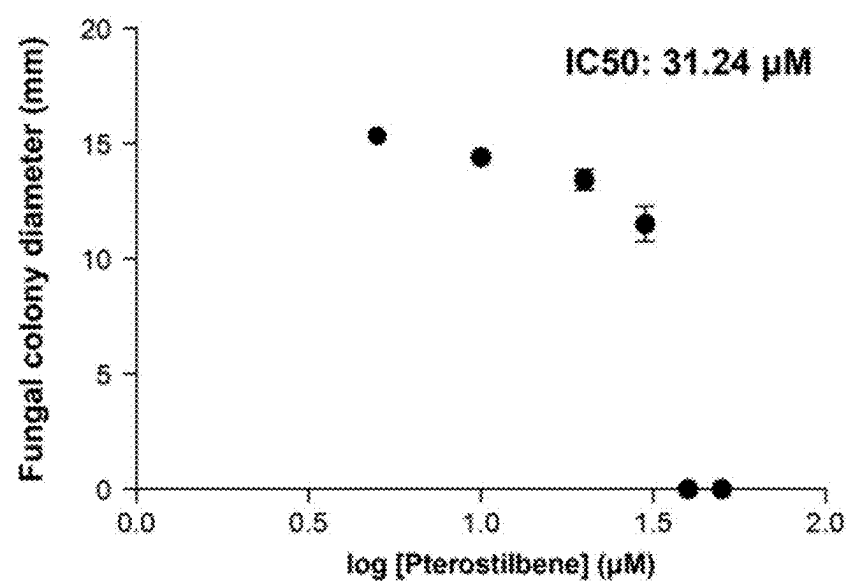


FIG. 27M

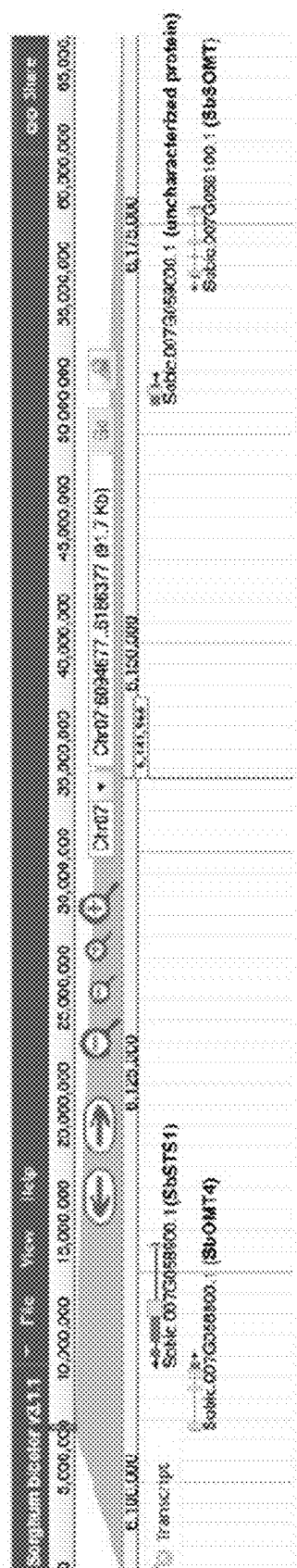


FIG. 28

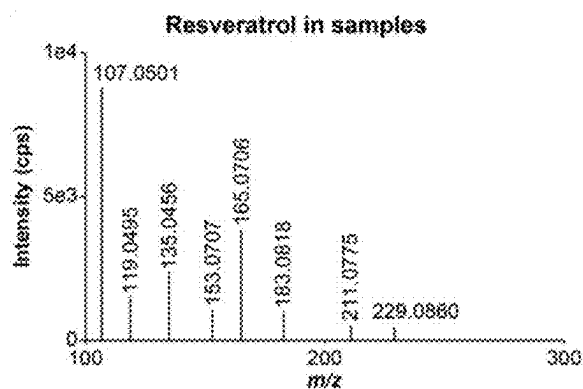


FIG. 29A

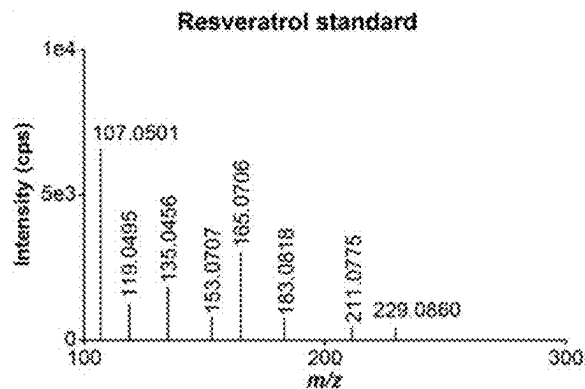


FIG. 29B

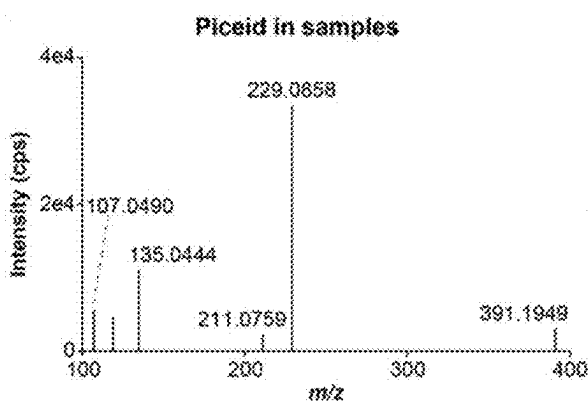


FIG. 29C

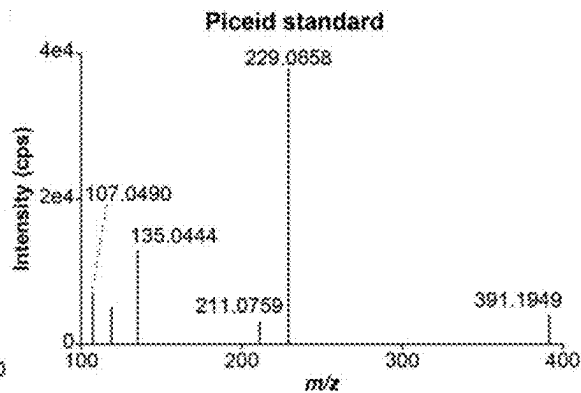


FIG. 29D

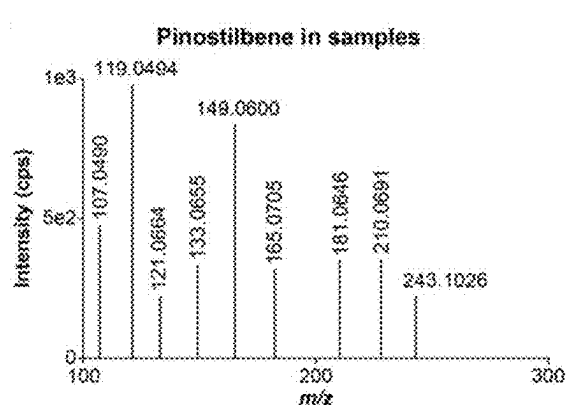


FIG. 29E

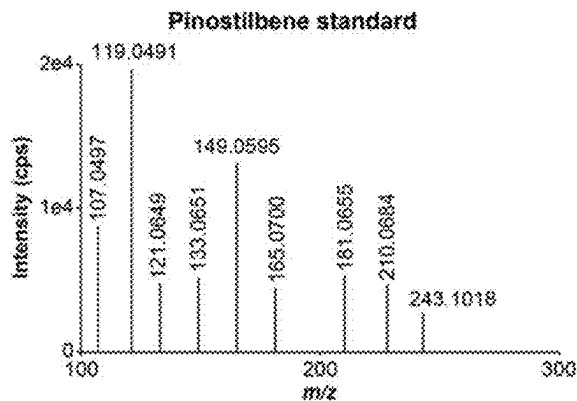


FIG. 29F

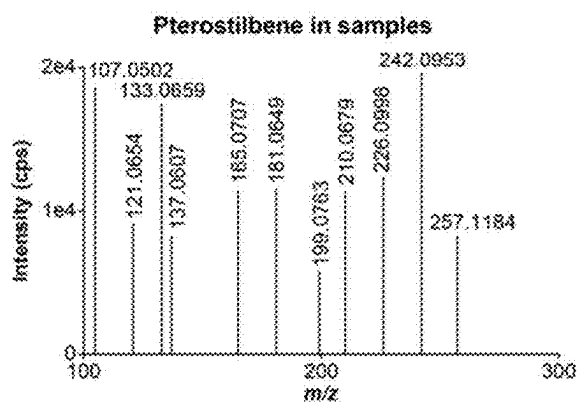


FIG. 29G

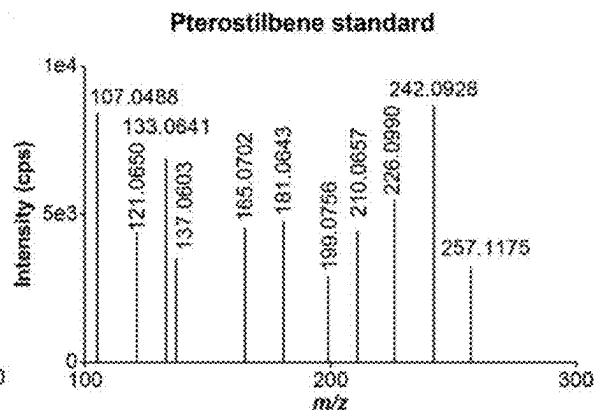


FIG. 29H

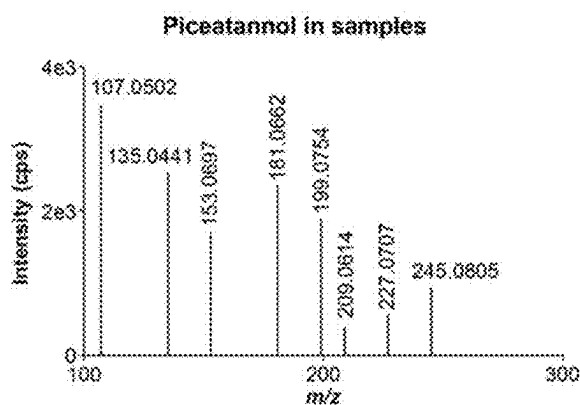


FIG. 29I

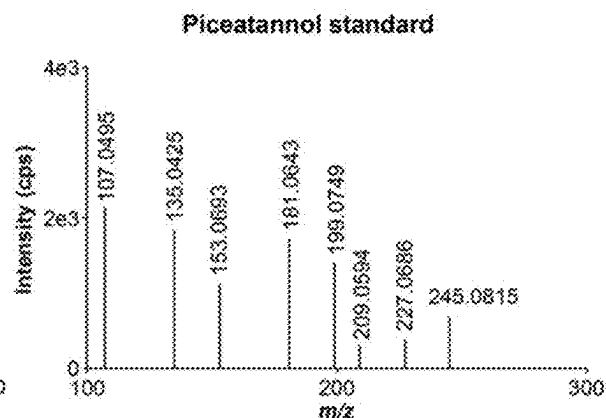


FIG. 29J

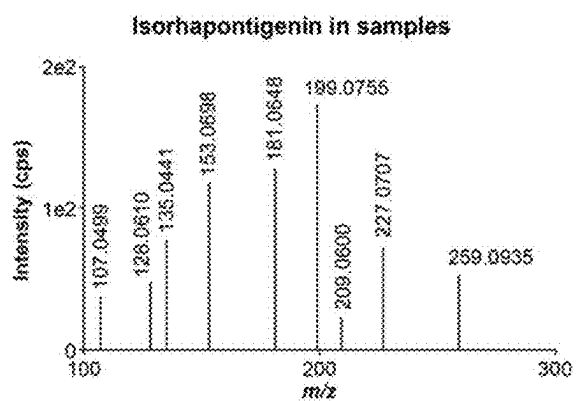


FIG. 29K

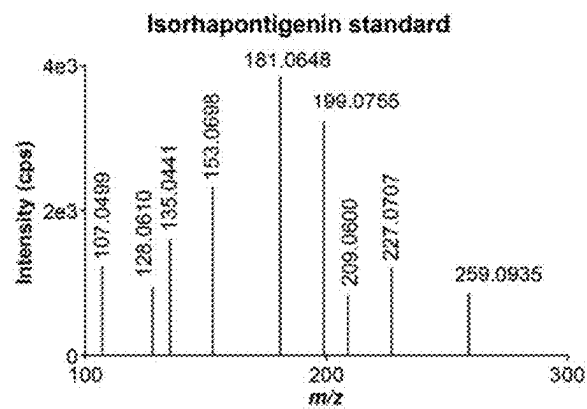


FIG. 29L

MODIFIED O-METHYLTRANSFERASES USEFUL FOR PRODUCTION OF O-METHYLATED PHENOLIC NATURAL PRODUCTS

REFERENCE TO SEQUENCE LISTING

[0001] The Sequence Listing XML submitted as a file named “UHK_01454_US_ST26.xml,” created on Mar. 6, 2024, and having a size of 109,775 bytes is hereby incorporated by reference pursuant to 37 C.F.R. § 1.834(c)(1).

FIELD OF THE INVENTION

[0002] The disclosed invention is generally in the field of O-methylation of natural products and specifically in the area of modified O-methyltransferases that methylate stilbenes, and related phenylpropanoids.

BACKGROUND OF THE INVENTION

[0003] Stilbenes are specialized metabolites sporadically distributed in the plant kingdom. Grape (*Vitis vinifera*, dicot), peanut (*Arachis hypogaea*, dicot), Scots pine (*Pinus sylvestris*, gymnosperm), *Sorghum* (*Sorghum bicolor*, monocot) and sugarcane (*Saccharum* spp., monocot) are iconic species producing various stilbene compounds upon (a)biotic challenges¹⁻⁵. They serve to maintain reactive oxygen species homeostasis and/or protect against microbial attacks³⁻⁶⁻⁸. Recently, stilbenes have emerged as prominent candidates for nutritional and pharmaceutical studies due to their anti-aging, anti-neurodegeneration, anti-diabetes and chemo-prevention properties⁹⁻¹¹. Over 200 stilbene-related clinical trials (internet site clinicaltrials.gov/) have been launched and stilbenes like resveratrol and pterostilbene are commercialized as dietary supplements. Despite the heightened interest in this minor, yet unique, class of phenylpropanoids as pharmaceuticals and nutraceuticals, in planta stilbene biosynthesis remains understudied, especially the downstream derivatization steps such as O-methylations and prenylations. Elucidation of the underlying enzymology would expedite bioengineering attempts to fortify crops with more or novel stilbenes.

[0004] O-Methyltransferases (OMTs) are indispensable in a diversity of metabolic pathways and hence biophysiological processes. For instance, *Sorghum* SbOMT3 catalyzes the biosynthesis of sorgoleones which are allelochemicals exuded by root hairs to suppress competing weeds¹². Two maize OMTs regioselectively methylate flavonoid phytoalexins which inhibit *Fusarium* spp. infection¹³. The degree of O-methylations of anthocyanidin pigments affects floral and fruit colors in flowering plants¹⁴⁻¹⁵. In *Arabidopsis* (*Arabidopsis thaliana*), disruption of a tapetum-specific OMT (AtTSM1) which methylates polyamine conjugates adversely affects silique development and seed setting¹⁶. Additionally, the degree of phenylpropanoid O-methylation determines lignin composition in tracheophytes¹⁷. For example, disruption of caffeic acid O-methyltransferase (COMT) and/or caffeoyl-CoA O-methyltransferase (CCoA-OMT) substantially alters lignin content and composition, potentially affecting plant growth and development¹⁸⁻²⁰. Biochemical and structural characterization of these phenylpropanoid OMTs have provided insights into their substrate enzyme interactions and catalytic mechanisms²¹⁻²⁴, hence facilitating genetic manipulations of lignin composition in biomass crop^{25, 26}.

[0005] Phylogenetically-unrelated plant species have independently recruited specific OMTs for stilbene O-methylation. In response to environmental stresses, grapevine resveratrol OMT (VvROMT) methylates resveratrol to pterostilbene whereas Scots pine pinosylvin OMT (PsPMT2) methylates pinosylvin to pinosylvin monomethyl ether^{27, 28}. In addition, a root hair-specific sorgoleone O-methyltransferase from *Sorghum*, SbOMT3, methylates resveratrol to pterostilbene in vitro or in transgenic models, albeit at poor efficiency^{12, 29}. Synthetic biology approaches have been initiated to generate O-methylated stilbenes of pharmaceutical significance such as pterostilbene and 3'-hydroxypterostilbene^{30, 31}, which are critically acclaimed for their potent cancer-chemopreventive³² and neuroprotective properties³³. In fact, their pharmacological properties are superior to the non-O-methylated analogs^{34, 35}. However, the lack of stilbene O-methyltransferase (SOMT) structural data greatly limits our understanding of their distinctive substrate specificities, regioselectivities and catalytic efficiencies, impeding bioengineering efforts in the production of high-value and/or novel O-methylated stilbenes.

[0006] Any discussion of documents, acts, materials, devices, articles or the like which has been included in the present specification is not to be taken as an admission that any or all of these matters form part of the prior art base or were common general knowledge in the field relevant to the present disclosure as it existed before the priority date of each claim of this application.

BRIEF SUMMARY OF THE INVENTION

[0007] Disclosed are modified Type 1 O-methyltransferases (OMTs) and methods of using such modified OMTs. In some forms of the modified OMTs, the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid. In some forms of the modified OMTs, the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

[0008] In some forms, the substituting polar amino acid is independently Asn, Gln, Thr, or Ser. In some forms, the substituting polar amino acid is Asn. In some forms, the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala. In some forms, the substituting hydrophobic amino acid is Phe.

[0009] In some forms, the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCAldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCAldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SiCOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis*

vinifera VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIOMT, *Planctopirus limnophila* PIOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

[0010] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93. In some forms, the modified OMT has at least 70% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0011] In some forms, the modified OMT has at least 75% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0012] In some forms, the modified OMT has at least 80% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0013] In some forms, the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0014] In some forms, the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0015] In some forms, the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0016] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

[0017] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

[0018] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

[0019] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13,

14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

[0020] In some forms, the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

[0021] Also disclosed are solutions including a modified OMT as disclosed. In some forms, the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions. Also disclosed are plants that express a modified OMT as disclosed.

[0022] Also disclosed are methods of producing an O-methylated stilbenoid or hydroxycinnamic acid. In some forms, the method involves bringing into contact, in a solution, a modified OMT as disclosed, a substrate for the modified OMT, and S-adenosylmethionine. In general, the O-methylated stilbenoid or hydroxycinnamic acid is produced by incubating the solution under suitable conditions.

[0023] In some forms, the method involves growing that express a modified OMT as disclosed and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

[0024] Additional advantages of the disclosed method and compositions will be set forth in part in the description which follows, and in part will be understood from the description, or can be learned by practice of the disclosed method and compositions. The advantages of the disclosed method and compositions will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIGS. 1A-1F are schematic and plots showing biosynthesis of stilbenes and flavonoids in *Sorghum* and stilbene profiles of *Colletotrichum sublineola*-infected *Sorghum* mesocotyls. FIG. 1A illustrate biosynthetic pathways in stilbene and flavonoid in *Sorghum*. Newly formed methoxy groups are enlarged. STS, stilbene synthase; SOMT, stilbene O-methyltransferase; COMT, caffeic acid O-methyltransferase; CHS, chalcone synthase; CHI, chalcone isomerase; FNR, flavanone 4-reductase; DFR, dihydroflavonol 4-reductase; FNSII, flavone synthase II; Sb, *Sorghum bicolor*. R=H, OH or OCH₃. FIG. 1B is a plot showing distribution of stilbene aglycones in *C. sublineola*-infected mesocotyls of *Sorghum* genotypes SC748-5 (resistant) and BTx623 (susceptible) 96 h after treatment. Values refer to means±SD (n=3). FIGS. 1C-1F are plots showing HPLC-QTOF-HRMS detection of resveratrol (FIG. 1C), piceid (FIG. 1D), pinostilbene (FIG. 1E) and pterostilbene (FIG. 1F) in *C. sublineola*-infected mesocotyls of *Sorghum* genotypes SC748-5 (resistant) and BTx623 (susceptible). inf, infected; ctrl, control; cps, counts per second.

[0026] FIGS. 2A-2D are plots showing catalytic activities and phylogeny of SbSOMT and SbCOMT. FIG. 2A is a pair of plot showing formation of pinostilbene and pterostilbene from resveratrol by SbSOMT and SbCOMT after two-hour incubation with 100 μM resveratrol. Values refer to means±SD (n=3). n.d.: not detected. Dots represent individual data points. FIG. 2B is plot showing formation of

ferulic acid from caffeic acid by SbSOMT and SbCOMT after two-hour incubation with 100 μ M phenylpropanoid substrates. Values refer to means $\pm$ SD (n=3). n.d.: not detected. Dots represent individual data points. FIG. 2C is a plot showing formation of sinapic acid from 5-hydroxyferulic acid by SbSOMT and SbCOMT after two-hour incubation. Values refer to means $\pm$ SD (n=3). n.d.: not detected. FIG. 2D demonstrates phylogenetic analysis of SbSOMT and SbCOMT. The unrooted phylogenetic tree was constructed by maximum likelihood using MEGA X⁹⁰. Bootstrapping with 1,000 replications was carried out. Bona fide stilbene O-methyltransferases from gymnosperm²⁷, dicot²⁸ and monocot (this study) are highlighted. *Sorghum* OMTs are bolded. Scale bar denotes 0.2 amino acid substitution per site.

[0027] FIGS. 3A-3E are plots showing genotypes, phenotypes and metabolite analysis of *Sorghum* sbsoMt mutants. FIG. 3A is a schematic showing genotypes of *Sorghum* sbsoMt mutants. Nucleotides corresponding to the proto-spacer adjacent motif (PAM) are bolded. CRISPR/Cas9-mediated mutations in sbsoMt mutant lines are underlined. Sequences are SEQ ID NO:94 (WT, exon 1), SEQ ID NO:95 (WT, exon 2), SEQ ID NO:96 (sbsoMt-a, exon 1), SEQ ID NO:97 (sbsoMt-a, exon 2), SEQ ID NO:98 (sbsoMt-b1, exon 1), SEQ ID NO:99 (sbsoMt-b1, exon 2), SEQ ID NO:100 (sbsoMt-b2, exon 1), and SEQ ID NO:101 (sbsoMt-b2, exon 2). FIGS. 3B-3E are plots showing HPLC-QTOF-HRMS detection of resveratrol (FIG. 3B), piceid (FIG. 3C), pinostilbene (FIG. 3D) and pterostilbene (FIG. 3E) in *C. sublineola*-infected mesocotyls of *Sorghum* sbsoMt mutants after 96 h. inf, infected; ctrl, control; cps, counts per second.

[0028] FIGS. 4A-4F demonstrate regioselective O-methylation of piceatannol and three-dimensional structure of SbSOMT and SbCOMT. FIG. 4A show structural comparison between resveratrol, caffeic acid, and piceatannol. Structural similarities are colored accordingly (pink: resveratrol-piceatannol; blue: caffeic acid-piceatannol). FIG. 4B show schematic of regioselective O-methylation of piceatannol catalyzed by SbSOMT (in pink) and SbCOMT (in blue). FIG. 4C Formation of different A- and B-ring O-methylated products from piceatannol by SbSOMT and SbCOMT after two-hour reaction with 100 μ M stilbenes. Values refer to means $\pm$ SD (n=3). n.d.: not detected. Dots represent individual data points. FIG. 4D represents a global structure of dimeric SbSOMT-resveratrol- β -NAD ternary complex solved at 1.72 Å resolution. In which, the two protomers are colored in white and grey, respectively. FIG. 4E highlights the functional domains, namely dimerization domain, substrate binding pocket and SAM-binding domain of a SbSOMT protomer and position of co-crystallized resveratrol (in black) bound to the substrate binding pocket of SbSOMT. FIG. 4F superimposed SbSOMT (in white) and SbCOMT (in grey; retrieved from PDB: 4PGH²¹) to show the high structural similarity between SbSOMT and SbCOMT (RMSD=2.322 Å) and the elongated threonine-rich loop (Val103-Cys114) unique to SbSOMT.

[0029] FIGS. 5A-5F are structural models showing interactions between stilbenes and SbSOMT catalytic residues, and docking positions of stilbenes with SbSOMT and SbCOMT. FIG. 5A represents a close-up view of SbSOMT substrate binding pocket revealed resveratrol conformation and resveratrol-interacting residues and water. Dashed line indicates hydrogen bond formation with the distance labelled. FIGS. 5B-5D are structural models that illustrate

close-up view on interactions between SbSOMT catalytic residues (His282, Asp283, Glu310 and Glu342) and resveratrol (FIG. 5B), pinostilbene (FIG. 5C), and pterostilbene (FIG. 5D). Dashed line indicates hydrogen bond formation with the distance labelled. FIG. 5E shows the docking conformation of piceatannol ligand (in black) in SbSOMT (left) and SbCOMT (right) along with the surrounding residues-of-interest and calculated binding energy. Dashed line indicates hydrogen bond formation with the distance labelled. FIG. 5F shows the two docking conformations of resveratrol in SbCOMT along with the surrounding residues-of-interest and calculated binding energy. The carbon backbone of non-productive conformation is coloured in black, and the productive conformation is coloured in grey, respectively. Dashed line indicates hydrogen bond formation with the distance labelled.

[0030] FIGS. 6A-6D are plots demonstrating validation of catalytic residues and key substrate-orientating residues of SbSOMT via site-directed mutagenesis. FIGS. 6A-6B are a pair of plots showing formation of pinostilbene (FIG. 6A) and pterostilbene (FIG. 6B) by catalytic residue mutant proteins over a one-hour incubation period. Data are expressed as yields relative to that of wild-type SbSOMT protein (WT). FIGS. 6C-6D are a pair of plots showing formation of pinostilbene (FIG. 6C) and pterostilbene (FIG. 6D) by key substrate-orientating residue mutant proteins over a one-hour incubation period. Data are expressed as yields relative to that of wild-type SbSOMT protein (WT). Values refer to means $\pm$ SD (n=3). Asterisks indicate significant differences between mutant protein and WT (Student's t-test, ***, p<0.001). Tricin was used as an internal standard for quantitation. Note that pterostilbene represents the major product for SbSOMT(WT)-resveratrol reactions at the 2-hour end point (FIG. 2A). n.d.: not detected. Dots represent individual data points.

[0031] FIGS. 7A-7C illustrate biosynthesis of stilbenes in wild sugarcane, stilbene profiles of mechanically-wounded wild sugarcane stalks, and in vitro enzyme activities of SsCOMT. FIG. 7A is a schematic showing proposed stilbene pathways in wild sugarcane. Newly formed methoxy groups are colored and bolded. STS, stilbene synthase; COMT: caffeic acid O-methyltransferase; Ss, *Saccharum spontaneum*. FIG. 7B is a plot showing distribution of stilbene aglycones in mechanically-wounded wild sugarcane stalks 120 h after treatment. Values refer to means $\pm$ SD (n=3). FIG. 7C is a plot showing formation of ferulic acid, sinapic acid, isorhapontigenin, pinostilbene, and pterostilbene from caffeic acid, 5-hydroxyferulic acid, piceatannol, and resveratrol respectively, by SsCOMT, after two-hour incubation with 100 μ M substrates. Tricin was used as an internal standard for quantitation. Values refer to means $\pm$ SD (n=3).

[0032] FIGS. 8A-8D are plots showing expression analysis of pterostilbene biosynthesis related candidate genes in *Sorghum*. FIG. 8A shows analysis of in silico gene expression. RNA-sequencing datasets derived from *Bipolaris sorghicola*-infected *Sorghum* leaves and *Colletotrichum sublineola*-infected *Sorghum* mesocotyls were obtained from literature^{40, 41}. FIGS. 8B-8D are plots showing quantitative RT-PCR gene expression analysis of SbSTS1 (FIG. 8B), SbOMT4 (FIG. 8C) and SbSOMT (FIG. 8D). Expression levels are expressed relative to *Sorghum bicolor* Eukaryotic Initiation Factor 4A-1 (SbEIF4 α)⁶⁸. Values refer to means $\pm$ SD (n=3). inf, infected; ctrl, control.

[0033] FIGS. 9A-9C demonstrate initial screening of catalytic activities of *Sorghum* OMTs. FIG. 9A represents chemical structures showing in vitro O-methylation of resveratrol by *Sorghum* OMTs. FIGS. 9B-9C are a pair of plots showing HPLC-QTOF-HRMS detection of pinostilbene (FIG. 9B), and pterostilbene (FIG. 9C) in *Sorghum* OMTs catalyzed reaction using resveratrol as a substrate. cps, counts per second; NC, negative control (reaction without enzyme).

[0034] FIGS. 10A-10D demonstrate co-overexpression of SbSTS1 and SbSOMT in *Nicotiana benthamiana* leaves. FIG. 10A is a schematic showing in planta generation of pterostilbene by *Sorghum* SbSTS1 and SbSOMT. FIGS. 10B-10D are plots showing HPLC-QTOF-HRMS detection of resveratrol (FIG. 10B), pinostilbene (FIG. 10C), and pterostilbene (FIG. 10D) in *Nicotiana benthamiana* leaves transiently expressing SbSTS1 or co-expressing SbSTS1 and SbSOMT. Cps, counts per second.

[0035] FIG. 11A-11F are plots showing genotyping of *Sorghum* sbSOMT mutants. FIGS. 11A-11F are representative chromatographs of direct sequencing of the target sites in the sbSOMT-a (FIGS. 11A-11B), sbSOMT-b1 (FIGS. 11C-11D), and sbSOMT-b2 (FIGS. 11E-11F) mutant lines (T_1 generation) are shown. CRISPR/Cas9-mediated mutations in sbSOMT mutant lines are underlined. Bold, protospacer adjacent motif (PAM) site. Sequences are SEQ ID NO:94 (reference, target 1), SEQ ID NO:95 (reference, target 2), SEQ ID NO:96 (sbSOMT-a, target 1), SEQ ID NO:97 (sbSOMT-a, target 2), SEQ ID NO:98 (sbSOMT-b1, target 1), SEQ ID NO:99 (sbSOMT-b1, target 2), SEQ ID NO:100 (sbSOMT-b2, target 1), and SEQ ID NO:101 (sbSOMT-b2, target 2).

[0036] FIG. 12 demonstrates predicted effects of mutations in SbSOMT in *Sorghum* sbSOMT mutant lines. The amino acid sequences of wild-type and mutant SbSOMT were aligned using Clustal Omega⁸⁸. First premature stop codons (*) are underlined and bolded. The catalytic residues of *Sorghum* SbSOMT are underlined and bolded. Amino acid residues contributing to the hydrophobic interactions with stilbene substrates are labelled with t above the corresponding residues. Sequences are SEQ ID NO:1 (WT), SEQ ID NO:102 (sbSOMT-a), SEQ ID NO:103 (sbSOMT-b1), and SEQ ID NO:104 (sbSOMT-b2).

[0037] FIG. 13 demonstrates multiple sequence alignment of bonafide stilbene O-methyltransferases (SOMTs). The amino acid sequence of SbSOMT, VvROMT and PsPMT2 were aligned using Clustal Omega⁸⁸. The catalytic residues of *Sorghum* SbSOMT are underlined and bolded, while those of other SOMTs are only underlined. Amino acid residues contributing to the hydrophobic interactions with stilbene substrates are labelled with t above the corresponding residue. Fully conserved amino acid residues across all three SOMTs are indicated by asterisks; those with strongly similar properties are indicated with colons; those with weakly similar properties are indicated with periods. Sequences are SEQ ID NO:1 (SbSOMT), SEQ ID NO:27 (VvROMT), and SEQ ID NO:14 (PsPMT2).

[0038] FIG. 14 is a schematic showing phylogeny of wild sugarcane OMTs. Phylogenetic analysis of wild sugarcane OMTs. The unrooted phylogenetic tree was constructed by maximum likelihood using MEGA X⁹⁰. Bootstrapping with 1,000 replications was carried out. Scale bar denotes 0.5 amino acid substitution per site. The sequences of the noted genes are hereby incorporated by reference.

[0039] FIGS. 15A-15X are titration thermograms and fitting curves of conducted ITCs. Each set of thermogram (above) and fitting curve (below) represents performed ITC of different combinations of protein (in cell) and ligand (in syringe) as titled.

[0040] FIGS. 16A-16C are structures obtained in this study. FIG. 16A is superimposed SbSOMT-resveratrol complex with two SbSOMT-stilbene- β -NAD complexes obtained in this study towards SbSOMT-resveratrol- β -NAD (top left). SbSOMT-resveratrol complex (top right; RMSD=0.468 Å), SbSOMT-pinostilbene- β -NAD, (bottom left; RMSD=0.995 Å) and SbSOMT-pterostilbene- β -NAD (bottom right; RMSD=0.730 Å) complexes all shares high structural similarity towards SbSOMT-resveratrol- β -NAD. FIG. 16B is a close-up view focusing the conformations of superimposed stilbene derivatives (in black) and correspondent surrounding residues. Leu28 (in grey) is constituted in adjacent protomer. FIG. 16C illustrate the FO-FC electron density omit maps of resveratrol and β -NAD ligands in SbSOMT-resveratrol- β -NAD ternary complex contoured at 3σ (mesh) and -3σ (blob).

[0041] FIG. 17 is a schematic showing reaction mechanism of consecutive stilbene O-methylations mediated by SbSOMT. Schematic of SbSOMT O-methylation cycle exemplified with S-adenosyl methionine (SAM, donor) and resveratrol (acceptor)^{21, 22, 43, 44}. Upon binding of SAM and resveratrol (Step I), the surrounding electrostatic interactions allow His282 N_ϵ to deprotonate 3-OH of resveratrol. Step II, deprotonated O^- catalyzes SN_2 attack towards methyl group of SAM sulfonium ylide to produce pinostilbene and S-adenosyl-homocysteine (SAH). Step III, the His282 protonated imidazole is regenerated by solvent and returns to the neutral state. Step IV, the SAH and pinostilbene are dissociated from SbSOMT.

[0042] FIGS. 18A-18C are plots showing stilbene profiles of mechanically-wounded wild sugarcane stalks. FIGS. 18A-18C demonstrate HPLC-QTOF-HRMS detection of resveratrol (FIG. 18A), piceatannol (FIG. 18B), and isorha-pontigenin (FIG. 18C) in mechanically-wounded wild sugarcane stalks 120 h after treatment. Cps, counts per second.

[0043] FIGS. 19A-19B are a pair of plots showing tentative identification of piceatannol-hexoside from mechanically wounded wild sugarcane stalks. FIG. 19A demonstrates HPLC-QTOF-HRMS detection of putative piceatannol-hexoside. FIG. 19B demonstrates MS² fragmentation pattern of putative piceatannol-hexoside. A neutral loss of 162 Da, potentially corresponding to an anhydro-hexose unit, was observed for this ion. Glc: glucose; cps, counts per second.

[0044] FIGS. 20A-20D are plots showing expression analysis of stilbene biosynthesis related genes in wild sugarcane. FIGS. 20A-B are a pair of plots showing quantitative RT-PCR gene expression analysis of SsSTS (FIG. 20A) and SsCOMT (FIG. 20B). Expression levels are expressed relative to *Saccharum spontaneum* Glyceraldehyde 3-Phosphate Dehydrogenase (SsGADPH)⁶⁸. Values refer to means $\pm$ SD (n=5). FIGS. 20C-20D are semi-quantitative RT-PCR gene expression analysis of SsSTS (FIG. 20C) and SsCOMT (FIG. 20D) relative to SsGADPH. Thirty cycles were used for amplification. Two biological duplicates from each time point were analyzed.

[0045] FIG. 21 represents multiple sequence alignment of SbSTS1 and SsSTS. The amino acid sequences of SbSTS1 and SsSTS were aligned using Clustal Omega⁸⁸. Fully

conserved amino acid residues are indicated by asterisks; those with strongly similar properties are indicated with colons; those with weakly similar properties are indicated with periods. The sequences are SEQ ID NO:23 (SbSTS1) and SEQ ID NO:33 (SsSTS).

[0046] FIGS. 22A-22E demonstrate catalytic activities of SsSTS. FIG. 22A is a schematic showing proposed STS reaction with feruloyl-CoA and malonyl-CoA as substrates. FIGS. 22B-22D are plots showing HPLC-QTOF-HRMS detection of isorhapontigenin (FIG. 22B), 3'-methoxybisnorangonin (FIG. 22C), and feruloyltriacetic acid lactone (FIG. 22D) in SsSTS catalyzed reaction using feruloyl-CoA and malonyl-CoA as substrates. The detection of these derailment products is consistent with a previous report showing that feruloyl-CoA is a poor substrate for STSs, at least under *in vitro* conditions⁵. FIG. 22E is a plot showing HPLC-QTOF-HRMS detection of piceatannol in SsSTS catalyzed reaction using caffeoyl-CoA and malonyl-CoA as substrates. cps, counts per second; NC, negative control (reaction without enzyme).

[0047] FIG. 23 represents multiple sequence alignment of SbCOMT, SbSOMT, and SsCOMT. The amino acid sequences of SbCOMT, SbSOMT, and SsCOMT were aligned using Clustal Omega⁸⁸. Fully conserved amino acid residues are indicated by asterisks; those with strongly similar properties are indicated with colons; those with weakly similar properties are indicated with periods. Residues of SbSOMT and SsCOMT corresponding to SbCOMT Asn128 and Asn323 were underlined and bolded. Sequences SEQ ID NO:1 (SbSOMT), SEQ ID NO:2 (SbCOMT), and SEQ ID NO:32 (SsCOMT).

[0048] FIGS. 24A-24I show catalytic activities of SsCOMT. FIG. 24A represents *in vitro* O-methylation of piceatannol by SsCOMT. FIG. 24B is a plot showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of isorhapontigenin in SsCOMT catalyzed reaction using resveratrol as a substrate. FIG. 24C represents *in vitro* O-methylation of caffeic acid by SsCOMT. FIG. 24D is a plot showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of ferulic acid in SsCOMT catalyzed reaction using caffeic acid as a substrate. FIG. 24E represents *in vitro* O-methylation of 5-hydroxyferulic acid by SsCOMT. FIG. 24F is a plot showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of sinapic acid in SsCOMT catalyzed reaction using 5-hydroxyferulic acid as a substrate. FIG. 24G represents *in vitro* O-methylation of resveratrol by SsCOMT. FIGS. 24H-24I are a pair of plots showing high-performance liquid chromatography-tandem MS (HPLC-MS/MS) detection of pinostilbene (FIG. 24H), and pterostilbene (FIG. 24I) in SsCOMT catalyzed reaction using resveratrol as a substrate. cps, counts per second; NC, negative control (reaction without enzyme).

[0049] FIG. 25 is a structural model that explains the basis of polarity pairing via the representative SbSOMT-resveratrol interaction. The structure shows that at the proximity of catalytic dyad (Hiscat and Aspcat), the non-polar Phe337 of SbSOMT is closely positioned to non-polar C4 (the vicinal group of reactive site of resveratrol) and demonstrate the coordination of substrate orientation via polarity pairing. In SbCOMT context (bottom right box), the polar Asn323 is positioned at the equivalent location of Phe337 of SbSOMT. Hence, SbCOMT coordinates via the opposite polar-to-polar polarity pairing.

[0050] FIGS. 26A-26X are structural models showing polarity pairing as a common regioselective mechanism across OMT-ligand complexes deposited to PDB. Twenty-four unique OMT-ligand complexes were retrieved from PDB (by 10 Aug. 2022), and their respective amino acid residue equivalent to SbCOMT^{Asn323}/SbSOMT^{Phe337}, catalytic dyad, and bound ligand were shown. Homolog search was used to retrieve ($\geq 30\%$ identity to SbCOMT or SbSOMT) and screened for complexes that are bound to ligand at substrate binding pocket. Further details are listed in Table 10. Appropriate polarity pairing between this residue's side chain and the functional group vicinal to the reactive site (methyl-accepting —OH group) was observed in all complexes, except for the three structures excluded from the analysis employed herein, either due to the use of non-reactive ligand (3I5U and 5ICF) or non-productive OMT-ligand conformation (5I2H). The carbon vicinal to the reactive site is circled in black.

[0051] FIGS. 27A-27M demonstrate *in vitro* toxicity of stilbenes towards *Colletotrichum sublineola*. FIGS. 27A-27I are images showing mycelial growth of *C. sublineola* on potato dextrose agar after a 72-hour incubation with 4% (v/v) DMSO (FIG. 27A), 50 μ M resveratrol (FIG. 27B), 50 μ M pinostilbene (FIG. 27C), 5 μ M pterostilbene (FIG. 27D), 10 μ M pterostilbene (FIG. 27E), 20 μ M pterostilbene (FIG. 27F), 30 μ M pterostilbene (FIG. 27G), 40 μ M pterostilbene (FIG. 27H) and 50 μ M pterostilbene (FIG. 27I). Scale bars denote 5 mm. FIGS. 27J-27L are images that show spore germination of *C. sublineola* on potato dextrose agar after a 12-hour incubation in darkness with 4% (v/v) DMSO (FIG. 27J), 40 μ M pterostilbene (FIG. 27K) and 50 μ M pterostilbene (FIG. 27L). Scale bars denote 500 μ m. FIG. 27M is a plot showing diameter of *C. sublineola* colonies after a 72-hour incubation with different concentrations of pterostilbene. IC50 was estimated by GraphPad Prism 6. Values refer to means $\pm$ SD (n=9). IC50, half-maximal inhibitory concentration.

[0052] FIG. 28 is an image showing arrangement of SbSTS1 (Sb07g004700; Sobic.007G058900), SbOMT4 (Sb07g004690; Sobic.007G058800), and SbSOMT (Sb07g004710; Sobic.007G059100) in *Sorghum* genome. Gene arrangement in Chr07:6094677 . . . 6186377 of *Sorghum* genome was obtained from Phytozome v13⁶⁵.

[0053] FIGS. 29A-29L are plots showing MSMS spectrum of stilbenes detected in plant samples and authentic standards. FIGS. 29A-29B are representative MSMS spectrum of resveratrol detected by HPLC-QTOF-HRMS in (FIG. 29A) *C. sublineola*-infected *Sorghum* mesocotyls or mechanically-wounded wild sugarcane, and (FIG. 29B) authentic resveratrol standard. FIGS. 29C-29D are representative MSMS spectrum of piceid detected by HPLC-QTOF-HRMS in (FIG. 29C) *C. sublineola*-infected *Sorghum* mesocotyls, and (FIG. 29D) authentic piceid standard. FIGS. 29E-29F are representative MSMS spectrum of pinostilbene detected by HPLC-QTOF-HRMS in (FIG. 29E) *C. sublineola*-infected *Sorghum* mesocotyls, and (FIG. 29F) authentic pinostilbene standard. FIGS. 29G-29H are representative MSMS spectrum of pterostilbene detected by HPLC-QTOF-HRMS in (FIG. 29G) *C. sublineola*-infected *Sorghum* mesocotyls, and (FIG. 29H) authentic pterostilbene standard. FIGS. 29I-29J are representative MSMS spectrum of piceatannol detected by HPLC-QTOF-HRMS in (FIG. 29I) mechanically-wounded wild sugarcane, and (FIG. 29J) authentic piceatannol standard. FIGS. 29K-29L

are representative MSMS spectrum of isorhapontigenin detected by HPLC-QTOF-HRMS in (FIG. 29K) mechanically-wounded wild sugarcane, and (FIG. 29L) authentic isorhapontigenin standard. cps, counts per second; m/z, mass-to-charge ratio.

DETAILED DESCRIPTION OF THE INVENTION

[0054] The disclosed method and compositions can be understood more readily by reference to the following detailed description of particular embodiments and the Example included therein and to the Figures and their previous and following description.

[0055] O-methyltransferases (OMTs) are ubiquitous enzymes in plants and are involved in a type of chemical modification, known as O-methylation, of phenolic natural products. Methylation of these compounds generally enhances their hydrophobicity, which confers higher bioavailability after intake and thus significantly increases the potency of their health-promoting properties. A biochemical mechanism termed compatible polarity pairing has been discovered that determines the catalytic regioselectivity in O-methylation of phenolic hydroxyl (—OH) groups in Type 1 OMTs. The discovery is based on the identification of a key amino acid position within Type 1 OMTs (corresponding or equivalent to Phe337 of SbSOMT), which, if the said amino acid is highly polar (or hydrophilic, e.g. asparagine) in nature, preferentially interacts with the polar functional group (e.g. —OH and —OMe) of an incoming aromatic ring, thus positioning the —OH group(s) adjacent to the polar functional group for O-methylation. In contrast, if the said amino acid is hydrophobic in nature (e.g. phenylalanine, valine, threonine), it preferentially interacts with an aromatic carbon in the phenolic ring of an incoming substrate, thus positioning the —OH group(s) adjacent to the aforementioned aromatic carbon for O-methylation. Thus, a structure-function relationship for the nature of the amino acid that corresponds to Phe337 of SbSOMT allows prediction and alteration of the regioselectivity of substrates for modified OMTs.

[0056] In the case of SbSOMT, Phe337 interacts with C4 of the stilbene A-ring and the C3-OH group is positioned for O-methylation. On the other hand, the equivalent residue in SbCOMT is Asn323 which interacts with 4'-OH group of the stilbene B-ring, hence t3'-OH group is positioned for O-methylation. Consistent with this, we discovered a panel of 24 Type 1 OMT-ligand complexes that supports this same compatible polarity pairing between the amino acid residue equivalent to SbSOMT Phe337 and SbCOMT Asn323 and their substrates. These Type 1 OMT enzymes all utilize the same catalytic mechanisms to catalyze O-methylation of phenolic substrates. Hence, our discovery allows the predictable engineering of Type 1 OMTs with an appropriate residue at the corresponding position for regiospecific O-methylation of phenolic substrates.

[0057] Previous attempts at biosynthesis of O-methylated phenolic compounds relied on the trial-and-error approach. They relied solely on the natural capability of Type 1 OMTs to catalyze O-methylation of phenolic substrates in vivo. Hence, both the OMTs involved and the resultant end products were unpredictable and had to be experimentally tested which is time- and resource-consuming. Optimization of Type 1 OMT-regioselectivity to tailor make the end products was very challenging and, again, relied on trial-

and-error. Thus, the mechanism underlying the catalytic regioselectivity of O-methyltransferases at the heart of our discovery had long been sought after.

[0058] With our discovery, biosynthesizing O-methylated phenolic natural products in a biosystem now becomes highly precise and predictable. Our discovery also allows the synthesis of novel O-methylated phenolic compounds.

[0059] Our discovery allows (1) straightforward prediction of Type 1 OMT(s) responsible for catalyzing certain —OH groups of phenolic substrates, (2) rational choice of Type 1 OMT(s) for targeted compound biosynthesis and synthetic biology, and (3) optimization of Type 1 OMT catalytic activity and regioselectivity by protein engineering.

[0060] The underlying biochemical mechanism determining the regioselectivity of Type 1 OMTs was largely unknown until our discovery here. Accordingly, the incorporation of a hydrophobic amino acid at position equivalent to SbSOMT Phe337 in any OMT will greatly favor the O-methylation of —OH group(s) adjacent to the interacting aromatic carbon of a phenolic ring in a substrate. In contrast, the incorporation of a hydrophilic amino acid at position equivalent to SbSOMT Phe337 in any OMT will greatly favor the O-methylation of —OH group(s) adjacent to an interacting polar functional group of a phenolic ring in a substrate.

[0061] Disclosed are modified Type 1 O-methyltransferases (OMTs) and methods of using such modified OMTs. In some forms of the modified OMTs, the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid. In some forms of the modified OMTs, the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid. In some forms of the modified OMTs, the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

[0062] In some forms, the substituting polar amino acid is independently Asn, Gln, Thr, or Ser. In some forms, the substituting polar amino acid is Asn. In some forms, the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala. In some forms, the substituting hydrophobic amino acid is Phe.

[0063] In some forms, the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCALDOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MslOMT, *Oryza sativa* OsCALDOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SlCOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* ClIEOMT, *Planctopirus limnophila*

PIOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeurp-tora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

[0064] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93. In some forms, the modified OMT has at least 70% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0065] In some forms, the modified OMT has at least 75% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0066] In some forms, the modified OMT has at least 80% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0067] In some forms, the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0068] In some forms, the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0069] In some forms, the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0070] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

[0071] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

[0072] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

[0073] In some forms, the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

[0074] In some forms, the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

[0075] Also disclosed are solutions including a modified OMT as disclosed. In some forms, the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions. Also disclosed are plants that express a modified OMT as disclosed.

[0076] Also disclosed are methods of producing an O-methylated stilbenoid or hydroxycinnamic acid. In some forms, the method involves bringing into contact, in a solution, a modified OMT as disclosed, a substrate for the modified OMT, and S-adenosylmethionine. In general, the O-methylated stilbenoid or hydroxycinnamic acid is produced by incubating the solution under suitable conditions.

[0077] In some forms, the method involves growing that express a modified OMT as disclosed and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

[0078] It is to be understood that the disclosed method and compositions are not limited to specific synthetic methods, specific analytical techniques, or to particular reagents unless otherwise specified, and, as such, can vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

[0079] The disclosed compositions and methods can be further understood through the following numbered paragraphs.

[0080] 1. A modified Type 1 O-methyltransferase (OMT), wherein the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid or wherein the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid.

[0081] 2. The modified OMT of paragraph 1, wherein the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid or wherein the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

[0082] 3. The modified OMT of paragraph 1 or 2, wherein the substituting polar amino acid is independently Asn, Gln, Thr, or Ser.

[0083] 4. The modified OMT of any one of paragraphs 1-3, wherein the substituting polar amino acid is Asn.

[0084] 5. The modified OMT of any one of paragraphs 1-4, wherein the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala.

[0085] 6. The modified OMT of any one of paragraphs 1-5, wherein the substituting hydrophobic amino acid is Phe.

[0086] 7. The modified OMT of any one of paragraphs 1-6, wherein the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCAldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCAldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAEOMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SiCOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum*

bicolor SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtlOMT, *Medicago truncatula* MtlOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIEOMT, *Planctopirus limnophila* PlOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

[0087] 8. The modified OMT of any one of paragraphs 1-7, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0088] 9. The modified OMT of any one of paragraphs 1-8, wherein the modified OMT has at least 70% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0089] 10. The modified OMT of any one of paragraphs 1-9, wherein the modified OMT has at least 75% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0090] 11. The modified OMT of any one of paragraphs 1-10, wherein the modified OMT has at least 80% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0091] 12. The modified OMT of any one of paragraphs 1-11, wherein the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0092] 13. The modified OMT of any one of paragraphs 1-12, wherein the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0093] 14. The modified OMT of any one of paragraphs 1-13, wherein the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

[0094] 15. The modified OMT of any one of paragraphs 1-14, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

[0095] 16. The modified OMT of any one of paragraphs 1-15, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino

acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

[0096] 17. The modified OMT of any one of paragraphs 1-16, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

[0097] 18. The modified OMT of any one of paragraphs 1-17, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

[0098] 19. The modified OMT of any one of paragraphs 1-18, wherein the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

[0099] 20. A solution comprising the modified OMT of any one of paragraphs 1-19, wherein the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions.

[0100] 21. A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising bringing into contact, in a solution, the modified OMT of any one of paragraphs 1-19, a substrate for the modified OMT and S-adenosylmethionine; and incubating under suitable conditions to produce the O-methylated stilbenoid or hydroxycinnamic acid.

[0101] 22. A plant that expresses the modified OMT of any one of paragraphs 1-19.

[0102] 23. A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising growing the plant of paragraph 22 and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

EXAMPLES

[0103] O-Methylated stilbenes are prominent nutraceuticals but rarely produced by crops. Here, the inherent ability of two Saccharinae grasses to produce regioselectively O-methylated stilbenes is reported. A stilbene O-methyltransferase, SbSOMT, is first shown to be indispensable for pathogeninducible pterostilbene (3,5-bis-O-methylated) biosynthesis in *Sorghum* (*Sorghum bicolor*). Phylogenetic analysis indicates the recruitment of genus-specific SOMTs from canonical caffeic acid O-methyltransferases (COMTs) after the divergence of *Sorghum* spp. from *Saccharum* spp. In recombinant enzyme assays, SbSOMT and COMTs regioselectively catalyze O-methylation of stilbene A-ring and B-ring respectively. Subsequently, SOMT-stilbene crystal structures are presented. Whilst SbSOMT shows global structural resemblance to SbCOMT, molecular characterizations illustrate two hydrophobic residues (Ile144/Phe337) crucial for substrate binding orientation leading to 3,5-bis-O-methylations in the A-ring. In contrast, the equivalent residues (Asn128/Asn323) in SbCOMT facilitate an opposite orientation that favors 3'-O-methylation in the B-ring. Consistently, a highly-conserved COMT is likely involved in isorhapontigenin (3'-O-methylated) formation in

wounded wild sugarcane (*Saccharum spontaneum*). Altogether, our work reveals the potential of Saccharinae grasses as a source of O-methylated stilbenes, and rationalize the regioselectivity of SOMT activities for bioengineering of O-methylated stilbenes.

[0104] Within the family Poaceae, the *Sorghum* and *Saccharum* genera evolved from a common ancestral grass and belong to the tribe Andropogoneae, subtribe Saccharinae. Species from both genera accumulate hydroxylated stilbenes (e.g. resveratrol, piceatannol) and their O-glycosides^{5, 7, 36}, but O-methylated stilbenes have not been reported in these species so far. Here, we report the identification of O-methylated stilbenes, pinostilbene (3-O-methylated) and pterostilbene (3,5-bis-O-methylated) in *Sorghum*, and isorhapontigenin (3'-O-methylated) in wild sugarcane (*S. spontaneum*). A stilbene OMT (SbSOMT) catalyzes sequential 3,5-bis-O-methylation of resveratrol to pterostilbene whereas a canonical COMT (SsCOMT) likely catalyzes 3'-O-methylation of piceatannol into isorhapontigenin in wild sugarcane. We generated *Sorghum* SbSOMT CRISPR/Cas9 mutants which were deficient in pinostilbene and pterostilbene, hence establishing the indispensable role of SbSOMT in resveratrol 3,5-bis-O-methylation. As indicated by phylogenetic analysis, divergence of *Sorghum* genus from other genera in the Saccharinae subtribe likely predated the recruitment of SbSOMT from a canonical COMT. By solving the high-resolution crystal structures of SbSOMT complexed with stilbenes, molecular features that differentiate the substrate binding mode and catalytic regioselectivity between SbSOMT and canonical COMTs were unveiled.

Methods

Plant Materials and Fungal Treatment Conditions

[0105] Two *Sorghum* (*Sorghum bicolor*) genotypes, BTx623 and SC748-5, were analyzed in this study. *Sorghum* seeds were germinated in aerated double-deionized water at 30° C. for 24 h. Wild sugarcane (*Saccharum spontaneum*) sets were obtained from Taiwan⁶⁶, and planted in soil until flowering. *C. sublineola* isolate TX430BB was propagated on 5% (w/v) oatmeal agar for 14 days prior to fungal infection experiments. Etiolated *Sorghum* seedlings were prepared as described previously⁶⁷, and were either sprayed with a 0.2% (w/v) bovine gelatin solution (control) or a conidial suspension (1.0×10⁶ spores ml⁻¹) of *C. sublineola* isolate TX430BB in the same gelatin solution (treatment). Both control and treatment groups were incubated under constant light at room temperature and 100% humidity.

Plant Metabolite Extraction

[0106] One hundred milligrams of *Sorghum* mesocotyls from fungal infection experiments or mechanically wounded wild sugarcane stalks were harvested for metabolite analysis. All plant tissues were frozen by liquid nitrogen and ground into fine powder using TissueLyser II (QIAGEN, Germany). Two hundred microliters of 80% (v/v) methanol (with 10 μM Apigenin-d₅ as internal standard) were added to the samples. Metabolites were extracted by ultra-sonicating the samples on an ice-water bath for 30 min. Samples were passed through a 0.22 μm PTFE membrane filter (Phenomenex, USA) prior to HPLC-QTOF-HRMS analysis described below.

Hplc-QTOF-HRMS and HPLC-MRM Analyses

[0107] To screen for OMT activities, purified stilbene products of enzyme assays were separated by a Synergi C18 column (Synergi 4μ Fusion RP 80 Å, 50×2 mm, Phenomenex, USA) under a flow rate of 0.5 ml min⁻¹ over a 6 min linear gradient of 10%-90% B (described below). Product detection was achieved with a quadruple-time-of-flight-high resolution mass spectrometer (QTOF-HRMS) X500R system (AB Sciex, China) operating under the information-dependent acquisition (IDA) mode. Meanwhile, purified phenylpropanoid products of enzyme assays were separated on the same column connected to a HP1100 series HPLC system (Agilent Technologies, USA) linked to an AP3200-QTRAP mass spectrometer (AB Sciex, China) operating under the multiple-reaction-monitoring (MRM) mode. To quantify these products, assay products were separated on a Kinetex C18 column (Kinetex 2.6 μm C18 100 Å, 100×2.1 mm, Phenomenex, USA) connected to a HP1100 series HPLC system (Agilent Technologies, USA) linked to an AP3200-QTRAP mass spectrometer (AB Sciex, China) operating under the multiple-reaction-monitoring (MRM) mode. A linear gradient of 10%-90% B under a flow rate of 0.3 ml min⁻¹ over 6 min was used to further enhance sensitivity. Total metabolites from all plant tissues were separated by the same Kinetex C18 column connected to the X500R QTOF-HRMS system. A linear gradient under a flow rate of 0.3 ml min⁻¹ over a 20 min linear gradient of 10%-90% B was used.

[0108] In all HPLC-MS analyses, the mobile phase consisted of 0.5% (v/v) formic acid/water (A) and 0.5% (v/v) formic acid/methanol (B). Stilbenes were detected using the positive ionization mode while phenolic acids were detected with the negative ionization mode. Compounds were quantified by integration of peak area using the quantification mode of SCIEX OS (for X500R) or Analyst software version 1.5.6 (for AP3200-QTRAP). Identification of compounds was achieved by comparing the retention time and MS/MS spectra (FIGS. 29A-29L) with authentic standards.

Gene Expression Analyses

[0109] Total RNA was extracted from *Sorghum* mesocotyls collected at a 24-hour interval for up to 96 h post fungal-treatment and wounded wild sugarcane stalks collected at different time points using the TRIzol method (Invitrogen, USA). Reverse transcription and qRT-PCR were performed using PrimeScript RT reagent kit with gDNA eraser (TaKaRa, Japan) and TB Green premix Ex Taq II kit (TaKaRa, Japan), respectively. Semi qRT-PCR was performed with GoTaq DNA polymerase (Promega, USA). Gene-specific primers used for qRT-PCR experiments were listed in Table 11. The housekeeping gene *Sorghum bicolor* Eukaryotic Initiation Factor 4A-1 (SbEIF4α⁶⁸; XM_002451491) and *Saccharum spontaneum* Glyceraldehyde 3-Phosphate Dehydrogenase (SsGADPH; Sspon. 08G0001560-1A)⁶⁹ were used as internal controls for *Sorghum* and wild sugarcane, respectively.

Cloning of SbSTS1, SbOMT4, SbSOMT, SbCOMT, SsSTS, and SsCOMT

[0110] The coding sequences (CDS) of SbSTS1, SbOMT4, SbSOMT and SbCOMT were amplified from cDNA prepared from fungal-infected mesocotyls of *Sorghum* genotype BTx623 using gene-specific primers (Table

11). The CDS of SsSTS and SsCOMT were cloned from cDNA prepared from wounded wild sugarcane stalk using gene-specific primers (Table 11). To generate recombinant proteins, the CDS of SsSTS (full length), SsCOMT (full length), SbCOMT (encoding residues 2-362), and SbSOMT (encoding residues 2-377) were cloned into pET-N-His-TEV (Beyotime) via HiFi DNA Assembly (New England BioLabs, USA) or Gibson Assembly (New England BioLabs, USA). Full length CDS of SbOMT4 was inserted between BamHI and HindIII restriction sites of pET23a(+) vector (Novagen, Germany). The full length CDSs of SbSTS1 and SbSOMT were individually subcloned into the binary vector pEAQ-HT by the Gibson Assembly method (New England BioLabs, USA) using gene-specific primers (Table 11).

Recombinant Protein Production and Site-Directed Mutagenesis

[0111] Desired mutations of SbSOMT were introduced into its construct using specific primers (Table 11). Expression constructs were transformed into Rosetta™ (DE3) competent *Escherichia coli* cells (Novagen, Germany). Protein expression was induced with 0.1 mM Isopropyl- β -thiogalactopyranoside (IPTG) overnight at 16° C. *E. coli* cells were harvested, resuspended in lysis buffer (20 mM Tris buffer pH 7.9, 300 mM NaCl, 2.5 mM β -mercaptoethanol), and lysed by sonication. Crude proteins were loaded onto a HisTrap HP column (Cytiva, USA) connected to a AKTA pure chromatography system (Cytiva, USA). Column was washed with 15 column volumes of lysis buffer with 50 mM imidazole. Proteins were eluted over a gradient of 50-500 mM imidazole. Eluate fractions with the target OMT were pooled, digested with TEV protease, and simultaneously dialyzed against a low salt buffer (20 mM Tris pH 7.9, 50 mM NaCl, 2.5 mM β -mercaptoethanol) overnight at 4° C. The dialyzed sample was filtered and flowed through HisTrap to remove His-tagged components. The flowthrough was then loaded to HiTrap Q FF columns (Cytiva, USA) over a gradient of 50-1000 mM NaCl. Polishing was achieved via size-exclusion chromatography using the storage buffer (100 mM HEPES pH 7.9, 100 mM NaCl) as mobile phase. SsSTS was purified under the same conditions except that the pH was adjusted to 7.0 for all buffers. Proteins were concentrated by ultrafiltration and flash frozen for storage at -80° C.

Enzyme Assays and Enzyme Kinetics

[0112] For initial screening of OMT activities, purified OMT enzymes (10 μ g) were incubated in 100 mM HEPES buffer (pH 7.9), 200 μ M S-adenosyl-L-methionine (SAM) and 100 μ M of phenylpropanoid or stilbene substrates (final volume 200 μ l) at 30° C. for 2 h. Enzyme kinetics were determined by incubating 1 μ g protein with 100 mM HEPES buffer (pH 7.9), 300 μ M SAM, and stilbene concentrations ranging from 5 to 200 μ M at 30° C. for 5 min. Similar conditions were used to determine enzyme kinetics of SbSOMT towards SAM, except that SAM concentrations ranging from 5 to 500 μ M (while stilbene substrates were kept at 1 mM) were included to accommodate its unexpectedly high Kd and Km values. For the SsSTS enzyme assays, 10 μ g of purified SsSTS enzymes were incubated in 100 mM HEPES buffer (pH 7.0), 100 μ M of caffeoyl-CoA or feruloyl-CoA, and 300 μ M of malonyl-CoA at 30° C. for 1 h. All enzyme assay reactions were quenched by snap-freezing the

reaction tubes in liquid nitrogen. Reaction products were extracted twice with ethyl acetate. The organic layers were then pooled, vacuum-dried, and resuspended in 50 μ l 80% (v/v) methanol with 10 μ M tricin as internal standard for HPLC-MRM quantification. All reactions were done in triplicates. Enzyme kinetics were calculated using the non-linear regression fitting by GraphPad Prism 6 software (GraphPad, USA).

Isothermal Titration Calorimetry (ITC)

[0113] Isothermal titration calorimetry (ITC) assays were conducted with MicroCal iTC200 (Malvern Panalytical, UK). Concentrated stocks of resveratrol, pinostilbene, pterostilbene, and piceatannol were prepared in 50% (v/v) PEG 400. Both cell sample and titrant were equilibrated to ITC buffer consisting of 100 mM HEPES pH 7.9 with 2.5% (v/v) PEG 400. For ITC assays involving pterostilbene, PEG 400 concentration was increased to 5% (v/v). Concentrations of protein (in cell) and ligand (in syringe) used in each run are listed in Table 11. Each run incorporated an initial delay of 60 s prior to the first injection (0.5 μ l) and spaced 150 s between the subsequent 19 injections (2.0 μ l) at 25° C. During the runs, the cell was continuously stirred at 750 rpm by flat paddle. Results were analysed based on the 2nd to 20th injections using the Origin (MicroCal Software, USA) and PEAQ-ITC software (Malvern Panalytical, UK).

Agroinfiltration of *N. benthamiana* Leaves

[0114] pEAQ constructs harboring either SbSTS1 or SbSOMT were transformed into *Agrobacterium tumefaciens* strain GV3101. To co-express SbSTS1 and SbSOMT in *N. benthamiana* leaves, two *Agrobacterium* cultures, each harboring one of the overexpression plasmids (individual OD600 at 0.8), were mixed in equal ratio and co-infiltrated to *N. benthamiana* leaves. Leaves were harvested five days after infiltration and were subjected to metabolite extraction and HPLC-QTOF-HRMS analysis as described above.

Constructs of CRISPR-SbSOMT and *Sorghum* Transformation

[0115] The CRISPR-SbSOMT construct was chemically synthesized (Gene Universal, USA). In brief, two gRNAs, including gRNA1: GTTGAACACGGTGTTCACG and gRNA2: GCACCGGACTACGCTGTGCG, were designed using CHOPCHOP v3⁷⁰, individually introduced to the downstream of SbU6 promoters, SbU62.3 and SbU63.1, respectively, and were further cloned into the plasmid that harbors a selective marker (NPTII) to generate CRISPR-SbSOMT construct as described previously⁵⁰. The CRISPR/Cas9 plasmid, pBUN411, was modified for *Sorghum* particle bombardment⁷¹. *Sorghum* tissue culture, transformation, and CRISPR-Cas9-mediated genome editing were performed as described previously⁴⁹. Briefly, *Sorghum* inbred line Tx430 were grown in a temperature-controlled (18-28° C.) glasshouse to provide the initial explant for transformation. Immature seeds were harvested 11-15 days post anthesis. Immature embryos were isolated and placed on callus induction medium for generating embryogenic calli. The CRISPR-SbSOMT construct and pBUN411 plasmid were co-transformed into embryogenic calli by particle bombardment as described previously⁷². After transformation, potential transgenic plantlets were grown in a temperature-con-

trolled (18–28° C.) glasshouse. Genome-edited plants (T0 and T1) were identified by PCR and direct sequencing using primers listed in Table 11.

X-Ray Crystallography

[0116] SbSOMT was prepared at 3.5 mg ml⁻¹ and pre-equilibrated with 0.5 mM of stilbene ligand in a final buffer of 20 mM HEPES pH 7.9, 150 mM NaCl, 4.5% dimethyl sulfoxide (DMSO) and 1 mM TCEP. Co-crystallization of SbSOMT and resveratrol were initially screened against commercial crystallization screens (Hampton Research, USA and QIAGEN, USA) with a sitting drop vapor diffusion approach incubated at 18° C. Two crystallizing conditions were further optimized. Crystals of SbSOMT-resveratrol- β -NAD ternary complex were obtained after two days of growing from the setup of 1 μ l of 5 mg ml⁻¹ SbSOMT, 0.15 μ l of 50 mM β -NAD and 0.5 μ l of reservoir solution (0.1 M sodium acetate pH 4.6, 0.2 M sodium acetate, 0.2 M NH₄Cl, 2.5% (w/v) polyethylene glycol 4000). SbSOMT co-crystallized with pinostilbene, piceatannol, or pterostilbene were obtained under the same condition. Crystal of SbSOMT-resveratrol binary complex was initially obtained after 2–3 weeks of growing from the setup of 1 μ l of 5 mg ml⁻¹ SbSOMT and 0.5 μ l of reservoir solution (0.1 M MES pH 6.5, 0.2 M ammonium sulphate, 30% (w/v) polyethylene glycol monomethyl ether 5000). Subsequent crystallization was accelerated by seeding and crystals were harvested after 1 week. Crystals collected for X-ray diffraction was cryo-protected by 20% (v/v) glycerol topped onto the reservoir solution.

[0117] X-ray diffraction data were collected from BL19U1 beamline (wavelength at 0.979 Å) at the Shanghai Synchrotron Radiation Facility and processed by either XDS or HKL3000 package^{73–74}. Data were analysed using pipelines provided in CCP4 suite⁷⁵. Data reduction was performed using AIMLESS⁷⁶. The initial structure of SbSOMT-resveratrol- β -NAD ternary complex was solved by molecular replacement; first through PHASER using a chimeric search model generated by MrBUMP, followed by model rebuilding from phase solution through BUCCANEER^{77–81}. Following structures were solved by molecular replacement with PHASER using protomer of SbSOMT-resveratrol- β -NAD ternary complex as a search model. The built models were iteratively refined through automated REFMAC5 refinement and manual refinement in COOT^{81, 82}. Validation and assessment of structure quality were performed using MolProbity and OneDep prior to finalization^{83, 84}. The statistics and metrics of reported structure are compiled in Table 13. Figures featuring protein structure were prepared using UCSF Chimera⁸⁵ or UCSF ChimeraX⁸⁶.

Ligand Docking with AutoDock Vina

[0118] Ligand docking was performed using the default setting⁸⁷. The determination of grid center was guided by superimposition of SbSOMT-resveratrol- β -NAD ternary complex, which the grid center was set in the binding pocket where the superimposed resveratrol is observed. For piceatannol docking in SbSOMT, the grid size was set to 18 Å×16 Å×12 Å, covering the entity of superimposed resveratrol. For docking in SbCOMT, a grid size of 24 Å×24 Å×24 Å was used with the similar grid center to cover the entire binding pocket. All docking were performed using global searching exhaustiveness of 8 and top 5 docked conformations were recorded for further analysis.

Phylogenetic Analysis

[0119] Multiple sequence alignment was done by ClustalW and Clustal Omega with default configurations^{88–89}. The unrooted phylogenetic trees were constructed by maximum likelihood method with 1,000 bootstrap replicates using MEGA X90.

In Vitro Fungitoxicity Assays

[0120] A conidial suspension of *Collectotrichum sublineola* isolate TX430BB at a concentration of 5×10⁴ spore ml⁻¹ was prepared in double deionized water. Stilbenes including resveratrol, pinostilbene, and pterostilbene were dissolved in 80% (v/v) methanol and supplemented to the potato dextrose agar at concentrations ranging from 0 (mock) to 50 μ M. Ten microliters of conidial suspensions were transferred onto the water agar plates, and 9 biological replicates were conducted for each concentration of stilbenes. These plates were sealed, incubated overnight at room temperature in darkness, followed by incubation under constant light for 3 days. Spore germination was observed under a Leica DM500 light microscope (Leica, USA) after incubation in dark. Mycelial growth, represented by the fungal colony diameter, was assessed after the 3-day incubation period under light using the same microscope and the ImageJ software⁹¹.

Accession Numbers

[0121] Protein sequences analyzed in this study and used in phylogenetic analysis can be found under accession numbers listed in Table 14⁶⁵.

Data Availability

[0122] Atomic coordinates and structure factors for the crystal structures reported in this work were deposited to the Protein Data Bank under accession numbers: 7VB8 (SbSOMT-resveratrol- β -NAD ternary complex), 7WAQ (SbSOMT-resveratrol binary complex), 7WAR (SbSOMT-pinostilbene- β -NAD ternary complex) and 7WAS (SbSOMT-pterostilbene- β -NAD ternary complex).

Results

Accumulation of O-Methylated Stilbenes in Pathogen-Infected *Sorghum*

[0123] The metabolic profiles of wild-type *Sorghum* upon infection of *Collectotrichum sublineola* (FIG. 1A) were first examined. Two *Sorghum* genotypes, BTx623 and SC748-5, susceptible and resistant to *C. sublineola* infection respectively^{37–38}, were analyzed. Accordingly, four stilbenes including resveratrol, piceid (resveratrol 3-O-glucoside), and resveratrol derivatives O-methylated at their A-ring: pinostilbene (3-O-methylated resveratrol), and pterostilbene (3,5-bis-O-methylated resveratrol) were identified in the infected *Sorghum* mesocotyls (FIGS. 1B–1F). In addition, flavones (apigenin, luteolin, chrysoeriol, tricetin; Table 3) and 3-deoxyanthocyanidins (orange-red pigments; apigeninidin, luteolinidin, diosmetinidin; Table 4), which are known *Sorghum* phytoalexins, were detected^{37–38}. Notably, the resistant genotype SC748-5 accumulated approximately 9-fold more pterostilbene than the susceptible genotype BTx623 starting from 48 h post infection (FIG. 1B & Table 5). Taken

together, stilbene accumulation in SC748-5 was more rapid, and at larger quantities than BTx623 during *C. sublineola* infection.

Sorghum Pathogen-Inducible SbSOMT Catalyzes Pterostilbene Biosynthesis

[0124] To identify potential OMTs involved in pterostilbene biosynthesis in *Sorghum*, an in silico expression dataset for *Bipolaris sorghicola*-infected *Sorghum* leaves (internet site matsui-lab.riken.jp/morokoshi)³⁹⁻⁴⁰ and in-house transcriptome dataset for *C. sublineola*-infected *Sorghum* mesocotyls⁴¹ were analyzed. Transcripts of SbOMT1 and SbOMT3, which were reported to methylate resveratrol in vitro and in transgenic plants, were not detected in either dataset (FIG. 8A). Meanwhile, SbCOMT, which encodes a bona fide canonical COMT^{18, 21, 22}, is constitutively expressed in *Sorghum* (FIG. 8A). Furthermore, two putative OMT genes, SbSOMT (Sb07g004710) and SbOMT4 (Sb07g004690), showed pathogen-inducible expression patterns similar to that of SbSTS1 (Sb07g004700) (FIG. 8A) encoding stilbene synthase (STS) which generates resveratrol from p-coumaroyl-CoA and malonyl-CoAs (FIG. 1A)⁵. Gene expression analyses confirmed their transcriptional upregulation in both *Sorghum* genotypes shortly after infection (FIGS. 8B-8D). In particular, the resistant genotype SC748-5 showed stronger expression for these genes than the susceptible genotype BTx623, consistent with its higher levels of pathogen-induced stilbene accumulation.

[0125] Subsequently, recombinant proteins expressed in *E. coli* were purified for in vitro enzyme assays. Both SbSOMT and SbCOMT generated pinostilbene and/or pterostilbene when incubated with resveratrol albeit at significantly different levels. Notably, SbSOMT converted about 60% of resveratrol into pterostilbene whereas SbCOMT displayed poor resveratrol 3,5-bis-O-methylation activities (FIG. 2A). Meanwhile, SbOMT4 showed minimal SOMT activities (FIGS. 9A-9C) and is thus unlikely to contribute to pterostilbene production in *Sorghum*. The catalytic activities of SbSOMT and SbCOMT towards hydroxycinnamic acids were also compared. Intriguingly, SbSOMT failed to methylate caffeic acid and 5-hydroxyferulic acid, whereas SbCOMT efficiently converted them to ferulic acid and sinapic acid, respectively (FIGS. 2B-2C).

[0126] Kinetic parameters of SbSOMT and SbCOMT were then determined using pinostilbene as a substrate since SbSOMT rapidly converts resveratrol into pterostilbene, thus preventing accurate quantification of pinostilbene produced. Results revealed the superior catalytic performance (kcat/Km and specific activity) of SbSOMT over SbCOMT and SbOMT4 towards pinostilbene, although SbSOMT showed a higher Km value (Table 1). These data indicated that SbSOMT, but not SbCOMT, is the primary SOMT for pterostilbene biosynthesis in *Sorghum*.

[0127] The biochemical function of SbSOMT was then examined via transient co-overexpression in *Nicotiana benthamiana* (FIGS. 10A-10D). Overexpression of both SbSTS1 and SbSOMT produced pterostilbene as the only stilbene product in agro-infiltrated leaves. By contrast, overexpression of SbSTS1 alone generated resveratrol and a small amount of pinostilbene, presumably due to endogenous promiscuous activities of tobacco OMT(s). These data are supportive for the role of SbSOMT in 3,5-bis-O-methylation of resveratrol to pterostilbene in planta.

Sorghum SbSOMT Mutants are Depleted in Pathogen-Inducible Pinostilbene and Pterostilbene

[0128] *Sorghum* sbSOMT mutants were generated via CRISPR/Cas9-mediated genome editing. Three homozygous sbSOMT mutant lines (sbSOMT-a, sbSOMT-b1, sbSOMT-b2; T1 generation) harboring different mutation pattern were isolated for metabolite analysis (FIG. 3A & FIGS. 11A-11F). In all mutant lines, the indels on exon 1 alone were sufficient to induce frameshift mutations and premature translation termination, resulting in knockout mutations (FIG. 12).

[0129] Metabolite profiles of sbSOMT mutants were analyzed upon *C. sublineola* infection. (FIGS. 3B-3E & Table 6). Consistent with the results above (FIGS. 1C-1F), wild-type Tx430 mesocotyls accumulated resveratrol, piceid, pinostilbene and pterostilbene 72 h after infection. By contrast, the sbSOMT-a/b1/b2 mutant mesocotyls accumulated resveratrol and piceid but not pinostilbene or pterostilbene even at 96 h after infection. These results established the indispensable role of SbSOMT in 3,5-bis-O-methylation of resveratrol to form pinostilbene and pterostilbene successively in *Sorghum*.

Sorghum Convergently Recruited SbSOMT Via Neofunctionalization of COMT

[0130] It has been discovered that SOMTs in grapevine (VvROMT) and Scots pine (PsPMT2) evolved convergently from non-COMT ancestors^{27, 28}. Multiple sequence alignment showed that SbSOMT shares only 31.9% and 29.9% sequence identity with VvROMT and PsPMT2, respectively (FIG. 13). Instead, SbSOMT is phylogenetically related to canonical COMTs in Poaceae (FIG. 2D). In addition, an SbSOMT ortholog (ShSOMT) is found in Johnsongrass (*Sorghum halapense*; FIG. 2D). Meanwhile, no potential SbSOMT orthologs could be retrieved from wild sugarcane (*Saccharum spontaneum*) proteome (internet site plants.ensembl.org/; all homologous proteins with <50% identity; FIG. 14) although *Sorghum* spp. and *Saccharum* spp. share a recent common ancestor (both belonging to the same subtribe). Furthermore, ShOMT1, ShSOMT, SbOMT1, SbOMT4, SbSOMT form a *Sorghum*-specific clade sister to Poaceae naringenin 7-O-methyltransferases (NOMTs) (FIG. 2D). Hence, these OMTs were most likely recruited via duplication and neofunctionalization of an ancestral grass COMT. Overall, SbSOMT represents an independent and genus-specific acquisition of SOMT activities distinct from those in grapevine and Scots pine.

Substrate Binding Affinity and Regioselectivity of SbSOMT and SbCOMT

[0131] So far, the results presented herein revealed that SbSOMT catalyzes the 3,5-bis-O-methylation of resveratrol upon fungal infection. Meanwhile, SbCOMT efficiently methylates hydroxycinnamic acids but demonstrates minimal SOMT activities towards resveratrol and pinostilbene which might be attributed to poor substrate binding. However, isothermal titration calorimetry (ITC) study did not corroborate this possibility (Table 2 & FIG. 15). In fact, SbCOMT exhibited strong, micromolar binding affinities (Kd) towards resveratrol (1.55 μ M), pinostilbene (1.79 μ M), and pterostilbene (3.99 μ M), comparable to those towards hydroxycinnamic acids²¹. Meanwhile, SbSOMT also showed strong binding affinities towards resveratrol and pinostilbene (4.69 μ M and 2.86 μ M respectively) but a

higher K_d to pterostilbene (11.20 μM), demonstrating SbSOMT favors its substrates over product for binding. The SAM binding affinity of SbSOMT was unexpectedly weak ($K_d=90.00\ \mu\text{M}$), whereas SbCOMT showed a K_d of 13.00 μM which is consistent with a previous study²¹. Collectively, SbSOMT and SbCOMT showed comparable stilbene-binding affinities, which would not account for their different catalytic activities towards resveratrol and pinostilbene.

[0132] Canonical COMTs including SbCOMT utilize the para-hydroxyl group of phenylpropanoids for proper substrate orientation^{21, 22} which concurs with results obtained herein of SbCOMT-catalyzed O-methylation of caffeic acid and 5-hydroxyferulic acid (FIG. 2B). On the other hand, SbSOMT-catalyzed O-methylation of resveratrol occurs in the A-ring which is not para-hydroxylated. Piceatannol (3'-hydroxylated resveratrol) was then used as an enzyme substrate since its stilbene B-ring is structurally identical to the phenolic ring in caffeic acid while its A-ring is identical to that in resveratrol (FIG. 4A). Both enzymes catalyzed in vitro O-methylation of piceatannol, but occurring in a regioselective manner (FIG. 4B-4C). SbSOMT methylated the 3- and 5-hydroxyl groups on the piceatannol A-ring to produce 3'-hydroxypinostilbene and 3'-hydroxypterostilbene successively. By contrast, SbCOMT converted a substantial amount of piceatannol (ca. 60%) into isorhapontigenin by 3'-O-methylation in the B-ring with a conversion rate similar to those of SbCOMT-caffeic acid and SbSOMT-resveratrol reactions (FIG. 2A-2B). A small amount of isorhapontigenin was further methylated at its A-ring to 3'-methoxypinostilbene by SbCOMT (FIG. 4B-4C) and the conversion rate was similar to that of SbCOMT-resveratrol reaction (FIG. 2A). Furthermore, ITC experiments showed that SbCOMT displayed significantly stronger binding affinity towards piceatannol when compared to SbSOMT (Table 2; $K_d=1.26\ \mu\text{M}$ and 37.50 μM , respectively). Overall, these experiments strongly demonstrated the different regioselective O-methylation properties of SbSOMT and SbCOMT.

Global Structure of SbSOMT Resembles that of SbCOMT **[0133]** To rationalize their different catalytic regioselectivities, structural analyses involving X-ray crystallography and computational methods were conducted. The crystal structure of an SbSOMT-resveratrol-nicotinamide adenine dinucleotide (β -NAD) ternary complex was solved (diffracted to 1.72 Å resolution, FIG. 4D; β -NAD was utilized as an additive to improve crystallization⁴²), an SbSOMT-resveratrol binary complex, and two ternary complexes reproduced with pinostilbene or pterostilbene (diffracted in the range of 2.10-2.56 Å, FIG. 16A). All complexes were in high structural resemblance and depicted as a homodimer with an open conformation. Superimposition of SbSOMT and SbCOMT (PDB: 4PGH²¹) revealed their high structural similarity (RMSD=2.322 Å), except that SbSOMT harbors an extended loop (Val103-Cys114) that forms a threonine-rich dimerization interface (FIG. 4D).

Structural Analyses and in Silico Docking Support Different Stilbene Orientations within the Substrate Binding Pockets of SbSOMT and SbCOMT

[0134] Close examination of the SbSOMT substrate binding pocket within the SbSOMT-resveratrol- β -NAD complex provided further insights into the interactions between SbSOMT and its stilbene substrates. The pocket mainly composed hydrophobic residues, and the stilbene backbone was secured by hydrophobic interactions with Met143,

Ile144, Met175, Phe189, Met193, Trp279, Tyr311, Leu329, Met333, Thr336, in addition to Leu28 of the adjacent protomer (FIG. 5A). The apparent binding cavity was confined to fit the stilbene backbone in a planar configuration with minimal rotational flexibility. Of the three hydroxyl groups on resveratrol capable of forming hydrogen bonds, only the methyl-accepting 3-OH group (A-ring) formed direct hydrogen bonds to SbSOMT residues (FIG. 5B), including the catalytic residues His282 (2.80 Å) and Asp283 (3.26 Å). Other hydroxyl groups were shielded by water molecules and did not directly interact with SbSOMT. The same binding mode was adopted for pinostilbene or pterostilbene in the corresponding SbSOMT co-crystals (FIG. 5C-5D & FIG. 16B). Overall, these structures depicted a productive substrate orientation within the substrate binding pocket which greatly favors A-ring O-methylation. Following 3-O-methylation of resveratrol, the pinostilbene intermediate will need to be dissociated and re-inserted in order to re-position the A-ring for 5-O-methylation to produce pterostilbene. Similarly, in silico docking of piceatannol with SbSOMT (FIG. 5E) conformed to such substrate orientation, with the A-ring positioned in proximity to the catalytic residues of SbSOMT, hence favoring the 3,5-bis-O-methylation to produce 3'-hydroxypinostilbene and 3'-hydroxypterostilbene (FIG. 4B-4C).

[0135] Docking stilbenes with SbCOMT (PDB: 4PGH²¹) revealed a contrasting substrate orientation to that of SbSOMT, suggesting that SbCOMT preferentially positions stilbenes with the B-ring close to the catalytic residues and the A-ring reaching inwardly into the pocket. SbCOMT-piceatannol docking showed a single favorable orientation of piceatannol among five hits which were docked at a sub-optimal position (FIG. 5E). In the top hit, the B-ring 3'- and 4'-hydroxyl groups were stabilized by hydrogen bonds with Asn323 while A-ring 5-hydroxyl group interacted with Asn128 (FIG. 5E). Consistent with the enzyme assay results (FIG. 2A), the 3'-hydroxyl group on the B-ring was orientated close to the catalytic residues and would thus predominantly favors the methylation of 3'-hydroxyl group (B-ring) over 3-/5-hydroxyl groups (A-ring) (FIG. 5E). Additionally, docking resveratrol with SbCOMT revealed two opposite orientations with similar free energies of binding (FIG. 5F), representing two competing binding modes which hampered its catalytic performance (FIG. 2A). In fact, the non-productive orientation with the B-ring closer to the catalytic residues was more energetically favorable (−7.3 kcal/mol, in blue), while the productive orientation with A-ring closer to the catalytic residues was slightly less stable (−7.1 kcal/mol, in grey). Collectively, these analyses fully corroborated with results of the biochemical assays and stilbene profiles of *Sorghum* presented herein, and provided a mechanistic rationale for the apparent differences in catalytic activity between SbCOMT and SbSOMT. Additionally, as the methyl-accepting —OH groups in piceatannol and hydroxycinnamic acids are both adjacent to a para-OH group (i.e., 4'-OH in piceatannol and 4-OH in hydroxycinnamic acids), this may be a prerequisite for efficient O-methylation by COMTs as suggested previously^{21, 22}.

Sbsomt and SbCOMT Employ the Same Catalytic Residues but Different Substrate Binding Orientations

[0136] The catalytic residues of SbSOMT are highly conserved with all reported COMTs, including His282 as the key nucleophile, whereas Asp283, Glu310, and Glu342

collectively charge the basicity or confine His282 in its reactive conformation (FIG. 5B)^{21, 22, 43, 44}. Accordingly, the surrounding electrostatic force allows NF of His282 to deprotonate the methyl-accepting —OH group, followed by a SN2 nucleophilic attack of the methyl-donor group on SAM by the oxyanion (—O—) (FIG. 17)^{21, 22, 43, 44}. The roles of His282 and Asp283 have been validated by mutagenesis, but not for those of Glu310 and Glu342 (PDB: 4PGG²¹, 5ICG⁴⁵, 1KYZ⁴⁴, and 6ISZ⁴⁶). Here, H282A, D283A, E310A, and E342A mutant proteins of SbSOMT were generated for catalytic and binding assays with resveratrol. H282A mutant protein was catalytically defective and unstable for ITC study (FIGS. 6A-6B). To further elucidate the role of His282 in resveratrol binding (FIG. 5B), a stable SbSOMT H282N/D283A double mutant protein was generated. While the D283A protein showed a slightly stronger affinity towards resveratrol (Kd=1.31 μ M), the H282N/D283A mutant protein showed a significantly weaker affinity (Kd=32.90 μ M), implying that His282 plays a moonlighting role in substrate binding in SbSOMT in addition to catalysis (Table 7). Meanwhile, D283A and E342A mutations resulted in significant reduction in pinostilbene yield by 94.7% and 91.9%, respectively (FIG. 6A). In addition, both of them failed to generate pterostilbene when incubated with resveratrol despite that they showed considerably stronger affinities to resveratrol (FIG. 6B & Table 7). Unexpectedly, E310A mutation only caused reduction of pterostilbene yield (by 70.1%), accompanying with reshuffled stilbene binding affinities (FIG. 6B & Table 7). It was reasoned that the E310A mutation decreased the rigidity and basicity of His282, thus weakening SbSOMT selectivity towards the reactive hydroxyl group against the inactive methoxy group, which is a vital feature allowing the subsequent O-methylation of the same stilbene substrate (FIGS. 5B-5D).

[0137] In SbCOMT, two key amino acid residues, Asn128 and Asn323, cooperatively mediate the productive orientation for catalysis²¹. Docking piceatannol with SbCOMT revealed that Asn323 forms a hydrogen bond with the para-OH group on the stilbene B-ring whose meta-OH group is positioned optimally for methylation, while Asn128 likely serves to stabilize their interaction via another hydrogen bond with 5-OH on the non-para-hydroxylated stilbene A-ring (FIG. 5E). In striking contrast, the equivalent positions in SbSOMT are occupied by two highly hydrophobic residues, Ile144 and Phe337, which are unfavorable for interactions with the para-OH group of an aromatic ring as predicted by in silico docking (FIG. 5E). On the other hand, these residues may contribute to accommodate the stilbene A-ring, which is not para-hydroxylated, for methylation of its meta-hydroxyl groups. I144N, F337N and I144N/F337N SbSOMT mutant proteins were then generated for catalytic and binding assays. All three SbSOMT mutant proteins showed significantly weakened binding for resveratrol, with F337N and I144N/F337N mutant proteins displaying undetectable binding affinities (Table 7). Correspondingly, both I144N and F337N produced substantially less pterostilbene (24.8% and 97.7% respectively), and accumulated slightly more pinostilbene when compared to the WT protein (28.4% and 50.8% respectively; FIGS. 6C-6D). Meanwhile, the I144N/F337N mutant protein showed complete abolishment of SOMT activities (FIGS. 6C-6D). As pterostilbene represents the major product generated by SbSOMT-resveratrol

reactions (FIG. 2A), these results established the important role of Ile144 and Phe337 in the resveratrol binding mechanism adopted by SbSOMT.

A Canonical COMT Likely Catalyzes Isorhapontigenin Biosynthesis in Wild Sugarcane

[0138] *Saccharum* spp. were reported to accumulate resveratrol and piceatannol (3'-hydroxylated resveratrol) upon mechanical wounding or fungal infection^{2, 36} (FIG. 7A). As *Saccharum* spp. are Saccharinae grasses like *Sorghum* spp., it was investigated whether O-methylated derivatives could also be detected in wild sugarcane whose genome is available⁴⁷. The slender stalks of flowering wild sugarcane (*S. spontaneum*) were sawn into 2-cm segments (excluding nodes) for metabolic profiling. No stilbenes were detected in freshly-cut segments whereas resveratrol, piceatannol, and isorhapontigenin (3'-O-methylated piceatannol) started to accumulate 72 h after wounding (FIG. 7B, FIG. 18A-18C, and Table 8). A miniscule amount of a hexoside of piceatannol was also tentatively identified (FIGS. 19A-19B). Overall, mechanically-wounded wild sugarcane produced piceatannol and its B-ring O-methylated derivative isorhapontigenin which constituted a small portion of the stilbene profile (FIG. 7B), while no A-ring O-methylated stilbene could be detected. Next, the understanding of isorhapontigenin biosynthesis was pursued in wild sugarcane. Substantial upregulation of SsSTS gene expression (Sspon. 06G0010290-2P) 72 h after wounding was detected (FIG. 20A, FIG. 20C & FIG. 21). Under in vitro condition, recombinant SsSTS generated piceatannol with caffeoyl-CoA as the starter substrate but failed to generate isorhapontigenin when feruloyl-CoA was used (FIGS. 22A-22E). Hence, SsSTS is a functional STS but an OMT is most likely required for 3'-O-methylation of piceatannol to produce isorhapontigenin. Meanwhile, a canonical SsCOMT (Sspon. 06G0010980-3C) highly conserved with SbCOMT (>94.0% protein sequence identity) was identified in wild sugarcane (FIG. 14 & FIG. 23). It also harbors the key Asn residues corresponding to Asn128 and Asn323 in SbCOMT (FIG. 23). In fact, recombinant enzyme assays demonstrated that SsCOMT and SbCOMT showed the same catalytic regioselectivity. Accordingly, isorhapontigenin, ferulic acid, and sinapic acid were generated when SsCOMT was incubated with piceatannol, caffeic acid, and 5-hydroxyferulic acid respectively (FIG. 7C & FIGS. 24A-24F). However, only small amounts of pinostilbene were generated when resveratrol was used as substrate (FIGS. 24G-24I). In addition, SsCOMT shares similar substrate-binding properties with SbCOMT (Table 9). Gene expression further revealed that SsCOMT is constitutively expressed in the wild sugarcane segments despite its downregulation after wounding (FIG. 20B & FIG. 20D). Taken together, SsCOMT is an important candidate for isorhapontigenin production through 3'-O-methylation of piceatannol in the B-ring.

Discussion

[0139] O-Methylated stilbenes are well-acclaimed for their health-promoting benefits and exceptional bioavailability³⁵⁻⁴⁸. *Sorghum* is a major staple crop and wild sugarcane is a genetic resource for sugarcane breeding. In this study, the potential of these Saccharinae grasses as biofactories for O-methylated stilbenes as well as the OMTs for regioselective stilbene O-methylations were revealed. The findings

presented herein also provide insights into bioengineering of specific O-methylated stilbenes via molecular breeding and transgenic approaches, representing a unique opportunity to improve dietary intake of these nutraceuticals which are scarcely present in natural food sources. Recent breakthroughs in *Sorghum* biotechnology, including stable transformation and

[0140] CRISPR/Cas9-mediated genome editing as well as its metabolic versatility, have warranted *Sorghum* as an emerging model for investigation and manipulation of specialized metabolism⁴⁹⁻⁵¹. Here, complete depletion of O-methylated stilbenes in infected *Sorghum* SbSOMT CRISPR-Cas9 mutants (FIGS. 3A-3E) firmly established SbSOMT as the primary SOMT for pathogen-inducible pterostilbene biosynthesis. Concomitantly, molecular, biochemical, and structural characterizations were integrated to elucidate the mechanistic details of SbSOMT-stilbene reactions. The first crystal structure of a bona fide stilbene O-methyltransferase, SbSOMT is reported herein. The enzyme utilizes the His282-Asp283-Glu310-Glu342 catalytic residues to generate 3,5-bis-O-methylated stilbenes (FIG. 2A, FIGS. 4B-4C, FIGS. 5B-5F & FIGS. 16A-16C). The same catalytic residues are found in canonical COMTs, indicating the highly conserved O-methylation catalysis irrespective of phenolic substrates involved^{13, 21, 22, 52, 53}. It was demonstrated that Glu310 positions the 5-OH group, instead of the more hydrophobic 3-OCH₃ group, in close proximity to His282 for the second O-methylation (FIGS. 5C-5D & FIG. 16B). As Glu310 is highly conserved in most OMTs (with some harboring an Asp), this feature may represent a common mechanism for mediating successive O-methylations of structurally different phenolic substrates.

[0141] Meanwhile, combinatorial analyses presented herein rationalized the regioselectivities of SbSOMT and canonical COMTs underlain by specific substrate binding modes instead of alterations in catalysis or substrate affinity (FIG. 2A, FIGS. 4B-4C, FIGS. 5B-5F, FIGS. 6A-6D, FIGS. 7A-7C, Table 1, Table 2, FIG. 17 & Table 7). The hydrophilic Asn128/Asn323 residues in SbCOMT, which are highly conserved in COMTs from diverse plant lineages, orchestrate productive coordination with their native substrates, i.e. hydroxycinnamic acids and their analogs, via hydrogen bonding^{21, 22}. The same binding mechanism is apparently crucial for attaining productive and energetically favorable orientation of piceatannol inside the binding pocket of Saccharinae COMTs including SbCOMTAsn128/Asn323 and SsCOMTAsn130/Asn323 for B-ring O-methylation (FIG. 5E & FIGS. 24A-24B). In marked comparison, the recruitment of more hydrophobic residues in SbSOMT like Ile144 and Phe337 during its emergence in the *Sorghum* genus would considerably favor hydrophobic interactions with the C4 and C3' of resveratrol (and piceatannol) and pinostilbene (and 3'-hydroxypinostilbene), hence favoring A-ring O-methylation (FIGS. 4B-4C, FIG. 5E & FIGS. 6C-6D).

[0142] Thorough mining of the Protein Data Bank (PDB) retrieved a diverse panel of 24 OMT-ligand complexes (from 13 OMTs with $\geq 30\%$ protein identity to SbCOMT/SbSOMT), supporting a general occurrence of compatible polarity pairing between the amino acid residue equivalent to SbCOMTAsn323/SbSOMTPhe337 in OMTs and their substrates (Table 10). It was reasoned that upon conformational transitions, this residue and the catalytic residues are

brought into close proximity, leading to compatible polarity pairing between 12 (out of 13) OMTs and their respective substrate (FIG. 25 & FIGS. 26A-26X). For example, six OMTs harboring an Asn residue at this position is paired to a polar-OH or -OCH₃ moiety of the ligand (Table 10 & FIGS. 26A-26X)^{43, 44, 46, 54-57}. For the others, the equivalent residue is a more hydrophobic residue (Thr, Val or Phe) which is paired to a non-polar group such as a methyl group or an aromatic ring of the ligand (Table 10 & FIGS. 26A-26X). Consistently, two previously characterized SOMTs, VvROMTPhe318 and SbOMT3-Ile336, harbor a hydrophobic residue at this equivalent position^{28, 29}. Meanwhile, naringenin OMTs, which harbor a hydrophobic Leu residue (OsNOMTLeu335; SbNOMTLeu324; ZmNOMTLeu325) paired to the aromatic carbon (C6), catalyze 7-O-methylation (A-ring) of naringenin to generate sakuranetin, but do not utilize hydroxycinnamic acid substrates⁵⁸. Collectively, the polarity of the amino acid residue in OMTs equivalent to SbCOMTAsn323/SbSOMTPhe337 may govern substrate selectivity through a compatible polarity pairing with the functional group vicinal to the methyl-accepting —OH group in a substrate (FIG. 25 & FIGS. 26A-26X). Consequently, replacement of Asn323 by a hydrophobic residue represents a key molecular event towards the functional divergence of regioselective OMT activities from canonical COMTs in plants. On the other hand, polarity pairing between ligand and SbCOMTAsn128 is only observed in some OMT-ligand complexes, as the equivalent residues in OMTs are diversified (Table 10). Overall, the findings presented herein underpin subsequent bio-engineering of OMTs, either by directed evolution or targeted site-directed mutagenesis, for production of regioselective O-methylated phenolic nutraceuticals/pharmaceuticals in vitro or in planta.

[0143] *Sorghum* and wild sugarcane are closely-related species but resveratrol is their only common stilbene detected by experimental conditions used herein. In addition to resveratrol, *Sorghum* produces pinostilbene and pterostilbene upon fungal infection while wild sugarcane accumulates resveratrol, piceatannol, and isorhapontigenin following mechanical wounding (FIGS. 1A-1F & FIGS. 7A-7C). The lack of B-ring O-methylated stilbenes (pinostilbene and pterostilbene) in wounded wild sugarcane stems primarily from the absence of SbSOMT orthologs, which are unique to *Sorghum* spp. On the other hand, the presence of isorhapontigenin (B-ring O-methylated) is resulting from canonical SsCOMT activities although genetic evidence is required for confirmation. Meanwhile, the underlying causes for the exclusivity of piceatannol (3'-hydroxylated) and related stilbenes in *Saccharum* spp. within the Saccharinae subtribe remain elusive. Elucidation of the piceatannol biosynthetic pathway, in combination with the aforementioned bio-engineering of OMTs for targeted stilbene O-methylation, would further unleash the potential of these Saccharinae crops for stilbene biofortification.

[0144] Apparently, *Sorghum*-specific SOMTs was originated from the canonical and ubiquitous COMTs in Poaceae (FIG. 2D). The COMT-to-SOMT evolution fits well into the generally-accepted Yeas-Jensen model which suggests that a promiscuous enzyme with poor activities towards specific substrates often serves as the basic scaffold undergoing sequence divergence (e.g. the aforementioned replacement of SbCOMTAsn323 by SbSOMTPhe337) to optimize catalytic efficiency^{59, 60}. Further nonsynonymous mutations

within an ancestral *Sorghum* COMT allowed neofunctionalization with acquisition of efficient resveratrol 3,5-bis-O-methylation activities in SbSOMT (FIG. 2A). Such molecular evolution might have been driven by the superior potency of pterostilbene over resveratrol and pinostilbene as a broad-spectrum phytoalexin (FIG. 27A-27M)^{6, 61}. Both SbSOMT and OsNOMT have lost the ability to O-methylate hydroxycinnamic acids (FIG. 2B), which are key in planta substrates of COMTs^{21, 22}. Further investigations will shed new light on whether compromised substrate recognition pattern is necessary, and to what extent, to achieve the highly efficient SOMT activities in SbSOMT. Since both SbSOMT and SbCOMT utilize the same catalytic residues and mechanism for O-methylation, evolution of pterostilbene biosynthesis primarily hinged on optimizing the stilbene binding mode with productive orientations. Furthermore, the recruitment of SbSTS1 from a chalcone synthase scaffold for resveratrol biosynthesis would logically predate that of SbSOMT from SbCOMT5. An increasing number of specialized metabolic pathways are considered to have evolved in such order as gene clusters⁶²⁻⁶⁴. Correspondingly, physically-linked SbSTS1 (Sb07g004700), SbOMT4 (Sb07g004690), and SbSOMT (Sb07g004710) (FIG. 28) were identified as constituents of a gene cluster for specialized metabolism (Cluster 408) in the *Sorghum* genome⁶³⁻⁶⁵.

TABLE 1		
Kinetic parameters of SbSOMT and SbCOMT with pinostilbene as a substrate.		
	SbCOMT	SbSOMT
K_m (μM)	3.14 ± 1.40	4.62 ± 0.46
k_{cat} (min ⁻¹)	0.14 ± 0.01	10.30 ± 0.20a

TABLE 1-continued		
Kinetic parameters of SbSOMT and SbCOMT with pinostilbene as a substrate.		
	SbCOMT	SbSOMT
k_{cat}/K_m (min ⁻¹ μM ⁻¹)	0.045	2.229
Specific activity (nmol min ⁻¹ mg ⁻¹)	3.46 ± 0.27	290.04 ± 5.64

Values refer to means ± SD (n = 3).

TABLE 2		
Dissociation constants of SbSOMT, SbCOMT with stilbenes.		
	K_d (μM)	
	SbCOMT	SbSOMT
Resveratrol	1.55 ± 0.13	4.69 ± 0.54
Pinostilbene	1.79 ± 0.09	2.86 ± 0.33
Pterostilbene	3.99 ± 0.24	11.20 ± 2.23
Piceatannol	1.26 ± 0.06	37.50 ± 4.33
SAM	13.00 ± 0.83	90.00 ± 9.59

Results are expressed as means of K_d ± standard error derived from curve fitting.

TABLE 3				
High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of flavone aglycones in control or <i>Colletotrichum sublineola</i> -infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes.				
Time (h)	BTx623		SC748-5	
	Control	Infected	Control	Infected
Apigenin [peak area (cps × 10 ⁶) g ⁻¹ FW]				
24	4.08 ± 0.19	5.48 ± 4.58	3.95 ± 0.57	4.10 ± 0.30
48	7.18 ± 2.34	12.41 ± 1.67	8.06 ± 0.83	52.27 ± 12.73
72	12.25 ± 1.93	105.55 ± 31.98	12.24 ± 2.55	33.61 ± 9.98
96	13.11 ± 1.04	272.56 ± 100.65	7.42 ± 1.71	20.59 ± 3.92
Luteolin [peak area (cps × 10 ⁶) g ⁻¹ FW]				
24	1.20 ± 0.25	0.11 ± 0.05	2.38 ± 0.57	2.57 ± 0.45
48	1.16 ± 0.11	1.66 ± 0.43	3.37 ± 1.35	158.69 ± 29.51
72	1.62 ± 0.09	21.66 ± 10.97	4.94 ± 1.60	512.03 ± 65.46
96	5.35 ± 0.82	65.99 ± 32.20	7.03 ± 2.30	497.52 ± 135.58
Chrysoeriol [peak area (cps × 10 ⁶) g ⁻¹ FW]				
24	3.19 ± 2.60	4.66 ± 0.87	4.75 ± 1.15	8.88 ± 0.86
48	6.18 ± 0.27	8.02 ± 0.64	9.54 ± 1.11	108.23 ± 14.69
72	10.26 ± 1.15	28.49 ± 5.22	13.31 ± 1.97	308.20 ± 24.77
96	12.91 ± 1.07	75.03 ± 21.31	14.01 ± 4.62	364.55 ± 100.89

TABLE 3-continued

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of flavone aglycones in control or <i>Colletotrichum sublineola</i> -infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes.				
Time (h)	BTx623		SC748-5	
	Control	Infected	Control	Infected
Tricin [peak area (cps × 10 ⁷) g ⁻¹ FW]				
24	2.58 ± 0.27	1.59 ± 1.17	41.14 ± 9.33	35.36 ± 2.76
48	2.75 ± 0.55	2.62 ± 0.20	59.66 ± 10.28	75.39 ± 16.11
72	3.80 ± 1.03	4.52 ± 0.53	88.92 ± 6.14	242.87 ± 31.12
96	4.58 ± 0.04	9.76 ± 1.42	73.18 ± 24.74	396.46 ± 22.25

Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of individual peaks with those from authentic standards. Apigenin-d₅ was used as an internal standard for quantitation. Values refer to means ± SD (n = 3). cps: count per second, FW: fresh weight.

TABLE 4

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of 3-deoxyanthocyanidin aglycones in mock or <i>Colletotrichum sublineola</i> -infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes.				
Time (h)	BTx623		SC748-5	
	Control	Infected	Control	Infected
Apigeninidin [peak area (cps × 10 ⁸) g ⁻¹ FW]				
48	n.d.	1.11 ± 0.25	n.d.	4.69 ± 1.65
72	n.d.	10.52 ± 5.66	n.d.	10.93 ± 1.72
96	n.d.	19.37 ± 2.09	n.d.	9.49 ± 1.24
Luteolinidin [peak area (cps × 10 ⁶) g ⁻¹ FW]				
48	n.d.	2.44 ± 0.65	n.d.	822.19 ± 271.66
72	n.d.	153.69 ± 94.92	n.d.	2735.93 ± 339.38
96	n.d.	246.27 ± 45.66	n.d.	2403.59 ± 761.50
Diosmetinidin [peak area (cps × 10 ⁶) g ⁻¹ FW]				
48	n.d.	0.97 ± 0.12	n.d.	167.38 ± 78.93
72	n.d.	21.13 ± 12.19	n.d.	803.91 ± 213.56
96	n.d.	46.81 ± 10.43	n.d.	1352.29 ± 185.15

Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of individual peaks with those from authentic standards. Apigenin-d₅ was used as an internal standard for quantitation. Values refer to means ± SD (n = 3). cps: counts per second, FW: fresh weight, n.d.: not detected.

TABLE 5

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of stilbenes in mock or <i>Colletotrichum sublineola</i> -infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes.				
Time (h)	BTx623		SC748-5	
	Control	Infected	Control	Infected
Resveratrol (μg g ⁻¹ FW)				
72	n.d.	0.03 ± 0.01	n.d.	0.31 ± 0.00
96	n.d.	0.47 ± 0.23	n.d.	0.91 ± 0.19
Piceid (μg g ⁻¹ FW)				
72	n.d.	2.63 ± 0.23	n.d.	7.99 ± 0.46
96	n.d.	6.27 ± 1.27	n.d.	19.60 ± 3.24
Pinostilbene (μg g ⁻¹ FW)				
72	n.d.	0.05 ± 0.00	n.d.	0.22 ± 0.00
96	n.d.	0.07 ± 0.02	n.d.	0.57 ± 0.14

TABLE 5-continued

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of stilbenes in mock or <i>Colletotrichum sublineola</i> -infected mesocotyls of resistant (SC748-5) and susceptible (BTx623) sorghum genotypes.				
Time (h)	BTx623		SC748-5	
	Control	Infected	Control	Infected
Pterostilbene (μg g ⁻¹ FW)				
48	n.d.	n.d.	n.d.	0.03 ± 0.02
72	n.d.	0.16 ± 0.06	n.d.	1.56 ± 0.13
96	n.d.	0.61 ± 0.23	n.d.	5.66 ± 1.89

Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of individual peaks with those from authentic standards. Apigenin-d₅ was used as an internal standard for quantitation. Values refer to means ± SD (n = 3). FW: fresh weight, n.d.: not detected.

TABLE 6

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of stilbenes in mock or <i>Colletotrichum sublineola</i> -infected mesocotyls of wild-type Tx430 and sbsomt mutants.				
Time (h)	Tx430	sbsomt-a	sbsomt-b1	sbsomt-b2
Resveratrol (μg g ⁻¹ FW)				
72	0.09 ± 0.12	0.09 ± 0.02	0.10 ± 0.03	0.11 ± 0.04
96	0.18 ± 0.03	0.31 ± 0.12	0.49 ± 0.07	0.53 ± 0.03
Piceid (μg g ⁻¹ FW)				
72	1.60 ± 0.25	2.60 ± 1.20	1.78 ± 1.06	1.97 ± 1.34
96	5.74 ± 0.70	8.03 ± 2.33	18.14 ± 2.32	17.14 ± 5.45
Pinostilbene (μg g ⁻¹ FW)				
72	0.03 ± 0.00	n.d.	n.d.	n.d.
96	0.09 ± 0.02	n.d.	n.d.	n.d.
Pterostilbene (μg g ⁻¹ FW)				
72	0.13 ± 0.02	n.d.	n.d.	n.d.
96	1.84 ± 0.32	n.d.	n.d.	n.d.

Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of individual peaks with those from authentic standards. Apigenin-d₅ was used as an internal standard for quantitation. Values refer to means ± SD (n = 3). FW: fresh weight, n.d.: not detected.

TABLE 7

Dissociation constant (K_d) of SbSOMT mutants with stilbenes			
Stilbene	SbSOMT mutant	Stoichiometry, N	K_d (μ M)
Resveratrol	I144N	0.6	151.6 ± 31.1
	D283A	0.58 ± 0.01	1.31 ± 0.13
	E310A	0.36 ± 0.04	18.5 ± 2.58
	F337N	—	No binding
	E342A	0.75 ± 0.01	11.1 ± 0.75
	I144N/F337N	—	No binding
Pinostilbene	H282N/D283A	0.61 ± 0.17	32.90 ± 5.93
	E310A	0.59 ± 0.01	3.91 ± 0.33
Pterostilbene	E310A	0.40 ± 0.08	6.04 ± 2.08

Results are expressed as means of $K_d \pm$ SE derived from curve fitting.

N of I144N was fixed at 0.6 (average to reported Ns in this table) to estimate the K_d .

TABLE 8

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of stilbenes in mechanically-wounded stalks wild sugarcane.			
Wounded wild sugarcane			
Time (h)	Resveratrol (μ g g ⁻¹ FW)	Piceatannol (μ g g ⁻¹ FW)	Isorhapontigenin (μ g g ⁻¹ FW)
0	n.d.	n.d.	n.d.
24	n.d.	n.d.	n.d.

TABLE 8-continued

High-performance liquid chromatography-quadrupole time-of-flight MS (HPLC-QTOF-MS) analysis of stilbenes in mechanically-wounded stalks wild sugarcane.			
Wounded wild sugarcane			
Time (h)	Resveratrol (μ g g ⁻¹ FW)	Piceatannol (μ g g ⁻¹ FW)	Isorhapontigenin (μ g g ⁻¹ FW)
72	2.55 ± 1.55	9.37 ± 5.98	1.03 ± 0.87
120	7.87 ± 3.03	33.79 ± 4.50	2.18 ± 1.77

Compound annotation was achieved by comparing the accurate mass and fragmentation pattern of individual peaks with those from authentic standards. Apigenin-d₅ was used as an internal standard for quantitation. Values refer to means $\pm$ SD (n = 3). FW: fresh weight, n.d.: not detected.

TABLE 9

Dissociation constants of SsCOMT with stilbenes.	
	K_d (μ M)
SsCOMT	
Resveratrol	2.78 ± 0.33
Pinostilbene	5.57 ± 0.77
Pterostilbene	9.93 ± 4.04
Piceatannol	25.50 ± 14.20
SAM	25.00 ± 12.90

Results are expressed as means of $K_d \pm$ standard error derived from curve fitting.

TABLE 10

List of OMT-ligand complexes deposited to PDB and the equivalent residues of SbCOMT ^{Asn128} /SbSOMT ^{Ile144} and SbCOMT ^{Asn323} /SbSOMT ^{Phe337} .							
		Residue at equivalent positions					
PDB ID	UNIPROT	SbCOMT Ile144	SbCOMT Phe337	Ligand identity	Vicinal functional group ^a and polarity	SEQ ID NO	Ref.
1FP1	P93324	Leu139	Val333	Substrate	—H and non-polar	82	[43]
1KYW	P28002	Asn131	Asn324	Substrate	—OH and polar	83	[44]
1KYZ	P28002	Asn131	Asn324	Product	—OH and polar		
1ZG3	Q29U70	Ala126	Phe326	Substrate	—H and non-polar	84	[54]
1ZGA	Q29U70	Ala126	Phe326	Substrate	—H and non-polar		
1ZGJ	Q29U70	Ala126	Phe326	Product	—H and non-polar		
2QYO	Q06YR3	Val123	Val319	Canonical product	—H and non-polar	85	NA
3I58	Q84HC8	Asp107	Phe294	Product	—H and non-polar	86	[55]
3I5U	Q84HC8	Asp107	Phe294	Substrate analogue (non-reactive)	Not compared		
3I64	Q84HC8	Asp107	Phe294	Substrate analogue (reactive)	Fused benzene ring and non-polar		
3P9I	Q9ZTU2	Asn128	Asn321	Product	—OH and polar	87	[22]
3P9K	Q9ZTU2	Asn128	Asn321	Product	—OH and polar		
3REO	O04385	Ala134	Asn327	Substrate	—OMe and polar	88	
3TKY	O04385	Ala134	Asn327	Substrate	—OMe and polar		
5CVU	O04385	Ala134	Asn327	Substrate	—OMe and polar		
5I2H	D5STZ7	Ala110	Thr322	Non-productive conformation	Not compared	89	
5ICE	Q5C9L7	Ile114	Thr311	Substrate	—H and non-polar	90	
5ICF	Q5C9L7	Ile114	Thr311	Non-reactive inhibitor	Not compared		

TABLE 10-continued

List of OMT-ligand complexes deposited to PDB and the equivalent residues of SbCOMT ^{Asn128} /SbSOMT ^{Ile144} and SbCOMT ^{Asn323} /SbSOMT ^{Phe337} .							
PDB ID	UNIPROT	Residue at equivalent positions		Ligand identity	Vicinal functional group [#] and polarity	SEQ ID NO Ref.	
		SbCOMT Asn128	SbCOMT Asn323				
		SbSOMT Ile144	SbSOMT Phe337				
5XOH	A0A166U5H3	His126	Val320	Substrate	Fused furan ring and non-polar	91	
6I6L	I3V6A7	Thr157	Asn351	Product	—OMe and polar	92	
6I6M	I3V6A7	Thr157	Asn351	Substrate	—OMe and polar		
6I72	Q9M602	Asn131	Asn324	Canonical substrate	—OH and polar	93	

[#]The vicinal functional group refers to the functional group vicinal to the reactive site positioned between catalytic dyads and SbCOMT^{Asn323}/SbSOMT^{Phe337} equivalent position as depicted in FIG. 25 & FIGS. 26A-26X.
NA, Not available (work to be published).

TABLE 11

Primers.		
Primer name	Sequence (5' to 3') (restriction sites added are in <i>italic</i>)	Purposes
SbEIF4a-qRT-F	CAACTTTGTCACCCGCGATGA (SEQ ID NO: 36)	qRT-PCR experiments for SbEIF4a
SbEIF4a-qRT-R	TCCAGAAACCTTAGCAGCCCA (SEQ ID NO: 37)	
SbSTS-qRT-F	TGCTACGGTGTGGCCATT (SEQ ID NO: 38)	qRT-PCR experiments for SbSTS1
SbSTS-qRT-R	ATCGACTTGTGGCATATCCTCTT (SEQ ID NO: 39)	
SbOMT4-qRT-F	CCTAAAGTGGATTCTTCACGATTG (SEQ ID NO: 40)	qRT-PCR experiments for SbOMT4
SbOMT4-qRT-R	GCTAGTTGCGATGTTAGTGTTC (SEQ ID NO: 41)	
SbSOMT-qRT-F	AGGTGGCAGCATGTTTCGATAA (SEQ ID NO: 42)	qRT-PCR experiments for SbSOMT
SbSOMT-qRT-R	ACGATCACCTTGCTCTCAC (SEQ ID NO: 43)	
SsGADPH-qRT-F	TTGGTTTCCACTGACTTCGTT (SEQ ID NO: 44)	qRT-PCR experiments for SsGADPH
SsGADPH-qRT-R	CTGTAGCCCCACTCGTTGT (SEQ ID NO: 45)	
SsSTS-qRT-F	AGAGCGAACACCTTACCGAC (SEQ ID NO: 46)	qRT-PCR experiments for SsSTS
SsSTS-qRT-R	GCCGAGTACGAGCTCATGTT (SEQ ID NO: 47)	
SsCOMT-qRT-F	GAGGACAAGGACGGCAAGTA (SEQ ID NO: 48)	qRT-PCR experiments for SsCOMT
SsCOMT-qRT-R	ACCGCGTCCTTGAGGTAGTA (SEQ ID NO: 49)	
SbOMT4-RE-F	CGCGGATCCATGGGCAGCTATACTAC CAG (SEQ ID NO: 50)	Cloning of SbOMT4 in pET23a(+) vector for recombinant protein

TABLE 11-continued

Primers.		
Primer name	Sequence (5' to 3') (restriction sites added are in <i>italic</i>)	Purposes
SbOMT4-RE-R	CCCAGCTTCTTTGTGAATTCAAGGG CCC (SEQ ID NO: 51)	expression in <i>E. coli</i>
SbSOMT-HF-F	TGAGAACCTGTACTTCCAAGGCAGC TACGACAGCAGCAGTAG (SEQ ID NO: 52)	HiFi Assembly of SbSOMT (2-377) in pET-N-His-TEV vector for recombinant
SbSOMT-HF-R	AGCCGGATCTCACTCGAGTTACTTTG TGAAGTCAAGGGCCAGAC (SEQ ID NO: 53)	protein expression in <i>E. coli</i>
SbCOMT-HF-F	TGAGAACCTGTACTTCCAAGGTCG ACGGCGGAGGAC (SEQ ID NO: 54)	HiFi Assembly of SbCOMT (2-362) in pET-N-His-TEV vector for recombinant
SbCOMT-HF-R	AGCCGGATCTCACTCGAGTTACTTGA TGAAGTCAAGGGCCAG (SEQ ID NO: 55)	protein expression in <i>E. coli</i>
SsSTS-Gib-F	CTTCTGCAGGAATTCGATATCATGAC TGGGAAGGTAACATTGGGG (SEQ ID NO: 56)	Gibson Assembly of SsSTS in pET-N- His-TEV vector for recombinant protein
SsSTS-Gib-R	GAGAGATCTGTCGACGATATCCTAC ACTGTGATGATGGGAACGC (SEQ ID NO: 57)	expression in <i>E. coli</i>
SsCOMT-Gib-F	CTTCTGCAGGAATTCGATATCATGGG CTCGACCGCC (SEQ ID NO: 58)	Gibson Assembly of SsCOMT in pET- N-His-TEV vector for recombinant
SsCOMT-Gib-R	GAGAGATCTGTCGACGATATCTTACT TGATGAAGTCAAGGGCCAG (SEQ ID NO: 59)	protein expression in <i>E. coli</i>
SbSOMT-1144N-F	GTGCTTCCGCTTGGGATGATGAACCT AAACAAGACATTCTGGACAGC (SEQ ID NO: 60)	Generation of SbSOMT 1144N and
SbSOMT-1144N-R	CATCATCCCAAGCGGAAGCAC (SEQ ID NO: 61)	I144N/F337N mutant proteins
SbSOMT-F337N-F	CCTCACCATGCTGGTCACGAACGGC AGTGGTAAAGAGAGGACACA (SEQ ID NO: 62)	Generation of SbSOMT F337N and
SbSOMT-F337N-R	CGTGACCAGCATGGTGAGG (SEQ ID NO: 63)	I144N/F337N mutant proteins
SbSOMT-H282A-F	TTCTGCTCAAGTGGATTCTTGCTGAT TGGGACGACAAGGCGTG (SEQ ID NO: 64)	Generation of SbSOMT H282A mutant
SbSOMT-H282A-R	AAGAATCCACTTGAGCAGAAGTCA (SEQ ID NO: 65)	protein
SbSOMT-H282N-F	GATTGGGACGACAAGGCGTG (SEQ ID NO: 66)	Generation of SbSOMT H282N mutant protein
SbSOMT-H282N-R	GTTAAGAATCCACTTGAGCAGAAC (SEQ ID NO: 67)	Generation of SbSOMT H282N and H282N/D283A mutant proteins
SbSOMT-D283A-F	GCGTGGGACGACAAGG (SEQ ID NO: 68)	Generation of SbSOMT D283A and H282N/D283A mutant proteins
SbSOMT-D283A-R	ATGAAGAATCCACTTGAGCAGAAC (SEQ ID NO: 69)	Generation of SbSOMT D283A mutant protein
SbSOMT-E310A-F	GCGTACGTTGTTCCGGATG (SEQ ID NO: 70)	Generation of SbSOMT E310A mutant

TABLE 11-continued

Primers.		
Primer name	Sequence (5' to 3') (restriction sites added are in <i>italic</i>)	Purposes
SbsOMT-E310A-R	CAGAACGATCACCTTGCCTC (SEQ ID NO: 71)	protein
SbsOMT-E342A-F	GCGAGGACACAGAGGGAGTTC (SEQ ID NO: 72)	Generation of SbsOMT E342A mutant
SbsOMT-E342A-R	TTTACCACTGCCAAACGTGAC (SEQ ID NO: 73)	protein
sbsomt-gPCR-F1	TGACTCATCAGCACGGAACG (SEQ ID NO: 74)	Genotyping of sbsomt CRISPR/Cas9
sbsomt-gPCR-R1	CACCCTCCTAGTAGCCGCAA (SEQ ID NO: 75)	mutants
sbsomt-gPCR-F2	GGCAAAACATTACGGATGCAGT (SEQ ID NO: 76)	Genotyping of sbsomt CRISPR/Cas9
sbsomt-gPCR-R2	TGACTTCACGGAGGCTCATAG (SEQ ID NO: 77)	mutants
SbSTS-PEAQ-F	ATCGGACCGGTATGACGACTGGGAA GGTAAC (SEQ ID NO: 78)	Cloning of SbSTSI in pEAQ-HT vector
SbSTS-PEAQ-R	CGATCCTCGAGTCATGCAGCCACTGT GGTGA (SEQ ID NO: 79)	for <i>N. benthamiana</i> expression
SbsOMT-PEAQ-F	TGCCCAAATTGCGAATGGGCAGCT ACGACAGC (SEQ ID NO: 80)	Cloning of SbsOMT in pEAQ-HT
SbsOMT-PEAQ-R	TGGTGATGGTGATGCATATTATTTT CAAATTGAGGATGAGACCACTTTGT GAACTCAAGGCCCCAG (SEQ ID NO: 81)	vector for <i>N. benthamiana</i> expression

TABLE 12

Protein and ligand concentrations used for ITC setup.			
Protein in cell	Protein concentration (μ M)	Ligand in syringe	Ligand concentration (μ M)
SbSOMT, WT	50	Resveratrol	750
	50	Pinostilbene	750
	25	Pterostilbene	500
	50	Piceatannol	1000
	50	SAM	2000
SbCOMT, WT	50	Resveratrol	750
	50	Pinostilbene	750
	25	Pterostilbene	500
	50	Piceatannol	750
	50	SAM	1500
<u>SbSOMT mutants:</u>			
I144N	25	Resveratrol	1000
D283A	50	Resveratrol	1000
	50	Resveratrol	1000
E310A	50	Pinostilbene	1000
	25	Pterostilbene	500
F337N	25	Resveratrol	1000
E342A	50	Resveratrol	1000
I144N/F337N	25	Resveratrol	1000
H282N/D283A	25	Resveratrol	1000
SsCOMT, WT	50	Resveratrol	750
	50	Pinostilbene	750

TABLE 12-continued

Protein and ligand concentrations used for ITC setup.			
Protein in cell	Protein concentration (μ M)	Ligand in syringe	Ligand concentration (μ M)
	25	Pterostilbene	500
	50	Piceatannol	1000
	50	SAM	2000

TABLE 13

X-ray data collection and refinement statistics of disclosed structures.				
	SbSOMT- Resveratrol- β -NAD	SbSOMT- Pinostilbene- β -NAD	SbSOMT- Pterostilbene- β -NAD	SbSOMT- Resveratrol
Data collection				
Wavelength ($\text{\AA}$)	0.9785	0.9784	0.9784	0.9785
Resolution ($\text{\AA}$) ^a	48.42-1.72 (1.76-1.72)	19.70-2.10 (2.14-2.10)	31.40-2.40 (2.44-2.40)	48.91-2.56 (2.65-2.56)
Space group	P 3 ₁ 2 1	P 3 ₁ 2 1	P 3 ₁ 2 1	P 2 ₁ 2 ₁ 2 ₁
Cell dimensions				
a, b, c ($\text{\AA}$)	96.8, 96.8, 168.0	96.7, 96.7, 166.9	96.5, 96.5, 169.4	97.5, 111.7, 131.1
α , β , γ ($^\circ$)	90, 90, 120	90, 90, 120	90, 90, 120	90, 90, 90
Total reflections	1882940	651200	709365	616499
Unique reflections	97225	101522	36363	46864
Multiplicity	19.4	6.4	19.5	13.2
Completeness (%) ^a	99.8 (97.1)	99.8 (99.0)	99.2 (99.9)	99.9 (100.0)
Mean I/sigma (I) ^a	30.2 (2.9)	9.2 (2.0)	7.5 (0.8)	16.3 (2.4)
Wilson B-factor ($\text{\AA}^2$)	22.6	31.43	36.78	52.2
R _{merge} (%) ^a	6.0 (94.1)	21.8 (72.4)	19.9 (95.5)	11.8 (115.6)
R _{meas} (%) ^a	6.3 (101.0)	23.8 (78.8)	20.4 (98.2)	12.7 (124.7)
R _{pin} (%) ^a	2.0 (35.7)	9.4 (31.0)	4.6 (22.8)	4.8 (46.8)
CC _{1/2} ^a	1.000 (0.849)	0.891 (0.790)	0.915 (0.871)	0.999 (0.902)

^a Value relative to the highest resolution shell are given in parentheses.

TABLE 13

X-ray data collection and refinement statistics of structures reported in this study (continued).				
	SbSOMT- Resveratrol- β -NAD	SbSOMT- Pinostilbene- β -NAD	SbSOMT- Pterostilbene- β -NAD	SbSOMT- Resveratrol
Refinement				
Reflections used in refinement	97159	53133	36012	46802
Reflections used for R-free	4899	2652	1899	2368
R-work	0.163	0.171	0.187	0.182
R-free	0.190	0.202	0.246	0.252
Number of atoms	6638	6216	6104	11441
Macromolecules	5841	5674	5721	11286
Ligands/Ions	229	274	214	92
Solvent	568	268	169	63
rmsd ^b bond lengths ($\text{\AA}$)	0.012	0.015	0.016	0.016
rmsd ^b bond angles ($^\circ$)	1.73	2.023	2.10	2.22
Ramachandran plot				
Favored (%)	99.03	97.36	95.57	95.72
Allowed (%)	0.97	2.22	3.60	3.31
Outlier (%)	0.00	0.42	0.83	0.97
Sidechain outlier (%)	0.64	2.64	1.96	4.56

TABLE 13-continued

X-ray data collection and refinement statistics of structures reported in this study (continued).				
	SbSOMT- Resveratrol- β -NAD	SbSOMT- Pinostilbene- β -NAD	SbSOMT- Pterostilbene- β -NAD	SbSOMT- Resveratrol
Molprobability Clash score	4.46	8.69	7.95	8.10
Mean B-value ($\text{\AA}^2$)	29.0	37.0	46.0	68.0
Macromolecules	28.38	36.72	45.54	62.15
Ligands/Ions ^c	53.31 (41.01)	61.19	78.06	52.78
Solvent	36.14	39.74	43.73	41.98
PDB ID code	7VB8	7WAR	7WAS	7WAQ

^b rmsd, root mean square deviation.^c The mean B-values ($\text{\AA}^2$) of ligands and ions are given separately, with value for ion given in parentheses.

TABLE 14

Accession numbers of sequences used in this study.		
Protein	GenBank accession number	SEQ ID NO
<i>Arabidopsis thaliana</i> AtOMT1	NP_200227	5
<i>Carthamus tinctorius</i> CtCaldOMT	BAG71895	6
<i>Catharanthus roseus</i> CrCOMT	Q8W013	7
<i>Hordeum vulgare</i> HvCOMT	ABQ58825	8
<i>Medicago sativa</i> MsCOMT	P28002	9
<i>Medicago sativa</i> MsIOMT	AAC49927	10
<i>Oryza sativa</i> OsCaldOMT1	Q6ZD89	11
<i>Oryza sativa</i> OsNOMT	QOIP69	12
<i>Panicum virgatum</i> PvCOMT	ADX98508	13
<i>Pinus sylvestris</i> PsPMT2	AQX17823	14
<i>Pinus taeda</i> PtAEOMT	Q43096	15
<i>Populus tremuloides</i> PtCOMT	AAB61731	16
<i>Rosa hybrid</i> RhOOMT	AAM23004	17
<i>Solanum lycopersicum</i> SlCOMT	XP_004235028	18
<i>Sorghum bicolor</i> SbCOMT	AAO43609	2
<i>Sorghum bicolor</i> SbNOMT	XP_002465645	19
<i>Sorghum bicolor</i> SbOMT1	ABP01563	20
<i>Sorghum bicolor</i> SbOMT3	ABP01564	21
<i>Sorghum bicolor</i> SbOMT4	XP_002443937	22
<i>Sorghum bicolor</i> SbSOMT	XP_021320201	1
<i>Sorghum bicolor</i> SbSTS1	XP_002445139	23
<i>Triticum aestivum</i> TaCOMT1	AAP23942	24
<i>Triticum aestivum</i> TaCOMT2	Q38J50	25
<i>Vitis vinifera</i> VvCOMT	NP_001268100	26
<i>Vitis vinifera</i> VvROMT	CAQ76879	27
<i>Zea mays</i> ZmCOMT	Q06509	28
<i>Zea mays</i> ZmNOMT	XP_020399799	29
Gene	Ensembl Plants Identifier	
<i>Saccharum spontaneum</i> SsSTS	Sspon.06G0010290-2P	3
<i>Saccharum spontaneum</i> SsCOMT	Sspon.06G0010980-3C	4
Gene	GenBank accession number	
<i>Sorghum halepense</i> ShOMT1	GGDZ01007645	30
<i>Sorghum halepense</i> ShSOMT	GGDZ01140964	31

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Additional Sequences

AtOMT1 (*Arabidopsis thaliana*; NP_200227)

(SEQ ID NO: 5)

MGSTAETQLT PVQVTDEAA LFAMQLASAS VLPMAKLSAL ELDLLEIMAK
 NGSPMSPTIE ASKLPTKNPE APVMLDRILR LLTSYSVLTC SNRKLSGDGV
 ERIYGLGPVC KYLTKNEDGV SIAALCLMNQ DKVLMESWYH LKDAILDGGI
 PFNKAYGMSA FEYHGTDPHF NKVENNGMSN HSTITMKKIL ETYKGFEGLT
 SLVDVGGGIG ATLMKIVSKY PNLKGINFDL PHVIEDAPSH PGIEHVGGDM
 FVSVPKGDIA FMKWICHDSW DEHCVKFLKN CYESLPEDGK VILAECILPE
 TPDSSLSTQK VVHVDCIMLA HNPGGKERTK KEFEALAKAS GFKGIKVVCD
 AFGVNLIELL KKL

CtCaldOMT (*Carthamus tinctorius*; BAG71895)

(SEQ ID NO: 6)

MAHLEEEEF LFAMQLASAS VLPMLKSAI DLDLLEIIAK AGPGAHVSPA
 YLAAQLPKAD NPEAAVMLDR ICRLLATYSV LTCTLKKLKH GDGDVERLYG
 LAPVCKFLVK NEDGVSNAFL LLMNQDKILM ESWYHLKDAV LDGGIPFNKA
 YGMSAFEYHG KDERFNKVPN SGMFNHSTMT MKKILDVYPG FNGVKTLDVD

-continued

GGGTGASLSM ILSKHPSIKG INFDLPHVIQ DATNYPGIEH VGGDMFESVP
 KGDAIFMKWI CHDWSDAHCL KFLKNCKYKAL PENGKVIVAE CILPETPDSS
 LATKNVVID VIMLAHNPBG KERTEKEFQA LAKAAGFKGF HKPCSALNTW
 LMEFTK

CrCOMT (*Catharanthus roseus*; Q8W013)

(SEQ ID NO: 7)

MGSANPDNKN SMTKEEEEAC LSAMRLASAS VLPMLKSAI ELDLLELIKK
 SGPGAYVSPS ELAAQLPTQN PDAPVMDLRI LRLLASYSVL NCTLKDLDPG
 GIERLYSLAP VCKFLTKNED GVSMAALLLM NQDKVLMESW YHLKDAVLEG
 GIPFNKAYGM TAFEYHGKDP RENKVENQGM SNHSTIIMKK ILEIYQGFQG
 LKTVVDVGGG TGATLNMIVS KYPSIKGINF DLPHVIEDAP SYPGVDHVGG
 DMFVSVPKGD AIFMKWICH DSDAHCLKFL KNCHEALPEN GKVILAECLL
 PEAPDSTLST QNTVHVVDIM LAHNPBGKER TEKEFEALAK GAGPRGFIKV
 CCAYNSWIME LLK

HvCOMT (*Hordeum vulgare*; ABQ58825)

(SEQ ID NO: 8)

MGSIAAGADE DACMYALQLV SSSILPMTLK NAIELGLLET LMSAGGKFLT
 PAEVAAKLPS TANPEAPDMV DRMLRLLASY KVVSCRTEES KDGRLSRRYG
 AAPVCKYLT NEDGVMSAL ALMNQDKVLM ESWYYLKDAV LDGGIPFNKA
 YGMSAFEYHG TDPFRNRFVN EGMKNHSIII TKKLLESYKG FEGLGTLVDV
 GGGVGATVGA IIRYPAVKG INFDLPHVIS EAPAFPGVTH IGGDMFQKVP
 SGDAILMKWI LHDWSDEHCA TLLKNCYDAL PAHGKVVLE CILPVNPEAT
 PEVQGVFHDV MIMLAHNPBG RERYEREFEA LAKGAGFAAM KTTYIYANAW
 AIEFTK

MsCOMT (*Medicago sativa*; P28002)

(SEQ ID NO: 9)

MGSTGETQIT PTHISDEEAN LFAMQLASAS VLPMLKSAL ELDLLEIIAK
 AGPGAQISPI EIASQLPTTN PDAPVMDLRI LRLACIYIIL TCSVRTQQDG
 KVQRLYGLAT VAKYLVKNED GVSISALNLM NQDKVLMESW YHLKDAVLGD
 GIPFNKAYGM TAFEYHGTDP RFNKVENKGM SDHSTITMKK ILETYTGFEG
 LKSLVDVGGG TGAVINTIVS KYPTIKGINF DLPHVIEDAP SYPGVEHVGG
 DMFVSIPKAD AVFMKWICH DSDAHCLKFL KNCYEALPDN GKVIVAECLL
 PVAPDSSLAT KGVVHIDVIM LAHNPBGKER TQKEFEDLAK GAGPQGFVKH
 CNAFNTYIME FLKKV

MsIOMT (*Medicago sativa*; AAC49927)

(SEQ ID NO: 10)

MASSINGRKP SEIFKAQALL YKHIYAFIDS MSLKWAVGMN IPNIIHNHGK
 PISLSNLVSI LQVPSSKIGN VRRLMRYLAH NGFFEITKE EESYALTVAS
 ELLVRGSDLC LAPMVECVLD PTLSGSYHEL KKWIYEEDLT LFGVTLGSGF
 WDFLDKNPEY NTSFNDAMAS DSKLINLALR DCDFVEDGLE SIVDVGGGTG

-continued

TTAKIICETF PKLKCIVEDR PQVVENLSGS NNLTYVGGDM FTSIPNADAV
LLKYILHNWT DKDCLRLKK CKEAVTNDGK RGKVTIIDMV INEKKDENQV
TQIKLLMDVN MACLNGKERN EEEWKLFIE AGFQHYKISP LTGFLSLIEI
YP

OsCaldOMT1 (*Oryza sativa*; Q6ZD89)
(SEQ ID NO: 11)
MGSTAADMAA AADEEACMYA LQLASSSILP MTLKNAIELG LLETLSAAV
AGGGGKAALL TPAEVADKLP SKANPAAADM VDRMLRLLAS YNVVRCMEE
GADGKLSRRY AAPVCKWLT PNEDGVSMMA LALMNQDKVL
MESWYYLKDA VLDGGIPFNK AYGMTAFEYH GTDARFNRFV NEGMKNHSVI
ITKLLDLTYT GFDAASTVVD VGGGVGATVA AVVSRHPHIR GINYDLPHVI
SEAPPPPGVE HVGGMDFASV PRGGDAILMK WILHDWSDEH CARLLKNCYD
ALPEHGKVVV VECVLPESD ATAREQGVFH VDMIMLAHNP GKKERYEREF
RELARAAGFT GFKATYIYAN AWAIEFTK

OsNOMT (*Oryza sativa*; QOIP69)
(SEQ ID NO: 12)
MVSPVVHRHA AGGGSGGDD DQACMYALEL LGGSVVSMTL KAAIELGLVD
ELLAAAGAAV TAEELAAARLR LPAAVAAAAA VDRMLRLLAS YGVVRCATEA
GPDGKARRSY AAPVCKWLA AGSSSGEGSM APLGLNLNDK VFMENWYYLK
EAVSEGGTAF DKAYGTSLFQ YLGQDGNES NTLFNQAMAS HSVVITNKLL
QFFRGFDAGA GVDVLVDVGG GVGATLRMIT ARHPLRGVN YDLPHVIAQA
PPVEGVEHIG GSMFDHVPSG SAILLKWILH LWGDEECVKI LKNCYKALPA
KGVVILVEYV LPASPEATLA AQEAFRLDVM MLNRLAGGKE RTQQEFTDLA
VDAGFSGDCK PTYIFTNVWA LEFTK

PvCOMT (*Panicum virgatum*; ADX98508)
(SEQ ID NO: 13)
MGSTATDVAA AADEEACMYA LQLASSSILP MTLKNAIELG LLEVLQKDPA
AALAPEEVVA QLPVAPANPD AAAMVDRMLR LLASYDVVRC QMEEGKDGRY
SRRYAAAPVC KWLTPNEDGV SMAALALMNQ DKVLMESWYY LKDAVLEGGI
PFNKAYGMTA FEYHGTDPFR NRVFNEGMKN HSVIITKKLL EBYAGFEGVG
TLVDVGGGVG ATLHAITSRY PGIRGVNFDL PHVISEAPPF PGVEHVGGDM
FKAVPAGDAI LMKWILHDWS DAHCAAILKN CYDALPAGGK VIAVECILPV
NPEATPKAQG VFHVDMIMLA HNPGGKERYE REFEELAKGA GFTGFKATYI
YANAWAIEFT K

PsPMT2 (*Pinus sylvestris*; AQX17823)
(SEQ ID NO: 14)
MNMQSVKDEE ALRASALGLA FSLETPFLK CAIRLKIPDI ISKAGPDVSL
SVHQIAAQLP SEDPDMGALS RILTYLSTMG ILQAIVPPEG VNAPMNIRYG
LTNLTKTYFT SEDISSRLV PFVLLQTHPL YVTAWDNIHE RVLHGGDNFK
NSSGNGKDFW NFAAGEPEFN AIPNAGMVSV TKATITYVLA VYDGFKDINT

-continued

LVDVGGGRGE ALSLITEAHP HIRAINFDLP QVIATAPTIP GVQHMSGNLF
ESAPSADAIF MKNFLHSWND EDCIKLLNNC HQALPEKGKL ILSEAILDLT
EGSDMIGSAN VLDVAVMLNCL PGGGERTRKQ WNDLLQAAGF SISKIVGRNG
TLTKVIEAIK S

PtAEOMT (*Pinus taeda*; Q43096)

(SEQ ID NO: 15)

MDSNMNGLAK SNGCEISRDG FFESEEEELQ GQAEAWKCTF APAESLAVKC
VVLGIPDMI AREGPRATLS LCEIVANLPT ESPDAACLFR IMRFLVAKGI
FPASKSARRR AFETRYGLTP ASKWLVKGRE LSMAPMLLMQ NDETTLAPWH
HFNECVLEGG VAFQKANGAE IWSYASDHPD FNNLFNNAMA CNARIVMKAI
LSKYQGPHSL NSLVDVGGGT GTAVAEIVRA YPFIRGINYD LPHVVATASS
LSGVQHVGGD MFETVPTADA IFMKWIMHDW NDEDCIKILK NCRKAIPDTG
KVIIVDVULD ADQGDNTDKK RKAVDPIVG TVFDLVMVAH SSGGKERTEK
EWKRILLEGG FSRYNIEIP ALQSVIEAFP R

PtCOMT (*Populus tremuloides*; AAB61731)

(SEQ ID NO: 16)

MGSTGETQMT PTQVSDEEAH LFAMQLASAS VLPMLKTAI ELDLLEIMAK
AGPGAFLSTS EIASHLPTKN PDAPVMLDRI LRLLASYSIL TCSLKDLDPDG
KVERLYGLAP VCKFLTKNED GVSVSPLCLM NQDKVLMESW YYLKDAILDG
GIPFNKAYGM TAFEYHGTDP RFNKVFNKGM SDHSTITMKK ILETYKGFEG
LTSLVDVGGG TGAVVNTIVS KYPSIKGINF DLPHVIEDAP SYPGVEHVGG
DMFVSVPKAD AVFMKWICH D WSDAHCLKFL KNCYDALPEN GKVILVECIL
PVAPDTSLAT KGVVHVDVIM LAHNPGGKER TEKEFEGLAK GAGFQGFVEM
CCAFNTHVIE PRKKA

RhOOMT (*Rosa hybrid*; AAM23004)

(SEQ ID NO: 17)

MERLNSFRHL NQKWSNGEHS NELLHAQAH WNHIFSFINS MSLKSAIQLG
IPDIINKHGY PMTLSEL TSA LPIHPTKSHS VYRLMRILVH SGFPAKKKLS
KTDEEGYTLT DASQLLLKDH PLSLTPYLTA MLDPVLTNPW NYLSTWFQND
DPTPFDTAHG MTFWDYGNHQ PSIAHLEND MASDARLVTS VIINDCKGVF
EGLESLVDVG GGTGTLAKAI ADAFPHIECT VLDLPHVVAD LQGSKNLKYT
GGDMFEAVPP ADTVLLKWIL HDWSDEECIK ILERSRVAIT GKEKKGKVII
IDMMENQKG DEESIETQLF FDMLMMALVG GKERNEKEWA KLFTDAGFSD
YKITPISGLR SLIEVYP

SICOMT (*Solanum lycopersicum*; XP_004235028)

(SEQ ID NO: 18)

MGSTSLTQTE DEAF LFAMQL ASASVLPMLV KSAVELELLE LMAKAGPGAS
ISPAELASQL PCKNPDAFVM LDRMLRLAA YSVLNCTLRT LPDGRVERLY
SLAPVCKFLT KNADGVSVAP LLLMNQDKVL MQSWYHLKDA VLDGGIPFNK
AYGMTAFEYH GTDPRFNKVF NRGMSDHTTL SIKKILEDYK GFEGLSIVD

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VGGGTGATVS MIVSKYPSIK GINFDLPHVI EDAPAYPGVE HIGGDMFVSV
PKADAIFMKW ICHDWSDEHC LKFLKNCYEA VPANGKVIIA ECLLPEVPDT
SSSTKNTVHV DVIMLAHNPV GKERTKEFE ALAKGAGENG FTKASCAYNT
WIMEFTK

SbNOMT (*Sorghum bicolor*; XP_002465645)

(SEQ ID NO: 19)

MACTTAASLQ HDKANDDEAC MYAQELLFSF IVPMTLKAVI ELGLIDYLLA
ADGRSVTPPE LAEWPQSAE AAAAVDRMMR LLASHSVVRC TTEVGPDKA
RRSYAAAPVC KWLATRNAGG QGSLAPMGLM NLNKAFMETW
YFMKEAVTEG ATPTEKAYGM PLFEHLGSDE ASNTLFNQAM AGHSEMIKK
LLEVYRGFEG VDVLDVGGG TGSTLRMVTA QYKHLRGVNY DLPHVIAQAP
PVQGVHEVGG SMFEYIPSGN AILLKWILHL WRDDECVKIL KNCHRALPAN
GKVIVVEYVL PASPEPTQVA QVSLLLDVAM LNRLRGAKER TEQEFAQLAA
EAGFSGGCRA TYVFASAWAL EFTK

SbOMT1 (*Sorghum bicolor*; ABP01563)

(SEQ ID NO: 20)

MASYSTSTSGQ FAVGKVAAAN QDDETCMHAL KLLGGLAVPF TIKAVIELGI
MDLLLAADRA MTAEALTAAL LCPAPAPAAA AAMVDRMLRF LASHGVVRCA
TESEELGSDD GKSCRRYAAA PVCKWFARGG GVESVVPMPG WMTSTTNMET
WHNIKDGVLG GETPFDKAYG MPVFEYLGAN GTMNTLFNEA MASHSMIITK
RLLEVFRGFE NYSVLVDVGG GNGTTQMIR SQYENISGIN YDLPHVIAQA
SPIEGVEHVA GNMEDNIPRG DAILKWILH NWGDKECVKI LKNCTALPV
NGTVIILEYI LPETPEETLA SQLAFDFDLG MMLFFGASGK ERTEKELLEL
AREAGFSGDY TATYIFANVW AHEFTK

SbOMT3 (*Sorghum bicolor*; ABP01564)

(SEQ ID NO: 21)

MVLISEDRE LLQAHVELWN QTYSPMKVA LAVALDLHIA DAIHRRGGAA
TLSQILGEIG VRPCKLPGLH RIMRVLTVSG TFTIVQPSAE TMSSESDGRE
PVYKLTASS LLVSSESSAT ASLSPMLNHV LSPFRDSPLS MGLTAWFRHD
EDEQAPGMCP FTLMYGTTW EVCRDDAIN ALFNNAMAAD SNFLMQILLK
EFSEVFLGID SLVDVAGGVG GATMAIAAAF PCLKCTVLDL PHVAKAPSS
SIGNVQFVGG DMFESIPPAN VVLLKWILHD WSNDECILK KNCKQAIPSR
DAGGKIIID VVVGSDSSDT KLETTQVIYD LHLMKIGGVE RDEQEWKKIF
LEAGFKDYKI MPILGLRSII ELYP

SbOMT4 (*Sorghum bicolor*; XP_002443937)

(SEQ ID NO: 22)

MGSYTTRVHA VEKAANNQDD ETCMHMLLL GGLAVPCTIK AVIELGIMDL
LLAADRAMTA EELTARLPCP AAATAAMVD RMLRFLASHG VVRCAAATKE
SSELGSDGGK SCRRYAAPV CRWFTRSGGV ESVVPMLWM TGKTVLETWY
NIKEAVLEGE TPFDRAYGQP FFEYLGNGT VNTLFDEAMA NHSTIITKRL

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VEVFRGFENY SVLVDVGGNK GTTLQMIRSQ YENISGINYD LPRVIAQAPP
 IEGVEHVGGN MFDNVPRGDA IILKWILHDW GDKDCVKILK NCYAALPVNG
 TMIILEYILP ETPEETLTSQ LAPNFDPGMM LMYGAKGKER TEKELSELAR
 EAGFSGDCTA TYIFASIWAL EFTK

SbSTS1 (*Sorghum bicolor*; XP_002445139)
 (SEQ ID NO: 23)

MTTGKVTLEA VRKAQRAEGP ATVLAIGTAT PANCVYQADY PDYYFRVTKS
 EHLLDLKEKF KRICKSMIR KRYMHLTEI LEENPNMSSY WAPSLDARQD
 ILIQEIPKLG AEAAEKALKE WGQPRSRITH LVFCTTSGVD MPGADYQLIK
 LLGLCPSVNR AMMYHQGCFA GGMVLRLLAKD LAENNRGARV LIVCSEITVV
 TFRGPSESHL DSLVGQALFG DGAAAVIVGA DPSEPAERPL FHLVSASQTI
 LPDSEGAIEG HLREVGILTFH LODRVPQLIS MNIERLLEDA FAPLGISDWN
 SIFWVAHPGG PAILNMVEAK VGLDKARMCA TRHILAEYGN MSSVCVLFIL
 DEMNRNSAKD GHTTTGEGME WGVLFGGFPG LTVETIVLHS VPITTVAA

TaCOMT1 (*Triticum aestivum*; AAP23942)
 (SEQ ID NO: 24)

MGSTAADMAA SADEEACMYA LQLVSSSILP MTLKNAIELG LLETLVAAGG
 KLLTPAEVAA KLPSTANPAA ADMVDRMLRL LASYNVVSCT MEEGKDGRLS
 RRYRAAPVCK FLTPNEDGVS MAALALMNQD KVLMEWYYL KDAVLGGIP
 FNKAYGMSAF EYHGTDPFRN RVFNEMGMKNH SIIITKKLLE VYKGFEGGLT
 IVDVGGGVGA TVGAITAAYP AIKGINFDLP HVISEAQPPF GVTHVGGDMF
 QKVPBGDAIL MKWILHDWSD EHCATLLKNC YDALPAHGKV VLVECILPVN
 PEATPKAQGV FHVDMIMLAH NPGGRERYER EFEALAKGAG FKAIKTTYIY
 ANAFAIEFTK

TaCOMT2 (*Triticum aestivum*; Q38J50)
 (SEQ ID NO: 25)

MGSAAGADE DACMYALQLV SSSILPMTLK NAIELGLLET LMAAGGKFLT
 PAEVAALKPS AANPEAPDMV DRMLRLLASY NVVSCRTEG KDGRLSRRYG
 AAPVCKYLTP NEDGVMSAL ALMNQDKVLM ESWYYLKDAV LDGGIPFNKA
 YGMSAFEYHG TDPFRNRFVN EGMKNHSIII TKKLLESYKG FEGLGTLVDV
 GGGVGATVAA ITAHYPTIKG INFDLPHVIS EAPPFPVGVTH VGGDMFQKVP
 SGDAILMKWI LHDWSDEHCA TLLKNCYDAL PAHGKVVLVE CILPVNPEAT
 PKAQGVFHV D MIMLAHNP GG RERYEREFEA LAKGAGFAAM KTTYIYANAW
 AIEFTK

VvCOMT (*Vitis vinifera*; NP_001268100)
 (SEQ ID NO: 26)

MESTLAFNSG SNSMNQSFSS SAEFNSPVPE TIPKSEEDTF VFATLLTSAS
 VLPMAKLSAL ELDLLEIIAK AGPGAFVSTS EIAAKITKRN PKAPVMLDRI
 LRLLATYDVV KCSLRDSPDG GVERLYGLGP VCKYFTTNE GVSAPLLLM
 NQDKVPMQSK RYHLKDAVLD GGIPFNKAYG MTDPEYHGTE PRENKVENNG
 VSGHPTITMK KILEAYKGFE GLTSIVDVGG GTGATLNMII SKYPTIKGIN

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FDLPHVIDDA PSYPGVEHVG GDMFVSVPKG DAIFMKWMCY EWDDAHCLKF
 LENCYQALPD NGKVIVAEI LPVVPDTSLA TKSAVHIDVI MLAYNTGGKA
 RTEKEFEALA KGAGFQGFKV VCCAFNSWIM EFCKTA

VvROMT (*Vitis vinifera*; CAQ76879)

(SEQ ID NO: 27)

MDLANGVISA ELLHAQAHVW NHIFNFIKSM SLKCAIQLGI PDIIHNHGKP
 MTLPELVAKL PVHPKRSQCV YRLMRILVHS GPLAAQRVQQ GKEEEGYVLT
 DASRLLLMDD SLSIRPLVLA MLDPIILTKPW HYLSAWFQND DPTPFHTAHE
 RSFWDYAGHE PQLNNSFNEA MASDARLLTS VLLKEGQGVF AGLNSLVDVG
 GGTGKVAKAI ANAPPHLNCT VLDLPHVVAG LQGSKNLNYF AGDMFEAIPP
 ADAILLKWIL HDWSDEECVK ILKRCREAIK SKENGGKVII IDMIMMKNQG
 DYKSTETQLF FDMTMMIFAP GRERDENWE KLFLDAGFSH YKITPILGLR
 SLIEVYP

ZmCOMT (*Zea mays*; Q06509)

(SEQ ID NO: 28)

MGSTAGDVAA VVDEEACMYA MQLASSSILP MTLKNAIELG LLEVLQKEAG
 GGKALAPEE VVARMPAAPG DPAAAAAMVD RMLRLLASYD
 VVRCQMEDRD GRYERRYSA PVCKWLTPNE DGVSMALAL MNQDKVLMES
 WYYLKDAVLD GGIPFNKAYG MTAFYHGTG SRFNRVFNKG MNHNSVIITK
 KLLDFYTGFV GVSTLVDVGG GVGATLHAIT SRHPHISGVN FDLPHVISEA
 PPPFGVRHVG GDMFASVPAG DAILMKWILH DWSDAHCAIT LKNCYDALPE
 NGKVIVVECV LPVNTEATPK AQGVFHVDMI MLAHNPGGKE RYEREFRELA
 KGAGFSGFKA TYIYANAWAI EFKI

ZmNOMT (*Zea mays*; XP_020399799)

(SEQ ID NO: 29)

MACTTAASLQ HDRANDDEAC MYAQELLSCF VVPMTLKAVI ELGLIDDLA
 ADGRFVTPEE LAARWARPAE AAAAVDRMLR FLASHSVVRC TTEAGPDGR
 ARRSYAAAPV CKWLIARNGT GQGSWAPIGL MNLKGFMT WYMKDAVAE
 GATPTEKAYG MPLFEHLGSD EALNTLFNQA MAGHSEIVIS KLEVVYRGFE
 GVDVLVDVGG GTGSTLRMVT AQYKHLRGVN YDLPHVIAQA PPVQGVHAG
 GSMFEYIPSG NAILLKWILH LWRDEECIKI LKNCHRALPA NGKVIVVECV
 LPASPEPTQV AQGALLLDV MLNRLRGAKE RTEREFTELA AEAGFSGGCR
 ATYVFTGAWA LEFTK

ShOMTI (*Sorghum halepense*; GGDZ01007645)

(SEQ ID NO: 30)

atgggcagct atactaccag agtccacgcc gtagagaaag cagctaacaa
 ccaagatgat gagacgtgca tgcattcttt gatgctcctt ggcggattgg
 ccgtaccttg taccatcaag geggctcag agctcggcat catggacctt
 ctctcgceg cggacegcgc catgaccgeg gaggagctca cggcaagggt
 gccatgtcct gctgctgcta ctgctgctgc tatggctgac cgcattgctc
 gettcctage ttgcgacggc gtggctcaggt gegccggcc caccaggag
 tcgtcggagc tgggctccga tgggtggcaag agctgccggc gctacgggc
 ggccggcggtg tgcaggtggt ttacaaggag cggcgccgtg gatcggtg

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ttccaatggg gttgtggatg accggcaaga cgtcttgga gacctggtac
 aacataaaag aagcagtgtt agaggagaa acaccatttg acagagcata
 cggccagcca ttttttgagt accttggtga aaacgggaca gtgaacacgt
 tgttcgatga ggcaatggcg aaccattcga caatcataac gaagaggctg
 gtcgaggtct tccgtggett tgagaattac agtgtgctcg tcgatgtcgg
 cggcaacaaa ggcaccacgt tgcaaatgat tagaagtcag tatgagaata
 ttagtggcat caactacgac ctctctcgtg taattgcgca ggcacctcca
 attgaagggtg tggagcatgt tggtggaac atgtttgata atgttcacg
 tggagatgca attatcctaa agtggattct tcacgattgg ggcgacaagg
 actgctcaa gatcctaaag aattgctatg cagctctccc ggtgaacggg
 acgatgatca ttctggagta catctctcca gagacaccag aagaaacact
 aacategcaa ctacggttca acttcgattt cgggatgatg ctcatgtacg
 gogccaaagg taaggagaga acagagaaaag agttgtcga gctcgcaga
 gaggccggct tctctggaga ctgcacggct acatacatct tegccagcat
 ctgggccctt gaattcacia agtaa

ShSOMT (*Sorghum halepense*; GGDZ01140964)

(SEQ ID NO: 31)

tcgtcctcta tctgtggcca ccaactgctg tgctatttga gtacacagat
 acagatgggc agctacgaca gcagcagtag tagtagtaat gactcatcag
 cacggaacga ggaggacgag tcgtgcatgt ttgctttgaa gctcctcggc
 gggttcgcg tacctttcac catcaaggcg gtgattgagc tgggcgtcat
 ggaccagctc ctgactgccg aacgcgcgat gagcgcggag gagctggtgg
 cagcagcagt ggcagcacag ctgccacggc ccgaggtagc ctgcaccatg
 gtggaccgcc tgctcctgtt ccttgctcg cacagcgtag tccggtgcac
 gaccaggtg gtggtgggca cggacgacgc caccaccacc acctgetgcc
 gccggagcta cgcegcgtca cccgtctgca agtggttcgc caggaaacggc
 gtcgaggatt cgggtcttcc gcttgggatg atgaccta acaagacatt
 cctggacagc tggcaaaaca ttacggatgc agtggtgaa ggagcagcac
 catttgagaa aacctacggg atgccaatgt tegagtacct aagtacaaac
 ggaccattga acacggtgtt ccacgaggca atggcaaatc attcgatgat
 tataaccaag aaactgctca agttcttccg cggcttcgaa ggccttgatg
 tgctggtcga cgtaggcggc ggcaacggca ccacgttgca aatgattaga
 ggtcaatata agaatatgag aggcataaac tacgacctc ctcatgtcat
 tgcgcaggct gcaccagttg aagggtgtga acatgtagg ggcagcatgt
 tcgataatat tccacgegga atgcagttc tgctcaagtg gattcttcat
 gattgggacg acaaggcgtg catcaagatc cttaaagaatt gctatacagc
 tctccatgtg agaggcaagg tgatcgttct ggagtacgtt gttccggatg
 aaccagaacc tactcttgca gtcagggtg ccttcgaaact ggacctcacc
 atgctggtca cgtttggcag tggtaaagag aggacacaga gggagttctc
 cgagctcgcc atggaggccg gcttctctag agagtttaag gctacgtata

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tctttgccaa cgtctgggcc cttgagttca caaagtaa

SsCOMT (*Saccharum spontaneum*)

(SEQ ID NO: 32)

MGSTAEDVAA VADEEACMYA MQLASASILP MTLKNALELG LLEVLQAEAP

AGKALAPEEV VARLPVAPTIN PDAADMVDRM LRLASYDVV KCQMEDKDGG

YERRYSAAPV GKWLTPNEDG VSMAALTLMN QDKVLMESRY YLKDAVLDGG

IPFNKAYGMT AFEYHGTDPN FNRVFNEMGM NHSVIITKKL LEFYTGFEVG

STLVDVGGGI GATLHAITSH HPQIKGINFD LPHVISEAPP FPGVQHVGGD

MPKSVPGADA ILMKWILHDW SDAHCATLLK NCYDALPENG KVIIVCEVLP

VNTEAVPKAQ GVFHVDMIML AHNPGGRERY EREFHDLAKG AGFSGFKATY

IYANAWAIEF IK

SsSTS (*Saccharum spontaneum*)

(SEQ ID NO: 33)

MTGKVTLGAV RKAQRAEGSA AVLAIQTATP ANCVYQADYP DYYPRVTKSE

HLTDLKEKFK RICHKSMITK RYMHLTEGFL QENPNMSSYS APSLDARQDI

LIEEVPKLGA AAAEKALKEW GQPRSQITHL VFCTTSGVDM PGADYQLIKL

LGLSLSVNRA MMYHQGCFAG GMVLRLLAKDL AENNRGARVL IVCSEITAVT

FRGPSESHLD SLVGQALFGD GAAAVIVGAD PSAAEWPLFQ LVSASQTILP

DSEGAIEGHL REVGLTFHLQ DRVPQLISTN IERLLEDAPT PLGISDWNSI

FWVAHPGPPA ILNMVEAKAG LDKARLCATR HILAEGNMS SACVLFILDE

MRNKAEDGH TTTGEGMEWG VLPFGPGPLT VETIVLQSVI IITV

ShOMT1 (*Sorghum halepense*)

(SEQ ID NO: 34)

MGSYTTRVHA VEKAANNQDD ETCMHMLLL GGLAVPCTIK AVIELGIMDL

LLAADRAMTA EELTARLPCP AAATAAMVD RMLRFLASHG VVRCAAATKE

SSELGSDGGK SCRRYAAPV CRWFTRSGGV ESVPVPMGLWM TGKTVLETWY

NIKEAVLEGE TPFDRAYGQP FFEYLGNGT VNTLFDEAMA NHSTIITKRL

VEVFRGFENY SVLVDVGGNK GTTLQMIRSQ YENISGINYD LPRVIAQAPP

IEGVEHVGGN MFDNVPRGDA IILKWILHDW GDKDCVKILK NCYAALPVNG

TMIILEYILP ETPEETLSQ LAFNFDGMM LMYGAKGKER TEKELSELAR

EAGFSGDCTA TYIFASIWAL EFTK

ShSOMT (*Sorghum halepense*)

(SEQ ID NO: 35)

MGSYDSSSSS SNDSSARNEE DESCMFALKL LGGFVAPFTI KAVIELGVMD

QLLTAERAMS AEELVAAVA AQLPRPEVAC TMVDRLLRFL ASHSVVRCTT

EVVVGTTDDAT TTTCCRRSYA ASPVCKWFAR NGVEDSVLPL GMMILNKTFI

DSWQNIIDAV LEGAAPFEKT YGMPMFYLS TNGPLNTVFH EAMANHSMII

TKKLLKFPRG FEGLDVLVDV GGGNGTTLQM IRGQYKNMRG INYDLPHVIA

QAAPVEGVEH VGGSMFDNIP RGNVLLKWI LHDWDDKACI KILKNCTAL

HVRGKVIVLE YVVPDEPEPT LAAQGAFELD LTMLVTFGSG KERTQREFSE

LAMEAGFSRE FKATYIFANV WALEFTK

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MsIOMT (*Medicago sativa*; P93324)

(SEQ ID NO: 82)

MGNSYITKED NQISATSEQT EDSACLSAMV LTTNLVYPAV LNAAIDLNLF
EIIAKATPPG AFMSPSEIAS KLPASTQHSD LPNRLDRMLR LLASYSVLTS
TTRTIEDGGA ERVYGLSMVG KYLVPDES RG YLASFTTFLC YPALLQVWMN
FKEAVVDEDI DLFKNVHGVT KYEFMGKDKK MNQIFNKSMV DVCATEMKRM
LEIYTGFEIGI STLVDVGGGS GRNLELIISK YPLIKGINFD LPQVIENAPP
LSGIEHVGGD MFASVPQGDA MILKAVCHNW SDEKCIEFLS NCHKALSPNG
KVIIVEFILP EEPNTSEESK LVSTLDNLMF ITVGGRETERE KQYEKLSKLS
GPSKQVACR AFNSLGVMEF YK

MsCOMT (*Medicago sativa*; P28002)

(SEQ ID NO: 83)

MGSTGETQIT PTHISDEEAN LFAMQLASAS VLPMLKSAL ELDLLEIIAK
AGPGAQISPI EIASQLPTTN PDAPVMLDRM LRLLACYIIL TCSVRTQQDG
KVQRLYGLAT VAKYLVKNED GVSISALNLM NQDKVLMESW YHLKDAVLGD
GIPFNKAYGM TAFEYHGTD P RFNKVENKGM SDHSTITMKK ILETYTGFEIG
LKSLVDVGGG TGAVINTIVS KYPTIKGINF DLPHVIEDAP SYPGVEHVGG
DMFVSIPKAD AVFMKWICH D WSEHCLKFL KNCYEALPDN GKVIVAECIL
PVAPDSSLAT KGVVHIDVIM LAHNPGGKER TQKEFEDLAK GAGFQGFVKH
CNAFNITYIME FLKKV

MtIOMT (*Medicago truncatula*; Q29U70)

(SEQ ID NO: 84)

MAFSTNGSEE SELYHAQIHL YKHVYNFVSS MALKSAMELG IADAIHNHGK
PMTLSELASS LKLHPSKVNI LHRFLRLLTH NGFFAKTIVK GKEGDEEEEI
AYSLTPPSKL LISGKPTCLS SIVKGALHPS SLDMWSSSKK WFNEDKEQTL
FECATGESFW DFLNKDSESS TLSMFQDAMA SDSRMFKLVL QENKRVFEGL
ESLVDVGGGT GGVTKLIHEI FPHLKCTVFD QPQVVGNTLG NENLNFVGGD
MFKSIPSADA VLLKWVLHDW NDEQSLKILK NSKEAISHKG KDGVVIIIDI
SIDETSDDRG LTELQLDYDL VMLTMFLGKE RTKQEWELI YDAGFSSYKI
TPISGFKSLI EVYP

MtIOMT3 (*Medicago truncatula*; Q06YR3)

(SEQ ID NO: 85)

MASSINNRKP SEIFKAQALL YKNMYAFVDS MSLKWSIEMN IPNIIHNHGK
PITLSNLVSI LQIPSTKVDN VQRLMRYLAH NGFPEIITNQ ELENEEAAYA
LTVASELLVK GTCLCLAPMV ECVLDPTLST SFHNLKKWVY EEDLTLFAVN
LGC DLWEFLN KNPEYNTLYN DALASDSKMI NLAMKDCNLV FEGLESIVDV
GGNGTTGKI ICETFPKLT C VVFD RPKVVE NL CGSNNLTY VGGDMFISVP
KADAVLLKAV LHDWTDKDCI KILKKCKEAV TSDGKRGKVI VIDMVINEKK
DENQLTQIKL LMNVTISCVN GKERNEEWK KLFIEAGFQD YKISPFTGLM
SLIEIYP

ScNOMT (*Streptomyces carzinostaticus*; Q84HC8)

(SEQ ID NO: 86)

MGKRAAHIGL RALADLATPM AVRVAATLRV ADHIAAGHRT AAETIASAAGA
HADSLDRLLR HLVAVGLFTR DGQGVYGLTE FGEQLRDDHA AGKRKWLDMN

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SAVGRGDLGF VELAHSIRTG QPAYPVRYGT SFWEDLGSDP VLSASFDTLM
SHHLELDYTG IAAKYDWAAL GHVVDVGGGS GGLLSALLTA HEDLSGTVLD
LQGPASAAHR RFLDTGLSGR AQVVVGSFFD PLPAGAGGYV LSAVLHDWDD
LSAVAILRRC AEAAGSGGVV LVIEAVAGDE HAGTGMDLFRM LTYFGGKERS
LAELGELAAQ AGLAVRAAHP ISYVSIVEMT AL

LpCOMT (*Loium perenne*; Q9ZTU2)

(SEQ ID NO: 87)

MGSTAADMAA SADEDACMFA LQLASSSVLP MTLKNAIELG LLEILVAAGG
KSLTPTEVAA KLPSAANPEA PDMVDRILRL LASYNVVTCL VEEGKDGRSL
RSYGAAPVCK FLTPNEDGVS MAALALMNQD KVLMSWYYL KDAVLDDGGIP
FNKAYGMSAF EYHGTDPREN RVFNEMGMKNH SIIITKKLLE LYHGFEGELGT
LVDVGGGVGA TVAAIAAHYP TIKGVNFDLP HWISEAPQFP GVTHVGGDMF
KEVPSGDTIL MKWILHDWSD QHCATLLKNC YDALPAHGKV VLVQCILPVN
PEANPSSQGV FHVDMIMLAH NPGGRERYER EFQALARGAG FTGVKSTYIY
ANAWAIEFTK

CHIEOMT (*Clarkia breweri*; 004385)

(SEQ ID NO: 88)

MGSTGNAIEQ IIPTHSSDEE ANLFAMQLAS AAVLPMALKA AIELDVLEIM
AKSVPPSGYI SPAEIAAQLP TTNPEAPVML DRVLRLASY SVVTTYTLREL
PSGKVERLYG LAPVCKFLTK NEDGVSLAPF LLTATDKVLL EPWFYLKDAI
LEGGIPFNKA YGMNEFDYHG TDHRFNKVFN KGMSNSTIT MKKILEMYNG
FEGLTITVDV GGGTGAVASM IVAKYPSINA INFDLPHVIQ DAPAFSGVEH
LGGDMFDGVP KGDAIFIKWI CHDWSDEHCL KLLKNCYAAL PDHGKVIVAE
YILPPSPDPS IATKVVIHTD ALMLAYNPGG KERTEKEFQA LAMASGFRGF
KVASCAFNTY VMEFLKTA

PIOMT (*Planctopirus limnophila*; D5STZ7)

(SEQ ID NO: 89)

MAVKDALRFP PTDVTPIFDL FRGNFATELL AASVAHLHVF DILNESPLSL
DELQRRVLVS ERATQVLVTG LCAMQLLTR LAGEIDLTP LARNHLVTTSP
FSVGGYISLA AQSAGTLALV ERLKSDRPEG AESEQAAFI FREGSESAMD
REDSARFLTL SLAGRAWNVA PRFADVLPAG QPGKILKDNS GSSGRVLLDV
AGSGGIYTMA VLQKYPTWRG IIFDRPEVLK IAAELAEQTG VRDRLELHAG
DMWVDPFPPA DDILLSNVLH DWDRPQCRL VAKATSGLPE GGRLLIHDVL
LNSDLTGPLE IALYSLALFS LTEGRAYSLE EYRGWIAGAD LKYVDCIPTS
AHGHLILSEK V

TINOMT (*Thalictrum flavum*; Q5C9L7)

(SEQ ID NO: 90)

MEMINKENLS SQAKLWNFIY GFADSLVLKS AVQLDLANII HNHGSPMTLS
ELSLHLPSPQ VNQDALYRVL RYLVMKLF T KSSIDGELRY GLAPPAKFLV
KGWDKCMLGA ILTITDKDFM APWHYLKEGI LNDGSTSTAF EKALGTNIWD
YMAEHPEKNQ LFNEGMANDT RLIMSALVKE CSSMFDGITT IVDVGGGTGT
AVRNIKAFPP HIKCTVYDLP HVIADSPGYT EINSIQGDMF KYIPNADAIM

-continued

MKCILHDWDD KECIEILKRC KDAVPRDGGK VIIIDIILDV KSEHPYTKMR
 LTLDLDMMLN TGGKERTEEE WKKLIHDAGY KGYKITHISA VQSVIEAYPY
 KpBOMT (*Kitagawia praeruptora*; A0A166U5H3) (SEQ ID NO: 91)
 MAGMKTSPSQ DEEACVLAIQ LATSTVLPMI LKSAIELDIL NTISKAGPGN
 YLSPSDLASK LLMSNPAPI MLERILRVLA TYKVLGCKPS ELSDGVEVEWL
 YCWTVPVCKFL SNNEDGASIA PLLLVHQDQV PMKSWYHLTD AILDGGTAFN
 KAYGMNIFDY ASQDPQFNKV FNRSMAGHST ITMKKILETY NGFEGLKSIV
 DVGGGSGATL NMIISKYPTI KGINFDLPHV VGDSPHHPGV EHVGGDMFAS
 VPKGDAIFLK WIFHSWDED CLRILKNCYE ALADNKKVIV AEFIPEVPG
 GSDDATKSVV HLDVMLAYV PGGKERTKE FEALATSAGF KSFRKVCCAF
 NTWIMEFSK
 PsSOMT (*Papaver somniferum*; I3V6A7) (SEQ ID NO: 92)
 MATNGEIFNT YGHNROTATV TKITASNESS NGVCYLSETA NLGLICIPM
 ALRAAMELNV FQLISKFGTD AKVSASEIAS KMPNAKNNPE AAMYLDRIIR
 LLGASSILSV STTKKSINRG GDDVVVHEKL YGLTNSSCCL VPRQEDGVSL
 VEELLFTSDK VVVDSPFKLK CVVEEKDVP FEVAHGAKIF EYAATEPRMN
 QVENDGMAVF SIVVFEAVFR FYDGFLDMKE LLDVGGGIGT SVSKIVAKYP
 LIRGVNFDLP HVISVAPQYP GVEHVAGDMF EEVPGQGNML LKWLHDWGD
 ERCVKLLKNC WNSLPVGGKV LIIEFVLPNE LGNNAESFNA LIPDLLLMAL
 NPGGKERTIS EYDDLGAAG FIKTIPIPI NGLHVIEPHK
 FaOMT (*Fragaria ananassa*; Q9M602) (SEQ ID NO: 93)
 MGSTGETQMT PTHVSDEEAN LFAMQLASAS VLPMLKAAI ELDLLEIMAK
 AGPGSFLSPS DLASQLPTKN PEAPVMLDRM LRLLASYSIL TCSLRTLDPG
 KVERLYCLGP VCKFLTCKED GVSIAALCLM NQDKVLVESW YHLKDAVLGD
 GIPFNKAYGM TAFDYHGTDP RENKVENKGM ADHSTITMKK ILETYKGFEG
 LKSIVDVGGG TGAVVNMIVS KYPSIKGINF DLPVHVEDAP QYPGVQHVGG
 DMFVSVPKGN AIFMKWICH DWSDEHCKFL KNCYAALPDD GKVILAECIL
 PVAPDTSLAT KGVVHMDVIM LAHNPGGKER TEQEFEALAK GSGFQGIRVC
 CDAFNITYVIE FLKKI

[0236] It is understood that the disclosed method and compositions are not limited to the particular methodology, protocols, and reagents described as these can vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims.

[0237] Disclosed are materials, compositions, and components that can be used for, can be used in conjunction with, can be used in preparation for, or are products of the disclosed method and compositions. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective combinations and permu-

tation of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. For example, if an OMT is disclosed and discussed and a number of modifications that can be made to a number of enzymes including the OMT are discussed, each and every combination and permutation of OMT and the modifications that are possible are specifically contemplated unless specifically indicated to the contrary. Thus, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F and an example of a combination molecule, A-D is disclosed, then even if each is not individually recited, each is individually and collectively contemplated. Thus, in this example, each of the combinations A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D,

E, and F; and the example combination A-D. Likewise, any subset or combination of these is also specifically contemplated and disclosed. Thus, for example, the sub-group of A-E, B-F, and C-E are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and F; and the example combination A-D. Further, each of the materials, compositions, components, etc. contemplated and disclosed as above can also be specifically and independently included or excluded from any group, sub-group, list, set, etc. of such materials. These concepts apply to all aspects of this application including, but not limited to, steps in methods of making and using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods, and that each such combination is specifically contemplated and should be considered disclosed.

[0238] It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural reference unless the context clearly dictates otherwise. Thus, for example, reference to “a OMT” includes a plurality of such OMTs, reference to “the OMT” is a reference to one or more OMTs and equivalents thereof known to those skilled in the art, and so forth.

[0239] Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude, for example, other additives, components, integers or steps.

[0240] “Optional” or “optionally” means that the subsequently described event, circumstance, or material may or may not occur or be present, and that the description includes instances where the event, circumstance, or material occurs or is present and instances where it does not occur or is not present.

[0241] Unless the context clearly indicates otherwise, use of the word “can” indicates an option or capability of the object or condition referred to. Generally, use of “can” in this way is meant to positively state the option or capability while also leaving open that the option or capability could be absent in other forms or embodiments of the object or condition referred to. Unless the context clearly indicates otherwise, use of the word “may” indicates an option or capability of the object or condition referred to. Generally, use of “may” in this way is meant to positively state the option or capability while also leaving open that the option or capability could be absent in other forms or embodiments of the object or condition referred to. Unless the context clearly indicates otherwise, use of “may” herein does not refer to an unknown or doubtful feature of an object or condition.

[0242] Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, also specifically contemplated and considered disclosed is the range from the one particular value and/or to the other particular value unless the context specifically indicates otherwise. Similarly, when values are expressed as approximations, by use

of the antecedent “about,” it will be understood that the particular value forms another, specifically contemplated embodiment that should be considered disclosed unless the context specifically indicates otherwise. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint unless the context specifically indicates otherwise. It should be understood that all of the individual values and sub-ranges of values contained within an explicitly disclosed range are also specifically contemplated and should be considered disclosed unless the context specifically indicates otherwise. Finally, it should be understood that all ranges refer both to the recited range as a range and as a collection of individual numbers from and including the first endpoint to and including the second endpoint. In the latter case, it should be understood that any of the individual numbers can be selected as one form of the quantity, value, or feature to which the range refers. In this way, a range describes a set of numbers or values from and including the first endpoint to and including the second endpoint from which a single member of the set (i.e. a single number) can be selected as the quantity, value, or feature to which the range refers. The foregoing applies regardless of whether in particular cases some or all of these embodiments are explicitly disclosed.

[0243] Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of skill in the art to which the disclosed method and compositions belong. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present method and compositions, the particularly useful methods, devices, and materials are as described. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such disclosure by virtue of prior invention. No admission is made that any reference constitutes prior art. The discussion of references states what their authors assert, and applicants reserve the right to challenge the accuracy and pertinency of the cited documents. It will be clearly understood that, although a number of publications are referred to herein, such reference does not constitute an admission that any of these documents forms part of the common general knowledge in the art.

[0244] Although the description of materials, compositions, components, steps, techniques, etc. can include numerous options and alternatives, this should not be construed as, and is not an admission that, such options and alternatives are equivalent to each other or, in particular, are obvious alternatives. Thus, for example, a list of different OMTs does not indicate that the listed OMTs are obvious one to the other, nor is it an admission of equivalence or obviousness.

[0245] Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the method and compositions described herein. Such equivalents are intended to be encompassed by the following claims.

SEQUENCE LISTING

Sequence total quantity: 104

SEQ ID NO: 1 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 1

MGSYDSSSSS	SNDSARNEE	DESCMPALKL	LGGFAVPFTI	KAVIELGVMD	QLLTAERAMS	60
AEELVAAAVA	AQLPRPEVAC	TMVDRLRLRFL	ASHSVVRCTT	EVVVGTTDDAT	TTTCRRRSYA	120
ASPVCKWFAR	NGVEDSVLPL	GMMILNKTFE	DSWQNTDAV	LEGAAPFEKT	YGMPMFEYLS	180
TNGPLNTVFH	EAMANHSIII	TKKLLKFFRG	FEGLDVLVDV	GGNGTTLQM	IRGQYKNMRG	240
INYDLPHVIA	QAAPVEGVEH	VGGSMFDNIP	RGNVALLKWI	LHDWDDKACI	KILKNCYTAL	300
HVRGKIVILE	YVVPDEPEPT	LAAQGAFFEL	LTMLVTFGSG	KERTQREFSE	LAMEAGFSRE	360
FKATYIFANV	WALEFTK					377

SEQ ID NO: 2 moltype = AA length = 362
 FEATURE Location/Qualifiers
 source 1..362
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 2

MGSTAEDVAA	VADEEACMYA	MQLASSSILP	MTLKNALELG	LLEVLQKDAG	KALAAEEVVA	60
RLPVAPTNP	AADMVDRMLR	LLASYDVVRC	QMEDKDGKYE	RRYSAAPVGK	WLTPNEDGVS	120
MAALALMNQD	KVLMESWYLL	KDAVLDDGGIP	FNKAYGMTAF	EYHGTDPFRN	RVFNEGMMKNH	180
SVIITKKLLE	FYTGFDESVS	TLVDVGGGIG	ATLHAITSHH	SHIRGINFDL	PHVISEAPPF	240
PGVQHVGGDM	FKSVPAGDAI	LMKWILHDWS	DAHCAITLKN	CYDALPEKGG	KVIVVECCLP	300
VTTDAVPKQA	GVFHVDMIML	AHNPPGGRERY	EREPRDLAKA	AGFSGFKATY	IYANAWAIEF	360
IK						362

SEQ ID NO: 3 moltype = length =
 SEQUENCE: 3
 000

SEQ ID NO: 4 moltype = length =
 SEQUENCE: 4
 000

SEQ ID NO: 5 moltype = AA length = 363
 FEATURE Location/Qualifiers
 source 1..363
 mol_type = protein
 organism = Arabidopsis thaliana

SEQUENCE: 5

MGSTAETQLT	PVQVTDDEAA	LFAMQLASAS	VLPMLKSAL	ELDLLEIMAK	NGSPMSPTTEI	60
ASKLPTKNPE	APVMLDRILR	LLTSYSVLTC	SNRKLSGDGV	ERIYGLGPVC	KYLTKNEDGV	120
SIAALCLMNQ	DKVLMESWYH	LKDAILDGGI	PFNKAYGMSA	FEYHGTDPFR	NKVFNNMGMSN	180
HSTITMKKIL	ETKGFEGELT	SLVDVGGGIG	ATLKMIIVSKY	PNLKGINFDL	PHVIEDAPSH	240
PGIEHVGGDM	FVSVPKGDAL	FMKWICHDS	DEHCVKFLKN	CYESLPEDGK	VILAECLLPE	300
TPDSSLSTKQ	VVHVDCIMLA	HNPGGKERT	KEFEALAKAS	GFKGIKVCCD	AFGVNLIELL	360
KKL						363

SEQ ID NO: 6 moltype = AA length = 356
 FEATURE Location/Qualifiers
 source 1..356
 mol_type = protein
 organism = Carthamus tinctorius

SEQUENCE: 6

MAHEEEEF	LFAMQLASAS	VLPMLKSAI	DLDLLEIIAK	AGPGAHVSPA	YLAAQLPKAD	60
NPEAAVMDLR	ICRLLATYSV	LTCTLKLDH	GDGDVERLYG	LAPVCKFLVK	NEDGVSNAPL	120
LLMNQDKILM	ESWYHLKDAV	LDGGIPFNKA	YGMSAFYHVG	KDERFNKVFN	SGMFNHSMT	180
MKKILDVYPG	FNGVKTLDV	GGGTGASLSM	ILSKHPSIKG	INFDLPHVIQ	DATNYPGIEH	240
VGGDMFESVP	KGDAIFMKWI	CHDWSDAHCL	KFLKNCKYKAL	PENGKIVIAE	CILPETPDSS	300
LATKNVVID	VIMLAHNP	GGKERTKEFQA	LAKAAGFKGF	HKPCSALNTW	LMEFTK	356

SEQ ID NO: 7 moltype = AA length = 363
 FEATURE Location/Qualifiers
 source 1..363
 mol_type = protein
 organism = Catharanthus roseus

SEQUENCE: 7

MGSANPDNKN	SMTKEEEAC	LSAMRLASAS	VLPMLKSAI	ELDLLELIKK	SGPGAYVSPS	60
ELAAQLPTQN	PDAPVMDLRI	LRLLASYSVL	NCTLKDLDPG	GIERLYSLAP	VCKFLTKNED	120
GVSMAALLLM	NQDKVLMESW	YHLKDAVLEG	GIPFNKAYGM	TAFYHKGKDP	RFNKVFNQGM	180
SNHSTIIMKK	ILEIYQGFQ	LKTVDVGGG	TGATLNMIIVS	KYPSIKGINF	DLPHVIEDAP	240
SYPGVDHVG	DMFVSVPKGD	AIFMKWICH	WSDAHLCKFL	KNCHEALPEN	GKVLAECLL	300

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PEAPDSTLST	QNTVHVDVIM	LAHNPGGKER	TEKEFEALAK	GAGPRGFIKV	CCAYNSWIME	360
LLK						363

SEQ ID NO: 8	moltype = AA length = 356
FEATURE	Location/Qualifiers
source	1..356
	mol_type = protein
	organism = Hordeum vulgare

SEQUENCE: 8						
MGSIAAGADE	DACMYALQLV	SSSILPMTLK	NAIELGLLET	LMSAGGKFLT	PAEVAAKLPS	60
TANPEAPDMV	DRMLRLLASY	KVVSCTEES	KDGRLSRRYG	AAPVCKYLTP	NEDGVMSAL	120
ALMNQDKVLM	ESWYLLKDAV	LDGGIPFNKA	YGMSAFYHVG	TDPRFNRVFN	EGMKNHSIII	180
TKKLLSYKYG	FEGLTLVDV	GGGVGATVGA	IIARYPAVKG	INFDLPHVIS	EAPAFPGVTH	240
IGGDMFQKVP	SGDAILMKWI	LHDWSDEHCA	TLLKNCYDAL	PAHGKVVVLE	CILPVNPEAT	300
PEVQGVFHV	DVIM	LAHNPGGKER	TEKEFEALAK	GAGPRGFIKV	CCAYNSWIME	360
						363

SEQ ID NO: 9	moltype = AA length = 365
FEATURE	Location/Qualifiers
source	1..365
	mol_type = protein
	organism = Medicago sativa

SEQUENCE: 9						
MGSTGETQIT	PTHISDEEAN	LFAQMLASAS	VLPMLKSAL	ELDLLEIIAK	AGPGAQISPI	60
ETASQLPTTN	PDAPVMDLDM	LRLLCYIIL	TCSVRTQQDG	KVQRLYGLAT	VAKYLVKNED	120
GVSISALNLM	NQDKVLMESW	YHLKDAVLDD	GIPFNKAYGM	TAFYHGTDP	RFNKVFNKGM	180
SDHSTITMKK	ILETYTGPEG	LKSLVDVGGG	TGAVINTIVS	KYPTIKGINF	DLPHVIEDAP	240
SYPGVEHVGG	DMFVSIPKAD	AVFMKWICH	WSEHCLKFL	KNCYEALPDN	GKIVVAECIL	300
PVAPDSSLAT	KGVVHIDVIM	LAHNPGGKER	TQKEFEDLAK	GAGPQGFVKH	CNAPNTYIME	360
FLKKV						365

SEQ ID NO: 10	moltype = AA length = 352
FEATURE	Location/Qualifiers
source	1..352
	mol_type = protein
	organism = Medicago sativa

SEQUENCE: 10						
MASSINGRKP	SEIFKAQALL	YKHIYAFIDS	MSLKWAVGMN	IPNIIHNHGK	PISLSNLVSI	60
LQVPSSKIGN	VRLMRYLAH	NGFFEITKE	EESYALTVAS	ELLVRGSDLC	LAPMVECVLD	120
PTLSGSYHEL	KKWIYEDLT	LFGVTLGSGF	WDFLDKNPEY	NTSFNDAMAS	DSKLINLALR	180
DCDFVFDGLE	SIVDVGGTSG	TTAKIICETF	PKLKCIVFDR	PQVVENLSGS	NNLTYVGGDM	240
FTSIPNADAV	LLKYLHNWT	DKDCLRLKK	CKEAVTNDGK	RGKVTIIDMV	INEKKDENQV	300
TQIKLLMDVN	MACLNGKERN	EEEWKKLFIE	AGFQHYKISP	LTGFLSLIEI	YP	352

SEQ ID NO: 11	moltype = AA length = 368
FEATURE	Location/Qualifiers
source	1..368
	mol_type = protein
	organism = Oryza sativa

SEQUENCE: 11						
MGSTAADMAA	AADEEACMYA	LQLASSSILP	MTLKNAIELG	LLETLQSAAV	AGGGGKAALL	60
TPAEVADKLP	SKANPAAADM	VDRMLRLLAS	YNVVRCEMEE	GADGKLSRRY	AAAPVCKWLT	120
PNEDGVSMMA	LALMNQDKVL	MESWYLLKDA	VLDGGIPFNK	AYGMTAFYEH	GTDARFNRVF	180
NEGMMKNHVI	ITKKLLDLYT	GFDAASTVVD	VGGGVGATVA	AVVSRHPHIR	GINYDLPHVI	240
SEAPPPPGVE	HVGDMPFASV	PRGGDAILMK	WILHDWSDEH	CARLLKNCYD	ALPEHGKVVV	300
VECVLPESSD	ATAREQGVFH	VDMIMLAHNP	GKERYEREF	RELARAAGFT	GFKATYIYAN	360
AWAIEFTK						368

SEQ ID NO: 12	moltype = AA length = 375
FEATURE	Location/Qualifiers
source	1..375
	mol_type = protein
	organism = Oryza sativa

SEQUENCE: 12						
MVSPVVRHHA	AGGGSGGDD	DQACMYALEL	LGGSVVSMTL	KAAIELGLVD	ELLAAAGAAV	60
TAEELAARLR	LPAAVAAAAA	VDRMLRLLAS	YGVVRCATEA	GPDGKARRSY	AAAPVCKWLA	120
AGSSSGEGSM	APLGLNLNDK	VFMENWYLLK	EAVSEGGTAF	DKAYGTSLFQ	YLGQDGNES	180
NTLNFQAMAS	HSVITNKLK	QFPRGFDAGA	GVDVLVDVGG	GVGATLRMIT	ARHPHLRGVN	240
YDLPHVIAQA	PPVEGVEHIG	GSMFDHVP	SGSAILLKWLH	LWGDEECVKI	LKNCYKALPA	300
KGKVILVEYV	LPASPEATLA	AQEAFLDVM	MLNRLAGGKE	RTQQEFTDLA	VDAGFSGDCK	360
PTYIFTNVWA	LEFTK					375

SEQ ID NO: 13	moltype = AA length = 361
FEATURE	Location/Qualifiers
source	1..361
	mol_type = protein
	organism = Panicum virgatum

-continued

SEQUENCE: 13
 MGSTATDVAA AADEEACMYA LQLASSSILP MTLKNAIELG LLEVLQKDP AALAPEEVVA 60
 QLPVAPANPD AAAMVDRMLR LLASYDVVRC QMEEGKDGRY SRRYAAAPVC KWLTPNEDGV 120
 SMAALALMNQ DKVLMESWY LKDAVLEGGI PFNKAYGMA FEYHGTDP RF NRVFNEG MKN 180
 HSVIITKKLL EFYAGFEGVG TLVDVGGGVG ATLHAITSRY PGIRGVNFDL PHVISEAPPF 240
 PGVEHVGGDM FKAVPAGDAI LMKWILHDWS DAHCAAILKN CYDALPAGGK VIAVECILPV 300
 NPEATPKAQG VFHVDIMILA HNPGGKERYE REFEELAKGA GFTGFKATYI YANAWAIEFT 360
 K 361

SEQ ID NO: 14 moltype = AA length = 361
 FEATURE Location/Qualifiers
 source 1..361
 mol_type = protein
 organism = Pinus sylvestris

SEQUENCE: 14
 MNMQSVKDEE ALRASALGLA FSLETPFLLK CAIRLKIPDI ISKAGPDVSL SVHQIAAQLP 60
 SEDPDMGALS RILTYLSTMG ILQAIVPPEG VNAPMNIRYG LTNLTKTYFT SEDISSRSLV 120
 PFVLLQTHPL YVTAWDNIHE RVLHGGDNFK NSSGNGKDFW NFAAGEPEFN AIFNAGMVSV 180
 TKATITTYVLA VYDGFKDINT LVDVGGGRGE ALSLITEAHP HIRAINFDLP QVIATAPTIP 240
 GVOHMSGNLF ESAPSADAIF MKNFLHSWND EDCIKLLNNC HQALPEKGL ILSEAILDLT 300
 EGSDMIGSAN VLDVMLNCL PGGGERTRKQ WNDLLQAAGF SISKIVGRNG TLTKVIEAIK 360
 S 361

SEQ ID NO: 15 moltype = AA length = 381
 FEATURE Location/Qualifiers
 source 1..381
 mol_type = protein
 organism = Pinus taeda

SEQUENCE: 15
 MDSNMNGLAK SNGCEISRDG FFEEEEELQ GQAEAWKCTF AFAESLAVKC VVLLGIPDMI 60
 AREGPRATLS LCEIVANLPT ESPDAACLF RMRFLVAKGI FPASKSARRR AFETRYGLTP 120
 ASKWLVKGRE LSMAPMLLMQ NDETTLAPWH HFNECVLEGG VAFQKANGAE IWSYASDHPD 180
 FNNLFNNAMA CNARIVMKAI LSKYQGFHSL NSLVDVGGGT GTAVAEIVRA YPFIRGINYD 240
 LPHVVATASS LSGVQHVGGD MFETVPTADA IFMKWIMHDW NEDCKIKLK NCRKAIPDTG 300
 KVIVDVVLD ADQGDNTDKK RKKAVDPVIG TVFDLVMVAH SSGGKERTEK EWKRILLEGG 360
 FSRYNIIIEIP ALQSVIEAPP R 381

SEQ ID NO: 16 moltype = AA length = 365
 FEATURE Location/Qualifiers
 source 1..365
 mol_type = protein
 organism = Populus tremuloides

SEQUENCE: 16
 MGSTGETQMT PTQVSDEEAH LFAMQLASAS VLPMLKTAI ELDLLEIMAK AGPGAFLSTS 60
 ETASHLPTKN PDAPVMDLRI LRLLASYSIL TCSLKDLDPG KVERLYGLAP VCKFLT KND 120
 GVSVPCLCLM NQDKVLMESW YYLKDAILDG GIPFNKAYGM TAFEYHGTDP RFNKVFNKGM 180
 SDHSTITMKK ILETYKGFEG LTSLVGVGGG TGAVVNTIVS KYPSIKGINF DLPHVIEDAP 240
 SYPGVEHVG DMFVSVPKAD AVFMKWICH WSDAHLKFL KNCYDALPEN GKVILVECIL 300
 PVAPDTSLAT KGVVHVDVIM LAHNPPGKER TEKEFEGLAK GAGFQGFVEM CCAFNTHVIE 360
 FRKKA 365

SEQ ID NO: 17 moltype = AA length = 367
 FEATURE Location/Qualifiers
 source 1..367
 mol_type = protein
 organism = Rosa sp.

SEQUENCE: 17
 MERLNSFRHL NQKWSNGEHS NELLHAQAH WNHIFSFINS MSLKSAIQLG IPDIINKHGY 60
 PMTLSELSTA LPIHPTKSHS VYRLMRILVH SGFFAKKKLS KTDEEGYTLT DASQLLLKDH 120
 PLSLTPYLTA MLDPVLTNPW NYLSTWFQND DPTPPDTAHG MTFWDYGNHQ PSIAHLFNDA 180
 MASDARLVTS VIINDCKGVF EGLESVDVVG GGTGTLAKAI ADAFPHIECT VLDLPHVVD 240
 LQGSKNLKYT GGDMEFVAPP ADTVLLKWIL HDWSDEECIK ILERSRVAIT GKEKKGVII 300
 IDMMENQKG DEESIEQTQF FDMLMMALVG GKERNEKEWA KLFTDAGFSD YKITPISGLR 360
 SLIEVYP 367

SEQ ID NO: 18 moltype = AA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = protein
 organism = Solanum lycopersicum

SEQUENCE: 18
 MGSTSLTQTE DEAFLFAMQL ASASVLPMLV KSAVELELLE LMAKAGPGAS ISPAELASQL 60
 PCKNPDAVPM LDRMLRLAA YSVLNCTLR LDPGRVERLY SLAPVCKFLT KNADGVSVA 120
 LLLMNQDKVL MQSWYHLKDA VLDGGIPFNK AYGMTAFHEYH GTDPRFNKVF NRGMSDHTTL 180
 STKILEDYK GFEGLSIVD VGGGTGATVS MIVSKYPSIK GINFDLPHVI EDAPAYPGVE 240
 HIGGDMFVSV PKADAFMKW ICHDWSDEHC LKFLKNCYEA VPANGKVI A ECLLPEVPDT 300

-continued

SSSTKNTVHV DVIMLAHNPG GKERTEKEFE ALAKGAGFNG FTKASCAYNT WIMEFTK 357

SEQ ID NO: 19 moltype = AA length = 364
FEATURE Location/Qualifiers
source 1..364
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 19
MACTTAASLQ HDKANDDEAC MYAQELLFSF IVPMTLKAVI ELGLIDYLLA ADGRSVTPPE 60
LAAEWPQSAE AAAAVDRMMR LLASHSVVRC TTEVGPDGKA RRSYAAAPVC KWLATRNAGG 120
QGSLAPMGLM NLNKAFFMETW YFMKEAVTEG ATPTEKAYGM PLFEHLGSDE ASNTLFNQAM 180
AGHSEMIKK LLEVYRGFEG VDLVDVGGG TGSTLRMTA QYKHLRGVNY DLPHVIAQAP 240
PVQGVHVGQ SMFEYIPSGN AILLKWILHL WRDDECVKIL KNCHRALPAN GKIVVVEYVL 300
PASPEPTQVA QVSLLLDVAM LNRLRGAKER TEQEFAQLAA EAGFSGGCRA TYVFASAWAL 360
EFTK 364

SEQ ID NO: 20 moltype = AA length = 376
FEATURE Location/Qualifiers
source 1..376
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 20
MASYTSTSGQ FAVGKVAAN QDDETCMHAL KLLGGLAVPF TIKAVIELGI MDLLAADRA 60
MTAEALTAAL LCPAPAPAAA AAMVDRMLRF LASHGVVRC TESEELGSDD GKSCRRYAAA 120
PVCKWFARGG GVESVPMGF WMTSTTMMET WHNIKDGVL GETPFDKAYG MPVFEYLGAN 180
GTMNTLFNEA MASHSMIITK RLLEVPRGFE NYSVLVDVGG GNGTTMQMIR SQYENISGIN 240
YDLPHVIAQA SPIEGVEHVA GNMFDNIPRG DAIILKWILH NWGDKCEVKI LKNCTALPV 300
NGTVIILEYI LPETPEETLA SQLAFDFDLG MMLFFGASGK ERTEKELLE AREAGFSGDY 360
TATYIFANVW AHEFTK 376

SEQ ID NO: 21 moltype = AA length = 374
FEATURE Location/Qualifiers
source 1..374
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 21
MVLISEDRE LLQAHVELWN QTSYFMKSA LVALDLHIA DAIHRRGGAA TLSQILGEIG 60
VRPCKLPGLH RIMRVLTSG TFTIVQPSAE TMSSESDGRE PVYKLTASS LLVSSSESSAT 120
ASLSPMLNHV LSPFRDPLS MGLTAWFRHD EDEQAPGMCP FTLMYGTTW EVCRDDAIN 180
ALFNNAMAAD SNFLMQILLK EFSEVFLGID SLVDVAGGVG GATMAIAAF PCLKCTVLDL 240
PHVVAKAPSS SIGNVQFVG DMPESIPPAN VVLKWLHLD WSNDECILK KNCKQAIPSR 300
DAGGKIIID VVVGSDSDT KLETTQVIYD LHLMKIGVE RDEQEWKIF LEAGFKDYKI 360
MPILGLRSII ELYP 374

SEQ ID NO: 22 moltype = AA length = 374
FEATURE Location/Qualifiers
source 1..374
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 22
MGSYTTRVHA VEKAANNQDD ETCMHMLML GGLAVPCTIK AVIELGIMDL LLAADRAMTA 60
EELTARLPCP AAATAAMVD RMLRFLASHG VVRCAAATKE SSELGSDGK SCRRYAAAPV 120
CRWFRTRSGV ESVPVPMGLWM TGKTVLETWY NIKEAVLEGE TPFDRAYGQ FFEYLGENG 180
VNTLFDEAMA NHSTIITKRL VEVFRGFENY SVLVDVGGNK GTTLQMIRSQ YENISGINYD 240
LPRVIAQAPP IEGVEHVGGN MFDNVPRGDA IILKWILHDW GDKDCVKILK NCYAALPVNG 300
TMIILEYILP ETPEETLTSQ LAFNFDGMM LMYGAKGKER TEKELSELAR EAGFSGDCTA 360
TYIFASIWAL EFTK 374

SEQ ID NO: 23 moltype = AA length = 398
FEATURE Location/Qualifiers
source 1..398
 mol_type = protein
 organism = Sorghum bicolor

SEQUENCE: 23
MTTGKVTLEA VRKAQRAEGP ATVLAIGTAT PANCYQADY PDYYFRVTKS EHLTDLKEKF 60
KRICKSMIR KRYMHLTEDI LEENPNMSSY WAPSLDARQD ILIQEIPKL AEAEEKALKE 120
WGQPRSRITH LVFCTTSGVD MPGADYQLIK LLGLCPSVNR AMMYHQGCFA GGMVLRLLAKD 180
LAENNRGARV LIVCSEITV TFRGPSESHL DSLVGQALFG DGAAAVIVGA DPSEPAERPL 240
FHLVSASQTI LPDSEGAIEG HLREVGTLTFH LQDRVPLIS MNIERLLEDA FAPLGISDWN 300
SIFWVAHPGG PAILNMVEAK VGLDKARMCA TRHILAEGN MSSVCVLFIL DEMRNRSAKD 360
GHTTTGEGME WGVLFGFGPG LTIVETIVLHS VPITTVAA 398

SEQ ID NO: 24 moltype = AA length = 360
FEATURE Location/Qualifiers
source 1..360
 mol_type = protein

-continued

organism = *Triticum aestivum*

SEQUENCE: 24

MGSTAADMAA	SADEEACMYA	LQLVSSSILP	MTLKNAIELG	LLETLVAAGG	KLLTPAEVAA	60
KLPSTANPAA	ADMVDRMLRL	LASYNVVST	MEEGKDGRLS	RRYRAAPVCK	FLTPNEDGVS	120
MAALALMNQD	KVLMESWYYL	KDAVLDDGIP	FNKAYGMSAF	EYHGTDPFRN	RVFNEGKMNH	180
SIITTKKLL	VYKGFEGELT	IVDVGGGVGA	TVGAITAAYP	AIKGINFDLP	HVISEAQPPF	240
GVTHTVGGDMF	QKVPSSGDAIL	MKWILHDWSD	EHCATLLKNC	YDALPAHGKV	VLVECILPVN	300
PEATPKAQGV	FHVDIMIMLAH	NPGRERYER	EFEALAKGAG	FKAIKTTYIY	ANAFIEFTK	360

SEQ ID NO: 25 moltype = AA length = 356
FEATURE Location/Qualifiers
source 1..356
 mol_type = protein
 organism = *Triticum aestivum*

SEQUENCE: 25

MGSIAAGADE	DACMYALQLV	SSSILPMTLK	NAIELGLLET	LMAAGGKFLT	PAEVAAKLPS	60
AANPEAPDMV	DRMLRLLASY	NVVSCTEDG	KDGRLSRRYG	AAPVCKYLTP	NEDGVSMASL	120
ALMNQDKVLM	ESWYLLKDAV	LDGGIPFNKA	YGMSAFEYHG	TDPRFNRVFN	EGMKNHSIII	180
TKKLLSYKGG	FEGLGTLDV	GGVGATVAA	ITAHYPTIKG	INFDLPHVIS	EAPPPGVTH	240
VGGDMFQKVP	SGDAILMKWI	LHDWSDEHCA	TLLKNCYDAL	PAHGKVVLLVE	CILPVNPEAT	300
PKAQGVFHV	MDLAHNPFG	RERYEREFEA	LAKGAGFAAM	KTTYIYANAW	AIEFTK	356

SEQ ID NO: 26 moltype = AA length = 386
FEATURE Location/Qualifiers
source 1..386
 mol_type = protein
 organism = *Vitis vinifera*

SEQUENCE: 26

MESTLAFNSG	SNSMNQSFSS	SAEFNSPVPE	TIPKSEEDTF	VFATLLTSAS	VLPMAKLSAL	60
EIDLLLEIIAK	AGPGAFVSTS	EIAAKITKRN	PKAPVMDRI	LRLLATYDVV	KCSLRDSPDG	120
GVERLYGLGP	VCKYFTTND	GVSVAPLLLM	NQDKVPMQSK	RYHLKDAVLD	GGIPFNKAYG	180
MTDFEYHGTE	PRPNKVPNNG	VSGHPTITMK	KILEAYKGFE	GLTSIVDVGG	GTGATLNMII	240
SKYPTIKGIN	FDLPHVIDDA	PSYPGVEHVG	GDMFVSVPKG	DAIFMKWMCY	EWDDAHCLKF	300
LENCYQALPD	NGKVIVAEIC	LPVVPDTSLA	TKSAVHIDVI	MLAYNTGGKA	RTEKEFEALA	360
KGAGFQGFVK	VCCAFNSWIM	EPCKTA				386

SEQ ID NO: 27 moltype = AA length = 357
FEATURE Location/Qualifiers
source 1..357
 mol_type = protein
 organism = *Vitis vinifera*

SEQUENCE: 27

MDLANGVISA	ELLHAQAHVW	NHIFNFIKSM	SLKCAIQGLI	PDIIHNHGKP	MTLPELVAKL	60
PVHPKRSQCV	YRLMRILVHS	GFLAAQRVQQ	GKEEEGYVLT	DASRLLLMDD	SLSIRPLVLA	120
MLDPILTKFW	HYLSAWFQND	DPTPFHTAHE	RSFWDYAGHE	PQLNNSFNEA	MASDARLLTS	180
VLLKEGGQVF	AGLNSLVDVG	GGTGKVAKAI	ANAPPHLNCT	VLDLPHVAVG	LQGSKNLNYF	240
AGDMFEAIPP	ADAILLKWIL	HDWSDEECVK	ILKRCREAIK	SKENGKGVII	IDMIMMKNQG	300
DYKSTETQLF	FDMTMMIFAP	GRERDENWE	KLFLDAGFSH	YKITPILGLR	SLIEVYP	357

SEQ ID NO: 28 moltype = AA length = 364
FEATURE Location/Qualifiers
source 1..364
 mol_type = protein
 organism = *Zea mays*

SEQUENCE: 28

MGSTAGDVAA	VVDEEACMYA	MQLASSSILP	MTLKNAIELG	LLEVLQKEAG	GGKAALAPEE	60
VVARMPAPAP	DPAAAAAMVD	RMLRLLASYD	VVRCQMEDRD	GRYERRYSAA	PVCKWLTPNE	120
DGVSMALAL	MNQDKVLMES	WYYLKDAVLD	GGIPFNKAYG	MTAFEYHGTD	SRFNRVFNEG	180
MKNHSVIITK	KLLDFYTGE	GVSTLVDVGG	GVGATLHAIT	SRHPHISGVN	FDLPHVISEA	240
PPFPGVRHV	GDMFASVPAG	DAILMKWILH	DWSDAHCATL	LKNCYDALPE	NGKVIVVECV	300
LPVNTTEATPK	AQGVFHVDMI	MLAHNPGGKE	RYEREFRELA	KGAGFSGFKA	TYIYANAWAI	360
EFIK						364

SEQ ID NO: 29 moltype = AA length = 365
FEATURE Location/Qualifiers
source 1..365
 mol_type = protein
 organism = *Zea mays*

SEQUENCE: 29

MACTTAASLQ	HDRANDDEAC	MYAQELLSCF	VVPMTLKAVI	ELGLIDDLA	ADGRFVTPEE	60
LAARWARPAE	AAAAVDRMLR	FLASHSVVRC	TTEAAGPDGR	ARRSYAAPV	CKWLIARNGT	120
GQGSWAPIGL	MNLNKGFMET	WYYMKDAVAE	GATPTEKAYG	MPLFEHLGSD	EALNTLFNQA	180
MAGHSEIVIS	KLLEVYRGFE	GVDVLVDVGG	GTGSTLRMVT	AQYKHLRGVN	YDLPHVIAQA	240
PPVQGVHEAG	GSMFEYIPSG	NAILLKWILH	LWRDEECIKI	LKNCHRALPA	NGKVIVVECV	300
LPASPEPTQV	AQGALLLDV	MLNRLRGAKE	RTEREFTELA	AEAGFSGGCR	ATYVFTGAWA	360
LEFTK						365

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SEQ ID NO: 30      moltype = DNA length = 1125
FEATURE           Location/Qualifiers
source            1..1125
                  mol_type = genomic DNA
                  organism = Sorghum halepense

SEQUENCE: 30
atgggcagct atactaccag agtccacgcc gtagagaaag cagctaacaa ccaagatgat 60
gagacgtgca tgcattgctt gatgctcctt ggccgattgg ccgtaccttg taccatcaag 120
gccgtcatcg agctcggcat catggacctt ctctcgcgcg cggaccgcgc catgaccgcg 180
gaggagctca cggcaagggt gccatgtcct gctgctgcta ctgctgctgc tatggtcgac 240
cgcatgctcc gcttcttagc ttccgacggc gtggtcaggt gcgcgcgcgc caccaaggag 300
tcgtcggagc tggggtccga tggtaggcaag agctgccggc gctacgcggc ggccgcgggtg 360
tgcagggtgt ttacaaggag cggcggcggt gagtcgggtg ttccaatggg gttgtggatg 420
accggcaaga ccgtcttgga gacctggtag aacataaaag aagcagtggt agaggagaga 480
acaccatttg acagagcata ggccagccca ttttttgagt acctgggtga aaacggggaca 540
gtgaacacgt tgttcgatga ggcaatggcg aaccattcga caatcataac gaagagggtg 600
gtcagaggtc tccgtggctt tgagaattac agtgtgctcg tcgatgtcgg cggcaacaaa 660
ggcaccacgt tgcaaatgat tagaagtcag tatgagaata ttagtggcat caactacgac 720
cttctctgtg taattgcaac ggccacctca attgaagggt tggagcatgt tggtaggcaac 780
atgtttgata atgttccacg tggagatgca attatcctaa agtggattct tcacgattgg 840
ggcgacaagg actcgcgtca gatcctaaa aattgctatg cagctctccc ggtgaacggg 900
acgatgatca tctcggagta catcctccca gagacaccag aagaaacact aacatcgcaa 960
ctagcgttca acttcgatgt cgggatgatg ctcatgtacg gcgccaagg taaggagaga 1020
acagagaaag agttgtcgga gctcgccaga gaggccggct tctctggaga ctgcacggct 1080
acatacatct tcgccagcat ctgggcccct gaattcacaa agtaa 1125

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SEQ ID NO: 31      moltype = DNA length = 1188
FEATURE           Location/Qualifiers
source            1..1188
                  mol_type = genomic DNA
                  organism = Sorghum halepense

SEQUENCE: 31
tcgtcctcta tctgtggcca ccaactgctg tgctatttga gtacacagat acagatgggc 60
agctacgaca gcagcagtag tagtagtaat gactcatcag caccgaacga ggaggacgag 120
tcgtgcattg ttgttttgaa gctcctcggc gggttcgcgcg tacctttcac catcaaggcg 180
gtgattgagc tggcgctcat ggaccagctc ctgactgcgc aacgcgcgat gagcgcggag 240
gagctgggtg cagcagcaggt ggccagcacg ctgccacggc ccgaggtagc ctgcaccatg 300
gtggaccgcc tgctcctgtt ccttgctcgc cacagcgtag tccgggtgca gaccgaggtg 360
gtggtgggca cggacgacgc caccaccacc acctgctgcc gccggagcta cgccgcgtca 420
ccgtctgca agtgggtcgc caggaacggc gtcgaggatt cggtgcttcc gcttgggatg 480
atgatcctaa acaagacatt cctggacagc tggcaaaaac ttacggatgc agtgttgga 540
ggagcagcac catttgagaa aacctacggg atgccaatgt tcgagtacct aagtacaaac 600
ggaccattga acacgggtgt ccacgaggca atggcaaatc attcgatgat tataaccaag 660
aaactgctca agttcttcgc cggcttcgaa ggccttgatg tgctgggtga cgtaggcgcc 720
ggcaacggca ccacgttgca aatgattaga ggtcaatata agaataatg aggcataaac 780
tacgaccttc ctcatgtcat tggcagggct gcaccagttg aagggttgga acatgtaggt 840
ggcagcatgt tcgataatat tcacgcggga aatgcagttc tgctcaagtg gattcttcat 900
gattgggacg acaaggcgtg catcaagatc ctacaagaat gctatacagc tctccatgtg 960
agaggcaagg tgatcgttct ggagtagctt gttccggatg aaccagaacc tactcttgca 1020
gctcaggggt ccttcgaact ggacctcacc atgctggcta cgtttgagc tggtaaagag 1080
aggcacaga gggagttctc cgagctcgcc atggaggccg gcttctctag agagttaaag 1140
gtacagtata tctttgcca cgtctgggccc cttgagttca caaagtaa 1188

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SEQ ID NO: 32      moltype = AA length = 362
FEATURE           Location/Qualifiers
source            1..362
                  mol_type = protein
                  organism = Saccharum spontaneum

SEQUENCE: 32
MGSTAEDVAA VADEEACMYA MQLASASILP MTLKNALELG LLEVLQAEAP AGKALAPEEV 60
VARLPVAPTIN PDAADMVDRM LRLASYDVV KCQMEDKDGK YERRYSAAPV GKWLTPNEDG 120
VSMALATLMN QDKVLMESRY YLKDAVLDDG IPFNKAYGMT AFEYHGTDPF FNRVFNEMGM 180
NHSVIITKKL LEFYTGFEV STLVDVGGGI GATLHAITSH HPQIKGINFD LPHVISEAPP 240
FPGVQHVGGD MFKSVPAQDA ILMKWILHDW SDAHCATLLK NCYDALPENG KVIIVCEVLP 300
VNTEAVPKAQ GVPHVDMIML AHNPGGRERY EREFHDLAKG AGFSGFKATY IYANAWAIEF 360
IK

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SEQ ID NO: 33      moltype = AA length = 394
FEATURE           Location/Qualifiers
source            1..394
                  mol_type = protein
                  organism = Saccharum spontaneum

SEQUENCE: 33
MTGKVTLGAV RKAQRAEGSA AVLAIGTATP ANCVYQADYP DYYFRVTKSE HLTDLKEKFK 60
RICKSKMITK RYMHLTEGFL QENPNMSSYS APSLDARQDI LIEEVPKLGAA AAEKALKEW 120

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GQPRSQITHL	VFCTTSGVDM	PGADYQLIKL	LGLSLSVNRA	MMYHQGCFAG	GMVLRLLAKDL	180
AENNRGARVL	IVCSEITAVT	FRGPSESHLD	SLVGQALFGD	GAAAVIVGAD	PSAAEWPLFQ	240
LVSASQTILP	DSEGAIEGHL	REVGLTFHLQ	DRVLPQLISTN	IERLLEDAT	PLGISDWNST	300
FWVAHPGGPA	ILNMVEAKAG	LDKARLCATR	HILAIEYGNMS	SACVLFILDE	MRNKSIEDGH	360
TTTGEEMEWG	VLFGFGPGLT	VETIVLQSV	IITV			394

SEQ ID NO: 34 moltype = AA length = 374
 FEATURE Location/Qualifiers
 source 1..374
 mol_type = protein
 organism = Sorghum halepense

SEQUENCE: 34

MGSYTTRVHA	VEKAANNQDD	ETCMHALMLL	GGLAVPCTIK	AVIELGIMDL	LLAADRAMTA	60
EELTARLP	AAATAAMVD	RMLRFLASHG	VVRCAATKE	SSELGSDGGK	SCRRYAAAPV	120
CRWFTRSGGV	ESVPMGLWM	TGKTVLETWY	NIKEAVLEGE	TPFDRAYGQP	FFEYLGENG	180
VNTLPDEAMA	NHSTIITKRL	VEVFRGFENY	SVLVDVGGNK	GTTLMIRSQ	YENISGINVD	240
LPRVIAQAPP	IEGVEHVGGN	MFDNVPRGDA	IILKWILHDW	GDKDCVKILK	NCYAALPVNG	300
TMILEYILP	ETPEETLSQ	LAFNFDGMM	LMYGAKGKER	TEKELSELAR	EAGFSGDCTA	360
TYIFASIWAL	EFTK					374

SEQ ID NO: 35 moltype = AA length = 377
 FEATURE Location/Qualifiers
 source 1..377
 mol_type = protein
 organism = Sorghum halepense

SEQUENCE: 35

MGSYDSSSSS	SNDSARNEE	DESCMPALKL	LGGFAVPFTI	KAVIELGVMD	QLLTAERAMS	60
AEELVAAAVA	AQLPRPEVAC	TMVDRLRLRL	ASHSVVRCCT	EVVVGTDAT	TTTCRRSYA	120
ASPVCKWFAR	NGVEDSVLPL	GMMILNKTF	DSWQNTDAV	LEGAAPFEKT	YGMMPFEYLS	180
TNGPLNTVFH	EAMANHSMII	TKKLLKFRRG	FEGLDVLVDV	GGGNGTTLQM	IRGQYKNMRG	240
INYDLPHVIA	QAAPVEGVEH	VGGSMFDNIP	RGNVLLKWI	LHDWDDKACI	KILKNCTAL	300
HVRGKIVILE	YVVPDEPEPT	LAAQGAFFLD	LTMLVTFGSG	KERTQREFSE	LAMEAGFSRE	360
FKATYIFANV	WALEFTK					377

SEQ ID NO: 36 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 36

caactttgtc	accgcgatg	a	21
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SEQ ID NO: 37 moltype = DNA length = 21
 FEATURE Location/Qualifiers
 source 1..21
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 37

tccagaaacc	ttagcagccc	a	21
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SEQ ID NO: 38 moltype = DNA length = 19
 FEATURE Location/Qualifiers
 source 1..19
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 38

tgctacggtg	ttggccatt	19
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SEQ ID NO: 39 moltype = DNA length = 23
 FEATURE Location/Qualifiers
 source 1..23
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 39

atcgacttgt	ggcatatcct	ctt	23
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SEQ ID NO: 40 moltype = DNA length = 24
 FEATURE Location/Qualifiers
 source 1..24
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 40

cctaaagtgg	attcttcacg	attg	24
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SEQ ID NO: 41 moltype = DNA length = 23
 FEATURE Location/Qualifiers

-continued

source	1..23 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 41		
gctagttgcg atgttagtgt ttc		23
SEQ ID NO: 42	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 42		
aggtggcagc atgttcgata a		21
SEQ ID NO: 43	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 43		
acgatcacct tgcctctcac		20
SEQ ID NO: 44	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 44		
ttggtttcca ctgacttcgt t		21
SEQ ID NO: 45	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 45		
ctgtagcccc actcgttgt		19
SEQ ID NO: 46	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 46		
agagcgaaca ccttaccgac		20
SEQ ID NO: 47	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 47		
gccgagtacg agctcatggt		20
SEQ ID NO: 48	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 48		
gaggacaagg acggcaagta		20
SEQ ID NO: 49	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 49		
accgcgtcct tgaggtagta		20
SEQ ID NO: 50	moltype = DNA length = 29	
FEATURE	Location/Qualifiers	
source	1..29 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 50		

-continued

cgcggatcca tgggcagcta tactaccag	29
SEQ ID NO: 51	moltype = DNA length = 29
FEATURE	Location/Qualifiers
source	1..29
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 51	
cccaagcttc tttgtgaatt caagggccc	29
SEQ ID NO: 52	moltype = DNA length = 42
FEATURE	Location/Qualifiers
source	1..42
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 52	
tgagaacctg tacttccaag gcagctacga cagcagcagt ag	42
SEQ ID NO: 53	moltype = DNA length = 45
FEATURE	Location/Qualifiers
source	1..45
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 53	
agccggatct cactcgagtt actttgtgaa ctcaagggcc cagac	45
SEQ ID NO: 54	moltype = DNA length = 37
FEATURE	Location/Qualifiers
source	1..37
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 54	
tgagaacctg tacttccaag ggtcgacggc ggaggac	37
SEQ ID NO: 55	moltype = DNA length = 43
FEATURE	Location/Qualifiers
source	1..43
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 55	
agccggatct cactcgagtt acttgatgaa ctcgatggcc cag	43
SEQ ID NO: 56	moltype = DNA length = 45
FEATURE	Location/Qualifiers
source	1..45
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 56	
cttctgcagg aattcgatat catgactggg aaggtaacat tgggg	45
SEQ ID NO: 57	moltype = DNA length = 44
FEATURE	Location/Qualifiers
source	1..44
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 57	
gagagatctg tcgacgatat cctacactgt gatgatggga acgc	44
SEQ ID NO: 58	moltype = DNA length = 36
FEATURE	Location/Qualifiers
source	1..36
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 58	
cttctgcagg aattcgatat catgggctcg accgcc	36
SEQ ID NO: 59	moltype = DNA length = 46
FEATURE	Location/Qualifiers
source	1..46
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 59	
gagagatctg tcgacgatat cttacttgat gaactcgatg gcccg	46
SEQ ID NO: 60	moltype = DNA length = 48
FEATURE	Location/Qualifiers

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source	1..48	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 60		
gtgcttccgc ttgggatgat gaacctaaac aagacattcc tggacagc		48
SEQ ID NO: 61	moltype = DNA length = 21	
FEATURE	Location/Qualifiers	
source	1..21	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 61		
catcatccca agcggagca c		21
SEQ ID NO: 62	moltype = DNA length = 45	
FEATURE	Location/Qualifiers	
source	1..45	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 62		
cctcaccatg ctggtcacga acggcagtgg taaagagagg acaca		45
SEQ ID NO: 63	moltype = DNA length = 19	
FEATURE	Location/Qualifiers	
source	1..19	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 63		
cgtgaccagc atggtgagg		19
SEQ ID NO: 64	moltype = DNA length = 43	
FEATURE	Location/Qualifiers	
source	1..43	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 64		
ttctgctcaa gtggattctt gctgattggg acgacaaggc gtg		43
SEQ ID NO: 65	moltype = DNA length = 25	
FEATURE	Location/Qualifiers	
source	1..25	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 65		
aagaatccac ttgagcagaa ctgca		25
SEQ ID NO: 66	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 66		
gattggggacg acaaggcgtg		20
SEQ ID NO: 67	moltype = DNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 67		
gttaagaatc cacttgagca gaac		24
SEQ ID NO: 68	moltype = DNA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 68		
gcgtggggacg acaagg		16
SEQ ID NO: 69	moltype = DNA length = 24	
FEATURE	Location/Qualifiers	
source	1..24	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 69		

-continued

atgaagaatc cacttgagca gaac	24
SEQ ID NO: 70	moltype = DNA length = 19
FEATURE	Location/Qualifiers
source	1..19
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 70	
gcgtacgttg ttccggatg	19
SEQ ID NO: 71	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 71	
cagaacgatac accttgccctc	20
SEQ ID NO: 72	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 72	
gcgaggacac agaggagatt c	21
SEQ ID NO: 73	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 73	
tttaccactg ccaaactgta c	21
SEQ ID NO: 74	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 74	
tgactcatca gcacggaacg	20
SEQ ID NO: 75	moltype = DNA length = 20
FEATURE	Location/Qualifiers
source	1..20
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 75	
caccctccta gtagccgcaa	20
SEQ ID NO: 76	moltype = DNA length = 22
FEATURE	Location/Qualifiers
source	1..22
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 76	
ggcaaaacat tacggatgca gt	22
SEQ ID NO: 77	moltype = DNA length = 21
FEATURE	Location/Qualifiers
source	1..21
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 77	
tgacttcacg gaggtcata g	21
SEQ ID NO: 78	moltype = DNA length = 31
FEATURE	Location/Qualifiers
source	1..31
	mol_type = other DNA
	organism = synthetic construct
SEQUENCE: 78	
atcggaaccgg tatgacgact gggaaggtaa c	31
SEQ ID NO: 79	moltype = DNA length = 31
FEATURE	Location/Qualifiers

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source                1..31
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 79
cgatcctcga gtcatgcagc cactgtggtg a                               31

SEQ ID NO: 80          moltype = DNA length = 33
FEATURE               Location/Qualifiers
source                1..33
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 80
tgcccaaatt cgcgaaatggg cagctacgac agc                             33

SEQ ID NO: 81          moltype = DNA length = 67
FEATURE               Location/Qualifiers
source                1..67
                        mol_type = other DNA
                        organism = synthetic construct

SEQUENCE: 81
tggtgatggt gatcatatt atttttcaaa ttgaggatga gaccactttg tgaactcaag 60
ggcccag                                                       67

SEQ ID NO: 82          moltype = AA length = 372
FEATURE               Location/Qualifiers
source                1..372
                        mol_type = protein
                        organism = Medicago sativa

SEQUENCE: 82
MGNSYITKED NQISATSEQT EDSACLSAMV LTTNLVYPVAV LNAAIDLNLF EIIAKATPPG 60
AFMSPSEIAS KLPASTQHS DLPNRLDRMLR LLASYSVLTS TTRTIEDGGA ERVYGLSMVG 120
KYLVPDES RG YLASFTTFLC YPALLQVWMN FKEAVVDEDI DLFKNVHGV T KYEFMGKDKK 180
MNQIFNKSMV DVCATEMKRM LEIYTGFE GI STLVDVGGGS GRNLELIISK YPLIKGINFD 240
LPQVIENAPP LSGIEHVGGD MFASVPQGD A MILKAVCHNW SDEKCI EFLS NCHKALSPNG 300
KVIVIEFILP EEPNTSEESK LVSTLDNLMP ITVGGRE RTE KQYEKLSKLS GFSKFPQVACR 360
AFNSLGVMEF YK                                                       372

SEQ ID NO: 83          moltype = AA length = 365
FEATURE               Location/Qualifiers
source                1..365
                        mol_type = protein
                        organism = Medicago sativa

SEQUENCE: 83
MGSTGETQIT PTHISDEEAN LFAQMLASAS VLPMLKSAL ELDLLEIIAK AGPGAQISPI 60
ETASQLPTTN PDAPVMLDRM LRLLCYIIL TCSVRTQQDG KVQRLYGLAT VAKYLVKNED 120
GVSISALNLM NQDKVLMESW YHLKDAVL DG GIPFNKAYGM TAFEYHGTDP RFNKVFENKGM 180
SDHSTITMKK ILETYTGFE G LKSLVDVGGG TGAVINTIVS KYPTIKGINF DLPHVIEDAP 240
SYPGVEHVGG DMFVSIPKAD AVFMKWICH D WSEHCLKFL KNCYEALPDN GKVIVAECIL 300
PVAPDSSLAT KGVVHIDVIM LAHNPGGKER TQKEFEDLAK GAGFQGFKVH CNAPNTYIME 360
FLKKV                                                       365

SEQ ID NO: 84          moltype = AA length = 364
FEATURE               Location/Qualifiers
source                1..364
                        mol_type = protein
                        organism = Medicago truncatula

SEQUENCE: 84
MAFSTNGSEE SELYHAQIHL YKHVYNFVSS MALKSAMELG IADAIHNHGK PMTLSELASS 60
LKLHPSKVNI LHRFLRL LTH NGFFAKTIVK GKEGDEEE EI AYSLTPPSKL LISGKPTCLS 120
SIVKGALHPS SLDMWSSSKK WFNEDKEQTL FECATGESFW DFLNKDSESS TLSMFQDAMA 180
SDSRMFKVLV QENKRVFEG L ESLVDVGGGT GGVTKLIHEI FPHLKCTVFD QPQVVGNLTG 240
NENLNFVGGD MFKSIPSADA VLLKWLHDW NDEQSLKILK NSKEAISHKG KDGKVIID I 300
SIDETSDDR G LTELQLDYDL VMLTMFLGKE RTKQEWELI YDAGFSSYKI TPISGFKSLI 360
EVYP                                                       364

SEQ ID NO: 85          moltype = AA length = 357
FEATURE               Location/Qualifiers
source                1..357
                        mol_type = protein
                        organism = Medicago truncatula

SEQUENCE: 85
MASSINNRKP SEIFKAQALL YKNMYAFVDS MSLKWSIEMN IPNIIHNHGK PITLSNLVSI 60
LQIPSTKVDN VQRLMRYL AH NGFFEIITNQ ELENEEEAYA LTVASELLVK GTCLCLAPMV 120
ECVLDP TLTST SFHNLKKVY EEDLT LFAVN LGCDLWEFLN KNPEYNTLYN DALADSKMI 180
NLAMKDCNLV FEGLESIVDV GGGNGTTGKI ICETFPKLTC VVFD RPKVVE NLCSNNLT Y 240
VGGDMFISVP KADAVLLKAV LHDWTDKDCI KILKKCKEAV TSDGKRGKVI VIDMVINEKK 300

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DENQLTQIKL LMNVTISCVN GKERNNEEWK KLFIEAGFQD YKISPFITGLM SLIEIYP 357
 SEQ ID NO: 86 moltype = AA length = 332
 FEATURE Location/Qualifiers
 source 1..332
 mol_type = protein
 organism = *Streptomyces* sp.
 SEQUENCE: 86
 MGKRAAHIGL RALADLATPM AVRVAATLRV ADHIAAGHRT AAEIASAAGA HADSLDRLLR 60
 HLVAVGLFTR DGQGVYGLTE FGEQLRDDHA AGKRKWLDMN SAVGRGDLGF VELAHSIRTG 120
 QPAYPVRYGT SFWEDLGSDP VLSASFDTLM SHHLELDYTG IAAKYDWAAAL GHVVDVGGGS 180
 GGLLSALLTA HEDLSGTVDL LQGPASAAHR RFLDTGLSGR AQVVVGSFFD PLPAGAGGYV 240
 LSAVLHDWDD LSAVAILRRC AEAAGSGGVV LVIEAVAGDE HAGTGMDLRM LTYFGGKERS 300
 LAELGELAAQ AGLAVRAAHP ISYVSIVEMT AL 332
 SEQ ID NO: 87 moltype = AA length = 360
 FEATURE Location/Qualifiers
 source 1..360
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 87
 MGSTAADMAA SADEADACMFA LQLASSSVLP MTLKNAIELG LLEILVAAGG KSLTPTEVAA 60
 KLPSEANPEA PDMVDRILRL LASYNVVTCL VEEGKDGRSL RSYGAAPVCK FLTPNEDGVS 120
 MAALALMNQD KVLMSWYLL KDAVLDDGGIP FNKAYGMSAF EYHGTDPFRN RVFNEMGMKNH 180
 SIIITKKLLE LYHGFEGELGT LVDVGGGVGA TVAAIAAHYP TIKGVNFDLP HVISEAPQFP 240
 GVTHVGGDMF KEVPSGDTIL MKWILHDWSD QHCATLLKNC YDALPAHGKV VLVQCILPVN 300
 PEANPSSQGV PHVDMIMLAH NPGGREERYER EFQALARGAG FTGVKSTYIY ANAWAIEFTK 360
 SEQ ID NO: 88 moltype = AA length = 368
 FEATURE Location/Qualifiers
 source 1..368
 mol_type = protein
 organism = *Clarkia breweri*
 SEQUENCE: 88
 MGSTGNAEQ IIPHTSSDEE ANLFAMQLAS AAVLPMALKA AIELDVLEIM AKSVPPSGYI 60
 SPAETIAAQPL TTNPEAPVML DRVLRLRLASY SVVYTLREL PSGKVERLYG LAPVCKFLTK 120
 NEDGVSLAPF LLTATDKVLL EPWFYLLKDAI LEGGIPFNKA YGMNEFDYHG TDHFRNKVFN 180
 KGMSSNSTIT MKKILEMYNG FEGLTTIVDV GGGTGAVASM IVAKYPSINA INFDLPHVIQ 240
 DAPAFSGVEH LGGDMFQGVV KGDALFIKWI CHDWSDEHCL KLLKNCYAAL PDHGKIVIAE 300
 YILPSPDPS IATKVVIHTD ALMLAYNPGG KERTEKEFQA LAMASGRGRF KVASCAFNTY 360
 VMEFLKTA 368
 SEQ ID NO: 89 moltype = AA length = 361
 FEATURE Location/Qualifiers
 source 1..361
 mol_type = protein
 note = *Planctopirus limnophila*
 organism = unidentified
 SEQUENCE: 89
 MAVKDALRFP PTDVTPIFDL FRGNFATELL AASVAHLHVF DILNESPLSL DELQRRVLVS 60
 ERATQVVLVT LCAMQQLTKR LAGEIDLTP L ARNHLVTTSP FSVGGYISLA AQSAGTLALV 120
 ERLKSDRPEG AESEQGAFFI FREGESAMD REDSARFLTL SLAGRAWNVA PRFADVLPAG 180
 QPGKILKDNS GSSGRVLLDV AGGSGIYTMA VLQKYPTWRG IIFDRPEVLK IAAELAEQTG 240
 VRDRLELHAG DMWVDPFPPA DDILLSNLVH DWDRPQCRL VAKATSGLPE GGRLLIHDVL 300
 LNSDLTGPLE IALYSLALFS LTEGRAYSL EYRGWIAGAD LKYVDCIPTS AHGHLILSEK 360
 V 361
 SEQ ID NO: 90 moltype = AA length = 350
 FEATURE Location/Qualifiers
 source 1..350
 mol_type = protein
 organism = *Thalictrum* sp.
 SEQUENCE: 90
 MEMINKENLS SQAKLWNFIY GFADSLVLKS AVQLDLANII HNHGSPMTLS ELSLHLPSPQ 60
 VNQDALYRVL RYLVMHMLFT KSSIDGELRY GLAPPAKFLV KGWDKCMGLA ILTITDKDFM 120
 APWHYLKEGI LNDGSTSTAF EKALGTNIWD YMAEHPEKNQ LFNEGMANDT RLIMSALVKE 180
 CSSMFDGITT IVDVGGGTGT AVRNIKAPP HIKCTVYDLP HVIADSPGYT EINSIQGDMF 240
 KYIPNADAIM MKCILHDWDD KECIEILKRC KDAVPRDGGK VIIIDIILDV KSEHPYTKMR 300
 LTLDLDMMLN TGGKERTEEE WKKLIHDAGY KGYKITHISA VQSVIEAYPY 350
 SEQ ID NO: 91 moltype = AA length = 359
 FEATURE Location/Qualifiers
 source 1..359
 mol_type = protein
 note = *Kitagawia praeruptora*
 organism = unidentified

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SEQUENCE: 91
MAGMKTSPSQ DEEACVLAIQ LATSTVLPMI LKSAIELDIL NTISKAGPGN YLSPSDLASK 60
LLMSNPAPI MLERILRVLA TYKVLGCKPS ELSDGVEVWL YCWTPVCKFL SNNEDGASIA 120
PLLLVHQDQV PMKSWYHLTD AILDGGTAFN KAYGMNIFY ASQDPQFNKV FNRS MAGHST 180
ITMKKILETY NGFEGLSKIV DVGGGSGATL NMIISKYPTI KGINFDLPHV VGDSPHPGV 240
EHVGGDMFAS VPKGDAIFLK WIFHSWDED CLRILKNCYE ALADNKKVIV AEFIIPEVPG 300
GSDDATKSVV HLDVAVMLAYV PGGKERTKE FEALATSAGF KSFRKVCCAF NTWIMEFSK 359

SEQ ID NO: 92 moltype = AA length = 390
FEATURE Location/Qualifiers
source 1..390
 mol_type = protein
 organism = Papaver somniferum

SEQUENCE: 92
MATNGEIPNT YGHNROTATV TKITASNESS NGVCYLSETA NLGKLICIPM ALRAAMELNV 60
FQLISKFGTD AKVSASEIAS KMPNAKNPE AAMYLDRI LR LLGASSILSV STTKKSINRG 120
GDDVVVHEKL YGLTNSSCCL VPRQEDGVSL VEELLFTSDK VVVDSPFKLK CVVEEKDSVP 180
FEVAHGAKIF EYAATEPRMN QVFNDGMAVF SIVVFEAVFR FYDGF LDMKE LLDVGGGIGT 240
SVSKIVAKYP LIRGVNFDLP HVISVAPQYP GVEHVAGDMF EEVPGQNM L LKWLHDWGD 300
ERCVKLLKNC WNSLPVGGKV LIIEFVLPNE LGNNAESFNA LIPDLLMAL NPGGKERTIS 360
EYDDLGAAG FIKTIPIPI NGLHVIEFHK 390

SEQ ID NO: 93 moltype = AA length = 365
FEATURE Location/Qualifiers
source 1..365
 mol_type = protein
 organism = Fragaria sp.

SEQUENCE: 93
MGSTGETQMT PTHVSDEEAN LFAQMLASAS VLPMVLKAAI ELDLLEIMAK AGPGSFLSPS 60
DLASQLPTN PEAPVMDLDRM LRLLASYSIL TCSLRTLDPG KVERLYCLGP VCKFLTKNED 120
GVSIAALCLM NQDKVLVESW YHLKDAVLDG GIPFNKAYGM TAFDYHGTDP RFNKVFNKGM 180
ADHSTITMKK ILETYKGFEG LKSIVDVGGG TGAVVNMIIVS KYPYIKGINF DLPHVIEDAP 240
QYPGVQHVGG DMFVSVPKGN AIFMKWICH D WSEHCIFKL KNCYALPDD GKVLAEICIL 300
PVAPDTSLAT KGVVHMDVIM LAHNPGGKER TEQEFELAK GSGFQGIRVC CDAFNITYVIE 360
FLKKI 365

SEQ ID NO: 94 moltype = DNA length = 23
FEATURE Location/Qualifiers
source 1..23
 mol_type = genomic DNA
 organism = Sorghum bicolor

SEQUENCE: 94
cctcgccacg cgtagtccgg tgc 23

SEQ ID NO: 95 moltype = DNA length = 23
FEATURE Location/Qualifiers
source 1..23
 mol_type = genomic DNA
 organism = Sorghum bicolor

SEQUENCE: 95
attgaacacg gtgttccacg agg 23

SEQ ID NO: 96 moltype = DNA length = 22
FEATURE Location/Qualifiers
source 1..22
 mol_type = genomic DNA
 organism = Sorghum bicolor

SEQUENCE: 96
cctcgccagc gtagtccggt gc 22

SEQ ID NO: 97 moltype = DNA length = 24
FEATURE Location/Qualifiers
source 1..24
 mol_type = genomic DNA
 organism = Sorghum bicolor

SEQUENCE: 97
attgaacacg gtgttccac gagg 24

SEQ ID NO: 98 moltype = DNA length = 18
FEATURE Location/Qualifiers
source 1..18
 mol_type = genomic DNA
 organism = Sorghum bicolor

SEQUENCE: 98
cctcgcgtag tccggtgc 18

SEQ ID NO: 99	moltype = DNA length = 24			
FEATURE	Location/Qualifiers			
source	1..24			
	mol_type = genomic DNA			
	organism = Sorghum bicolor			
SEQUENCE: 99				
attgaacacg gtgttcccac gagg			24	
SEQ ID NO: 100	moltype = DNA length = 22			
FEATURE	Location/Qualifiers			
source	1..22			
	mol_type = genomic DNA			
	organism = Sorghum bicolor			
SEQUENCE: 100				
cctcgcaagc gtagtccggt gc			22	
SEQ ID NO: 101	moltype = DNA length = 24			
FEATURE	Location/Qualifiers			
source	1..24			
	mol_type = genomic DNA			
	organism = Sorghum bicolor			
SEQUENCE: 101				
attgaacacg gtgttcccac gagg			24	
SEQ ID NO: 102	moltype = AA length = 371			
FEATURE	Location/Qualifiers			
source	1..371			
	mol_type = protein			
	organism = Sorghum bicolor			
SEQUENCE: 102				
MGSYDSSSSS SNDSSARNEE	DESCMFALKL	LGGFAVPPTI	KAVIELGVMD QLLTAERAMS	60
AEELVAAAVA AQLPRPEVAC	TMVDRLLRFL	ASPASGARPR	WWWARTTPP PPAAAGATPR	120
HPSASGSPGT ASRIRCRLG	STRHSWTAGK	TLRMQCWKEQ	HHLRKPTGCQ CSSTVQTDHT	180
RCSHEAMANH SMIITKKLLK	FPRGFEGLDV	LVDVGGGNGT	TLQMIRGQYK NMRGINYDLP	240
HVIAQAAPVE GVEHVGGSMF	DNIPRGNNAV	LKWILHDWDD	KACIKILKNC YTALHVRGKV	300
IVLEYVVPDE PEPTLAAQGA	FELDLTMLVT	FGSGKERTQR	EFSELAAMEAG FSREFKATYI	360
FANVWALEFT K				371
SEQ ID NO: 103	moltype = AA length = 365			
FEATURE	Location/Qualifiers			
source	1..365			
	mol_type = protein			
	organism = Sorghum bicolor			
SEQUENCE: 103				
MGSYDSSSSS SNDSSARNEE	DESCMFALKL	LGGFAVPPTI	KAVIELGVMD QLLTAERAMS	60
AEELVAAAVA AQLPRPEVAC	TMVDRLLRFL	ASRSPVHNRG	GGGHGRRHHH HLLPPELRRV	120
TRLQVVRQER RRGFGASAWD	DDPKQDIPGQ	LAKHYGCSVG	RSSTIENLRD ANVRVPKYKR	180
TIEHGVPTQR WQIIRLPRNC	SSSSAASKAL	MCWSTAAATA	PRCKLEVNIR IEATTTFLMS	240
LRRLHQLKVW NMVAACSIIF	HAEMQPCSSG	PFMIGTTRRA	SRSRIAQLS MEARSFWSTL	300
FRMNQNLLLQ LRVPSNWTSP	CWSRLAVVKR	GHRGSSPSSP	WRPASLES LRISLPTSGP	360
LSSQS				365
SEQ ID NO: 104	moltype = AA length = 371			
FEATURE	Location/Qualifiers			
source	1..371			
	mol_type = protein			
	organism = Sorghum bicolor			
SEQUENCE: 104				
MGSYDSSSSS SNDSSARNEE	DESCMFALKL	LGGFAVPPTI	KAVIELGVMD QLLTAERAMS	60
AEELVAAAVA AQLPRPEVAC	TMVDRLLRFL	ASQASGARPR	WWWARTTPP PPAAAGATPR	120
HPSASGSPGT ASRIRCRLG	STRHSWTAGK	TLRMQCWKEQ	HHLRKPTGCQ CSSTVQTDHT	180
RCSHEAMANH SMIITKKLLK	FPRGFEGLDV	LVDVGGGNGT	TLQMIRGQYK NMRGINYDLP	240
HVIAQAAPVE GVEHVGGSMF	DNIPRGNNAV	LKWILHDWDD	KACIKILKNC YTALHVRGKV	300
IVLEYVVPDE PEPTLAAQGA	FELDLTMLVT	FGSGKERTQR	EFSELAAMEAG FSREFKATYI	360
FANVWALEFT K				371

We claim:

1. A modified Type 1 O-methyltransferase (OMT), wherein the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) is substituted with a polar amino acid or wherein the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) is substituted with a hydrophobic amino acid.

2. The modified OMT of claim 1, wherein the amino acid corresponding to Ile144 of SbSOMT is substituted with a polar amino acid or wherein the amino acid corresponding to Asn128 of SbCOMT is substituted with a hydrophobic amino acid.

3. The modified OMT of claim 1, wherein the substituting polar amino acid is independently Asn, Gln, Thr, or Ser.

4. The modified OMT of claim 1, wherein the substituting polar amino acid is Asn.

5. The modified OMT of claim 1, wherein the substituting hydrophobic amino acid is independently Phe, Leu, Trp, Tyr, Ile, Met, Val, or Ala.

6. The modified OMT of claim 1, wherein the substituting hydrophobic amino acid is Phe.

7. The modified OMT of claim 1, wherein the modified OMT is a modified form of *Arabidopsis thaliana* AtOMT1, *Carthamus tinctorius* CtCAldOMT, *Catharanthus roseus* CrCOMT, *Hordeum vulgare* HvCOMT, *Medicago sativa* MsCOMT, *Medicago sativa* MsIOMT, *Oryza sativa* OsCAldOMT1, *Oryza sativa* OsNOMT, *Panicum virgatum* PvCOMT, *Pinus sylvestris* PsPMT2, *Pinus taeda* PtAE-OMT, *Populus tremuloides* PtCOMT, *Rosa hybrid* RhOOMT, *Saccharum spontaneum* SsSTS, *Saccharum spontaneum* SsCOMT, *Solanum lycopersicum* SlCOMT, *Sorghum bicolor* SbCOMT, *Sorghum bicolor* SbNOMT, *Sorghum bicolor* SbOMT1, *Sorghum bicolor* SbOMT3, *Sorghum bicolor* SbOMT4, *Sorghum bicolor* SbSOMT, *Sorghum bicolor* SbSTS1, *Sorghum halepense* ShOMT1, *Sorghum halepense* ShSOMT, *Triticum aestivum* TaCOMT1, *Triticum aestivum* TaCOMT2, *Vitis vinifera* VvCOMT, *Vitis vinifera* VvROMT, *Zea mays* ZmCOMT, *Zea mays* ZmNOMT, *Medicago truncatula* MtIOMT, *Medicago truncatula* MtIOMT3, *Streptomyces carzinostaticus* ScNOMT, *Loium perenne* LpCOMT, *Clarkia breweri* CbIEOMT, *Planctopirus limnophila* PlIOMT, *Thalictrum flavum* TfNOMT, *Kitagawia praeruptora* KpBOMT, *Papaver somniferum* PsSOMT, or *Fragaria ananassa* FaOMT.

8. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

9. The modified OMT of claim 1, wherein the modified OMT has at least 85% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

10. The modified OMT of claim 1, wherein the modified OMT has at least 90% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19,

20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

11. The modified OMT of claim 1, wherein the modified OMT has at least 95% sequence identity with SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93.

12. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Phe337 of *Sorghum bicolor* stilbene OMT (SbSOMT) substituted with a polar amino acid.

13. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn323 of *Sorghum bicolor* caffeic acid OMT (SbCOMT) substituted with a hydrophobic amino acid.

14. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Ile144 of SbSOMT substituted with a polar amino acid.

15. The modified OMT of claim 1, wherein the modified OMT is a modified form of SEQ ID NO:1, 2, 32, 33, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 34, 35, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, or 93 with the amino acid corresponding to Asn128 of SbCOMT substituted with a hydrophobic amino acid.

16. The modified OMT of claim 1, wherein the modified OMT is further modified to improve or alter the production of an O-methylated stilbenoid or hydroxycinnamic acid by the modified OMT.

17. A solution comprising the modified OMT of claim 1, wherein the solution is composed to allow production of one or more O-methylated forms of stilbenoid or hydroxycinnamic acid from a substrate for the modified OMT under suitable conditions.

18. A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising bringing into contact, in a solution, the modified OMT of claim 1, a substrate for the modified OMT and S-adenosylmethionine; and

incubating under suitable conditions to produce the O-methylated stilbenoid or hydroxycinnamic acid.

19. A plant that expresses the modified OMT of claim 1.

20. A method of producing an O-methylated stilbenoid or hydroxycinnamic acid, the method comprising growing the plant of claim 19 and extracting the O-methylated stilbenoid or hydroxycinnamic acid from the grown plant.

* * * * *